AP PHYSICS 2



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UNIT 9

Thermodynamics

15–18%
AP EXAM WEIGHTING

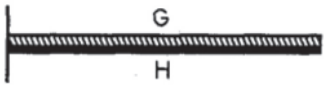
~10–16
CLASS PERIODS

Conceptual Questions

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) The temperature in your classroom is closest to 1) _____
A) 295 K. B) 68 K. C) 68°C. D) 50°C.

2) As shown in the figure, a bimetallic strip, consisting of metal G on the top and metal H on the bottom, is rigidly attached to a wall at the left. The coefficient of linear thermal expansion for metal G is greater than that of metal H. If the strip is uniformly heated, it will 2) _____



A) curve downward.
B) bend in the middle.
C) curve upward.
D) remain horizontal, but get longer.
E) remain horizontal, but get shorter.

3) The coefficient of linear expansion for aluminum is $1.8 \times 10^{-6}\ \text{K}^{-1}$. What is its coefficient of volume expansion? 3) _____
A) $5.8 \times 10^{-18}\ \text{K}^{-1}$
B) $0.60 \times 10^{-6}\ \text{K}^{-1}$
C) $9.0 \times 10^{-6}\ \text{K}^{-1}$
D) $5.4 \times 10^{-6}\ \text{K}^{-1}$
E) $3.6 \times 10^{-6}\ \text{K}^{-1}$

4) Two containers of equal volume each hold samples of the same ideal gas. Container A has twice as many molecules as container B. If the gas pressure is the same in the two containers, the correct statement regarding the absolute temperatures T_A and T_B in containers A and B, respectively, is 4) _____
A) $T_A = \frac{1}{2} T_B$.
B) $T_A = T_B$.
C) $T_A = \frac{1}{4} T_B$.
D) $T_A = 2T_B$.
E) $T_A = \frac{1}{\sqrt{2}} T_B$.

5) Consider two equal-volume flasks of gas at the same temperature and pressure. One gas, oxygen, has a molecular mass of 32. The other gas, nitrogen, has a molecular mass of 28. What is the ratio of the number of oxygen molecules to the number of nitrogen molecules in these flasks? 5) _____
A) $\sqrt{\frac{32}{28}}$ B) $\sqrt{\frac{28}{32}}$ C) $\frac{1}{1}$ D) $\frac{28}{32}$ E) $\frac{32}{38}$

6) An ideal gas is held in a container of volume V at pressure p. The rms speed of a gas molecule under these conditions is v. If now the volume and pressure are changed to $2V$ and $2p$, the rms speed of a molecule will be 6) _____

A) $2v$ B) $v/2$ C) $v/4$ D) $4v$ E) v

7) A mole of diatomic oxygen molecules and a mole of diatomic nitrogen molecules are at STP. Which statements are true about these molecules? (There could be more than one correct choice.) 7) _____

A) Both gases have the same average molecular speeds.
B) Both gases have the same average momentum per molecule.
C) Both gases have the same number of molecules.
D) Both gases have the same average kinetic energy per molecule.

8) The root–mean–square speed of the molecules of an ideal gas is v. The gas is now slowly compressed to one–half its original volume with no change in temperature. What is the root–mean–square speed of the molecules now? 8) _____

A) $v/\sqrt{2}$ B) $2v$ C) $4v$ D) v E) $v/2$

9) A sample of an ideal gas is heated and its Kelvin temperature doubles. If the root–mean–square speed of its molecules was originally v, what is the new root–mean–square speed? 9) _____

A) $2v$ B) $v/\sqrt{2}$ C) $4v$ D) $v\sqrt{2}$ E) $v/4$

10) The absolute temperature of an ideal gas is directly proportional to which of the following quantities? 10) _____

A) the mass of its molecules
B) the average kinetic energy of its molecules
C) the average momentum of its molecules
D) the average speed of its molecules
E) It is proportional to all of the above quantities.

11) The absolute temperature of a gas is T. In order to double the rms speed of its molecules, what should be the new absolute temperature? 11) _____

A) $4T$ B) $2T$ C) $16T$ D) $T\sqrt{2}$ E) $8T$

12) Oxygen molecules are 16 times more massive than hydrogen molecules. At a given temperature, the average molecular kinetic energy of oxygen molecules, compared to that of hydrogen molecules, 12) _____

A) is less.
B) is greater.
C) is the same.
D) cannot be determined without knowing the pressure and volume.

13) Oxygen molecules are 16 times more massive than hydrogen molecules. At a given temperature, how do their average molecular speeds compare? The oxygen molecules are moving 13) _____

A) at $1/\sqrt{2}$ the speed of the hydrogen molecules.
B) at 1/4 the speed of the hydrogen molecules.
C) sixteen times faster than the hydrogen molecules.
D) at 1/16 the speed of the hydrogen molecules.
E) four times faster than the hydrogen molecules.

14) A fixed container holds oxygen and helium gases at the same temperature. Which of the following statements are correct? (There could be more than one correct choice.) 14) _____
A) The oxygen molecules have the greater speed.
B) The helium molecules have the greater speed.
C) The oxygen molecules have the greater average kinetic energy.
D) The helium molecules have the same average kinetic as the oxygen molecules.
E) The helium molecules have the greater average kinetic energy.

15) 15) _____

A) B) C) D)

16) If you wanted to know how much the temperature of a particular piece of material would rise when a known amount of heat was added to it, which of the following quantities would be most helpful to know? 16) _____
A) specific heat
B) initial temperature
C) density
D) thermal conductivity
E) coefficient of linear expansion

17) A thermally isolated system is made up of a hot piece of aluminum and a cold piece of copper, with the aluminum and the copper in thermal contact. The specific heat capacity of aluminum is more than double that of copper. Which object experiences the greater magnitude gain or loss of heat during the time the system takes to reach thermal equilibrium? 17) _____
A) the aluminum
B) the copper
C) Neither one; both of them experience the same size gain or loss of heat.
D) It is impossible to tell without knowing the masses.
E) It is impossible to tell without knowing the volumes.

18) A thermally isolated system is made up of a hot piece of aluminum and a cold piece of copper, with the aluminum and the copper in thermal contact. The specific heat capacity of aluminum is more than double that of copper. Which object experiences the greater temperature change during the time the system takes to reach thermal equilibrium? 18) _____
A) the copper
B) the aluminum
C) Neither one; both of them experience the same size temperature change.
D) It is impossible to tell without knowing the masses.
E) It is impossible to tell without knowing the volumes.

19) Which one of the following quantities is the *smallest* unit of heat energy? 19) _____
A) Btu B) calorie C) kilocalorie D) joule

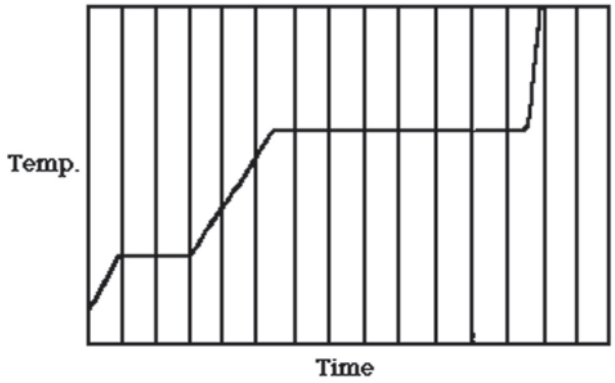
20) Object 1 has three times the specific heat capacity and four times the mass of Object 2. The two objects are given the same amount of heat. If the temperature of Object 1 changes by an amount ΔT, the change in temperature of Object 2 will be 20) _____

A) $6\Delta T$. B) $\frac{4}{3}\Delta T$. C) $12\Delta T$. D) ΔT. E) $\frac{3}{4}\Delta T$.

21) Object 1 has three times the specific heat capacity and four times the mass of Object 2. The two objects are heated from the same initial temperature, T_0, to the same final temperature T_f. During this process, if Object 1 absorbs heat Q, the amount of heat absorbed by Object 2 will be 21) _____

A) $\frac{1}{12}Q$. B) $\frac{3}{4}Q$. C) $\frac{4}{3}Q$. D) $12Q$. E) 6Q.

22) The figure shows a graph of the temperature of a pure substance as a function of time as heat is added to it at a constant rate in a closed container. If L_F is the latent heat of fusion of this substance and L_V is its latent heat of vaporization, what is the value of the ratio L_V/L_F? 22) _____

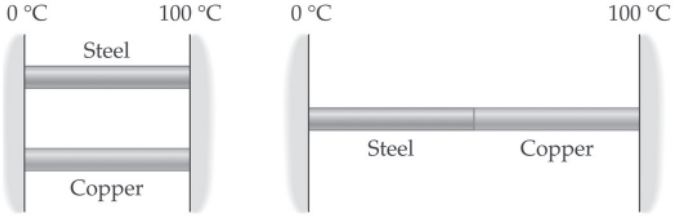


A) 5.0 B) 1.5 C) 7.2 D) 3.5 E) 4.5

23) A solid cylindrical bar conducts heat at a rate of 25 W from a hot to a cold reservoir under steady state conditions. If both the length and the diameter of this bar are doubled, the rate at which it will conduct heat between these reservoirs will be 23) _____

A) 200 W B) 12.5 W C) 25 W D) 100 W E) 50 W

24) Two metal rods are to be used to conduct heat from a region at 100°C to a region at 0°C as shown in the figure. The rods can be placed in parallel, as shown on the left, or in series, as on the right. When steady state flow is established, the heat conducted in the series arrangement is 24) _____



A) less than the heat conducted with the rods in parallel.
B) the same as the heat conducted with the rods in parallel.
C) greater than the heat conducted with the rods in parallel.

25) On a cold day, a piece of metal feels much colder to the touch than a piece of wood. This is due to the difference in which one of the following physical properties of these materials? 25) _____

A) thermal conductivity
B) emissivity
C) specific heat
D) mass
E) density

26) If, with steady state heat flow established, you double the thickness of a wall built from solid uniform material, the rate of heat loss for a given temperature difference across the thickness will 26) _____

A) become $1/\sqrt{2}$ of its original value.
B) become one–half its original value.
C) also double.
D) become four times its original value.
E) become one–fourth its original value.

27) The process in which heat flows by the mass movement of molecules from one place to another is known as 27) _____

A) convection. B) radiation. C) conduction.

28) An object having a fixed emissivity of 0.725 radiates heat at a rate of 10 W when it is at an absolute temperature T. If its temperature is doubled to $2T$, at what rate will it now radiate? 28) _____

A) 80 W B) 40 W C) 160 W D) 20 W E) 320 W

29) If the absolute temperature of an object is tripled, the thermal power radiated by this object (assuming that its emissivity and size are not affected by the temperature change) will 29) _____

A) increase by a factor of 18.
B) increase by a factor of 27.
C) increase by a factor of 81.
D) increase by a factor of 9.
E) increase by a factor of 3.

30) By what primary heat transfer mechanism does the sun warm the earth? 30) _____

A) conduction
B) convection
C) radiation
D) All of the above processes are equally important in combination.

31) 31) _____

A) B) C) D)

32) When a gas expands adiabatically, 32) _____

A) work is done on the gas.
B) it does no work.
C) the internal (thermal) energy of the gas increases.
D) the temperature of the gas remains constant.
E) the internal (thermal) energy of the gas decreases.

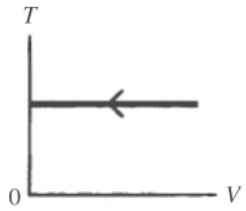
33) An ideal gas is compressed isothermally to one–third of its initial volume. The resulting pressure will be 33) _____

A) less than three times as large as the initial value.
B) more than three times as large as the initial value.
C) equal to the initial value.
D) three times as large as the initial value.
E) impossible to predict on the basis of this data.

34) In a given reversible process, the temperature of an ideal gas is kept constant as the gas is compressed to a smaller volume. Which one of the following statements about the gas is correct? 34) _____

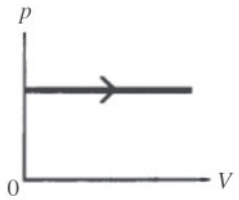
A) The gas must absorb heat from its surroundings.
B) The pressure of the gas also stays constant.
C) The process is adiabatic.
D) The gas must release heat to its surroundings.
E) It is impossible to predict on the basis of this data.

35) The process shown on the *TV* graph in the figure is an 35) _____



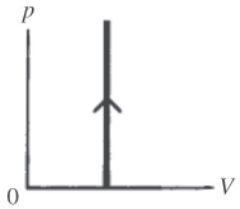
A) isobaric compression.
B) isochoric compression.
C) adiabatic compression.
D) isothermal compression.

36) The process shown on the *pV* diagram in the figure is an 36) _____



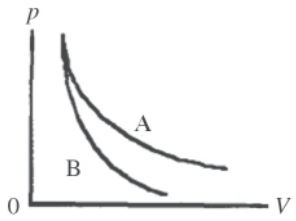
A) adiabatic expansion.
B) isometric expansion.
C) isobaric expansion.
D) isothermal expansion.

37) The process shown on the *pV* diagram in the figure is 37) _____



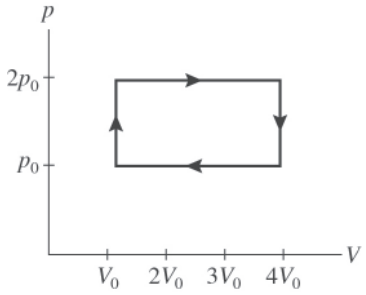
A) adiabatic.
B) isochoric.
C) isobaric.
D) isothermal.

38) Two processes are shown on the *pV* diagram in the figure. One of them is an adiabat and the other one is an isotherm. Which process is the isotherm? 38) _____



A) process A
B) process B
C) The processes shown are neither isotherms nor adiabats.
D) It is not possible to tell without knowing if the gas is monatomic or diatomic.

39) A gas is taken through the cycle shown in the *pV* diagram in the figure. During one cycle, how much work is done by the gas? 39) _____



A) p_0V_0 B) 3 p_0V_0 C) 2 p_0V_0 D) 4 p_0V_0

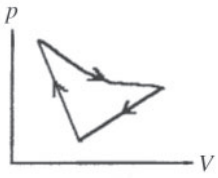
40) A 10–L flask and a 1–L flask each contain two moles of ideal diatomic gas (but not the *same* gas) at 25°C. Which of the following statements about these gases *must* be true? (There could be more than one correct choice.) 40) _____

A) The molecules in the larger flask have the same root–mean–square speed as those in the smaller flask.
B) The internal (thermal) energy of the gas in both flasks is the same.
C) The internal (thermal) energy of the gas in the larger flask is greater than the internal (thermal) energy of the gas in the smaller flask.
D) The molecules in the smaller flask have the same average kinetic energy per molecule as those in the larger flask.
E) The internal (thermal) energy of the gas in the smaller flask is greater than the internal (thermal) energy of the gas in the larger flask.

41) A certain gas is compressed adiabatically. The amount of work done on the gas is 800 J. What is the change in the internal (thermal) energy of the gas? 41) _____

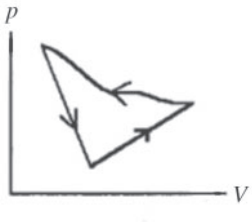
A) 400 J
B) –800 J
C) 800 J
D) 0 J
E) More information is needed to answer this question.

42) A cyclic process is carried out on an ideal gas such that it returns to its initial state at the end of a cycle, as shown in the *pV* diagram in the figure. If the process is carried out in a clockwise sense around the enclosed area, as shown on the figure, then the change of internal energy over the full cycle 42) _____



A) is negative.
B) is zero.
C) is positive.
D) cannot be determined from the information given.

43) A cyclic process is carried out on an ideal gas such that it returns to its initial state at the end of a cycle, as shown in the *pV* diagram in the figure. If the process is carried out in a counter-clockwise sense around the enclosed area, as shown on the figure, then the magnitude of the enclosed area represents 43) _____



A) the heat added to the gas.
B) the heat that flows out of the gas.
C) the work done on the gas.
D) the work done by the gas.

44) Which one of the following is a *true* statement? 44) _____
A) It is possible for heat to flow spontaneously from a hot body to a cold one or from a cold one to a hot one, depending on whether or not the process is reversible or irreversible.
B) It is not possible to convert work entirely into heat.
C) The second law of thermodynamics is a consequence of the first law of thermodynamics.
D) It is impossible to transfer heat from a cooler to a hotter body.
E) All of these statements are false.

45) The second law of thermodynamics leads us to conclude that 45) _____
A) it is theoretically possible to convert heat into work with 100% efficiency.
B) the entropy of the universe remains constant.
C) the total energy of the universe is constant.
D) disorder in the universe is increasing with the passage of time.
E) the average temperature of the universe is increasing with the passage of time.

46) If the efficiency of a Carnot engine were to be 100%, the heat sink would have to be 46) _____
A) at absolute zero.
B) infinitely hot.
C) at 0°C.
D) at 100°C.



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47) An important feature of the Carnot cycle is that 47) _____
A) its efficiency depends only on the absolute temperature of the hot reservoir used.
B) no engine can be more efficient than a Carnot engine operating between the same two temperatures.
C) its efficiency can be 100%.
D) it is an example of an irreversible process that can be analyzed exactly without approximations.
E) its efficiency is determined by the temperatures of the hot and cold reservoirs between which it works and by the properties of the working substance used, and on nothing else.

48) A Carnot cycle consists of 48) _____
A) four isotherms.
B) four adiabats.
C) two adiabats and two isobars.
D) two adiabats and two isotherms.
E) two isobars and two isotherms.

49) Which of the following is a *false* statement? 49) _____
A) Entropy can be measured in units of J/K.
B) The total entropy change in one cycle of a Carnot engine is zero.
C) The entropy of an isolated system must be conserved (it remains constant).
D) Entropy is a quantitative measure of disorder.

50) An ideal gas undergoes an isothermal expansion. During this process, its entropy 50) _____
A) increases.
B) remains unchanged.
C) decreases.
D) cannot be predicted from the data given.

51) When water at 0°C freezes, the entropy of the water 51) _____
A) remains constant.
B) decreases.
C) increases.
D) could either increase or decrease; it depends on other factors.



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1. Which of the following statement is NOT a correct assumption of the classical model of an ideal gas?

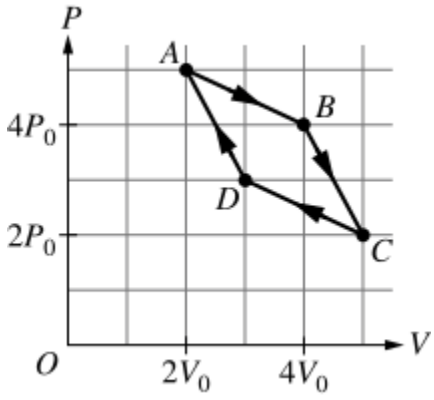
(A) The molecules are in random motion

(B) The volume of the molecules is negligible compared with the volume occupied by the gas.

(C) The molecules obey Newton's laws of motion.

(D) The collisions between molecules are inelastic.

(E) The only appreciable forces on the molecules are those that occur during collisions.



An ideal gas in a closed container undergoes the cyclical process shown in the above graph of pressure P as a function of volume V.

2. In which state is the temperature of the gas the greatest?

(A) A

(B) B

(C) C

(D) D

3. For which of the following paths is the greatest amount of work done on the gas?

(A) $A \rightarrow B$

(B) $B \rightarrow C$

(C) $C \rightarrow D$

(D) $D \rightarrow A$

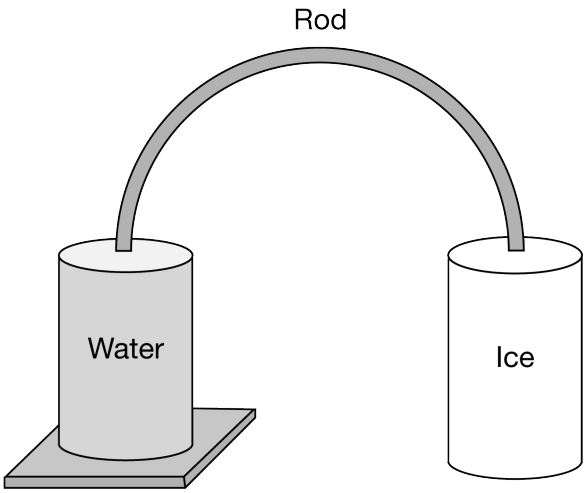
An empty container is filled with helium to a pressure P at a temperature T. Neon, which has atoms that are 5 times more massive than those of helium, is then added to the container until the new pressure in the container is $2P$ at the same temperature T.

4. The ratio of the average speed of the helium atoms to the average speed of the neon atoms is

(A) 25 to 1
(B) 5 to 1
(C) $\sqrt{5}$ to 1
(D) $\sqrt{2}$ to 1
(E) 1 to 1

5. The ratio of the number of helium atoms in the container to the number of neon atoms in the container is

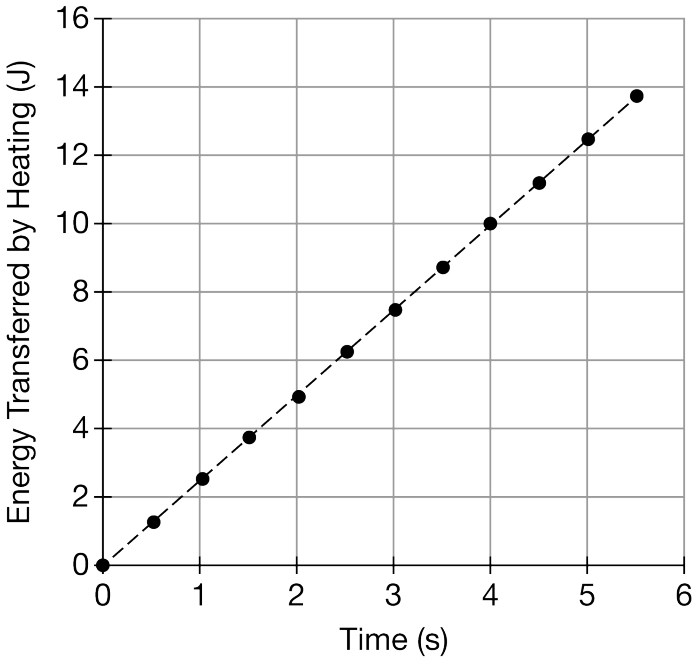
(A) 25 to 1
(B) 5 to 1
(C) $\sqrt{5}$ to 1
(D) $\sqrt{2}$ to 1
(E) 1 to 1



Students design an experiment with a curved, insulated rod of an unknown thermally conducting material, as shown in the figure. The rod is 1 m long and 2 cm in diameter. One end of the rod is touching the surface of water kept at $350\ \mathrm{K}$ by a hot plate, and the other end is just touching the surface of a well-insulated block of ice.

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6.



The figure shows a graph of the energy transferred by heating from the water through the rod and to the ice over time. If the temperature of the ice is held constant at $250\ \mathrm{K}$, what is the thermal conductivity of the rod?

(A) 0.03 $\mathrm{W/(m \cdot K)}$

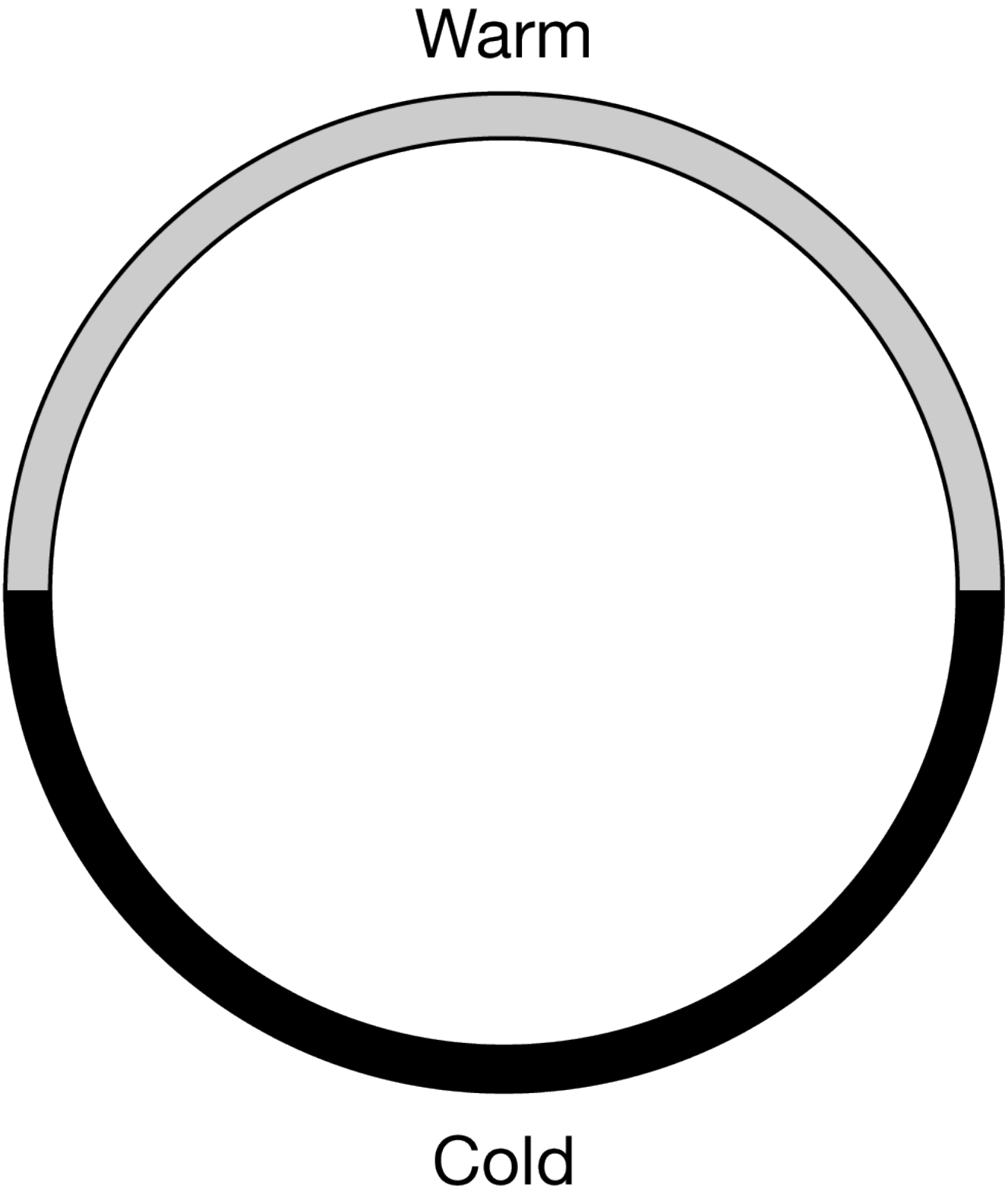
(B) 2.5 $\mathrm{W/(m \cdot K)}$

(C) 80 $\mathrm{W/(m \cdot K)}$

(D) 314 $\mathrm{W/(m \cdot K)}$

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7.



The rod is removed from the apparatus and placed in a warm room until it reaches thermal equilibrium. Its ends are then positioned to touch the ends of an identical rod with a much colder temperature, as shown in the figure. Which

of the following is the best explanation of how energy is transferred between the rods as they approach thermal equilibrium?

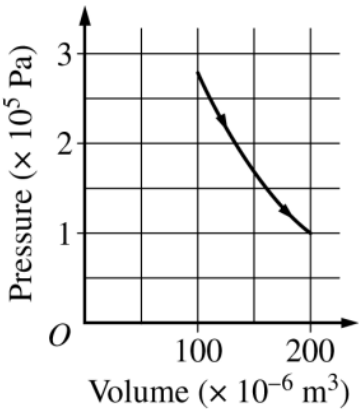
(A) Quickly moving molecules in the warm rod move into the cold rod, and slowly moving molecules in the cold rod move into the warm rod.

(B) Quickly vibrating molecules in the warm rod collide with slowly vibrating molecules in the cold rod, and on average the slower molecules begin to vibrate faster.

(C) Molecules of the warm rod collide with molecules of the cold rod, causing a net motion of molecules which results in positive work being done on the molecules of the cold rod.

(D) Energy cannot be transferred directly across the interface of the rods, but must first be transferred to the air molecules around the rods and then transferred back to the rods.

8.



The graph above of pressure as a function of volume illustrates a thermodynamic process for an ideal gas in an engine. Which of the following best describes the change in the internal energy of the gas and the work done during the process?

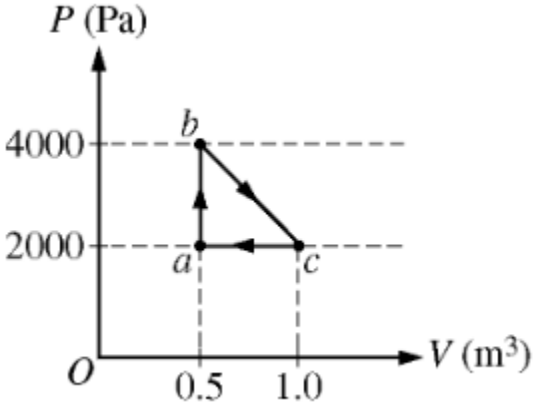
(A) The internal energy increases, and positive work is done by the gas.

(B) The internal energy decreases, and positive work is done on the gas.

(C) The internal energy decreases, and positive work is done by the gas.

(D) The internal energy increases, and positive work is done on the gas.

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A sample of an ideal gas is taken through the cycle *abca*, as shown in the *PV* diagram above.

9. What is the change in internal energy of the gas for the process *bc* ?

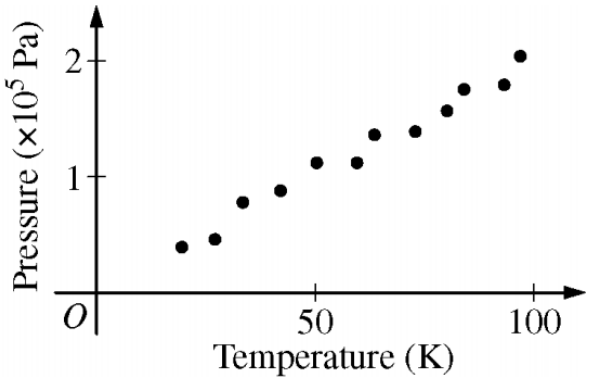
(A) -1500 J
(B) -500 J
(C) 0 J
(D) +500 J
(E) +1500 J

10. If 1000 J of heat is added to the gas during process *ab*, what is the change in internal energy of the gas for process *ab* ?

(A) -2000 J
(B) -1000 J
(C) 0 J
(D) +1000 J
(E) +2000 J

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11.



A sample of gas is in a container of fixed volume. The pressure of the gas is measured as its absolute temperature is increased. A graph of pressure measurements as a function of temperature is shown above. According to the ideal gas law, which of the following will be true if the container volume is reduced to half the original value and the experiment is repeated?

(A) The graph will still show a linear relationship that extrapolates to the origin, but with a smaller slope.

(B) The graph will still show a linear relationship that extrapolates to the origin, but with a larger slope.

(C) The graph will still show a linear relationship, but it will not extrapolate to the origin.

(D) The graph will not show a linear relationship, but it will still extrapolate to the origin.

12. A student conducts an experiment in which two solid rods, X and Y, are each held vertically with the bottom end just submerged in a bath of boiling water. The temperature of the room is $21^{\circ}\mathrm{C}$, and the rods are each $80\ \mathrm{cm}$ long. After each rod has been in the water for the same amount of time, the student measures the temperature at $20\ \mathrm{cm}$ intervals along each rod. The student's data are shown below. Which of the following correctly describes an analysis of the data that can be used to compare the thermal conductivity of the rods?

Height Above Water (cm)	Temperature of Rod $\mathrm{X}(^{\circ}\mathrm{C})$	Temperature of Rod $\mathrm{Y}(^{\circ}\mathrm{C})$
0	100	100
20	86.7	97.8
40	73.3	95.6
60	60.0	93.3
80	46.7	91.1

(A) Compare the temperatures of the rods at $0\ \mathrm{cm}$.

(B) Compare the temperatures of the rods at each height.

(C) Determine which rod has differences in temperatures between adjacent positions that are most similar for all pairs of positions.

(D) The data cannot be used to compare the thermal conductivities because the top ends of the rods are not at room temperature.

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13. Two samples of ideal gas in separate containers have the same number of molecules and the same temperature, but the molecular mass of gas X is greater than that of gas *Y.* Which of the following correctly compares the average speed of the molecules of the gases and the average force the gases exert on their respective containers?

(A)

Average speed of Molecules	Average Force on container
Greater for gas X	Greater for gas X

(B)

Average speed of Molecules	Average Force on container
Greater for gas X	The forces cannot be compared without knowing the volumes of the gases

(C)

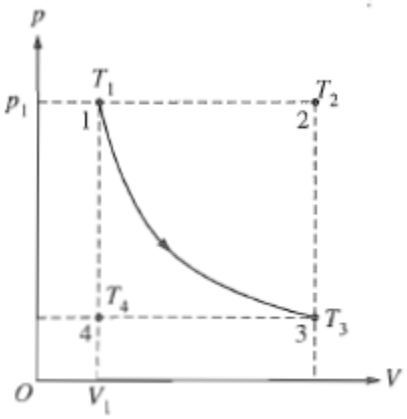
Average speed of Molecules	Average Force on container
Greater for gas Y	Greater for gas Y

(D)

Average speed of Molecules	Average Force on container
Greater for gas Y	The forces cannot be compared without knowing the volumes of the gases.

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14.



An ideal gas is initially in a state that corresponds to point 1 on the graph above, where it has pressure p_1 volume V_1, and temperature T_1. The gas undergoes an isothermal process represented by the curve shown, which takes it to a final state 3 at temperature T_3. If T_2 and T_4 are the temperatures the gas would have at points 2 and 4, respectively, which of the following relationships is true?

(A) $T_1 < T_3$

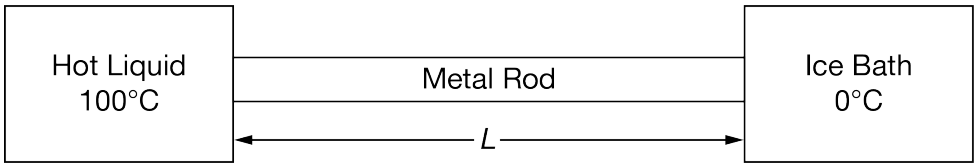
(B) $T_1 < T_2$

(C) $T_1 < T_4$

(D) $T_1 = T_2$

(E) $T_1 = T_4$

15.



A student wants to compare the thermal conductivities of several metal rods with different lengths L and cross-sectional areas A. The student connects each rod to two reservoirs, one containing a hot liquid at temperature 100°C and the other containing an ice bath at 0°C, as shown in the figure. Each rod is kept in contact with the reservoirs for the same fixed time interval Δt. Let Q be the amount of energy transferred through a rod in the time Δt. Which of the following lists all the additional quantities that must be measured to be able to compare the thermal conductivities?

(A) Q only

(B) L only

(C) L and A only

(D) Q, L, and A

16. A sealed container of air has been sitting on a table in a dark room for a very long time. The room is always kept at a constant temperature. Which of the following best describes what will happen to the speeds and the average kinetic energy of the molecules of the air in the container as the container continues to sit on the table?

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(A) Each molecule will continue to move at constant speed, keeping the average kinetic energy of the molecules constant.

(B) Some molecules will speed up and others will slow down, keeping the average kinetic energy of the molecules constant.

(C) The molecules will gradually slow down, decreasing their average kinetic energy.

(D) The molecules will gradually speed up, increasing their average kinetic energy.

17. Two identical containers, X and Y, each contain one mole of an ideal monatomic gas. The atoms in container X have a higher average kinetic energy than those in container Y. The containers are thermally insulated from the external environment and placed into thermal contact with each other at time $t = 0$. Which of the following accurately describes the interactions between containers X and Y ?

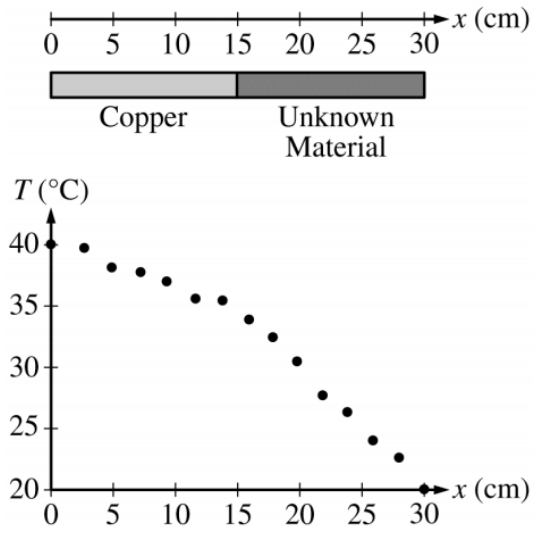
(A) The net result is that energy is transferred from container X to container Y.

(B) The amount of energy transferred between the containers depends on the type of atoms each contains.

(C) After equilibrium is reached, the atoms in container X will still have a different average kinetic energy than those in container Y.

(D) All the atoms in container X slow down, and all the atoms in container Y speed up.

18.



A student designs an experiment intended to determine the thermal conductivity of an unknown material. Two 15 cm long bars with the same cross-sectional area are joined together, as shown in the figure above. The bar on the left is made of copper, and the bar on the right is made of the unknown material. The left and right ends of the two-bar combination are maintained at 40°C and 20°C , respectively, and the bars reach equilibrium. The student then measures the temperature T along the bars and produces the data shown in the graph, where x = 0 is the left end and 30 cm is the right end. The experimental uncertainty in the data is negligible. Which of the following can the student conclude about k_c, the thermal conductivity of copper, compared with k_u , the thermal conductivity of the unknown material?

(A) $k_c < k_u$

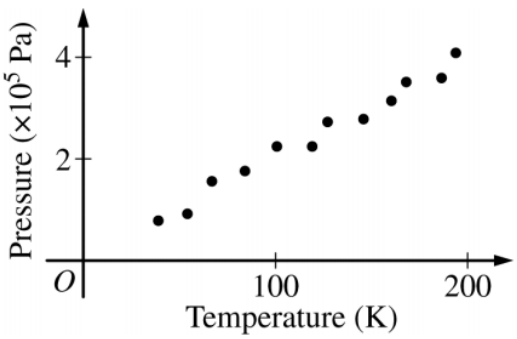
(B) $k_c = k_u$

(C) $k_c > k_u$

(D) The relationship cannot be determined from the data given.

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19.



A sample of gas is in a sealed container whose volume is held fixed. The pressure of the gas is measured as its absolute temperature is increased. A graph of pressure measurements as a function of temperature is shown above. According to the ideal gas law, which of the following is expected if the sealed container's volume is increased to three times the original value and the experiment is repeated?

(A) The graph will show a linear relationship that still extrapolates to the origin, but with a smaller slope.

(B) The graph will show a linear relationship that still extrapolates to the origin, but with a larger slope.

(C) The graph will show a linear relationship with the same slope, but it will extrapolate to a point on the horizontal axis to the right of the origin.

(D) The graph will show a linear relationship with the same slope, but it will extrapolate to a point on the vertical axis above the origin.

20. Three different gas samples that have the same number of molecules and are at room temperature are kept at different pressures. A lab technician has determined the molecular mass of each gas and recorded the pressure and molecular mass of each sample in the table below.

Gas	Molecular Mass (u)	Pressure (¥100 kPa)
X	2.0	6.0
Y	4.0	12
Z	40	1.0

Which of the following ranks the density ρ of the gas samples?

(A) $P_x = P_y > P_z$

(B) $P_y > P_z > P_x$

(C) $P_z > P_x = P_y$

(D) $P_z > P_y > P_x$

21. A small, well-insulated building is kept cool on a hot day. A person in the building holds a hand close to, but not touching, a solid metal door to the outside and feels warmth coming from the metal door. Which statement best describes how most of the energy is transferred from the outside to the person's hand?

(A) Thermal energy from the outside is convected through the metal door. Thermal energy from the door is conducted by the air and is detected by the person's hand.

(B) Thermal energy from the outside is convected through the metal door. The door radiates energy that is detected by the person's hand.

(C) Thermal energy from the outside is conducted through the metal door. The door radiates energy that is detected by the person's hand.

(D) Thermal energy from the outside is radiated through the metal door. The door radiates energy that is detected by the person's hand.

22. A student writes the following information for a process that involves a fixed quantity of ideal gas.

$W = -P\Delta V$

$\Delta U = Q + W$

$P = 2.0x10^{5}$ Pa

$\Delta V = -2.0x10^{-3}$ m^3

$\Delta U = -600J$

Which of the following descriptions best represents the process?

(A) The gas expands at a constant pressure of 200 kPa.

(B) The gas is cooled at constant volume until its pressure falls to 200 kPa.

(C) The gas is compressed at a constant pressure of 200 kPa.

(D) The gas is heated and its pressure increases at constant volume.

23. An ideal monatomic gas is in an insulated container with an insulated, movable piston at the top. Which of the following correctly describes energy changes of the gas as the piston moves downward?

(A) Work is done on the gas and no energy escapes through heating, so the temperature of the gas increases.

(B) Work is done on the gas and energy escapes through heating, so the temperature of the gas remains the same.

(C) Work is done by the gas and energy escapes through heating, so the temperature of the gas decreases.

(D) Work is done by the gas and no energy escapes through heating, so the temperature of the gas remains the same.

24. A group of students performs an experiment to determine the thermal conductivity k of a material. They have several cylindrical rods made of the material, and the rods have different cross-sectional areas A and lengths L. They place one end of each rod in an ice bath and the other end in a hot bath and measure the rate $\frac{Q}{\Delta t}$ of heat transfer. They plot their data with $\frac{Q}{\Delta t}$ on the vertical axis. Which of the following quantities, when plotted on the horizontal axis, will yield a linear best-fit line from which the thermal conductivity of the material could be determined?

(A) $\frac{\Delta T}{L}$

(B) $\frac{A}{\Delta t}$

(C) $\frac{k\Delta T}{L}$

(D) $\frac{A}{L}$

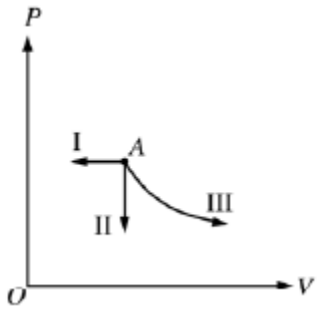
25. In the kinetic theory of gases, it is assumed that collisions between particles in an ideal gas are perfectly elastic. Which of the following claims is true based on this assumption?

(A) The average kinetic energy of the gas particles does not change due to interactions between particles.

(B) The temperature of the gas does not depend on the motion of the gas particles.

(C) No energy is transferred between gas particles when they collide.

(D) All the gas particles have the same speed.



A monatomic ideal gas is initially in state *A*, as shown in the *PV* diagram above. The gas undergoes a transition from state *A* by the three different processes shown, where process III is isothermal.

26. Energy is added to the gas by heating in which of the following processes?

(A) I only

(B) III only

(C) I and II only

(D) II and III only

(E) I, II, and III

27. Positive work is performed on the gas in which of the following processes?

(A) I only

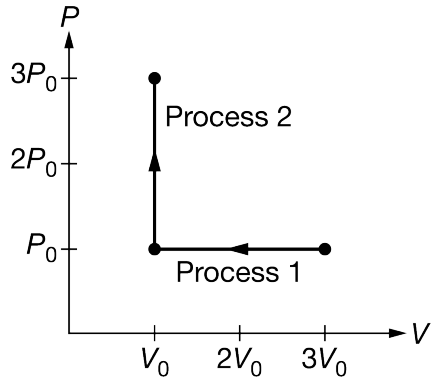
(B) III only

(C) I and II only

(D) II and III only

(E) I, II, and III

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A sealed cylinder with a movable piston contains N molecules of an ideal gas. The gas is initially in the state with pressure P_0 and volume $3V_0$ shown on the graph of pressure P as a function of volume V. The gas is then taken through the two processes shown.

28. Which of the following correctly describes the change in average speed of the molecules, if any, as the gas is taken through each process?

(A)

Process 1	**Process 2**
It increases.	It increases.

(B)

Process 1	**Process 2**
It increases.	It remains the same.

(C)

Process 1	**Process 2**
It decreases.	It increases.

(D)

Process 1	**Process 2**
It decreases.	It remains the same.

29. Which of the following claims correctly describes the average force the gas exerts on the piston as the gas is taken through one of the processes?

(A) The average force increases during process 1 because the volume of the gas decreases.

(B) The average force decreases during process 1 because the temperature decreases.

(C) The average force increases during process 2 because the pressure of the gas increases.

(D) The average force remains the same during process 2 because the volume of the gas does not change.

30. How much energy is exchanged between the system and its surroundings by heating as the gas is taken through the two processes, and is the energy added to or removed from the gas?

(A) $2P_0V_0$ added

(B) $2P_0V_0$ removed

(C) $4P_0V_0$ added

(D) $4P_0V_0$ removed

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31. Two containers, each holding a different gas, are brought into thermal contact. The gas atoms in container *X* have twice the mass and half the average speed of the gas atoms in container *Y*. Which of the following best describes the energy transfer by thermal conduction, if any, between the two containers?

(A) There is a net energy transfer from container *X* to container *Y*.

(B) There is a net energy transfer from container *Y* to container *X*.

(C) There is no net energy transfer between the containers, because no atoms move between them.

(D) There is no net energy transfer between the containers, because energy is transferred at approximately equal rates in each direction.

32. A cold metal sphere is placed inside a box with room-temperature air. The box and sphere are then isolated from their surroundings. Over time, condensation begins to form on the metal sphere as water vapor in the air becomes liquid water droplets on the surface of the sphere. This causes the entropy of the water molecules inside the box to decrease. Which of the following correctly indicates whether the entropy of the entire system consisting of the air and the sphere increases or decreases, and provides a valid justification?

(A) Increases, because the total energy of the system increases

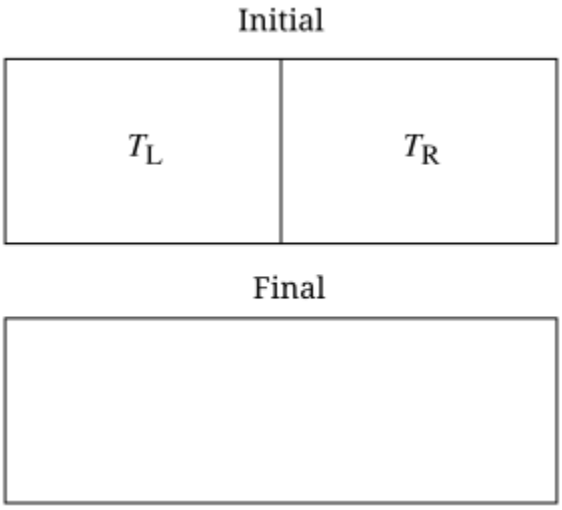
(B) Increases, because the overall increase in the entropy of the metal sphere is greater in magnitude than the decrease in entropy of the water molecules

(C) Decreases, because the total energy of the system decreases

(D) Decreases, because the overall increase in the entropy of the metal sphere is less in magnitude than the decrease in entropy of the water molecules

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33.



A system consists of two gases in separate containers isolated from their surroundings, as shown. The gas on the left side of the container has a temperature T_L and the gas on the right side of the container has a greater temperature T_R. The gases are initially separated by an insulating barrier. The barrier between the gases is then removed and the system is allowed to reach thermal equilibrium. Which of the following correctly compares the initial entropy of the system to the final entropy of the system?

(A) Initial entropy $>$ Final entropy

(B) Initial entropy $=$ Final entropy

(C) Initial entropy $<$ Final entropy

(D) The comparison cannot be made without knowing the values of the initial temperatures of the gases.

34. An ice cube is placed on a table in a warm room. As the ice cube melts, how does the entropy of the water in the ice cube and the entropy of the entire system of the ice cube and its surroundings change?

	Entropy of the Water	**Entropy of the Entire System**
A	Stays the same	Decreases
B	Stays the same	Increases
C	Increases	Decreases
D	Increases	Increases

(A) A

(B) B

(C) C

(D) D

35. A closed system consists of two adjacent chambers containing the same ideal gas. Initially, one chamber is hot and the other chamber is cold. Which of the following best explains the change in entropy of this closed system over time?

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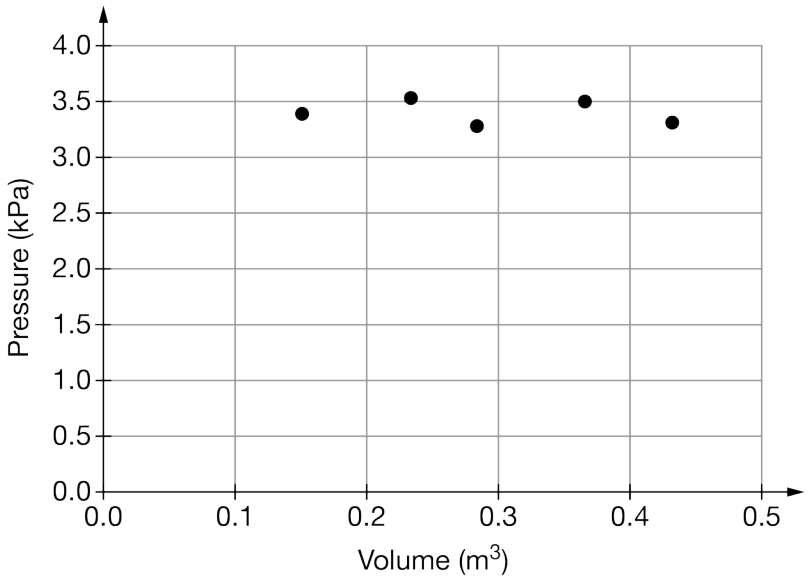
(A) The entropy increases over time because the process is irreversible.

(B) The entropy stays the same because it is a conserved quantity.

(C) The entropy decreases because the system moves toward a more orderly state of uniform temperature.

(D) Nothing can be determined about the change in entropy without knowing how many moles of gas are in each container.

36.



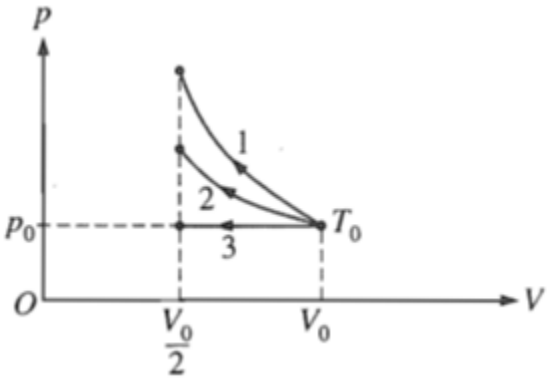
A student heats a sample of gas in a cylinder with a movable piston and a pressure gauge. The student measures the volume and pressure of the gas as the gas is heated. The student's data is plotted in the graph above. Which of the following gives the best estimate of the work done by the gas on its surroundings during the motion of the piston?

(A) -1.0×10^3 J

(B) 1.0×10^3 J

(C) -1.1×10^4 J

(D) 1.1×10^4 J



A certain quantity of an ideal gas initially at temperature T_0, pressure p_0, and volume V_0 is compressed to one-half its initial volume. As shown above, the process may be adiabatic (process 1), isothermal (process 2), or isobaric (process 3).

37. Which of the following is true of the final temperature of this gas?

(A) It is greatest for process 1.

(B) It is greatest for process 2.

(C) It is greatest for process 3.

(D) It is the same for processes 1 and 2.

(E) It is the same for processes 1 and 3.

38. Which of the following is true of the mechanical work done on the gas?

(A) It is greatest for process 1.

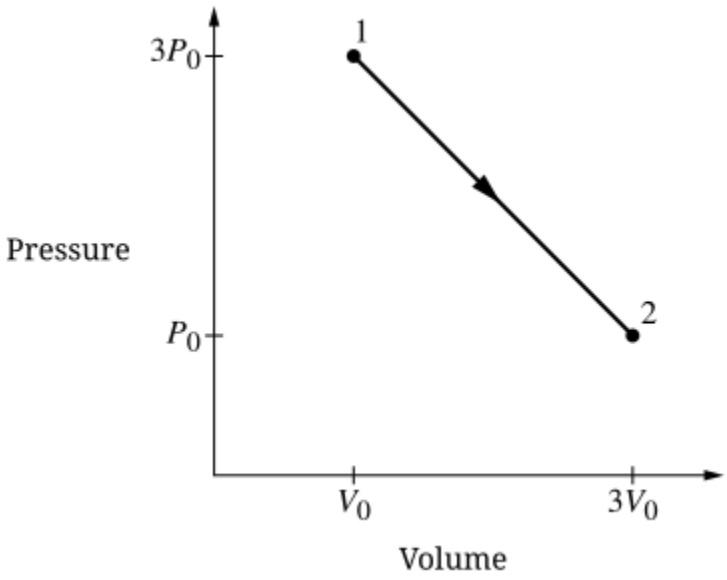
(B) It is greatest for process 3.

(C) It is the same for processes 1 and 2 and less for process 3.

(D) It is the same for processes 2 and 3 and less for process 1.

(E) It is the same for all three processes.

39.



A monatomic ideal gas is initially in State 1 at a pressure of $3P_0$ and volume V_0. The gas goes through the thermodynamic process shown in the PV diagram until it is in State 2 with a pressure of P_0 and volume $3V_0$. Which of the following is equal to the energy transferred to the gas through heating during this process?

(A) 0

(B) $2P_0V_0$

(C) $3P_0V_0$

(D) $4P_0V_0$

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40. Two identical containers hold identical samples of an ideal gas in the same initial state. In Process 1, an amount of energy Q is transferred to one of the containers through heating during an isothermal expansion. In Process 2, the same amount of energy Q is transferred to the other container during an isovolumetric heating process. Which of the following correctly compares ΔU_2 and ΔU_1, the change in internal energy for processes 1 and 2, respectively?

(A) $\Delta U_2 > (\Delta U_1 = 0)$

(B) $\Delta U_2 > \Delta U_1 > 0$

(C) $\Delta U_2 = \Delta U_1$

(D) $\Delta U_2 < \Delta U_1$

41. A container is filled with an ideal gas at a high gauge pressure and sealed with a cork of radius R that has been inserted into a circular opening of radius R. The force exerted on the cork by the gas has magnitude F. If the radii of the opening and cork are increased to $2R$, what will be the magnitude of the new force exerted on the cork by the gas?

(A) $F/4$

(B) $F/2$

(C) $2F$

(D) $4F$

42. A sheet of paper in a classroom has an area of $0.060\ \text{m}^2$. The classroom has a volume of $300\ \text{m}^3$ and contains $1.3 \times 10^4\ \text{mol}$ of air molecules at atmospheric pressure. How much force does the air exert on one side of the paper?

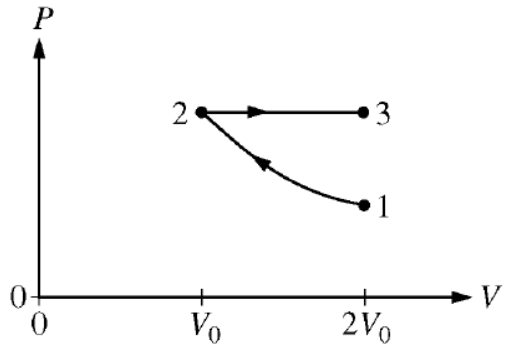
(A) $6.0 \times 10^{-7}\ \text{N}$

(B) $0.60\ \text{N}$

(C) $6000\ \text{N}$

(D) $1.7 \times 10^6\ \text{N}$

43.



An ideal gas is initially in state 1 at a temperature of 200 K. The gas is taken through the two reversible thermodynamic processes shown in the PV diagram above. The process from state 1 to state 2 is isothermal. The process from state 2 to state 3 is isobaric. What is the temperature of the gas when it is in state 3?

(A) 800 K

(B) 400 K

(C) 200 K

(D) 100 K

44. A student wants to increase the temperature of an ideal gas in a cylinder that is initially at room temperature. The cylinder has a movable piston with negligible friction. Which of the following correctly indicates an action the student could take to increase the temperature of the gas, and why the temperature increases?

(A) Pull the piston out rapidly. The gas will expand violently and increase in temperature.

(B) Pull the piston out slowly. The gas will have time to absorb energy from the surroundings by heating as it expands.

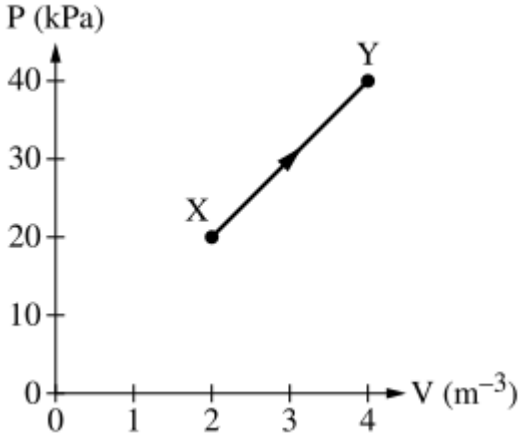
(C) Push the piston in rapidly. The temperature of the gas will increase because no energy will be transferred to the surroundings by cooling.

(D) Push the piston in slowly. The gas will have time to absorb energy from the surroundings by heating as it is compressed.

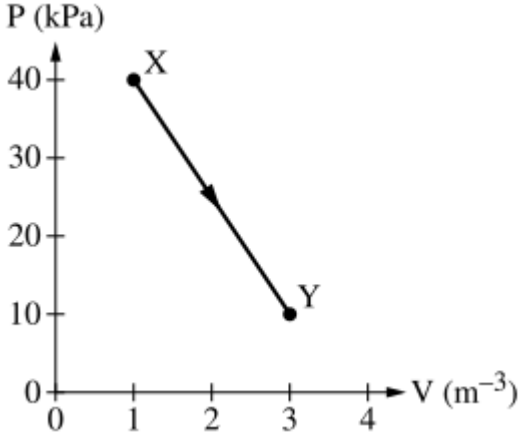
45. A sample of an ideal gas in a closed container with a moveable piston is taken from State X to State Y by a thermodynamic process in which the gas expands, and the temperature of the gas at State Y is less than the temperature of the gas at State X. Which of the following graphs of pressure as a function of volume corresponds to this process?

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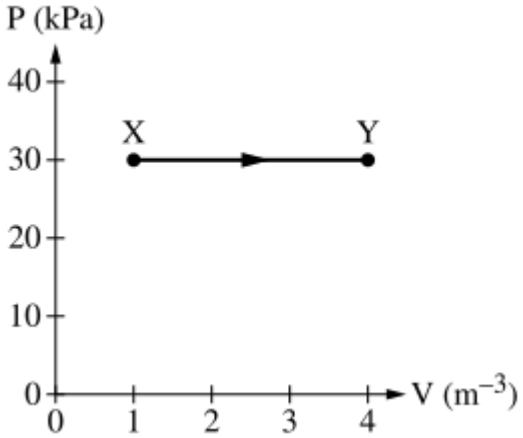
(A)



(B)

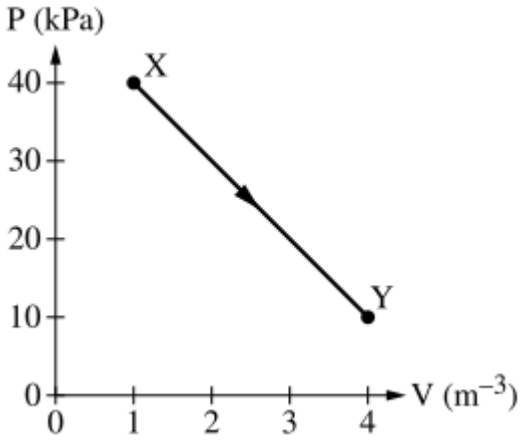


(C)

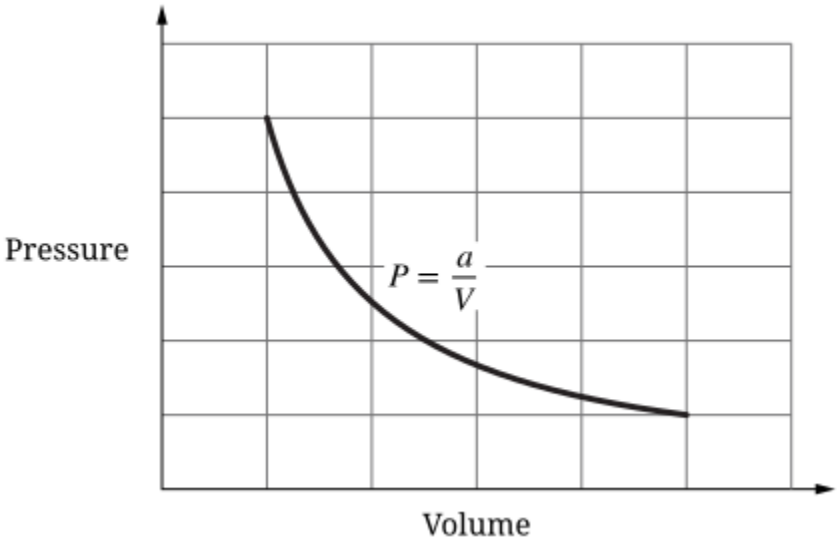


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(D)



46.



A sample of an ideal gas is in a sealed container with a moveable piston. During a thermodynamic process, the pressure P and volume V of the gas can be described by the equation $P = \frac{a}{V}$, where a is a positive constant with appropriate units. The graph shows P as a function of V during the process. Which of the following claims best describes how the temperature of the gas changes, if at all, during the process?

(A) The temperature decreases.

(B) The temperature increases.

(C) The temperature remains constant.

(D) The temperature could decrease, increase, or stay the same depending on how many atoms are in the gas.

47. An ideal monatomic gas is compressed while its temperature is held constant. What happens to the internal energy of the gas during this process, and why?

(A) It decreases because the gas does work on its surroundings.

(B) It decreases because the molecules of an ideal gas collide.

(C) It does not change because the internal energy of an ideal gas depends only on its temperature.

(D) It increases because work is done on the gas.

(E) It increases because the molecules travel a shorter path between collisions.

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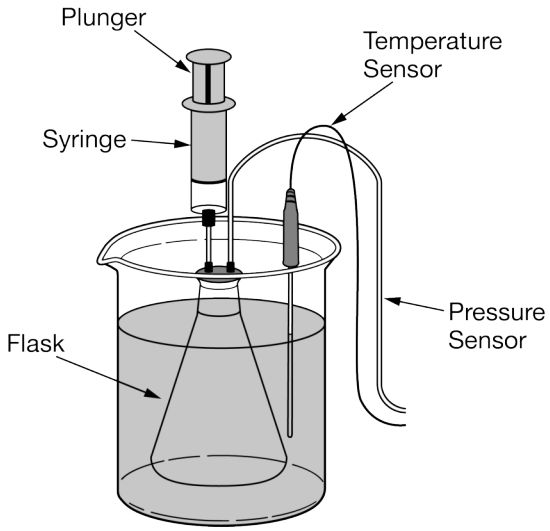
48. A container filled with gas is sealed on the top with a piston of negligible mass and area $0.007\ \text{m}^2$. The container and gas are allowed to reach thermal equilibrium with the surrounding air, and a block on top of the piston keeps the piston in equilibrium. If the pressure inside the container is $2 \times 10^5\ \text{Pa}$, what is the mass of the block?

(A) $70\ \text{kg}$

(B) $140\ \text{kg}$

(C) $700\ \text{kg}$

(D) $1400\ \text{kg}$



Students perform an experiment to determine the relationship between the pressure and temperature of an ideal gas. They have a flask with a stopper through which a syringe and a pressure sensor are inserted, as shown in the figure. The flask is in a water bath whose temperature can be adjusted. There is a temperature sensor to measure the temperature of the water.

49. The water bath is initially hotter than the gas. The plunger will eventually reach an equilibrium position. Which of the following statements describing the microscopic model of the gas correctly explains the process by which equilibrium is reached? Assume the mass of the plunger is negligible.

(A) The force exerted by the gas molecules and the force exerted by the air molecules on the plunger are always equal. The plunger is always in equilibrium so it does not move.

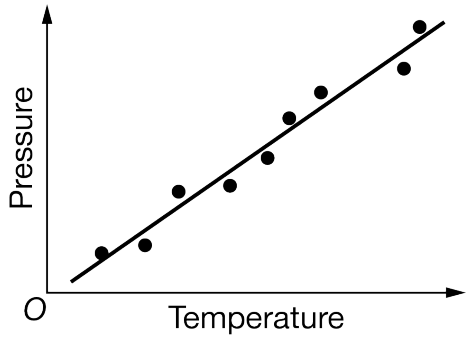
(B) The gas molecules make elastic collisions with the plunger. These collisions exert a force on the plunger. The plunger moves up until the temperature of the gas is the same as the temperature of the water and the pressure of the gas is the same as the air pressure on the plunger.

(C) The gas molecules make elastic collisions with the plunger. These collisions exert a force on the plunger. The plunger moves up until the volume of gas in the syringe is the same as the volume of the gas in the flask.

(D) The temperature inside the container makes the gas less dense than the air. The gas molecules rise to the top of the flask and exert an upward force on the plunger. The plunger moves upward until the density of the gas near it equals the density of the air.

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50.



The students fix the plunger in place and take measurements of temperature and pressure. They graph the data and draw a best-fit line, as shown. Which of the following is equal to the slope of the line?

(A) nRV

(B) $\frac{V}{nR}$

(C) $\frac{nR}{V}$

(D) $\frac{1}{nRV}$

51. The students now press the plunger inward rapidly, decreasing the volume of the gas. Which of the following describes the effect this has on the pressure and the temperature of the gas?

(A) The volume of the gas decreases, so the gas molecules hit the walls of the flask more often. This increases the pressure and thus the average force exerted on the gas molecules. The speed of the gas molecules increases, which means greater kinetic energy, and thus an increase in the temperature.

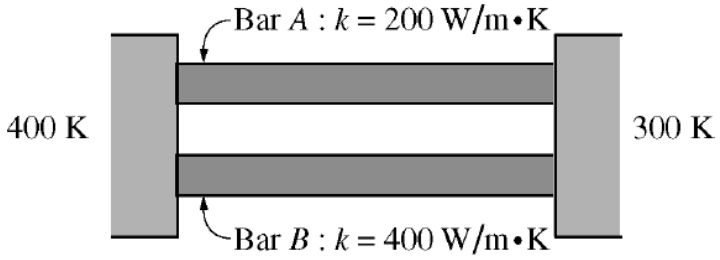
(B) The volume of the gas decreases, so the pressure inside the flask increases. The gas molecules slow down due to the increased pressure, which means their kinetic energy increases. Collisions of molecules with the walls involve less force. As a result, temperature decreases.

(C) The volume of the gas decreases. The pressure inside the flask increases and so does the temperature. As a result the molecules hit the walls with a greater force and thus exert a larger pressure. The gas reaches instant equilibrium with the surroundings.

(D) The volume of the gas decreases. The pressure inside the flask increases; however, there is no change in the temperature of the gas.

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52.



Two metal bars with the same length and same cross-sectional area are placed between two tanks with temperatures of 400 K and 300 K, as shown above. The thermal conductivity of the top bar is 200 W/ m • K, and that of the bottom bar is 400 W/ m • K . If the net energy transferred through the top bar in a given time interval is Q, what is the net energy transferred through the bottom bar during the same time interval?

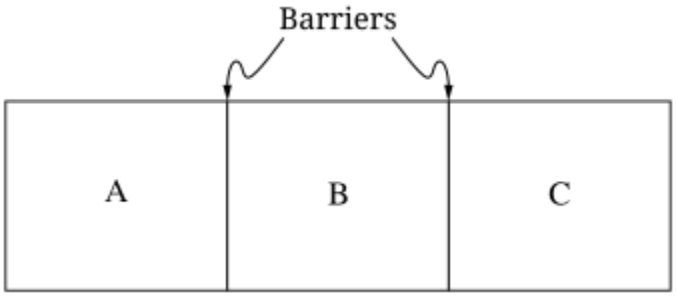
(A) 4Q

(B) 2Q

(C) Q

(D) Q/2

53.



A container with three sections initially contains samples of three different monatomic ideal gases, which are separated by removable barriers, as shown. The container is thermally isolated from its surroundings. The masses of gas atoms in gases A, B, and C are m_A, m_B, and m_C, respectively, where $m_A > m_B > m_C$. The initial temperatures gases A, B, and C are T_A, T_B, and T_C, respectively, where $T_C > T_B > T_A$. Both barriers are removed at the same time and the gases mix. Which of the following statements about the average kinetic energy of the gas atoms after the barriers are removed must be true?

(A) The average kinetic energy of Gas A atoms will increase.

(B) The average kinetic energy of Gas B atoms will not change.

(C) The average kinetic energy of Gas C atoms will increase.

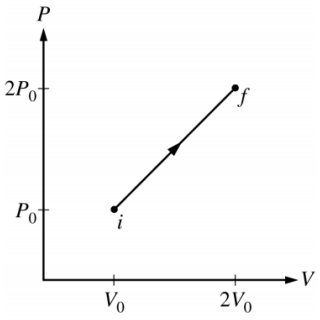
(D) The average kinetic energy of Gas C atoms will not change.

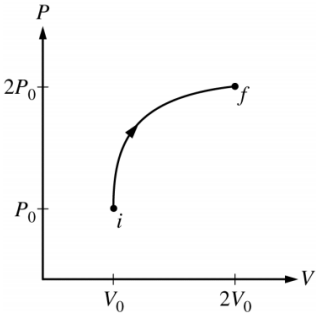
54. A partially evacuated vertical cylindrical container is covered by a circular lid that makes an airtight seal. The pressure in the room surrounding the container is 1.01×10^5 N/m^2, and the pressure inside the container is 0.41×10^5 N/m^2. The lid has radius 0.20 m and weight 200 N. The minimum upward applied force required to lift the lid is most nearly

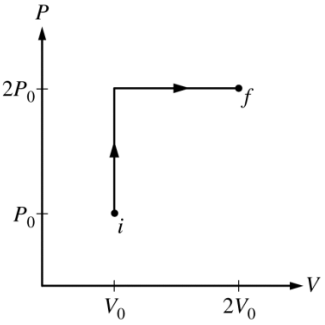
WorkBook Unit 9

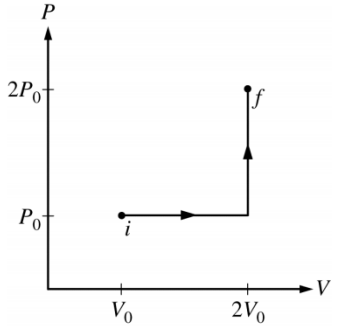
(A) 7.5×10^3 N
(B) 7.7×10^3 N
(C) 1.3×10^4 N
(D) 1.8×10^4 N

55. An ideal gas is taken from initial state i at pressure P_0 and volume V_0 to final state *f* at pressure $2P_0$ and volume $2V_0$. The figures below represent four possible processes by which the gas can be taken from state i to state *f*. For which process is the work done by the gas greatest?

(A)


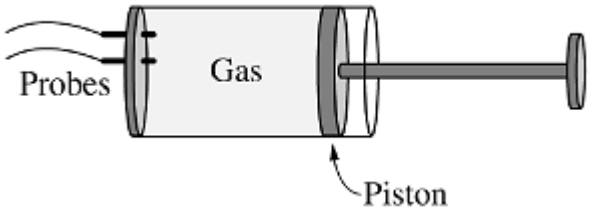
(B)


(C)


(D)


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56. Directions: This question is a long free-response question. Show your work for each part of the question.



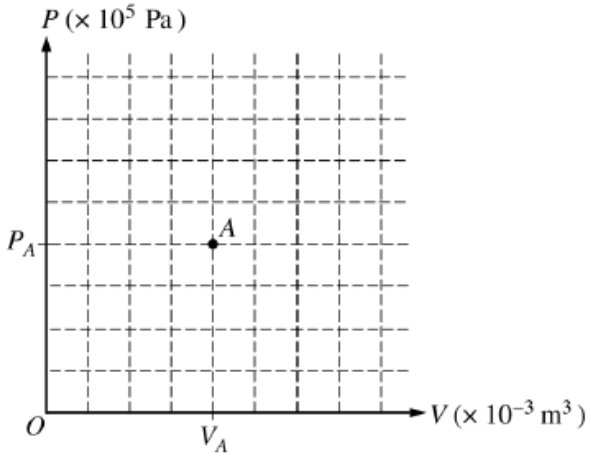
2. (12 points, suggested time 25 minutes)

A cylinder with a movable piston contains a sample of ideal gas. A temperature probe and a pressure probe are inserted into the cylinder, as shown above. The sample of gas is taken through a three-step cycle, *ABCA*.

- In process *AB*, the volume is decreased to $1/4$ its original value while constant pressure is maintained.
- In process *BC*, the gas expands back to its original volume isothermally.
- In process *CA*, the gas returns to its original state without any work being done on or by the gas.

(a)

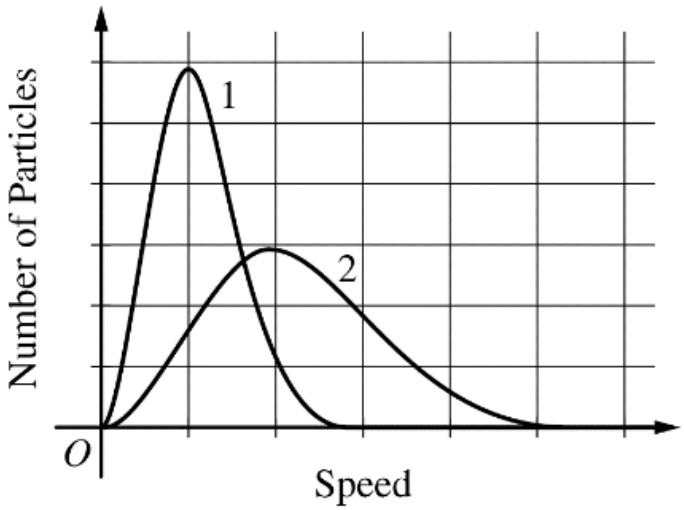
i. On the axes below, draw a graph of pressure *P* as a function of volume *V* for the three-step process *ABCA*, labeling states *B* and *C*. The pressure and volume of state *A* are P_A and V_A, respectively. Label the axes with the values of pressure and volume for states *B* and *C* in terms of P_A and V_A.



ii. For process *BC*, is energy added to the gas by heating or removed from the gas by cooling? Support your claim using physics principles.

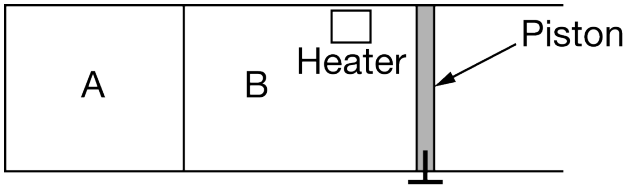
(b) For process *CA*, is the net flow of energy into the gas or out of the gas? Explain your reasoning using physics principles. Describe one method by which this flow of energy could be accomplished in the laboratory.

(c) Curves 1 and 2 below show the distributions of particle speeds when the sample of gas is in two of the states, with curve 1 a state that occurs before curve 2 as the gas goes through the cycle. Which states correspond to curve 1 and 2? Support your claim using physics principles.



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57. This question is a short free-response question. Show your work for each part of the question.



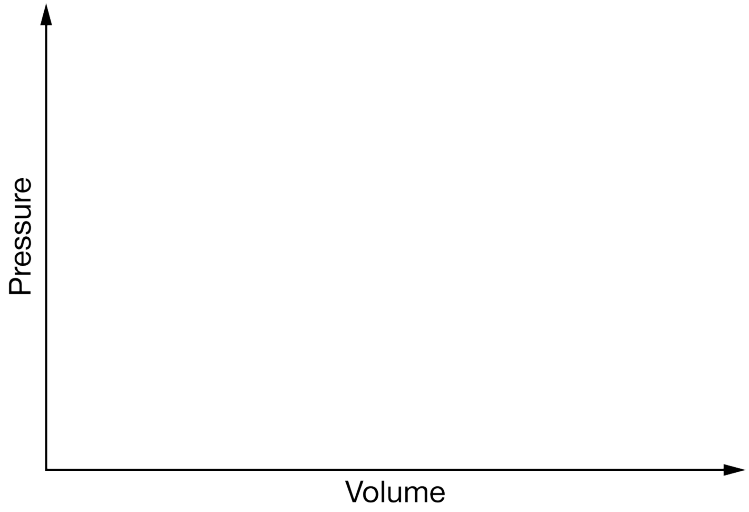
(10 points, suggested time 20 minutes)

Two sealed containers each contain a sample of the same ideal gas. Container A with gas at initial temperature $293\ \text{K}$ is placed into thermal contact with container B, containing gas that is maintained at a temperature $540\ \text{K}$ by a heater, as shown above. Contain Berhas a piston that is locked in place. The system of the containers and Container heater is insulated from its surroundings.

(a) In which direction will energy flow when the containers are initially put into thermal contact? Identify the behavior of the gas particles in the two containers that is different and is responsible for this energy flow.

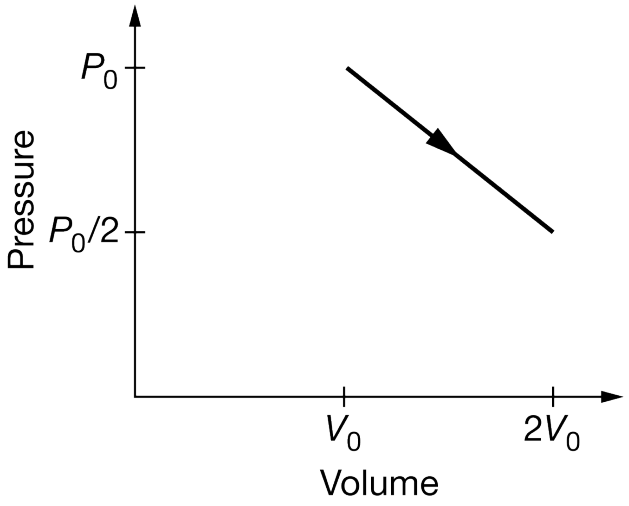
(b) Explain how energy is transferred between the containers on a microscopic scale.

(c) Container A has a fixed volume. As the gas in container A gets hotter, it eventually reaches equilibrium at $540\ \text{K}$. On the axes below, sketch the process the gas in container A goes through as it gets hotter. Indicate the direction of the process with an arrow.



(d) The piston in container B is unlocked, and the gas is allowed to expand through the process shown in the graph below of pressure P as a function of volume V.

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i. What is the change in the internal energy of the gas in container B as a result of the process shown? Support your answer.

ii. Calculate the work done on the gas as it expands.

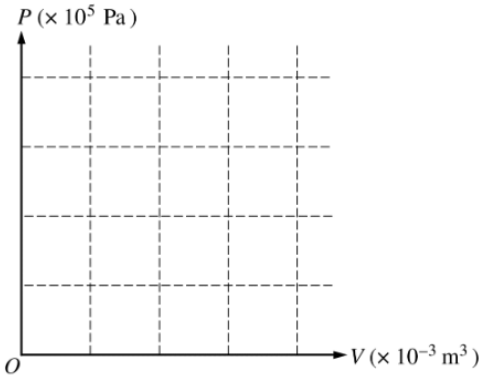
WorkBook Unit 9

58. A student gives the following description of a thermodynamic cycle through which a sample of an ideal monatomic gas is taken.

"Initially, the gas has a volume of 1×10^{-3} m^3 and a pressure of 1×10^5 Pa. The gas is in a closed cylinder that has a movable piston on one end. Call the initial state of the gas state *A*. Process *AB* takes the gas at constant pressure from state *A* to state *B*, where *B* has three times the volume of state *A*. The gas is then compressed isothermally until it is back to its initial volume, reaching state *C*. Finally, a single process *CA* returns the gas to its initial state *A*."

a. On the axes below, plot states *A, B*, and *C*, and sketch a graph of pressure *P* as a function of volume *V* for cycle *ABCA*. Label states *A*, *B*, and *C*, and label the axes with numerical values.

i. On the axes below, plot states *A*, *B*, and *C*, and sketch a graph of pressure *P* as a function of volume *V* for cycle *ABCA*. Label states *A*, *B*, and *C*, and label the axes with numerical values.



ii. Let the temperature for state *A* be T_0. In terms of T_0, determine the temperature for state *B*.

iii. Compare the temperature at state *C* to the temperature at state *A*. Explain how your comparison is consistent with your graph above.

b. i. Briefly describe an experimental procedure that would take the gas from state *C* to state *A*.

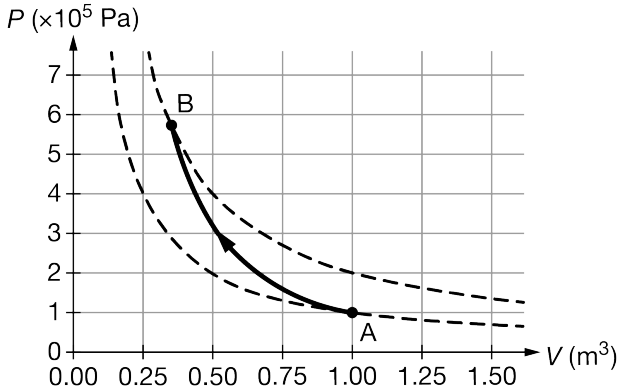
ii. Relate the temperature of the gas to the microscopic processes involved in the energy transfer as the gas goes from state *C* to state *A* for the procedure you described in part (b)(i).

c. i. Use the student's verbal description or your graph in part (a)(i) to calculate a numerical value for the work done on the gas in process *AB*.

ii. The magnitude of the change in internal energy for process *AB* is $|\Delta U| = 300$ J. Calculate a numerical value for the energy added to or removed from the gas by heating in process *AB*.

WorkBook Unit 9

59. This question is a long free-response question. Show your work for each part of the question.



(12 points, suggested time 25 minutes)

A sample of ideal gas in a thermally insulated container with a moveable piston is initially in state A. As shown in the graph of pressure P as a function of volume V, the gas is taken from state A to state B by an adiabatic process. The dashed lines represent isotherms.

(a) Let W be the work done on the gas, Q be the energy transferred to the gas by heating, and ΔU be the change in the internal energy of the gas during the process shown.

i. Is W greater than, less than, or equal to zero? Briefly explain your answer.

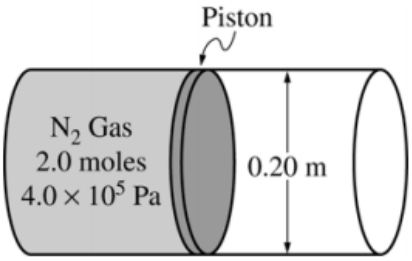
ii. Is Q greater than, less than, or equal to zero? Briefly explain your answer.

iii. Is ΔU greater than, less than, or equal to zero? Briefly explain your answer.

(b) If the temperature of the gas in state A is $200\ \text{K}$, what is the approximate temperature of the gas in state B ?

(c) Is your numerical result for part (b) consistent with your explanation for part (a)(iii)? Briefly justify your answer.

60.



The figure above shows a 0.20 m diameter cylinder fitted with a frictionless piston, initially fixed in place. The cylinder contains 2.0 moles of nitrogen gas at an absolute pressure of 4.0 x 10^5 Pa. Nitrogen gas has a molar mass of 28 g/mole and it behaves as an ideal gas.

(a) Calculate the force that the nitrogen gas exerts on the piston.

(b) Calculate the volume of the gas if the temperature of the gas is 300 K.

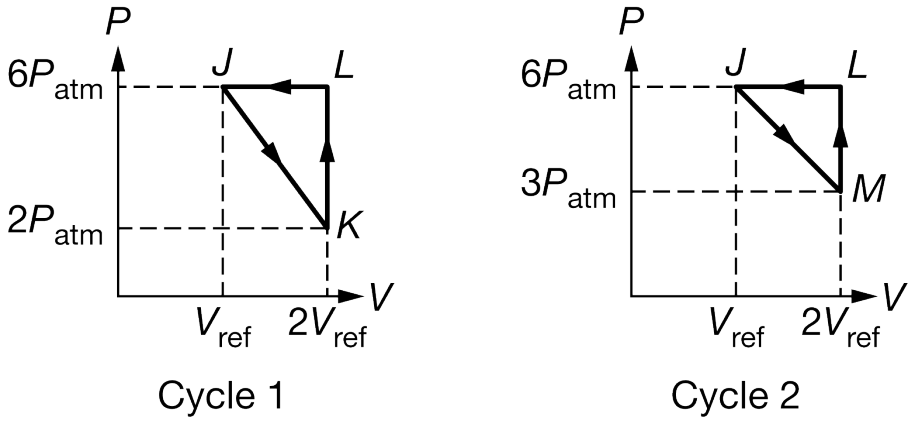
(c) In a certain process, the piston is allowed to move, and the gas expands at constant pressure and pushes the piston out 0.15 m. Calculate how much work is done by the gas.

(d) Which of the following is true of the heat energy transferred to or from the gas, if any, in the process in part (c)?

_____ Heat is transferred to the gas.
_____ Heat is transferred from the gas.
_____ No heat is transferred in the process.
Justify your answer.

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61. This question is a short free-response question. Show your work for each part of the question. (10 points, suggested time 15 minutes)



A sample of an ideal gas can be taken through the two cycles shown in the graphs above of pressure P as a function of volume V. In each cycle the gas starts and ends in state J. The gas is contained in a cylindrical container with a movable piston.

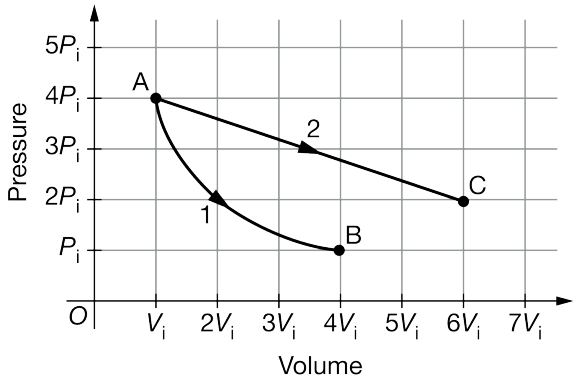
(a) Rank the temperature T of the gas for each labeled state in cycle 1. Clearly indicate the order of ranking and if any states have the same temperature. Justify your ranking using information from the graph.

(b) Calculate the net work on the gas during cycle 1. Justify your answer.

(c) Given that the net change in internal energy for a closed cycle is zero, determine whether energy is added to the gas by heating or removed by cooling in each cycle. Support your claim in a coherent paragraph-length response that qualitatively compares the amounts of energy added to the gas by heating or removed by cooling in the two cycles and refers to the work done on the gas and the change in internal energy of the gas.

WorkBook Unit 9

62. This question is a long free-response question. Show your work for each part of the question. (15 points, suggested time 25 minutes)

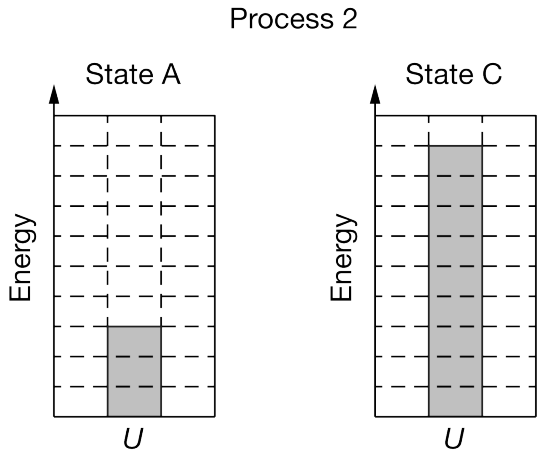


A sample of a monatomic ideal gas is confined in a cylinder with a frictionless piston. The gas is taken through two thermal processes (process 1 and process 2), as shown above.

(a) Does the graph indicate that the process that takes the gas from state A to state B is a constant temperature process? Justify your answer using details from the graph.

(b) Is the internal energy of the gas in state A greater than, less than, or equal to the internal energy of the gas in state B? Justify your answer.

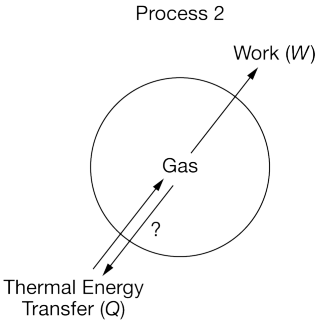
(c) Consider process 2, where the gas is taken from state A to state C. At which of the labeled states, A or C, is the average kinetic energy of the molecules larger? Justify your answer using specific values from the graph.



A student draws the correct bar charts shown above representing the internal energy U of the gas at states A and C. The charts are drawn to the same scale.

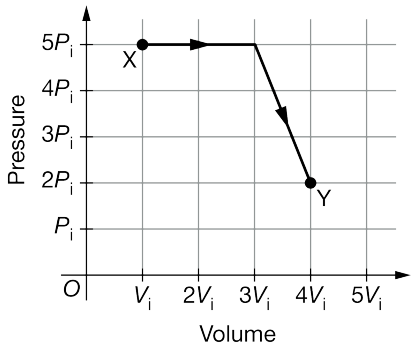
(d) Describe how the bar charts support or contradict your answer to part (c), including reference to the size of the

bars and the specific values from part (c).



The student also draws a diagram to show the direction (not the magnitude) of the net transfers of energy into and out of the gas in process 2. The direction of the work is correct, but the student is not sure which way the thermal energy arrow should point.

(e) What is the correct direction of thermal energy transfer, into or out of the gas? Justify your answer.



A sample of the gas is taken from state X to state Y by the processes shown above. The product of P_i and V_i has a value equal to M.

(f) What is the magnitude of the work done by the gas as it goes from state X to state Y? Describe your reasoning.

(g) If the net change in internal energy as the gas goes from state X to state Y is $4.5M$, how much energy is transferred by heating or cooling during the processes? Describe your reasoning.

AP PHYSICS 2

UNIT 10

Electric Force, Field, and Potential

15–18%
AP EXAM WEIGHTING

~14–21
CLASS PERIODS

Conceptual Questions

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

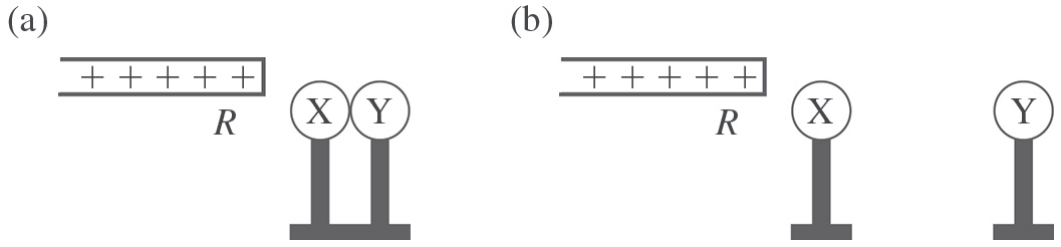
1) If two objects are electrically attracted to each other, 1) _____
A) the objects could be electrically neutral.
B) both objects must be positively charged.
C) both objects must be negatively charged.
D) one object must be negatively charged and the other object must be positively charged.
E) None of the above statements are absolutely true.

2) Electrically neutral objects cannot exert an electrical force on each other, but they can exert a gravitational force on each other. 2) _____
A) True B) False

3) A plastic rod is charged up by rubbing a wool cloth, and brought to an initially neutral metallic sphere that is insulated from ground. It is allowed to touch the sphere for a few seconds, and then is separated from the sphere by a small distance. After the rod is separated, the rod 3) _____
A) is attracted to the sphere.
B) is repelled by the sphere.
C) feels no force due to the sphere.

4) A negatively–charged plastic rod is brought close to (but does not touch) a neutral metal sphere that is connected to ground. After waiting a few seconds, the ground connection is removed (without touching the sphere), and after that the rod is also removed. The sphere is now 4) _____
A) neutral. B) negatively charged. C) positively charged.

5) 5) _____



X and Y are two initially uncharged metal spheres on insulating stands, and they are in contact with each other. A positively charged rod *R* is brought close to X as shown in part (a) of the figure. Sphere Y is now moved away from X, as shown in part (b). What are the final charge states of X and Y?
A) Both X and Y are neutral.
B) Both X and Y are negative.
C) X is negative and Y is positive.
D) X is neutral and Y is positive.
E) X is positive and Y is neutral.



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人丑就要多读书，特别是电磁学这本书。

6) Two uncharged metal spheres, #1 and #2, are mounted on insulating support rods. A third metal sphere, carrying a positive charge, is then placed near #2. Now a copper wire is momentarily connected between #1 and #2 and then removed. Finally, sphere #3 is removed. In this final state 6) _____

A) sphere #1 carries positive charge and #2 carries negative charge.
B) spheres #1 and #2 are still uncharged.
C) spheres #1 and #2 both carry positive charge.
D) spheres #1 and #2 both carry negative charge.
E) sphere #1 carries negative charge and #2 carries positive charge.

7) A negatively–charged rod is brought close to (but does not touch) two neutral spheres that are in contact with each other but insulated from the ground. If the two spheres are then separated, what kind of charge will be on the spheres? 7) _____

A) Both spheres become positive.
B) The sphere near the charged rod becomes positive and the other one becomes negative.
C) Both spheres become negative.
D) The spheres do not get any charge.
E) The sphere near the charged rod becomes negative and the other one becomes positive.

8) Two very small plastic balls of equal mass are released from rest. One of them carries +10 μC of excess charge and the other one carries +1μC of charge. No other charges or fields are present. Which of the following statements are true about them as they move away from each other? (There may be more than one correct choice.) 8) _____

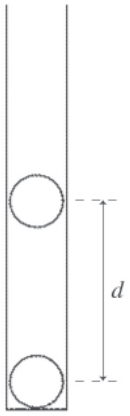
A) The acceleration of the 1–μC ball is 10 times that of the 10–μC ball.
B) The speed of the balls keeps increasing.
C) The acceleration of the 10–μC ball is 10 times that of the 1–μC ball.
D) The balls always have accelerations of equal magnitude.
E) The acceleration of the balls keeps increasing.

9) In outer space, a positive charge q is released near a positive fixed charge Q, as shown in the figure. As q moves away from Q, what is true about the motion of q? (There may be more than one correct choice.) 9) _____



A) It will move with decreasing speed.
B) It will move with constant acceleration.
C) It will move with increasing speed.
D) It will move with increasing acceleration.
E) It will move with decreasing acceleration.

10) A small charged plastic ball is vertically above another charged small ball in a frictionless test tube as shown in the figure. The balls are in equilibrium a distance d apart. If the charge on each ball is doubled, the equilibrium distance between the balls in the test tube would become 10) _____



A) $8d$. B) $4d$. C) $d/4$. D) $\sqrt{2}d$. E) $2d$.

11) A hydrogen nucleus, which has a charge $+e$, is situated to the left of a carbon nucleus, which has a charge $+6e$. Which statement is true? 11) _____

A) The electrical force experienced by the hydrogen nucleus is to the left, and the magnitude is less than the force exerted on the carbon nucleus.
B) The electrical force experienced by the hydrogen nucleus is to the left, and the magnitude is greater than the force exerted on the carbon nucleus.
C) The electrical force experienced by the hydrogen nucleus is to the right, and the magnitude is equal to the force exerted on the carbon nucleus.
D) The electrical force experienced by the hydrogen nucleus is to the left, and the magnitude is equal to the force exerted on the carbon nucleus.

12) Two identical small charged spheres are a certain distance apart, and each one initially experiences an electrostatic force of magnitude F due to the other. With time, charge gradually diminishes on both spheres by leaking off. When each of the spheres has lost half its initial charge, what will be the magnitude of the electrostatic force on each one? 12) _____

A) 1/4 F B) 1/16 F C) $1/\sqrt{2}\,F$ D) 1/8 F E) 1/2 F

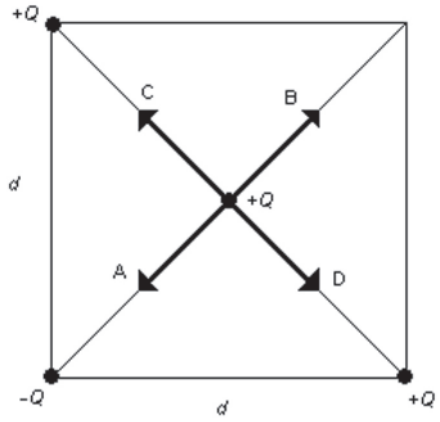
13) Two point charges, Q_1 and Q_2, are separated by a distance R. If the magnitudes of both charges are doubled and their separation is also doubled, what happens to the electrical force that each charge exerts on the other one? 13) _____

A) It is reduced by a factor of $\sqrt{2}$.
B) It remains the same.
C) It increases by a factor of $\sqrt{2}$.
D) It increases by a factor of 2.
E) It increases by a factor of 4.

14) Two point charges, Q_1 and Q_2, are separated by a distance R. If the magnitudes of both charges are halved and their separation is also halved, what happens to the electrical force that each charge exerts on the other one? 14) _____

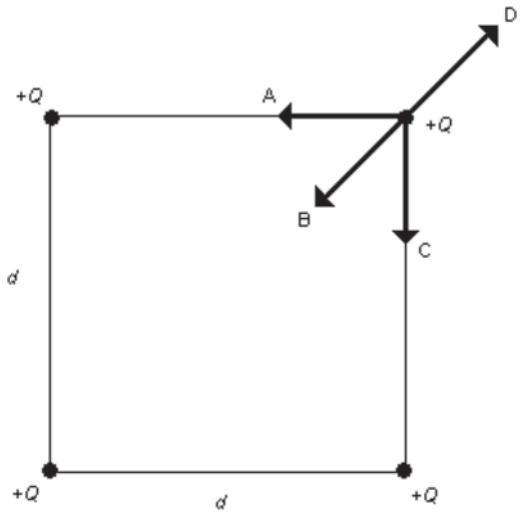
A) It remains the same.
B) It increases by a factor of 16.
C) It increases by a factor of 8.
D) It increases by a factor of 2.
E) It increases by a factor of 4.

15) Four point charges of equal magnitudes but with varying signs are arranged on three of the corners and at the center of the square of side d as shown in the figure. Which one of the arrows shown represents the net force acting on the center charge? 15) _____



A) A B) B C) C D) D

16) Four point charges Q of equal magnitude and sign are arranged on three of the corners of the square of side d as shown in the figure Which one of the arrows shown represents the net force acting on the charge at the upper right hand corner of the square? 16) _____

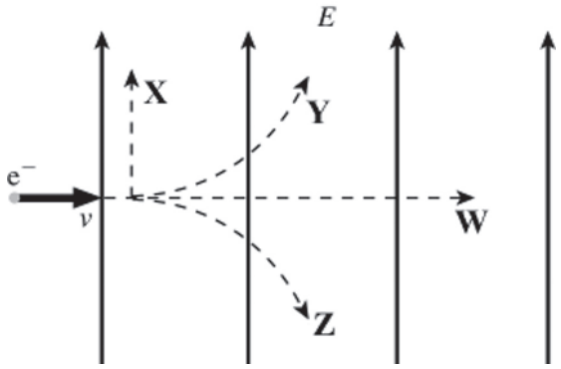


A) A B) B C) C D) D

17) An electron and a proton are released simultaneously from rest and start moving toward each other due to their electrostatic attraction, with no other forces present. Which of the following statements are true just before they are about to collide? (There could be more than one correct choice.) 17) _____

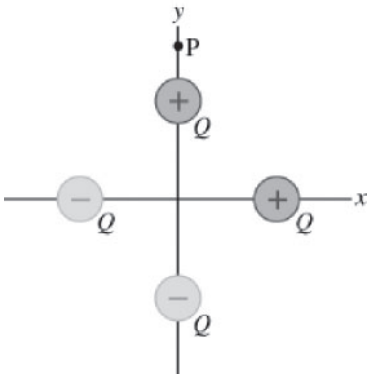
A) They both have the same speed.
B) The electrostatic force on the proton is greater than the electrostatic force on the electron.
C) They are closer to the to the initial position of the electron that to the initial position of the proton.
D) They are at the midpoint of their initial separation.
E) They are closer to the to the initial position of the proton that to the initial position of the electron.

18) An electron is initially moving to the right when it enters a uniform electric field directed upwards, as shown in the figure. Which trajectory (**X**, **Y**, **Z**, or **W**) will the electron follow in the field? 18) _____



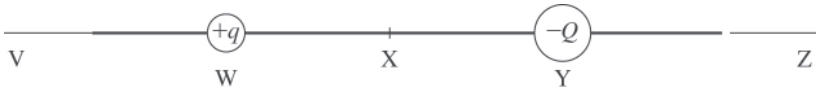
A) trajectory **W** B) trajectory **X** C) trajectory **Y** D) trajectory **Z**

19) Four tiny charged particles (two having a charge +Q and two having a charge −Q) are distributed on the x− and y−axes as shown in the figure. Each charge is equidistant from the origin. In which direction is the net electric field at the point P on the y−axis? 19) _____



A) upwards and towards the right
B) directly left (−x direction)
C) upwards and towards the left
D) directly up (in the +y direction)
E) The net field is zero, so there is no direction.

20) The figure shows two unequal charges, $+q$ and $-Q$. Charge $-Q$ has greater magnitude than charge $+q$. Point X is midway between the charges. In what section of the line will there be a point where the resultant electric field is zero? 20) _____



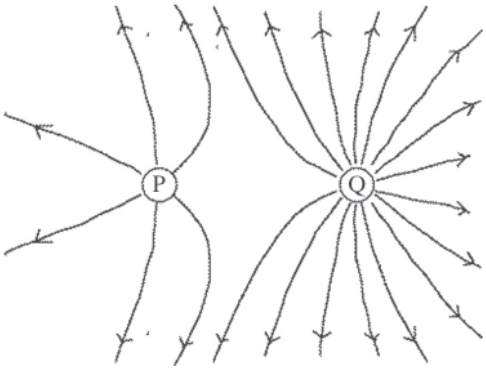
A) VW B) WX C) XY D) YZ

21) The electric field at point P due to a point charge Q a distance R away from P has magnitude E. In order to double the magnitude of the field at P, you could 21) _____

A) double the charge to $2Q$ and at the same time reduce the distance to $R/2$.
B) double the distance to $2R$.
C) double the charge to $2Q$.
D) reduce the distance to $R/4$.
E) reduce the distance to $R/2$.

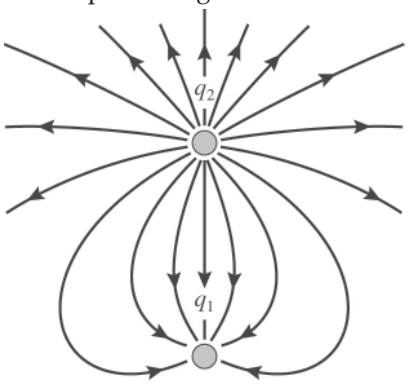
22) The figure shows electric field lines arising from two small charged particles P and Q. Consider the following two statements: 22) _____

(i) The charge on P is smaller than the charge on Q.
(ii) The electrostatic force on P is smaller than the force on Q.
Which of the above statements are true?



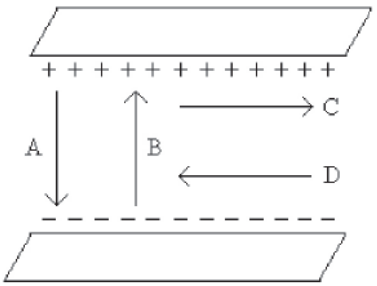
A) Only (i) is true. B) Only (ii) is true.
C) Both (i) and (ii) are true. D) Neither (i) nor (ii) is true.

23) Two stationary point charges q_1 and q_2 are shown in the figure along with a sketch of some field lines representing the electric field produced by them. What can you deduce from the sketch? 23) _____



A) q_1 is positive and q_2 is negative; the magnitude of q_1 is greater than the magnitude of q_2.
B) q_1 is negative and q_2 is positive; the magnitudes are equal.
C) q_1 is negative and q_2 is positive; the magnitude of q_1 is less than the magnitude of q_2.
D) q_1 and q_2 have the same sign; the magnitude of q_1 is greater than the magnitude of q_2.
E) q_1 and q_2 have the same sign; the magnitudes are equal.

24) Which one of the arrows shown in the figure best represents the direction of the electric field between the two uniformly charged metal plates? 24) _____



A) A
B) B
C) C
D) D
E) None of the above

25) 25) _____

A) B) C) D)

26) If the result of your calculation of a quantity has SI units kg • m^2/(s^2 • C), that quantity could be 26) _____
A) an electric field strength.
B) a dielectric constant.
C) a capacitance
D) an electric potential energy.
E) an electric potential difference.

27) If the result of your calculation of a quantity has SI units of kg • m/(s^2 • C), that quantity could be 27) _____
A) an electric field strength.
B) a dielectric constant.
C) a capacitance
D) an electric potential energy.
E) an electric potential difference.

28) As an electron moves in the direction the electric field lines 28) _____
A) it is moving from low potential to high potential and gaining electric potential energy.
B) it is moving from low potential to high potential and losing electric potential energy.
C) it is moving from high potential to low potential and gaining electric potential energy.
D) it is moving from high potential to low potential and losing electric potential energy.
E) both its electric potential and electric potential energy remain constant.

29) As a proton moves in a direction perpendicular to the electric field lines 29) _____
A) it is moving from low potential to high potential and gaining electric potential energy.
B) it is moving from low potential to high potential and losing electric potential energy.
C) it is moving from high potential to low potential and gaining electric potential energy.
D) it is moving from high potential to low potential and losing electric potential energy.
E) both its electric potential and electric potential energy remain constant.

30) If the electric potential at a point in space is zero, then the electric field at that point must be 30) _____
A) uniform.
B) zero.
C) positive.
D) negative.
E) impossible to determine based on the information given.

31) The electron–volt is a unit of 31) _____
A) electric force.
B) energy.
C) electric potential.
D) electric field.
E) charge.

32) The electric potential at a distance of 4 m from a certain point charge is 200 V relative to infinity. What is the potential (relative to infinity) at a distance of 2 m from the same charge? 32) _____
A) 400 V B) 100 V C) 50 V D) 600 V E) 200 V

33) The potential (relative to infinity) at the midpoint of a square is 3.0 V when a point charge of +*Q* is located at one of the corners of the square. What is the potential (relative to infinity) at the center when each of the other corners is also contains a point charge of +*Q*? 33) _____
A) 3.0 V B) 12 V C) 0 V D) 9.0 V

34) Which statements must be true about the surface of a charged conductor in which no charge is moving? (There could be more than one correct choice.) 34) _____

A) The electric field is perpendicular to the surface.
B) The electric potential of the surface is zero.
C) The electric field is zero at the surface.
D) The electric field is constant at the surface.
E) The electric potential is constant over the surface.

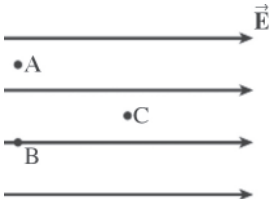
35) Two ideal parallel–plate capacitors are identical in every respect except that one has twice the plate area of the other. If the smaller capacitor has capacitance *C*, the larger one has capacitance 35) _____

A) *C*. B) 4*C*. C) *C*/2. D) 2*C*.

36) An ideal parallel–plate capacitor has a capacitance of *C*. If the area of the plates is doubled and the distance between the plates is halved, what is the new capacitance? 36) _____

A) 4C B) *C*/4 C) 2C D) *C*/2

37) A region of space contains a uniform electric field, directed toward the right, as shown in the figure. Which statement about this situation is correct? 37) _____



A) The potential at point A is the highest, the potential at point B is the second highest, and the potential at point C is the lowest.
B) The potential at points A and B are equal, and the potential at point C is lower than the potential at point A.
C) The potential at all three locations is the same.
D) The potentials at points A and B are equal, and the potential at point C is higher than the potential at point A.

38) If the result of your calculations for a quantity has SI units of $C^2 \cdot s^2/(kg \cdot m^2)$, that quantity could be 38) _____

A) an electric potential difference.
B) an electric potential energy.
C) a dielectric constant.
D) a capacitance.
E) an electric field strength.

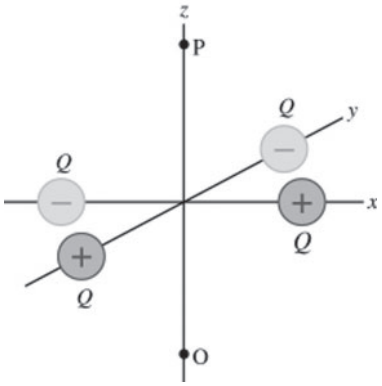
39) A proton and an electron are released from rest, with only the electrostatic force acting. Which of the following statements must be true about them as they move toward each other? (There could be more than one correct choice.) 39) _____

A) Their kinetic energy keeps decreasing.
B) Their electric potential energy keeps decreasing.
C) Their acceleration keeps decreasing.
D) Their electric potential energy keeps increasing.
E) Their kinetic energy keeps increasing.

40) Two protons are released from rest, with only the electrostatic force acting. Which of the following statements must be true about them as they move apart? (There could be more than one correct choice.) 40) _____

A) Their acceleration keeps decreasing.
B) Their kinetic energy keeps decreasing.
C) Their kinetic energy keeps increasing.
D) Their electric potential energy keeps increasing.
E) Their electric potential energy keeps decreasing.

41) Four charged particles (two having a charge +Q and two having a charge −Q) are arranged in the xy−plane, as shown in the figure. These particles are all equidistant from the origin. The electric potential (relative to infinity) at point P on the z−axis due to these particles, is 41) _____



A) negative.
B) positive.
C) zero.
D) impossible to determine based on the information given.

42) Which of the following will increase the capacitance of a parallel−plate capacitor? (There could be more than one correct choice.) 42) _____

A) an increase in the plate area and a decrease in the plate separation
B) an increase in the potential difference between the plates
C) a decrease in the potential difference between the plates
D) an increase in the charge on the plates
E) a decrease in the plate area and an increase in the plate separation

43) An ideal parallel−plate capacitor consists of two parallel plates of area A separated by a distance d. This capacitor is connected across a battery that maintains a constant potential difference between the plates. If the separation between the plates is now doubled, the magnitude of the charge on the plates will 43) _____

A) double.
B) be cut in fourth.
C) quadruple.
D) not change.
E) be cut in half.

44) An capacitor consists of two large parallel plates of area A separated by a very small distance d. This capacitor is connected to a battery and charged until its plates carry charges $+Q$ and $-Q$, and then disconnected from the battery. If the separation between the plates is now doubled, the potential difference between the plates will 44) _____
A) be cut in half.
B) quadruple.
C) be cut in fourth.
D) double.
E) not change.

45) The plates of a parallel-plate capacitor are maintained with constant voltage by a battery as they are pulled apart. What happens to the strength of the electric field between the plates during this process? 45) _____
A) It decreases.
B) It remains constant.
C) It increases.
D) It cannot be determined from the information given.

46) Doubling the potential across a given capacitor causes the energy stored in that capacitor to 46) _____
A) reduce to one-fourth.
B) double.
C) quadruple.
D) reduce to one-half.

47) Doubling the capacitance of a capacitor that is holding a constant charge causes the energy stored in that capacitor to 47) _____
A) decrease to one-fourth.
B) quadruple.
C) double.
D) decrease to one-half.

48) An ideal parallel-plate capacitor consists of two parallel plates of area A separated by a distance d. This capacitor is connected to a battery that maintains a constant potential difference across the plates. If the separation between the plates is now doubled, the amount of electrical energy stored on the capacitor will 48) _____
A) be cut in half.
B) not change.
C) quadruple.
D) double.
E) be cut in fourth.

49) An ideal parallel-plate capacitor consists of two parallel plates of area A separated by a distance d. This capacitor is connected to a battery and charged until its plates carry charges $+Q$ and $-Q$, and the battery is then disconnected. If the separation between the plates is now doubled, the electrical energy stored in the capacitor will 49) _____
A) be cut in fourth.
B) quadruple.
C) not change.
D) be cut in half.
E) double.

WorkBook Unit 10

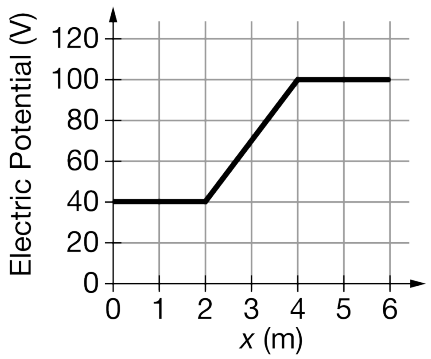
1. What is the acceleration of an object of mass 4.5 x 10^{-5} kg with a charge of 9 x 10^{-9}C under the influence of an electric field of magnitude 5000 V/m, assuming gravity is negligible?

(A) 0.05 m/s^2
(B) 0.5 m/s^2
(C) 0.9 m/s^2
(D) 1.0 m/s^2
(E) 5.0 m/s^2

2. An air-filled parallel-plate capacitor of capacitance *C* is connected to a battery and charged to a voltage *V*. The capacitor is then disconnected from the battery. If the distance between the capacitor plates is halved while the charge on the capacitor remains the same, which of the following is true?

(A) Neither *C* nor *V* will change.
(B) Both *C* and *V* will be halved.
(C) *C* will be doubled and *V* will be halved.
(D) *C* will be halved and *V* will be doubled.
(E) Both C and *V* will be doubled.

3.



The graph shows the electric potential along the $x-$axis in a region of space. What is the magnitude of the average electric field between $x = 2$ m and $x = 4$ m?

(A) $10\ \mathrm{V/m}$
(B) $30\ \mathrm{V/m}$
(C) $50\ \mathrm{V/m}$
(D) $70\ \mathrm{V/m}$

4. Charges *Q* of equal magnitude and opposite sign are on the plates of an isolated parallel-plate capacitor of capacitance *C*. The voltage between the plates is *V*, and the energy stored in the capacitor is *U*. If *Q* is doubled and *C* is not changed, how are *V* and *U* affected?

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(A)

V	U
Halved	Halved

(B)

V	U
Halved	Quartered

(C)

V	U
Not changed	Not changed

(D)

V	U
Doubled	Doubled

(E)

V	U
Doubled	Quadrupled

5.

Capacitance C (arbitrary units)	Plate Separation S (arbitrary units)
20	80.2
40	39.8
50	32.4
70	22.8

The table above shows experimental data for the capacitance C of a parallel-plate capacitor as a function of the plate separation S. The students who took the data neglected to record their units. Do the data support the accepted relationship between capacitance and plate separation? Why or why not?

WorkBook Unit 10

(A) No, because the data show that C is proportional to S.

(B) No, because the data show that C is proportional to S^2.

(C) Yes, because the data show that C is proportional to $1/S$.

(D) Yes, because the data show that C is proportional to $1/S^2$.

6.



The figures show two different capacitors made of identical parallel conducting plates. In Capacitor 1, the plates are separated by a distance d and a block of material with dielectric constant κ is inserted between the plates. In Capacitor 2, the plates are separated by a distance $2d$ and two identical blocks of material with dielectric constant κ are inserted between the plates. A student claims that the capacitance of Capacitor 2 is the same as that of Capacitor 1. Which of the following indicates whether the claim is correct or incorrect, and provides a valid justification?

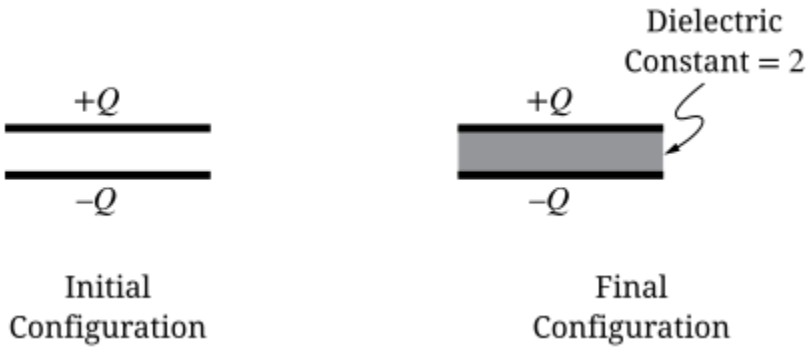
(A) Correct, because the effects of doubling the distance between the plates and doubling the volume of the dielectric cancel each other.

(B) Correct, because the capacitance is a property of the plates and the plates have not changed.

(C) Incorrect, because doubling the thickness of the dielectric material will cause the capacitance to double.

(D) Incorrect, because doubling the distance between the plates will reduce the capacitance by half.

7.



Two parallel plates have charges of magnitude Q and opposite sign, as shown in the figure labeled "Initial Configuration." The capacitor plates are isolated and there is an electric field of magnitude E_i in between the plates. An insulating slab of material with a dielectric constant of 2 is then inserted into the capacitor, completely filling the space between the plates, as shown in the figure labeled "Final Configuration." How does the magnitude E_f of the electric field between the plates in the final configuration compare to E_i, and what evidence or reasoning supports the claim?

WorkBook Unit 10

(A) $E_f < E_i$. Some of the charge from each plate moves into the dielectric, resulting in less overall charge on the plates.

(B) $E_f < E_i$. The dielectric induces a field in the opposite direction of the initial field, reducing the net field magnitude.

(C) $E_f = E_i$. Insulators have no effect on the electric field.

(D) $E_f = E_i$. A dielectric constant of 2 induces a field of magnitude $2E_1$ in the opposite direction of the initial field, for a net field of the same magnitude.

8.

	Area of Each Plate	Distance Between Plates	Potential Difference
Capacitor 1	$10\ \text{cm}^2$	$0.10\ \text{cm}$	$4.0\ \text{V}$
Capacitor 2	$10\ \text{cm}^2$	$0.20\ \text{cm}$	$2.0\ \text{V}$
Capacitor 3	$20\ \text{cm}^2$	$0.10\ \text{cm}$	$4.0\ \text{V}$

Three air-filled, parallel-plate capacitors have the dimensions shown in the table. The table also indicates the potential difference applied between the plates of each capacitor. Which of the following correctly ranks the electric potential energies U_1, U_2, and U_3 stored in capacitors 1, 2, and 3, respectively?

(A) $U_1 = U_2 = U_3$

(B) $(U_1 = U_3) > U_2$

(C) $U_3 > (U_1 = U_2)$

(D) $U_3 > U_1 > U_2$

9. Two parallel conducting plates, separated by a distance *d*, are connected to a battery of emf ε. Which of the following is correct if the plate separation is doubled while the battery remains connected?

(A) The electric charge on the plates is doubled.

(B) The electric charge on the plates is halved.

(C) The potential difference between the plates is doubled.

(D) The potential difference between the plates is halved.

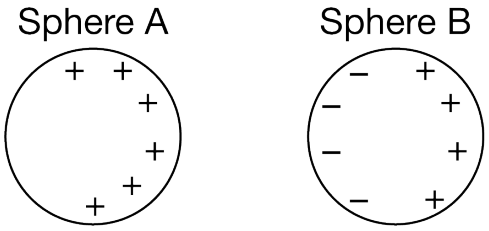
(E) The capacitance is unchanged.

Spheres A and B are identical in size and made from the same conducting material. Sphere A is attached to an insulating rod, and sphere B is suspended from an insulating thread. Sphere A initially has a charge of $+10\ \text{nC}$, and sphere B is initially uncharged.

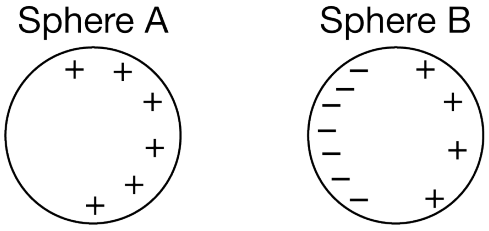
10. Which of the following best represents the distribution of charge on the surfaces of spheres A and B when sphere A is brought near sphere B?

WorkBook Unit 10

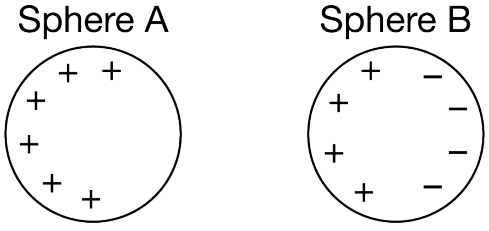
(A)



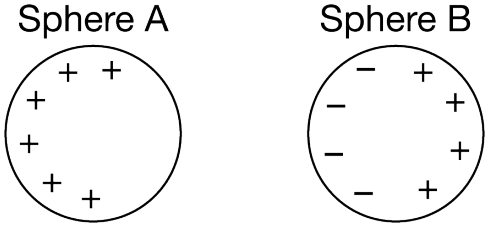
(B)



(C)



(D)



11. Spheres A and B are put briefly into contact. Which of the following best explains the resulting charge on spheres A and B?

(A) A charge that is equal in magnitude and sign to the charge on sphere A will be attracted onto sphere B, so both spheres will have a charge of $+10\ \mathrm{nC}$.

(B) A charge that is equal in magnitude and opposite in sign to the charge on sphere A will be attracted onto sphere B, so sphere A will still have a charge of $+10\ \mathrm{nC}$ and sphere B will have a charge of $-10\ \mathrm{nC}$.

(C) The net charge on identical conductors is shared equally when they are put into contact, so both spheres will have a charge of $+5\ \mathrm{nC}$.

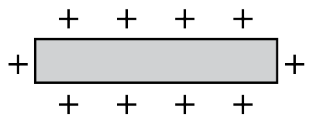
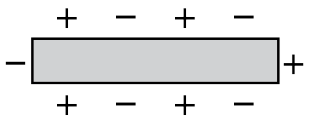
(D) Before sphere A and sphere B are brought into contact, sphere A will polarize sphere B, resulting in a net charge of $-5\ \mathrm{nC}$ on the near side of sphere B and $+5\ \mathrm{nC}$ on the far side of sphere B. When sphere A and sphere B are in contact, the charge $-5\ \mathrm{nC}$ on the near side of sphere B will move onto sphere A, resulting in sphere A having a net charge of $+5\ \mathrm{nC}$ and sphere B having a net charge of $-5\ \mathrm{nC}$.

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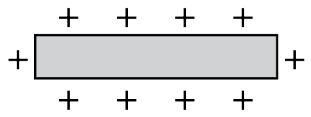
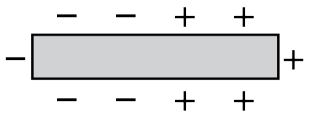
12. An insulating rod with uniformly distributed positive charge is brought near the end of a neutral conducting rod. Which of the following most accurately represents the distribution of charge on the insulating and conducting rods?

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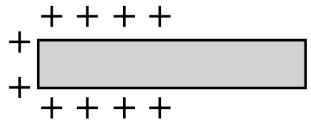
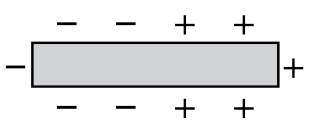
(A)

Insulating Rod	Conducting Rod
	

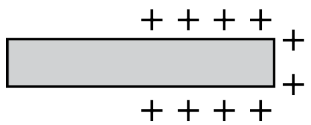
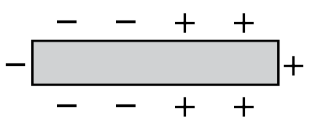
(B)

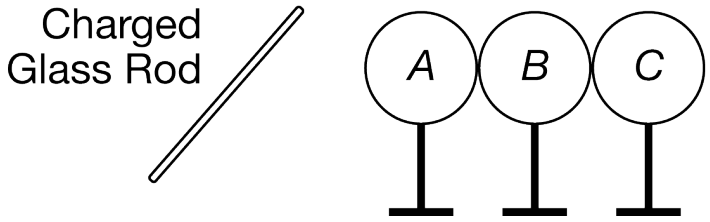
Insulating Rod	Conducting Rod
	

(C)

Insulating Rod	Conducting Rod
	

(D)

Insulating Rod	Conducting Rod
	



Three identical uncharged conducting spheres, A, B, and C, are on insulating stands. The spheres are arranged in a straight line, touching each other as shown in the figure above. A positively charged glass rod is held near the spheres, resulting in an induced charge of $-Q$ on sphere A. Spheres B and C are moved far away from sphere A and the rod while remaining in contact with each other, and are then separated.

13. The rod is now moved far away from sphere A. What is the charge remaining on sphere A ?

(A) $-Q/2$
(B) $-Q$
(C) $+Q$
(D) Zero

14. After spheres B and C are separated, what is the charge on sphere C ?

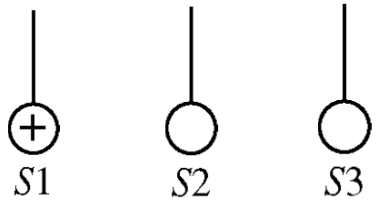
(A) $+Q$
(B) $+Q/2$
(C) $-Q/2$
(D) Zero

---

15. A student is performing an experiment that involves the charge on a metal cup that is attached to an uncharged electroscope. A charged rod is lowered into the cup without touching the sides of the cup. As a result the leaves of the electroscope separate. The rod is then removed from the cup, and the leaves return to their unseparated position. The student repeats the procedure, but this time the electroscope is grounded and the ground is removed before the rod is removed from the cup. The leaves then remain separated. What can be concluded about the charge on the separated leaves of the electroscope?

(A) The leaves have a positive charge in the first trial and a negative charge in the second trial.
(B) The leaves have a negative charge in the first trial and a positive charge in the second trial.
(C) The leaves have a positive charge in both trials.
(D) The leaves have a charge in each trial, but the sign of the charge cannot be determined.

16.



Three identical conducting spheres, *S*1, *S*2, and *S*3, are supported by insulating thread, as shown above. Initially, sphere *S*1 has a net positive charge and the other two spheres are uncharged. Spheres *S*1 and *S*2 are brought into contact and then separated. Next, spheres *S*2 and *S*3 are brought into contact and then separated. Which of the following shows the signs of the final net charges on the spheres?

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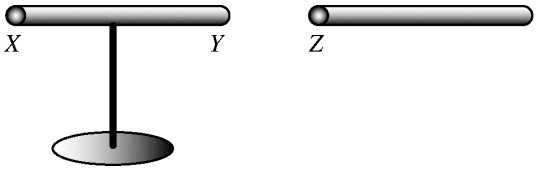
(A) ⊕ ⊕ ⊕

(B) ⊕ ○ ○

(C) ○ ○ ⊕

(D) ○ ⊕ ⊕

17.



In the figure above, the initially uncharged insulating rod on the left is free to rotate on an insulating stand. End *X* is then rubbed with a piece of fur. End *Z* of a second insulating rod is also rubbed with fur and then brought near the first rod. As the second rod is moved around, it is found that end *Y* of the first rod is attracted to end *Z,* and end *X* is repelled. Which of the following correctly describes the signs of the charges on ends *X* and *Y*, if any?

(A)

End X	End Y
Opposite Z's charge	Same as Z's charge

(B)

End *X*	**End Y**
Opposite *Z*'s charge	No net charge

(C)

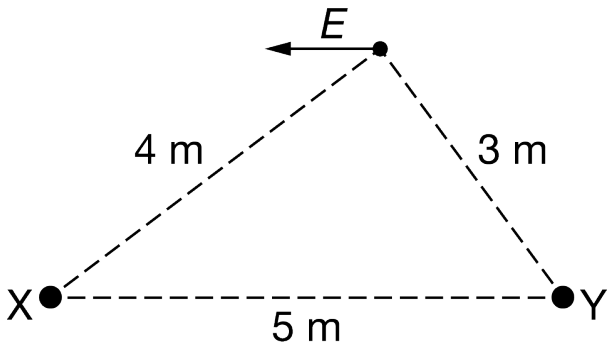
End *X*	**End Y**
Opposite *Z*'s charge	No net charge

(D)

End *X*	**End Y**
Same as *Z*'s charge	No net charge

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18.



Two charged spheres, X and Y, are held fixed at two vertices of a triangle, as shown above. The direction of the electric field E at the third vertex due to the two spheres is also shown. Which of the following correctly indicates the sign of the charges on the spheres?

(A)

Sphere X	Sphere Y
Positive	Positive

(B)

Sphere X	Sphere Y
Positive	Negative

(C)

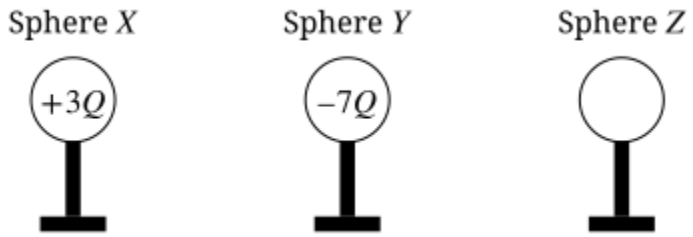
Sphere X	Sphere Y
Negative	Positive

(D)

Sphere X	Sphere Y
Negative	Negative

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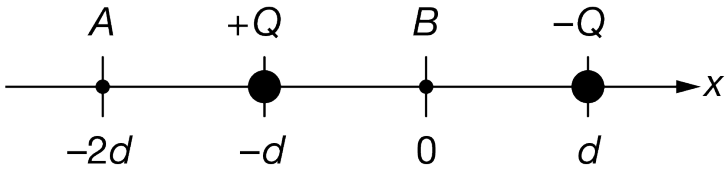
19.



The figure above shows three identical solid conducting spheres on insulating stands. Spheres X and Y initially have the charges indicated. First, Sphere X is momentarily touched to Sphere Y and then moved far away. Next, Sphere Y is momentarily touched to Sphere Z. If the final charge on Sphere Z is zero, what was its initial charge?

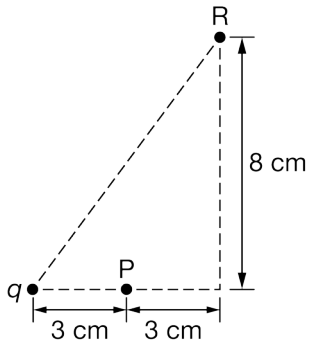
(A) $-4Q$
(B) $-2Q$
(C) $+2Q$
(D) $+4Q$

20.



Two small spheres with charges of $+Q$ and $-Q$ are held fixed along the x-axis, as shown above. Points A and B on the axis are also shown. Which of the following correctly compares the electric potential V and the magnitude E of the electric field at the two points?

(A) $V_A > V_B$ and $E_A > E_B$
(B) $V_A > V_B$ and $E_B > E_A$
(C) $V_B > V_A$ and $E_A > E_B$
(D) $V_B > V_A$ and $E_B > E_A$



A sphere with charge q is fixed at the bottom left corner of the right triangle shown in the figure. Points P and R are at the locations shown.

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21. A test charge is placed at point P, and the charge-sphere system has potential energy U_0. If the test charge is then moved to point R, what will the potential energy of the charge-sphere system be?

(A) $\frac{10}{3}U_0$

(B) $\frac{8}{3}U_0$

(C) $\frac{3}{8}U_0$

(D) $\frac{3}{10}U_0$

22. If the magnitude of the electric field due to the sphere at point P is E_P, what is the magnitude of the field at point R, and why?

(A) $\frac{9E_\mathrm{P}}{100}$, because the electric field is inversely proportional to the square of the distance from the charge.

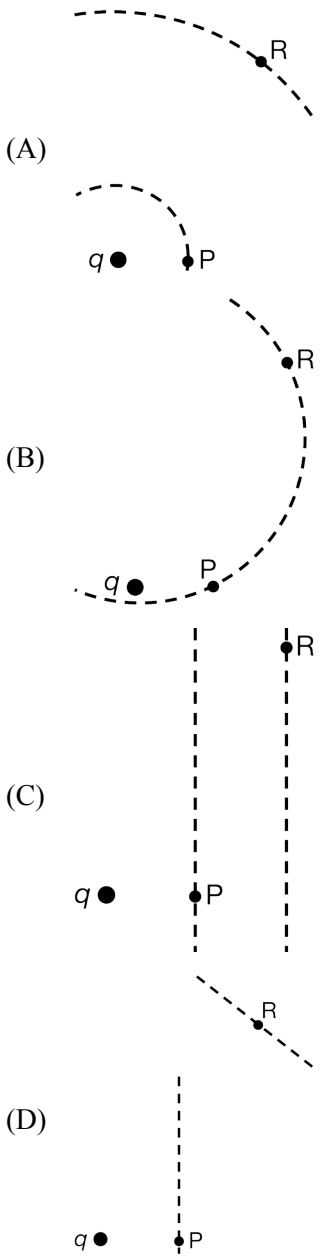
(B) $\frac{9E_\mathrm{P}}{25}$, because the electric field is inversely proportional to the square of the distance from the charge.

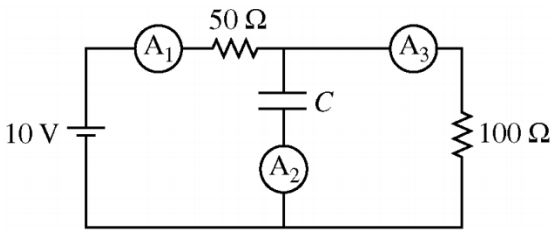
(C) E_P, because the electric field is proportional to the magnitude of the charge.

(D) $\frac{10E_\mathrm{P}}{3}$, because the electric field is proportional to the distance from the charge.

23. Which of the following diagrams best shows isolines of electric potential in the vicinity of points P and R?

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The figure above shows a 10 V battery connected in a circuit with two resistors, a parallel-plate capacitor of capacitance *C*, and three ammeters. The circuit has been connected for a long time.

24. Assume the capacitance C of the capacitor is known. The gap between the capacitor's plates is now filled with an unknown insulating material, and the circuit is again left connected for a long time. Measuring which of the following provides enough information to determine the unknown material's dielectric constant k ?

(A) The potential difference across the capacitor and the current in each branch of the circuit

(B) The potential difference across the capacitor and the charge on one of its plates

(C) The potential difference across the capacitor and the charge on the dielectric material

(D) The separation between the capacitor plates and the area of the plates

25. Let I_i be the current in ammeter A_i . Which of the following correctly ranks the currents in the three ammeters?

(A) $(I_1 = I_3) > I_2$

(B) $I_1 > (I_3 = I_2)$

(C) $I_1 > I_3 > I_2$

(D) $I_3 > (I_1 = I_2)$

26. Which of the following is correct about the voltage V_C across the capacitor?

(A) $V_C = 0$

(B) $0 < V_C < 10$ V

(C) $V_C = 10$ V

(D) $V_C > 10$ V

---

27. A small amount of charge is placed on both an isolated conducting sphere and an isolated insulating sphere. For both spheres, the charge is added at a small area at the top of the sphere. After a few seconds, where on each of the spheres is the charge that was added?

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(A)

Conducting	Insulating
At the top	At the top

(B)

Conducting	Insulating
At the top	Spread over the surface

(C)

Conducting	Insulating
Spread over the surface	At the top

(D)

Conducting	Insulating
Spread over the surface	Spread over the surface

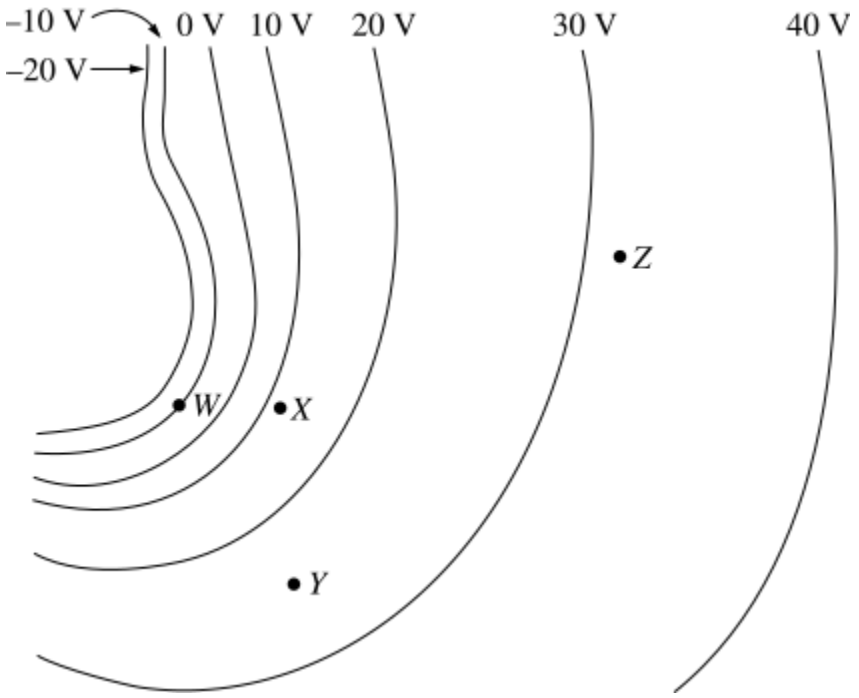
28. Which of the following describes a difference in the behavior of an electrically conducting sphere and that of an insulating sphere?

(A) A conducting sphere can be charged by friction, but an insulating sphere cannot.

(B) An uncharged object can be charged by touching it to a charged conducting sphere, but not by touching it to a charged insulating sphere.

(C) When a conducting sphere is brought near a positively charged object, some of the sphere's electrons move closer to that object. No polarization occurs in the atoms of an insulating sphere.

(D) Excess charge placed on a conducting sphere becomes distributed over the entire surface of the sphere. Excess charge placed on an insulating sphere can remain where it is placed.



The figure above shows equipotential lines in a region of space. The potential of each of the lines is indicated. Four points within the region are labeled.

29. A negative point charge is placed at point X. Which of the following best indicates the direction of the electric force exerted on the particle?

(A) 

(B) 

(C) 

(D) 

30. Which of the following lists the four points in order from greatest magnitude of the electric field to least?

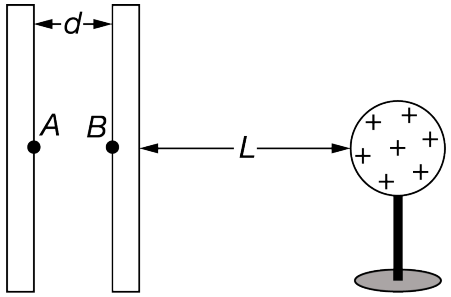
(A) $W,\ X,\ Y,\ Z$

(B) $Z,\ W,\ Y,\ X$

(C) $Y,\ X,\ W,\ Z$

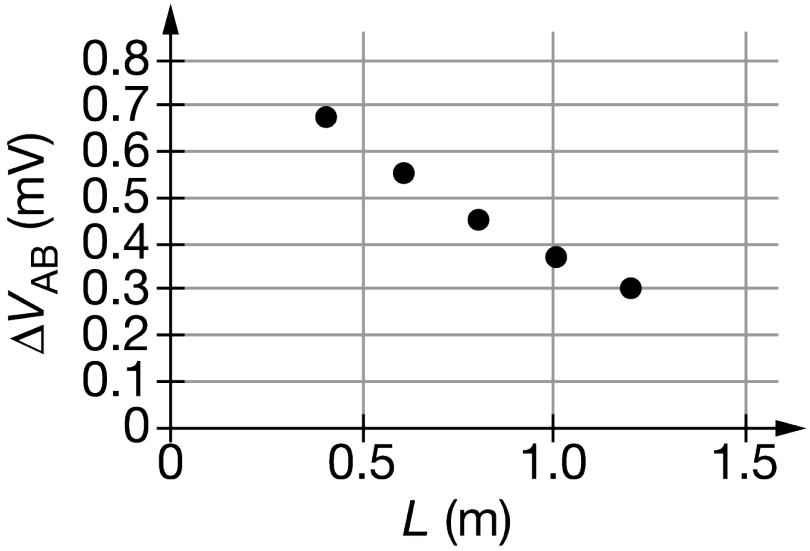
(D) $Z,\ Y,\ X,\ W$

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Students perform an experiment using the setup shown in the figure. A sphere with uniform charge distribution on an insulating stand is placed a distance from parallel uncharged conducting plates separated by a fixed distance d. Points A and B are labeled.

31.



The students predict that the closer the sphere gets to the plates, the larger the electric field between points A and B . They measure the potential difference ΔV_{AB} between points A and B for several values of L, and the results are shown in the graph. Do the results support their prediction, and why or why not?

(A) Yes. The distance between the plates is fixed, so the potential difference between points A and B is proportional to the electric field. The increasing potential difference with decreasing distance to the sphere indicates an increasing electric field.

(B) Yes. The electric field is given by $\frac{\Delta V_{AB}}{L}$, and this ratio increases for decreasing values of L. Therefore, the electric field increases as the sphere gets closer.

(C) No. The electric field depends on the distance between the points. The prediction cannot be confirmed without knowing the numerical value of d.

(D) No. The electric field between two plates is uniform. It will not change regardless of the potential difference between the points.

32. Which of the following best explains how and why the charges will distribute in the plates?

(A) The sphere will induce the net charge of the right plate to become negative, which will in turn induce the net charge of the left plate to become positive.

(B) The sphere will initially induce the net charge of the right plate to become negative, but then electrons from the right plate will conduct to the left plate until the net charge of the right plate returns to zero.

(C) The sphere will induce charge separation in the right plate, which will in turn induce charge separation in the left plate, and the net charges of the plates will remain zero.

(D) The sphere will induce charge separation in the right plate, which will in turn induce charge separation in the left plate, and the net charge of the right plate will be slightly negative, while the net charge of the left plate will be slightly positive.

33. A small particle with a charge of magnitude q is released from rest at point B and moves until it hits point A. If the potential difference ΔV between the plates is known, which of the following correctly explains a method to determine the increase in kinetic energy of the particle?

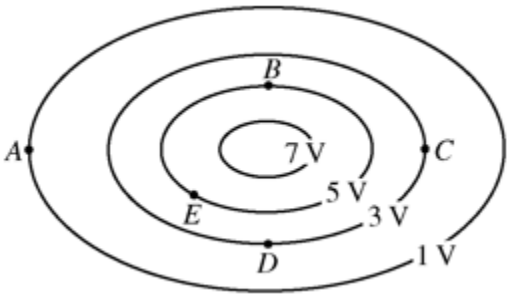
(A) Use $q\Delta V$ to determine the decrease in the potential energy of the plates-particle system. That is equal to the increase in the kinetic energy of the particle.

(B) Use $\frac{q\Delta V}{d}$ to determine the decrease in the potential energy of the plates-particle system. That is equal to the increase in the kinetic energy of the particle.

(C) Use $q\Delta V$ to determine the work done by an external force on the plates-particle system. That is equal to the increase in the kinetic energy of the particle.

(D) Use $\frac{q\Delta V}{d}$ to determine the work done by an external force on the plates-particle system. That is equal to the increase in the kinetic energy of the particle.

34.



The elliptical equipotential lines resulting from a particular charge distribution are shown in the figure above. At which of the labeled points will a proton experience an electric force directed toward the top of the page?

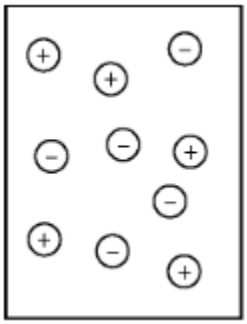
(A) *A*

(B) *B*

(C) *C*

(D) *D*

(E) *E*

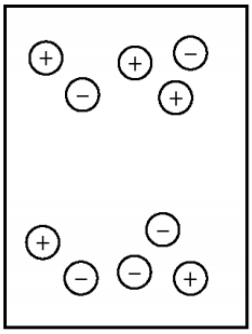


A very hot gas composed of equal numbers of positive and negative ions is in a closed, thermally insulated container and is in thermal equilibrium at temperature T_0. The figure above represents the initial distribution of the ions. A strong uniform electric field directed toward the bottom of the page is now created in and around the container, and the gas is allowed to reach a new thermal equilibrium at temperature T_1.

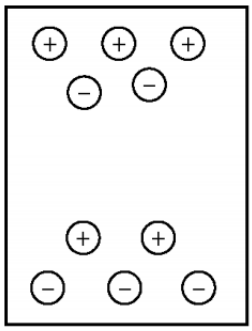
35. Which of the following best represents a possible distribution of the ions at temperature T_1?

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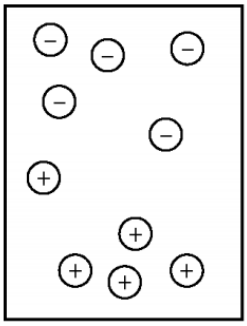
(A)



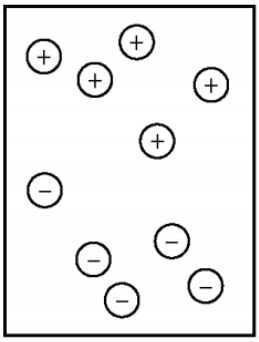
(B)



(C)



(D)



36. The electric field is now turned off, and the gas is allowed to reach a final equilibrium temperature T_2. How does T_2 compare to T_1, and why?

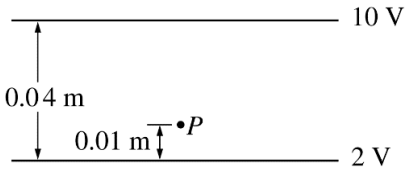
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(A) T_2 is less, because the electric potential energy added by the field to separate the ions is gone once the field is turned off.

(B) T_2 is less, because pairs of positive and negative ions bond to become molecules once the field is turned off, and energy is required to create the bonds.

(C) T_2 is greater, because work is done on the ions in the process of turning off the field.

(D) T_2 is greater, because the electric potential energy of the separated ions is converted to kinetic energy as they mix and collide.



Two large, flat, parallel, conducting plates are 0.04 m apart, as shown above. The lower plate is at a potential of 2 V with respect to ground. The upper plate is at a potential of 10 V with respect to ground. Point *P* is located 0.01 m above the lower plate.

37. The electric potential at point *P* is

(A) 10 V

(B) 8 V

(C) 6 V

(D) 4 V

(E) 2 V

38. The magnitude of the electric field at point *P* is

(A) 800 V/m

(B) 600 V/m

(C) 400 V/m

(D) 200 V/m

(E) 100 V/m



An isolated, hollow aluminum sphere is positively charged. A cross section through the center of the sphere is shown above.

39. The magnitude of the electric field is greatest at point

(A) *A*
(B) *B*
(C) *C*
(D) *D*
(E) *E*

40. The direction of the electric field is correctly indicated in which of the following?

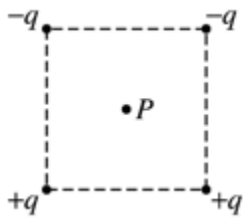
I. To the left at point *A*

II. To the right at point *D*

III. To the right at point *E*

(A) I only
(B) II only
(C) III only
(D) I and III
(E) II and III

41.

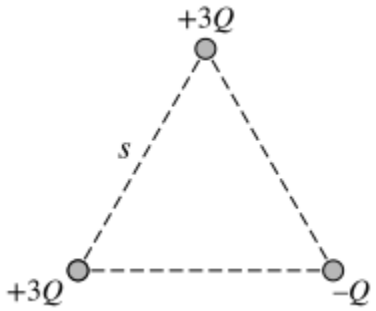


The four charged particles shown above are held fixed at the corners of a square. The electric field at point *P*, the center of the square, is directed

(A) toward the left
(B) toward the right
(C) out of the plane of the page
(D) toward the bottom of the page
(E) toward the top of the page

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42.



Three charged particles are held in place at the vertices of an equilateral triangle of side length s, as shown in the figure. How much work must be done by external forces in order to move the particles from rest very far apart into the configuration shown in the figure?

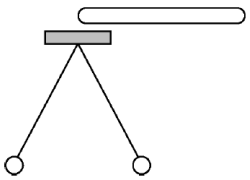
(A) $+\frac{3kQ^2}{s}$

(B) $+\frac{5kQ^2}{s}$

(C) $+\frac{9kQ^2}{s}$

(D) $+\frac{15kQ^2}{s}$

43.



An initially uncharged electroscope consists of two thin, 50 cm long conducting wires attached to a cap, with a 25 g conducting sphere attached to the other end of each wire. When a charged rod is brought close to but not touching the cap, as shown above, the spheres separate a distance of 30 cm. What can be determined about the induced charge on each sphere from this information?

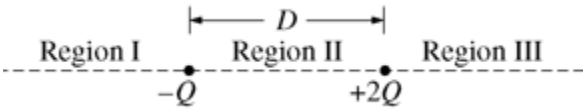
(A) The magnitude but not the sign

(B) The sign but not the magnitude

(C) Both the magnitude and the sign

(D) Nothing can be determined about the induced charges.

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A particle with a charge -Q is located a distance D from a particle with a charge +2Q, as shown in the figure above.

44. In which of the regions is there a location, other than at infinity, at which the electric potential is zero?

(A) Region I only
(B) Region II only
(C) Region III only
(D) Region I and region II
(E) Region II and region III

45. Which of the following best describes the location of a point, other than at infinity, at which the net electric field created by the two charges is equal to zero?

(A) In region I
(B) In region II, closer to -Q
(C) In region II, midway between the charges
(D) In region II, closer to +2Q
(E) In region III

---

46. When charges Q of equal magnitude and opposite sign are placed on the plates of a parallel-plate capacitor of capacitance C, the voltage between the plates is V and the energy stored in the capacitor is U. If C is doubled and Q is not changed, how are V and U affected?

WorkBook Unit 10

(A)

V	U
Halved	Halved

(B)

V	U
Halved	Quartered

(C)

V	U
Not changed	Not changed

(D)

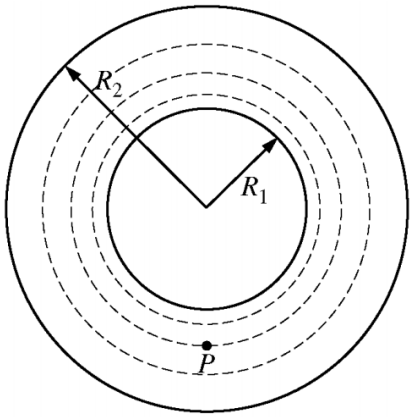
V	U
Doubled	Doubled

(E)

V	U
Doubled	Quadrupled

WorkBook Unit 10

47.



The figure above represents two charged, concentric conducting shells with radii R_1 and R_2 . The space between the shells is filled with air. The dashed lines represent equipotentials. The electric field just outside the inner shell has magnitude E_1, and the electric field just inside the outer shell has magnitude E_2. Which of the following statements is true of the magnitude E of the electric field at point P?

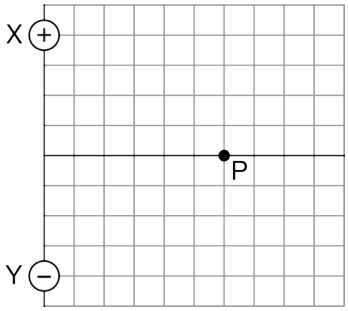
(A) $E = 0$

(B) $E_1 > E > E_2$

(C) $E = E_1$

(D) $E_1 < E < E_2$

48.



Two charged particles, X and Y, are fixed on a grid as shown in the figure. The particles have the same amount of charge but with opposite signs, as shown. Which of the following best represents the direction of the net electric field at point p due to the two particles?

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(A) 

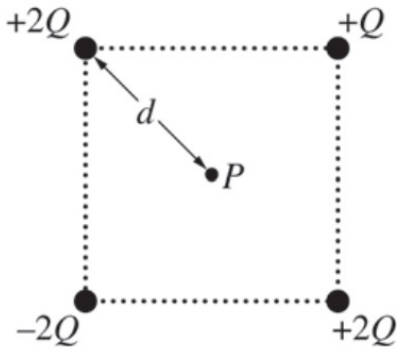
(B) 

(C) 

(D)

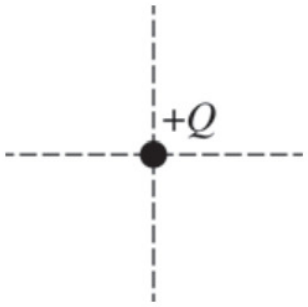
WorkBook Unit 10

49.



The figure above represents four objects, with charges as shown, that are held in place at the corners of a square. Point *P* is at the center of the square, a distance d from each of the objects. Express all algebraic answers to the following in terms of *Q, d*, and physical constants.

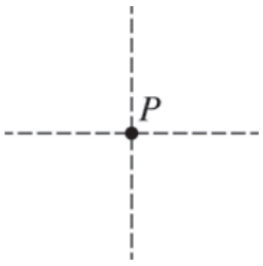
a. On the dot below, draw an arrow that represents the direction of the net electric force exerted on the object with charge +*Q* by the other three objects.



b. i. Calculate the magnitude of the electric field at point *P* due to all four objects. On the dot below, draw an arrow to indicate the direction of the net field at point *P*

Draw Arrow

Calculate Electric Field

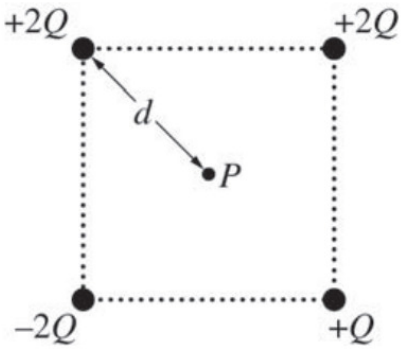


ii. Calculate the electric potential at point *P* due to all four objects

c. In a coherent, paragraph-length response, briefly describe the meaning of electric potential energy and explain qualitatively how electric potential energy can be related to work. Also explain qualitatively how

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the electric potential energy of the four-object system would change if the $+Q$ and $+2Q$ objects on the right side of the square now switch positions as shown in the figure below. Support your explanation using appropriate physics principles.



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50. This question is a long free-response question. Show your work for each part of the question. (15 points, suggested time 25 minutes)



The figure above shows two conducting spheres that are very far apart. The spheres have the same radius R and the different negative charges shown. Outside the spheres, the charge on each sphere can be treated as if it were at the center of the sphere.

(a) Compare the magnitude of the electric field at the surface of sphere A to the magnitude of the electric field at the surface of sphere B. Justify your answer.

Consider two scenarios, one in which an individual electron is moved from far away to a position outside one of the spheres at a distance d from the surface, and the other in which an individual proton is moved in the same manner.

(b) An external force must do non-zero work to move each particle to the distance d. Indicate whether the sign of the work is the same for both the electron and the proton. Justify your answer using the concepts of electric fields or forces.

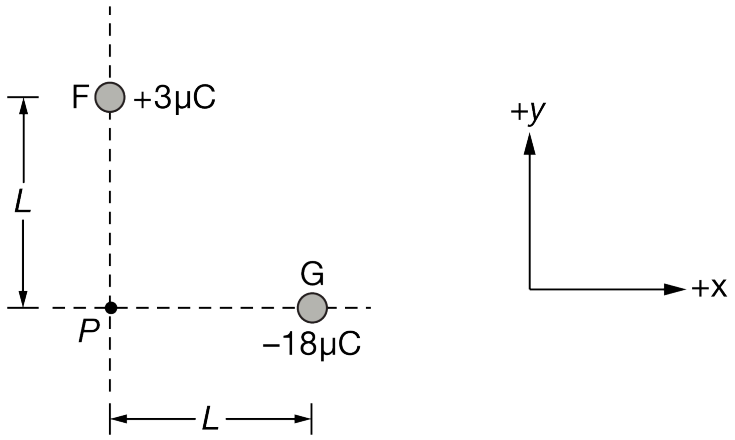
(c) Compare the magnitude of the work required to move an electron to a distance d from the surface of sphere A to the magnitude of the work required to move an electron to a distance d from the surface of sphere B. Justify your answer using the concepts of electric fields or forces.

(d) When an electron with charge of magnitude e is brought from far away to a distance d from the center of one of the spheres, the electron-sphere system experiences a change in potential energy of magnitude ΔU. The equation for this change is $\Delta U = \frac{keQ}{d}$, where $k = \frac{1}{4\pi\varepsilon_0}$. Use this equation to compare the magnitude of the

change in potential energy of the electron-sphere A system to that of the electron-sphere B system when the electron is brought from far away to a distance d from the center of each sphere. Explain your reasoning.

(e) Is your answer from part (d) consistent with your answer from part (c)? Justify your answer using physics principles.

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Two point charges, F and G, have the charges shown in the figure above. Each charge is a distance L from point P. The electric field at point P due to charge F has magnitude E_{F}.

(f) Describe the magnitude and direction of the net electric field at point P due to both charges in terms of E_{F} and with respect to the coordinate axes shown. If the net electric field is not directed along an axis, indicate to which axis the direction of the net electric field is closest. If the direction of the net electric field is exactly in the middle (45° from each axis), state this explicitly. Explain your reasoning using relevant physics principles.

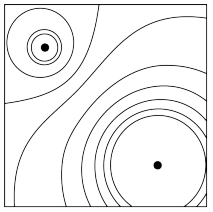


Figure 1

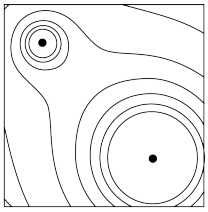


Figure 2

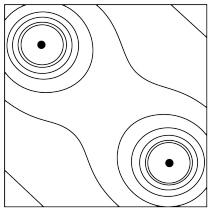


Figure 3

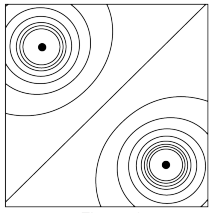


Figure 4

(g) The diagrams shown above depict isolines of electric potential. Which diagram best depicts the isolines of electric potential of charges F and G? Explain your reasoning.

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51. This question is a long free-response question. Show your work for each part of the question.

(12 points, suggested time 25 minutes)

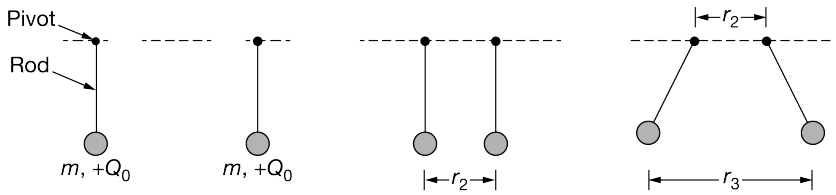


Figure 1 Figure 2 Figure 3

Note: Figure 1 not drawn to scale.

Two insulating spheres with mass m and charge $+Q_0$ are near Earth's surface, suspended vertically from identical light, stiff rods with pivots at the top, as shown in Figure 1. The spheres are initially far apart. The pivots are locked so the rods remain vertical while the spheres are moved together by external forces until the spheres are a distance r_2 from each other, as shown in Figure 2. Finally, the pivots are held a distance r_2 apart and unlocked, and the rods are allowed to swing out until the spheres momentarily come to rest a distance r_3 apart, as shown in Figure 3. The pivots are always at the same height.

(a) Consider the system that includes only the two spheres, with Earth external to the system.

i. Write an expression for the work done moving the spheres horizontally from far apart to a separation of r_2. Express your answer in terms of Q_0, r_2, and physical constants, as appropriate.

ii. The total energy of the two-sphere system (with Earth still external to the system) changes as the system goes from the configuration in Figure 2 to the configuration in Figure 3. Explain how the energy changes in terms of the forces exerted on the system.

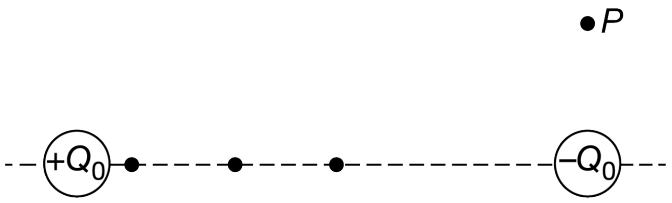
(b) Now consider the system that contains both Earth and the two spheres. What is the work done on this new system for the changes in sphere position described in parts (a)(i) and (ii) ? Explain your reasoning.

Part (a)(i) Part (a)(ii)

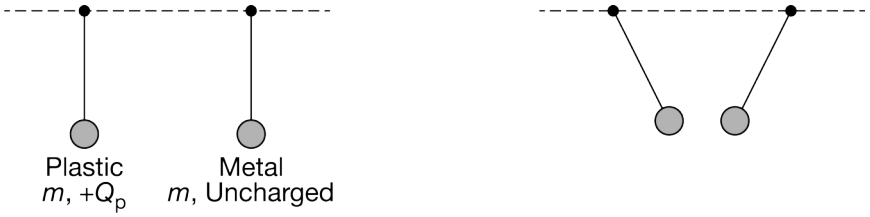
(c) The rods are now held fixed at the positions shown in Figure 3, and negative charge is added to the right sphere so its net charge is $-Q_0$. The circles in the figure below represent the spheres. On the figure, do the following.

- Draw a single vector originating on point P that represents the net electric field at that point.
- For each of the dots on the horizontal daashed line, draw an isoline of electric potential that goes through the dot.

WorkBook Unit 10

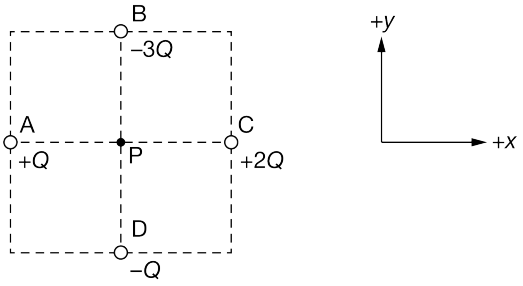


(d) In a new situation, the two spheres still have the same mass m, but the left sphere is made of plastic and has charge $+Q_{\mathrm{p}}$ and the right sphere is made of metal and is uncharged. The spheres are brought near but do not touch, as shown below on the left. When the rods are allowed to pivot, the spheres move toward each other, as shown on the right. In terms of charge distributions, explain why the force between the spheres is attractive.



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52. This question is a short free-response question. Show your work for each part of the question. (10 points, suggested time 15 minutes)

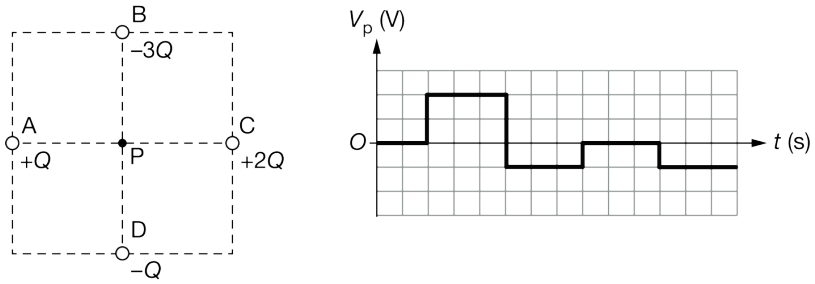


Four point charges, A, B, C, and D, are located at the centers of the sides of a square, as shown above. The charges have the values indicated in the figure. Each charge is held in place by an external force. Point P is at the center of the square.

(a) The magnitude of the electric field at point P due to charge A is E. Describe the magnitude and direction of the net electric field resulting from all four charges at point P in terms of E and the coordinate axes shown. Briefly explain your reasoning.

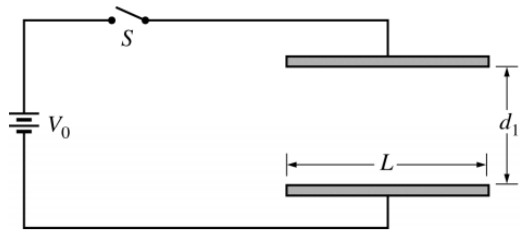
(b) The magnitude of the forces required to hold charges A and C in place are F_{A} and F_{C} respectively. Is F_{A} greater than, less than, or equal to F_{C} ? Briefly justify your answer.

The configuration of charges was assembled by placing individual charges one at a time onto the square. The configuration is shown again below. The graph below shows the electric potential V_{P} at point P as the charges were placed onto the square.



(c) In a clear, coherent paragraph-length response, describe the order in which the charges were assembled into their final configuration. Justify your claims by citing relevant physics principles and features of the graph.

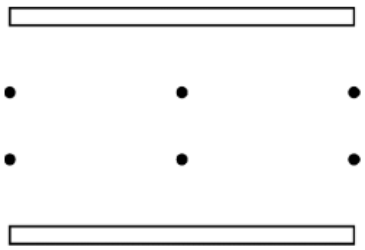
53.



An uncharged air-filled capacitor has square plates of side L whose separation can be varied. The initial plate separation is d_1, which is comparable to L. The capacitor is connected to a battery with potential difference V_0,and a switch S, as shown above. Assume the dielectric constant of the air is 1.0

a. The switch is closed and remains closed for a long time

i. On the diagram below, draw a vector at each dot to represent the electric field created by the charge on the capacitor plates



.

ii.The plates are now moved to a new separation $d_2 << d_1$ and allowed to reach equilibrium. Calculate the magnitude of the charge on each plate of the capacitor in terms of V_0, L, d_2, and physical constants, as appropriate.

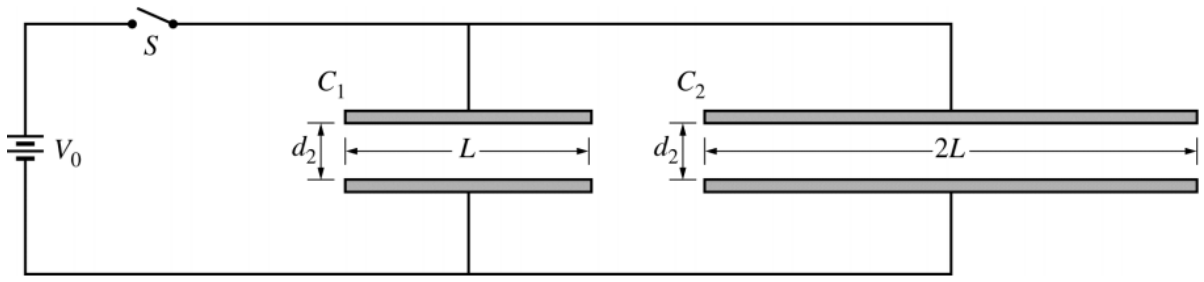
b. The plates are pulled apart to a separation of $2d_2$, where $2d_2 << d_1$.

i. Suppose the switch remains closed while the plates are pulled apart. Compare the magnitude of the new electric field between the plates when their separation is $2d_2$ to the magnitude of the field for plate separation d_2. Support your comparison using appropriate physics principles.

ii. Suppose the switch is opened before the plates are pulled apart. Compare the magnitude of the new electric field between the plates when their separation is $2d_2$ to the magnitude of the field for plate separation d_2 . Support your claim using appropriate physics principles.

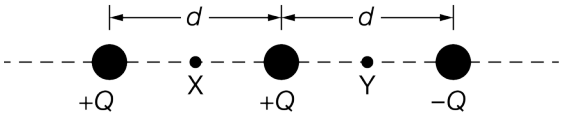
c. The new circuit shown below is now assembled. The plates of the original capacitor, C_1 , are at separation d_2 . A second capacitor, C_2 , with square plates of side 2L and plate separation d_2 , is connected in parallel with the

original capacitor. Both capacitors are initially uncharged. The switch is closed and the circuit reaches equilibrium. Let Q_0 be the magnitude of the charge on each plate of capacitor C_1 in the previous circuit, as determined in part (a)(ii). In terms of Q_0 calculate the magnitude of the charge on each plate of the two capacitors in the new circuit.



WorkBook Unit 10

54. This question is a long free-response question. Show your work for each part of the question. (15 points, suggested time 25 minutes)



Three point charges of equal magnitudes but different signs are arranged in a line as shown. The distance d between each charge is the same. Point X is exactly halfway between the two positive charges, and point Y is exactly halfway between the negative charge and the positive charge in the middle of the line.

(a) Compare the magnitude of the electric field at point X to the magnitude of the electric field at point Y. Justify your answer.

Consider two scenarios, one in which an individual proton is moved from far away to point X, and the other in which an individual proton is moved from far away to point Y.

(b) An external force must do non-zero work to move the proton to point X. Indicate whether the work is positive or negative. Justify your answer using the concepts of electric fields or forces.

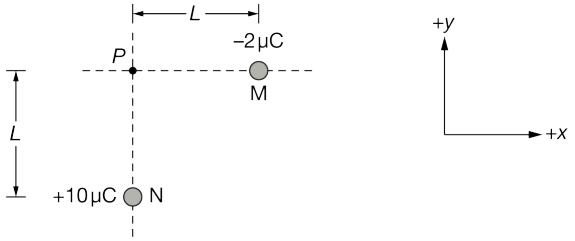
(c) The work required to move an individual proton from far away to point X is W_{X}. The work required to move an individual proton from far away to point Y is W_{Y}. Compare the magnitudes of W_{X} and W_{Y}. Justify your answer using the concept of electric potential.

When a proton with a charge of magnitude e is brought from far away to either point X or point Y, the proton-charges system experiences a change in potential energy of magnitude ΔU_{X} and ΔU_{Y}, respectively. Expressions that represent the change in potential energy as a proton is brought from far away to either point X or point Y are given below.

Expression 1	Expression 2
$\Delta U = \frac{2keQ}{3d}$	$\Delta U = \frac{4keQ}{d} - \frac{2keQ}{3d}$

(d) Which expression represents the change in potential energy ΔU_{Y}? Justify your answer using physics principles.

(e) Is your answer from part (d) consistent with your answer from part (c)? Justify your answer using physics principles.



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Two point charges, M and N, have the charges shown in the figure above. Each charge is a distance L from point P. The electric field at point P due to charge M has magnitude E_{M}.

(f) Describe the magnitude and direction of the net electric field at point P due to both charges in terms of E_{M} and with respect to the coordinate axes shown. If the net electric field is not directed along an axis, indicate to which axis the direction of the net electric field is closest. If the direction of the net electric field is exactly in the middle (45° from each axis), state this explicitly. Explain your reasoning using relevant physics principles.

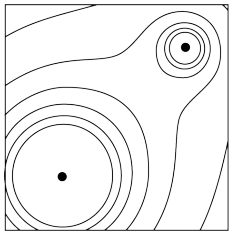


Figure 1

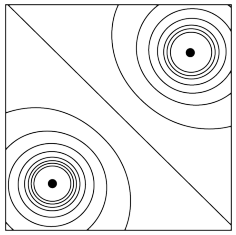


Figure 2

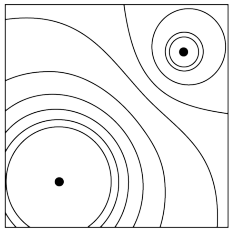


Figure 3

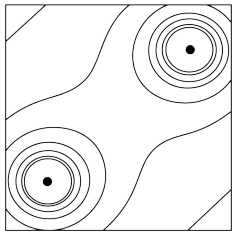


Figure 4

(g) The diagrams shown above depict isolines of electric potential. Which diagram best depicts the isolines of electric potential of charges M and N? Explain your reasoning.

AP PHYSICS 2

UNIT 11

Electric Circuits

15–18%
AP EXAM WEIGHTING

~12–20
CLASS PERIODS

Conceptual Questions

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

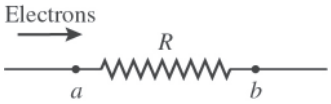
1) If a quantity you calculated has units of A • s, what is that quantity? 1) _____
A) capacitance
B) resistance
C) charge
D) potential
E) resistivity

2) When a current flows through a metal wire, the moving charges are 2) _____
A) negative metal ions.
B) only protons.
C) only electrons.
D) positive metal ions.
E) both protons and electrons.

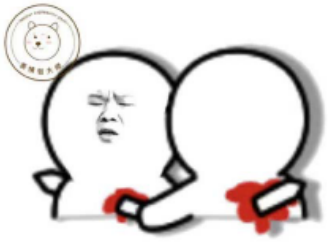
3) When a current flows through an ionic liquid such as salty water, the moving charges are 3) _____
A) both positive and negative ions.
B) only electrons.
C) only protons.
D) only negative ions.
E) only positive ions.

4) If a quantity you calculated has units of $\frac{\text{kg} \cdot \text{m}^2}{\text{s}^2 \cdot \text{C}}$, what is that quantity? 4) _____
A) resistance
B) charge
C) potential
D) resistivity
E) capacitance

5) The figure shows electrons passing through a resistor. The arrow shows the direction in which the electrons are moving. Which of the following statements are correct? (There could be more than one correct choice.) 5) _____

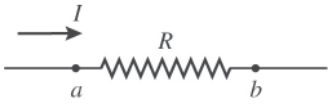


A) The speed of the electrons at point *b* is the same as it is at point *a*.
B) The electrons are moving slower at point *b* than at point *a*.
C) The electrons are losing electric potential energy as they move through the resistor from *a* to *b*.
D) The electric potential is higher at point *b* than at point *a*.
E) The electric potential is lower at point *b* than at point *a*.



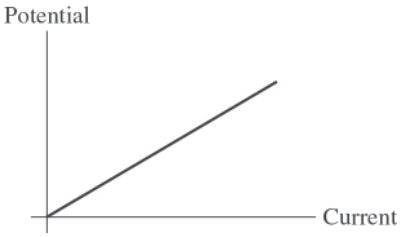
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6) The figure shows conventional current passing through a resistor. The arrow shows the direction in which this conventional current is flowing. Which of the following statements are correct? (There could be more than one correct choice.) 6) _____



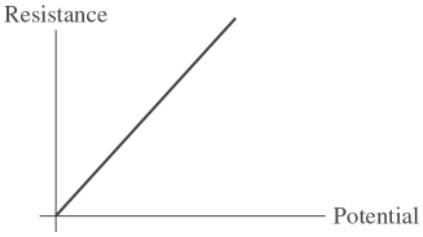
A) The electric potential is lower at point *b* than at point *a*.
B) The current at point *b* is the same as the current at point *a*.
C) The electric potential at point *b* is the same as it is at point *a*.
D) The electric potential is higher at point *b* than at point *a*.
E) The charges are moving slower at point *b* than at point *a*.

7) For the graph shown in the figure, what physical quantity does the slope of the graph represent for ohmic material? 7) _____



A) power
B) resistance
C) resistivity
D) 1/(resistance)
E) 1/(resistivity)

8) For the graph shown in the figure, what physical quantity does the slope of the graph represent for ohmic material? 8) _____

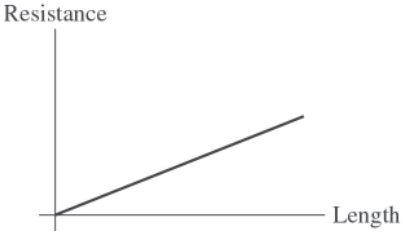


A) 1/(current)
B) resistivity
C) power
D) current
E) 1/(resistivity)

9) Copper wire #1 has a length *L* and a radius *b*. Copper wire #2 has a length 2*L* and a radius 2*b*. Which statement about the resistance across the ends of the wires is true? 9) _____

A) The resistance of wire #1 is four times higher than that of wire #2.
B) The resistance of wire #1 is equal to that of wire #2.
C) The resistance of wire #1 is half that of wire #2.
D) The resistance of wire #1 is twice as high as that of wire #2.

10) The figure shows a graph of the resistance of a wire as a function of its length. What physical quantities does the slope of this graph represent? 10) _____



A) the reciprocal of the resistivity of the material
B) the resistivity of the material divided by the cross–sectional area of the wire
C) the resistivity of the material of which the wire is made
D) the cross–sectional area of the wire
E) the product of the resistivity and the cross–sectional area of the wire

11) A wire of resistivity ϱ must be replaced in a circuit by a wire of the same material but four times as long. If, however, the total resistance is to remain as before, the diameter of the new wire must 11) _____
A) be one–fourth the original diameter.
B) be four times the original diameter.
C) be one–half the original diameter.
D) be the same as the original diameter.
E) be two times the original diameter.

12) The length of a certain wire is kept same while its radius is doubled. What is the new resistance of this wire? 12) _____
A) It is reduced by a factor of 4.
B) It is increased by a factor of 4.
C) It is reduced by a factor of 8.
D) It is reduced by a factor of 2.
E) It is increased by a factor of 2.

13) Which one of the following quantities is equivalent to 1 Ω? 13) _____
A) 1 V/A B) 1 V • A C) 1 J/s D) 1 A • s E) 1 W/A

14) Which one of the following quantities is equivalent to 1 W? 14) _____
A) 1 A • s B) 1 Ω • m C) 1 V • A D) 1 V/A E) 1 V/Ω

15) A kilowatt–hour is equivalent to 15) _____
A) 3600 J.
B) 3,600,000 J.
C) 3600 J/s
D) 1000 W.
E) 3,600,000 J/s.

16) If the resistance in a constant voltage circuit is doubled, the power dissipated by that circuit will 16) _____
A) decrease to one–half its original value.
B) increase by a factor of four.
C) increase by a factor of two.
D) decrease to one–fourth its original value.

17) If the current flowing through a circuit of constant resistance is doubled, the power dissipated by that circuit will 17) _____

A) decrease to one–fourth of what it was. B) double in magnitude.
C) decrease to one–half of what it was. D) quadruple in magnitude.

18) Over ordinary temperature ranges, the resistance of most metals such as copper 18) _____

A) is independent of the temperature of the metal.
B) increases as the metal gets hotter.
C) decreases as the metal gets hotter.
D) is a maximum at 20°C.

19) When two or more different capacitors are connected in series across a potential source, which of the following statements must be true? (There could be more than one correct choice.) 19) _____

A) The potential difference across each capacitor is the same.
B) Each capacitor carries the same amount of charge.
C) The total voltage across the combination is the algebraic sum of the voltages across the individual capacitors.
D) The equivalent capacitance of the combination is less than the capacitance of any of the capacitors.
E) The capacitor with the largest capacitance has the most charge.

20) Three identical capacitors are connected in series across a potential source (battery). If a charge of Q flows into this combination of capacitors, how much charge does each capacitor carry? 20) _____

A) $Q/9$ B) $3Q$ C) $Q/3$ D) Q

21) Three identical capacitors are connected in parallel to a potential source (battery). If a charge of Q flows into this combination, how much charge does each capacitor carry? 21) _____

A) $3Q$ B) $Q/3$ C) Q D) $Q/9$

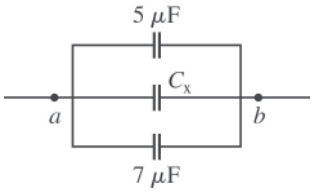
22) When two or more different capacitors are connected in parallel across a potential source (battery), which of the following statements must be true? (There could be more than one correct choice.) 22) _____

A) The potential difference across each capacitor is the same.
B) Each capacitor carries the same amount of charge.
C) The capacitor with the largest capacitance has the most charge.
D) The capacitor with the largest capacitance has the largest potential difference across it.
E) The equivalent capacitance of the combination is less than the capacitance of any one of the capacitors.

23) A 5–μF, a 7–μF, and an unknown capacitor C_X are connected in series between points a and b. What do you know about the equivalent capacitance C_{ab} between a and b? (There could be more than one correct choice.) 23) _____

A) $5\ \mu F < C_{ab} < 7\ \mu F$
B) $5\ \mu F < C_{ab} < 12\ \mu F$
C) $C_{ab} < C_X$
D) $C_{ab} > 12\ \mu F$
E) $C_{ab} < 5\ \mu F$

24) A 5–μF, a 7–μF, and an unknown capacitor C_X are connected in parallel between points *a* and *b* as shown in the figure. What do you know about the equivalent capacitance C_{ab} between *a* and *b*? (There could be more than one correct choice.) 24) _____



A) $C_{ab} > C_X$
B) $C_{ab} > 12$ μF
C) 5 μF $< C_{ab} <$ 12 μF
D) $C_{ab} < C_X$
E) $C_{ab} <$ 5 μF

25) Suppose you have two capacitors and want to use them to store the *maximum* amount of energy by connecting them across a voltage source. You should connect them 25) _____
A) in series across the source.
B) in parallel across the source.
C) It doesn't matter because the stored energy is the same either way.

26) Four unequal resistors are connected in series with each other. Which one of the following statements is correct about this combination? 26) _____
A) The equivalent resistance is less than that of the smallest resistor.
B) The equivalent resistance is equal to that of any one of the resistors.
C) The equivalent resistance is less than that of the largest resistor.
D) The equivalent resistance is equal to average of the four resistances.
E) The equivalent resistance is more than the largest resistance.

27) When unequal resistors are connected in parallel in a circuit, 27) _____
A) the power generated in each resistor is the same.
B) the largest resistance has the largest current through it.
C) the same current always runs through each resistor.
D) the potential drop is always the same across each resistor.

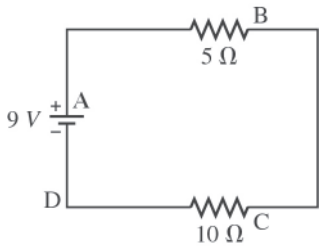
28) As more resistors are added in series to a constant voltage source, the power supplied by the source 28) _____
A) does not change.
B) increases.
C) increases for a time and then starts to decrease.
D) decreases.

29) As more resistors are added in parallel across a constant voltage source, the power supplied by the source 29) _____
A) increases for a time and then starts to decrease.
B) does not change.
C) increases.
D) decreases.

30) The lamps in a string of decorative lights are connected in parallel across a constant-voltage power source. What happens if one lamp burns out? (Assume negligible resistance in the wires leading to the lamps.) 30) _____

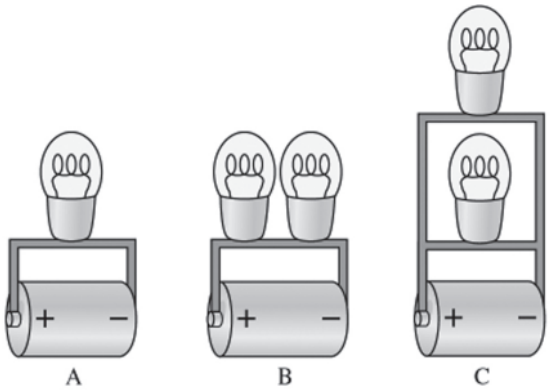
A) The brightness of the lamps will not change appreciably.
B) The other lamps get dimmer equally.
C) The other lamps get brighter, but some get brighter than others.
D) The other lamps get dimmer, but some get dimmer than others.
E) The other lamps get brighter equally.

31) A 9-V battery is hooked up to two resistors in series using wires of negligible resistance. One has a resistance of 5 Ω, and the other has a resistance of 10 Ω. Several locations along the circuit are marked with letters, as shown in the figure. Which statements about this circuit are true? (There could be more than one correct choice.) 31) _____



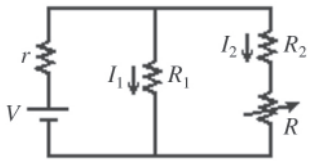
A) The current is exactly the same at points A, B, C, and D.
B) The potential at B is equal to the potential at C.
C) The current at A is greater than the current at B, which is equal to the current at C, which is greater than the current at D.
D) The potential at D is equal to the potential at C.
E) The current at A is greater than the current at B, which is greater than the current at C, which is greater than the current at D.

32) Identical light bulbs can be attached to identical ideal batteries in three different ways (A, B, or C), as shown in the figure. The ranking (from lowest to highest) of the *total* power produced by the battery is 32) _____



A) B, A, C B) C, A, B C) A, B, C D) C, B, A E) A, C, B

33) In the circuit shown in the figure, the resistor R has a variable resistance. As R is decreased, what happens to the currents? 33) _____



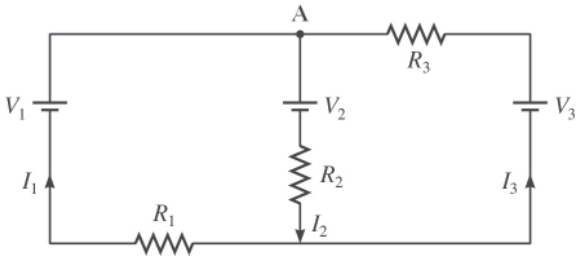
A) I_1 decreases and I_2 decreases.
B) I_1 increases and I_2 decreases.
C) I_1 remains unchanged and I_2 increases.
D) I_1 increases and I_2 increases.
E) I_1 decreases and I_2 increases.

34) Kirchhoff's junction rule is a statement of 34) _____
A) Newton's second law.
B) the law of conservation of momentum.
C) the law of conservation of angular momentum.
D) the law of conservation of energy.
E) the law of conservation of charge.

35) Kirchhoff's loop rule is a statement of 35) _____
A) the law of conservation of angular momentum.
B) the law of conservation of momentum.
C) the law of conservation of charge.
D) Newton's second law.
E) the law of conservation of energy.

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

36) For the circuit shown in the figure, write the Kirchhoff current equation for the node labeled A. Notice the directions of the currents! 36) _____

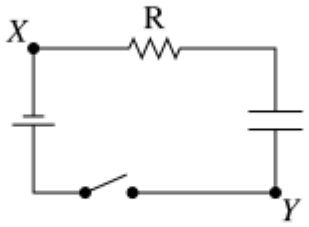


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

37) An ideal ammeter should 37) _____
A) introduce a very large series resistance into the circuit whose current is to be measured.
B) consist of a galvanometer in series with a large resistor.
C) introduce a very small series resistance into the circuit whose current is to be measured.
D) have a high coil resistance.

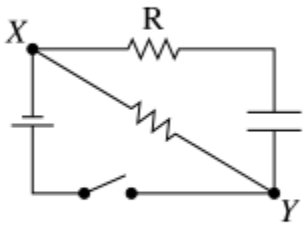
38) A galvanometer can be converted to an ammeter by the addition of a 38) _____
A) small resistance in series. B) large resistance in series.
C) small resistance in parallel. D) large resistance in parallel.

WorkBook Unit 11



The figure above shows Resistor R and an initially uncharged capacitor connected in a circuit with a switch and a battery.

1.



The switch is open and the capacitor is uncharged. A second resistor is added to the circuit, connected between X and Y as shown above. How does the potential difference across Resistor R immediately after the switch is closed compare to that in the original circuit, immediately after the switch is closed?

(A) The potential difference is greater with the new resistor.

(B) The potential difference is less with the new resistor.

(C) The potential difference is the same with the new resistor.

(D) The potential differences cannot be compared without knowing the relative values of the resistances.

2. A long time after the switch is closed, the potential differences across the battery, the resistor, and the capacitor are constant. Which of the following correctly indicates whether the potential differences are zero or nonzero?

(A) The potential difference across the battery is zero, the potential difference across the resistor is nonzero, and the potential difference across the capacitor is nonzero.

(B) The potential difference across the battery is nonzero, the potential difference across the resistor is zero, and the potential difference across the capacitor is zero.

(C) The potential difference across the battery is nonzero, the potential difference across the resistor is zero, and the potential difference across the capacitor is nonzero.

(D) The potential difference across the battery is nonzero, the potential difference across the resistor is nonzero, and the potential difference across the capacitor is zero.

---

3. A simple circuit has a $12.0\ \text{V}$ battery and three resistors of $6.0\ \Omega$ connected in parallel. If the current through the battery is $4.0\ \text{A}$, what is the internal resistance of the battery?

(A) $0.4\ \Omega$

(B) $1.0\ \Omega$

(C) $2.0\ \Omega$

(D) $3.0\ \Omega$

4. A group of students wants to determine the internal resistance of a battery. They connect the battery to a variable resistor. The students measure the potential difference across the battery as a function of the current through the battery as they vary the resistance. Which of the following analyses of the data could be used to determine the internal resistance of the battery?

(A) Divide the potential difference across the battery by the current through it for each data point. The average of these calculations gives the internal resistance of the battery.

(B) Graph the potential difference across the battery as a function of the current through it. Extrapolate to find the y-intercept and divide this by the average of the current measurements to find the internal resistance of the battery.

(C) Find the best-fit straight line for a graph of potential difference across the battery as a function of the current through it. The absolute value of the slope represents the internal resistance of the battery.

(D) This data cannot be analyzed to give the internal resistance of the battery, because the potential difference across the battery does not depend on the current.

5.

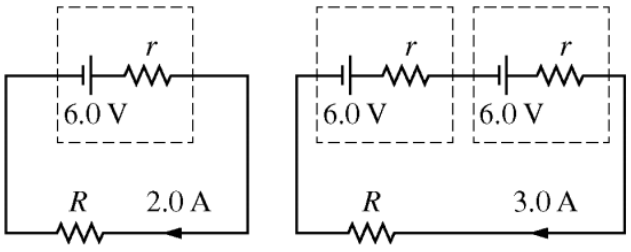


Figure 1 Figure 2

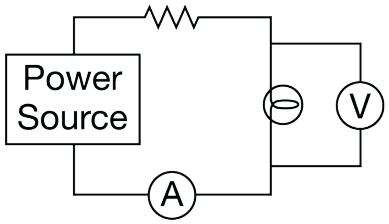
A battery with emf 6.0 V and an internal resistance r is connected to a resistor of resistance R, as shown in Figure 1 above. The current in the circuit is 2.0 A. When an identical battery is added to the circuit in series, as shown in Figure 2, the current in the circuit is 3.0 A. Which of the following statements about the resistances in the circuit is true?

(A) r is very small compared to R.

(B) r is very large compared to R.

(C) r is equal to R.

(D) r and R are comparable in magnitude, but they are not equal.



A student is given a box of identical lightbulbs and is asked to design a procedure to determine if the lightbulbs are ohmic or nonohmic. The student plans to connect the circuit shown above and use the meters to measure a potential difference and current.

6. Which of the following is the best procedure to use to determine if the lightbulbs are ohmic or nonohmic?

(A) Use one potential difference setting on the power source and take one set of measurements for each lightbulb.

(B) Use two potential difference settings on the power source and take a set of measurements for one lightbulb.

(C) Use one lightbulb and take measurements for a number of different potential difference settings on the power source.

(D) Use each lightbulb and take measurements for the same two different potential difference settings on the power source.

7. Do the results of the experiment depend on whether the power source has some internal resistance, and why or why not?

(A) No, because the measurements apply to the lightbulb only.

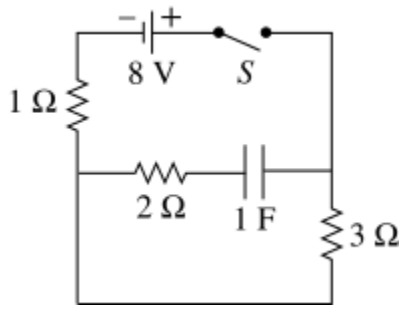
(B) No, because the resistor in the circuit ensures that the correct current is delivered to the lightbulb.

(C) Yes, because the current in the circuit depends on the total resistance of the circuit.

(D) Yes, because the emf provided by the power source will be different if it has internal resistance.

WorkBook Unit 11

8.



A capacitor and three resistors are wired in a circuit with a switch S, as shown above. What is the current through the $3\ \Omega$ resistor a long time after the switch has been closed?

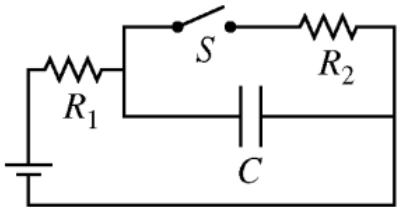
(A) $0\ \text{A}$

(B) $1.3\ \text{A}$

(C) $2.0\ \text{A}$

(D) $2.7\ \text{A}$

9.



In the circuit represented above, resistors R_1 and R_2, capacitor C, and open switch S are connected to a battery. The circuit reaches equilibrium. The switch is then closed, and the circuit is allowed to come to a new equilibrium. Which of the following is a true statement about the charge q_f stored on the capacitor after the switch is closed compared with the charge q_i stored on the capacitor before the switch is closed?

(A) $q_f > q_i$

(B) $q_f = q_i$

(C) $q_f < q_i$

(D) The relationship between the charges cannot be determined unless the resistances of the resistors are known.

10. A 20 kΩ resistor is connected in series with an initially uncharged 100 μF capacitor and a 5 V battery. What is the charge on the capacitor when the circuit has reached steady state?

(A) 5×10^{4} C

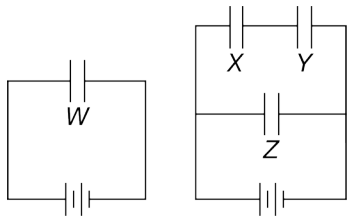
(B) 2.5×10^{-1} C

(C) 2.5×10^{-4} C

(D) 5×10^{-4} C

WorkBook Unit 11

11.



When capacitor W is connected to a battery, as shown above on the left, and allowed to reach steady state, the charge stored on its positive plate is $36\ \mu C$. Three other capacitors, X, Y, and Z, which are identical to each other, are connected to the same battery in the circuit shown above on the right. The equivalent capacitance of the three capacitors is equal to that of capacitor W. What is the charge on the positive plate of each of the three capacitors after that circuit has reached a steady state?

(A)

Capacitor X	**Capacitor Y**	**Capacitor Z**
12 µC	12 µC	24 µC

(B)

Capacitor X	**Capacitor Y**	**Capacitor Z**
12 µC	12 µC	12 µC

(C)

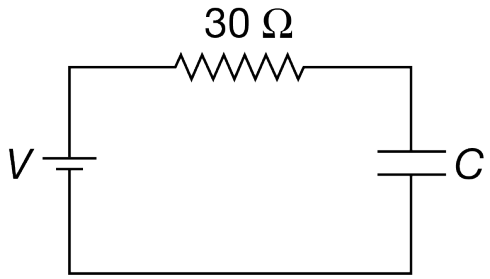
Capacitor X	**Capacitor Y**	**Capacitor Z**
9 µC	9 µC	18 µC

(D)

Capacitor X	**Capacitor Y**	**Capacitor Z**
6 µC	6 µC	24 µC

WorkBook Unit 11

12.



A parallel-plate capacitor is connected in series with a battery and a $30\ \Omega$ resistor, as shown above. The capacitor consists of two circular metal plates with radius $1.0\ \text{cm}$ that are separated only by a circular piece of paper with the same radius as the plates. When the capacitor is fully charged, each plate has an excess charge of magnitude $9.0 \times 10^{-11}\ \text{C}$. The capacitor is replaced by a new capacitor made from two circular metal plates with radius $2.0\ \text{cm}$ that are now separated by two circular pieces of paper with the same radius as the plates. What is the magnitude of the charge on each plate of the new capacitor after it is fully charged?

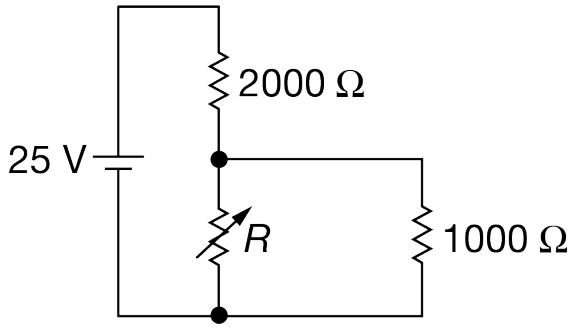
(A) $36.0 \times 10^{-11}\ \text{C}$

(B) $18.0 \times 10^{-11}\ \text{C}$

(C) $9.0 \times 10^{-11}\ \text{C}$

(D) $4.5 \times 10^{-11}\ \text{C}$

13.



In the circuit shown above, R is a variable resistor. What resistance must R have so that the voltage across the 1000Ω resistor is $5.0\ \text{V}$?

(A) 400Ω

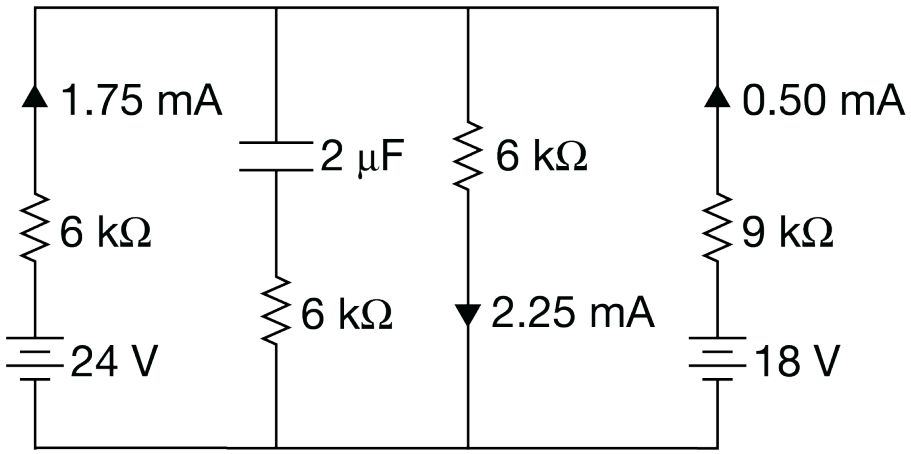
(B) 500Ω

(C) 1000Ω

(D) 5000Ω

WorkBook Unit 11

14.



The circuit shown above has been connected for a long time. What is the charge on one of the plates of the capacitor?

(A) 1.20×10^{-5} C

(B) 1.35×10^{-5} C

(C) 2.70×10^{-5} C

(D) 8.40×10^{-5} C

15.

Voltmeter Reading (V)	**Ohmmeter Reading** (Ω)
9.9	100
9.8	50
9.1	10
8.3	5
6.7	2

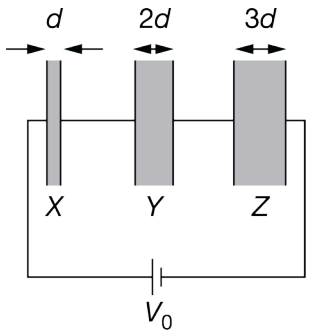
A student has a nonideal 10 V battery, some resistors, a voltmeter, and an ohmmeter and wants to determine the internal resistance of the battery and whether the resistors are ohmic. The student uses the ohmmeter to measure the resistance of each resistor. The student then connects the voltmeter across the battery terminals and one at a time connects each resistor to the battery. The table above shows the student's data. Which of the following indicates what can be concluded about the battery and resistors?

(A) The internal resistance of the battery is negligible, and the resistors are nonohmic.

(B) The internal resistance of the battery is negligible, and the ohmic nature of the resistors cannot be confirmed using only one potential difference measurement each.

(C) The internal resistance of the battery is about 1Ω, and the resistors are ohmic.

(D) The internal resistance of the battery is about 1Ω, and the ohmic nature of the resistors cannot be confirmed using only one potential difference measurement each.



A student is performing experiments to determine the dielectric constant of various materials. The figure above shows one of the student's circuits, containing three parallel-plate capacitors arranged in a circuit with a battery of emf V_0. The capacitors have the same plate area but have different separations, as shown, and are filled with dielectrics of unknown dielectric constant. The circuit has reached steady state.

16. The student has no way to directly measure the capacitances but measures the potential difference across each capacitor and finds that they are the same. What can be concluded about the dielectric constants κ of the materials?

(A) $\kappa_X = \kappa_Y = \kappa_Z$

(B) $\kappa_X < \kappa_Y < \kappa_Z$

(C) $\kappa_X > \kappa_Y > \kappa_Z$

(D) Nothing can be concluded about the dielectric constants without knowing the capacitances of the capacitors.

17. The dielectric is now removed from capacitor Z, and the circuit is again allowed to reach steady state. Which of the following statements about the potential difference across each capacitor and the charge on each capacitor is true?

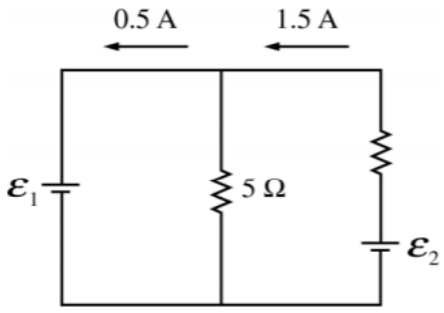
(A) Neither will change, because the total potential difference across the capacitors is the same.

(B) Only the charge on each capacitor will change, because the ratio of potential differences across the capacitors will remain the same.

(C) Only the potential difference across each capacitor will change, because the charges on the capacitors must still be equal to each other.

(D) Both will change, because the charge on each capacitor will change due to the change in total capacitance.

18.

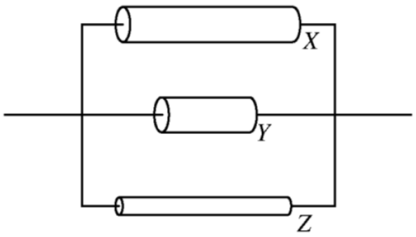


What are the value and the direction of the current through the 5 Ω resistor in the circuit shown above?

WorkBook Unit 11

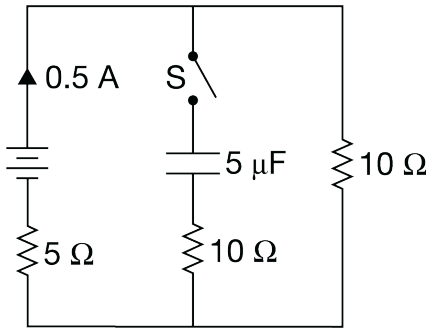
(A) 1.0 A toward the top of the page
(B) 1.0 A toward the bottom of the page
(C) 2.0 A toward the top of the page
(D) 2.0 A toward the bottom of the page

19.



The figure above represents a section of a circuit containing three resistors, *X*, *Y*, and *Z*, of different sizes but made of the same material. Which of the following correctly ranks the current in the resistors?

(A) $I_Z > I_X > I_Y$
(B) $I_Z = I_X > I_Y$
(C) $I_Y = I_X = I_Z$
(D) $I_Y > I_X > I_Z$



In the circuit shown above, the switch S is open, the capacitor is uncharged, and the current through the battery is 0.5 A.

20. At what time is the current through the resistor in the middle branch 0.5 A?

(A) Immediately after the switch is closed
(B) When the circuit has reached steady state
(C) Sometime between when the switch is closed and steady state is reached
(D) At no time

21. What is the charge on the capacitor a long time after the switch is closed?

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(A) $1.0\ \mu C$

(B) $1.5\ \mu C$

(C) $25\ \mu C$

(D) $37.5\ \mu C$

22. In which switch position, open or closed, can the current be the same in the left and right branches, and why?

(A) Neither position, because the resistors in these branches have different resistances.

(B) Both positions, because the current in the middle branch will be zero at some time for both positions.

(C) Only when the switch is open, because then the resistors are in series.

(D) Only immediately after the switch is closed, because then the equivalent resistance of the middle and right branches equals that of the left branch.

23.

Wire	**1**	**2**	**3**	**4**	**5**
Current (A)	**0.5**	**1.0**	**2.0**	**2.5**	**3.0**

A student was given five wires of the same length and diameter. The student connected the wires to the same battery one by one and measured the current through each wire. The table above shows the data collected. Which of the following can be concluded from the data?

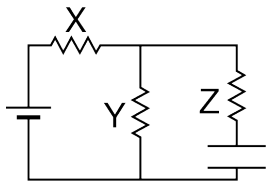
(A) The wires are ohmic.

(B) The wires are made of different materials.

(C) The resistance of the wires depends on their size and shape.

(D) The battery has internal resistance.

24.



A student wants to determine the potential difference across the capacitor in the circuit shown above, which has reached equilibrium. The student only has an ammeter to collect data about the circuit. The student measures the current in resistor Y, of known resistance, and calculates the potential difference across Y. Which of the following explains why this will allow the potential difference across the capacitor to be determined?

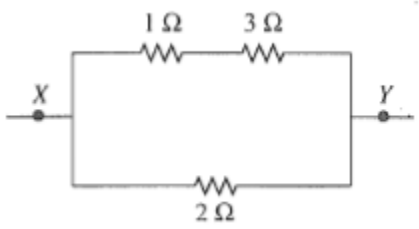
(A) By the loop rule, this allows the emf of the battery to be determined, which is equal to the potential difference across the capacitor.

(B) There is no current in the capacitor branch, so by the loop rule, the potential difference across Y equals the potential difference across the capacitor.

(C) This allows the potential difference across resistor X to be determined, which is equal to the potential difference across the battery.

Y Z

25. The following diagram shows part of a closed electrical circuit.



The electrical resistance of the part of the circuit shown between point *X* and point *Y* is

(A) $1\frac{1}{3}\Omega$

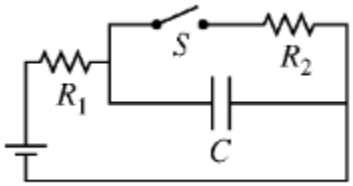
(B) 2Ω

(C) $2\frac{3}{4}\Omega$

(D) 4Ω

(E) 6Ω

26.



In the circuit represented above, two resistors (R_1 and R_2), a capacitor *C,* and an open switch *S* are connected to a battery. The circuit reaches equilibrium. The switch is then closed, and the circuit is allowed to come to a new equilibrium. Which of the following is a true statement about the energy stored in the capacitor after the switch is closed compared with the energy stored in the capacitor before the switch is closed?

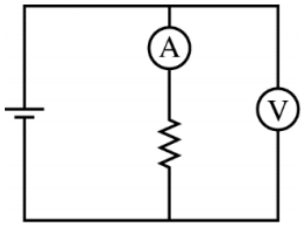
(A) The energy is greater.

(B) The energy is less.

(C) The energy is the same.

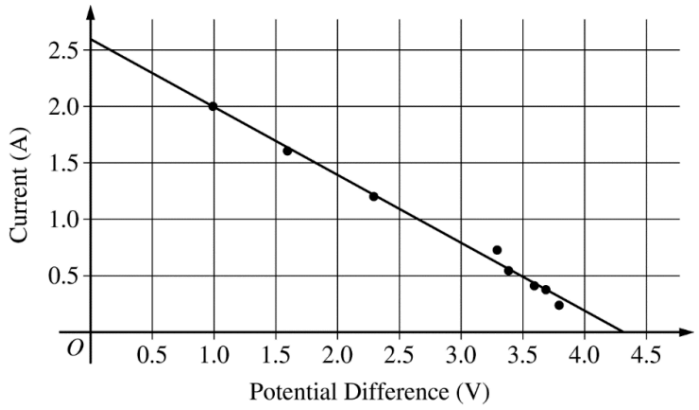
(D) The energy cannot be determined without knowing the resistances of the resistors.

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A student performs an experiment using the circuit shown above, with the goal of determining the characteristics of the battery, which has nonnegligible internal resistance. The student has many resistors with resistances ranging from 0.5 Ω to 15 Ω. The student puts different resistors in the circuit and records the current and the potential difference measured by the ammeter A and voltmeter V. The student's data table and graph of current as a function of potential difference are shown below.

Resistance (Ω)	0.5	1.0	2.0	4.0	6.0	8.0	10	15
Potential Difference (V)	1.0	1.6	2.3	3.3	3.4	3.6	3.7	3.8
Current (A)	2.0	1.6	1.2	0.72	0.55	0.41	0.35	0.23



27. Which of the following is most nearly the battery's internal resistance?

(A) 0.4 ohm

(B) 0.6 ohm

(C) 1.6 ohm

(D) 2.6 ohm

28. The student now replaces the battery with a different one that has the same emf but a significantly smaller internal resistance. The student runs the experiment again and graphs the new data. Which of the following correctly describes how the new graph compares to the plot shown above?

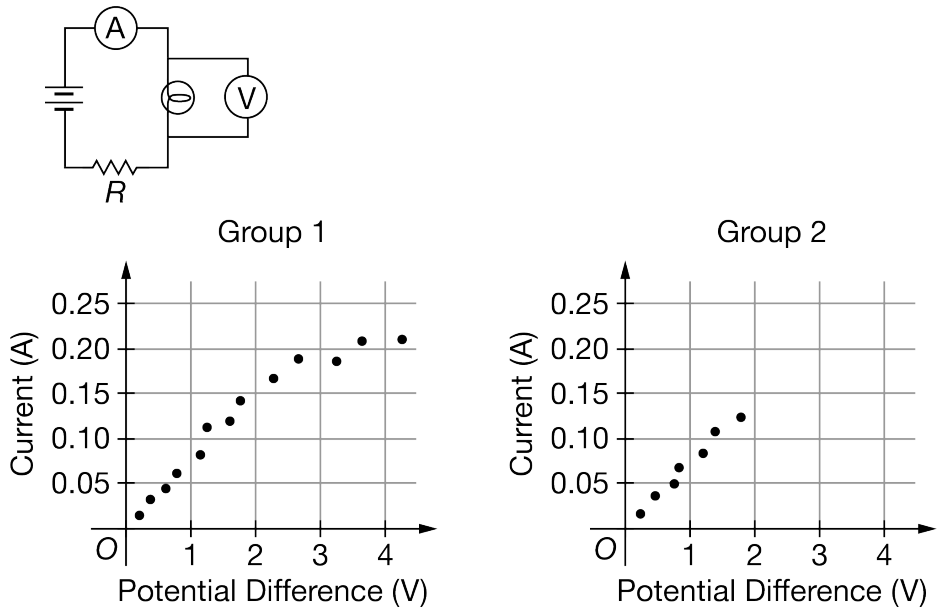
(A) The new graph will be the same as the original graph.

(B) The vertical intercept will have a greater value, but the slope will be the same.

(C) The vertical intercept will be the same, but the magnitude of the slope will be greater.

(D) Both the vertical intercept and the magnitude of the slope will be greater.

29.



Two groups of students use the circuit shown above to measure the potential difference across and current through the same lightbulb. The results of the two groups' measurements are shown in the graphs. What can the students correctly conclude about the ohmic behavior of the lightbulb?

(A) Both groups can conclude that the lightbulb is ohmic.

(B) Group 2 can conclude that the lightbulb is always ohmic, but group 1 cannot.

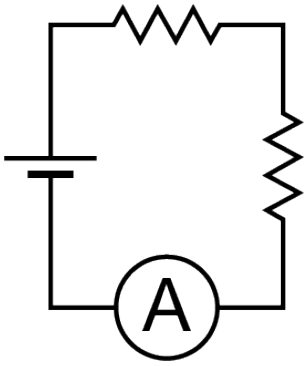
(C) Group 2 can conclude that the lightbulb shows ohmic behavior within the range of values measured. Group 1 can conclude that the lightbulb is not ohmic.

(D) Neither group can make any claims about the ohmic behavior of the lightbulb.

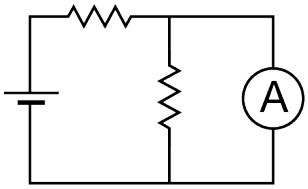
30. A student wants to investigate how different values of resistance affect the total current in a circuit when the resistors are connected in parallel. Which of the following circuits will allow the student to do this?

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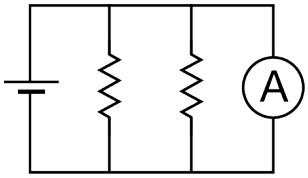
(A)



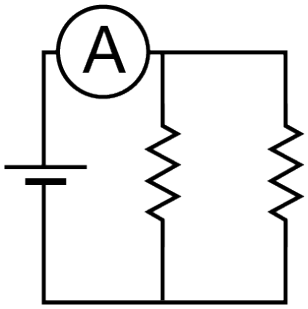
(B)



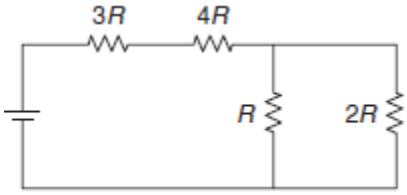
(C)



(D)



31.



The figure above shows four resistors connected in a circuit with a battery. Which of the following correctly ranks the potential difference, ΔV, across the four resistors?

(A) $\Delta V_{4R} > \Delta V_{3R} > \Delta V_{2R} > \Delta V_R$
(B) $\Delta V_{4R} > \Delta V_{3R} > \Delta V_{2R} = \Delta V_R$
(C) $\Delta V_{4R} = \Delta V_{3R} > \Delta V_R > \Delta V_{2R}$
(D) $\Delta V_{2R} = \Delta V_R > \Delta V_{3R} > \Delta V_{4R}$

32. A resistor and an initially uncharged capacitor are connected in series with a battery. A long time after the circuit is connected, which of the following is true?

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(A) The voltage across the resistor is at its maximum.

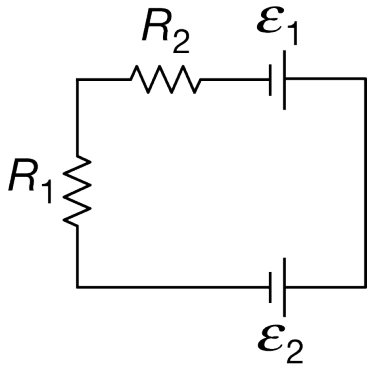
(B) The voltage across the capacitor is at its minimum.

(C) The charge on the capacitor is at its minimum.

(D) The current in the circuit is at its maximum.

(E) The current in the circuit is zero.

33.



The current in the circuit shown above is I. Which of the following correctly expresses the sum of the electric potential differences around the loop?

(A) $-\mathcal{E}_1 + \mathcal{E}_2 + IR_1 - IR_2 = 0$

(B) $-\mathcal{E}_1 - \mathcal{E}_2 - IR_1 + IR_2 = 0$

(C) $\mathcal{E}_1 + \mathcal{E}_2 - IR_1 - IR_2 = 0$

(D) $\mathcal{E}_1 - \mathcal{E}_2 - IR_1 - IR_2 = 0$

34. A voltmeter must be connected in parallel with a circuit component to correctly measure the potential difference across the component. Which of the following explains why the voltmeter should have a very large resistance?

(A) A large resistance is needed to make sure the voltmeter has the same current through it as the circuit component, so the potential difference across the voltmeter is the same as that across the component.

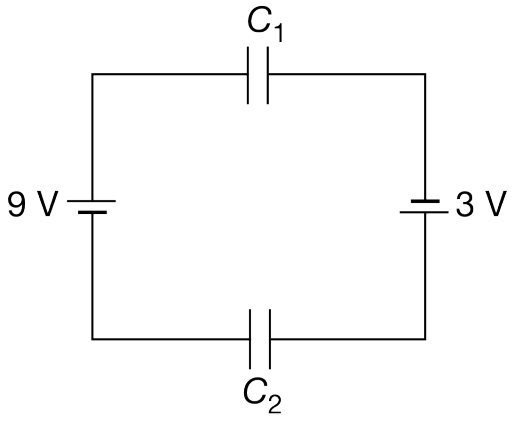
(B) A large resistance is needed to make sure a large current passes through the voltmeter, so the rest of the circuit is not affected.

(C) A large resistance is needed to make sure negligible current passes through the voltmeter, so the rest of the circuit is not affected.

(D) A large resistance is needed to make sure the current through the voltmeter has the same nonnegligible value whenever it is used, regardless of the current in the rest of the circuit.

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35.



Two capacitors and two batteries are connected in the circuit shown in the diagram above. The potential difference across C_2 is two times the potential difference across C_1. What is the potential difference across C_2?

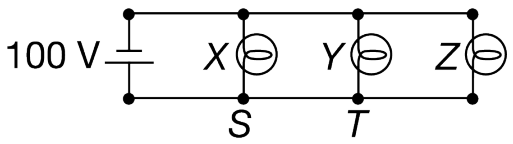
(A) $2\ \text{V}$

(B) $4\ \text{V}$

(C) $6\ \text{V}$

(D) $8\ \text{V}$

36.



The figure above represents three identical lightbulbs and a 100 V power source, connected at the dots to six wires with the same nonzero resistance. A student measures the potential difference across each bulb, getting 68.1 V for bulb X, 49.8 V for bulb Y, and 41.5 V for bulb Z. What potential difference should the student expect to measure between points S and T ?

(A) 4.1 V

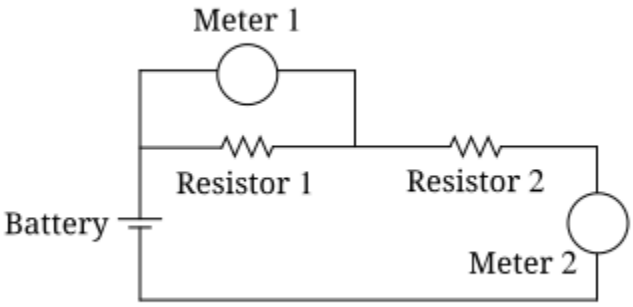
(B) 8.3 V

(C) 9.1 V

(D) 18.3 V

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37.



A student has a collection of different resistors, a nonideal battery, a voltmeter, and an ammeter. The student wants to use the circuit design shown to determine if one of the resistors is ohmic. Which of the following describes a procedure that would produce data that could be graphed to achieve this goal?

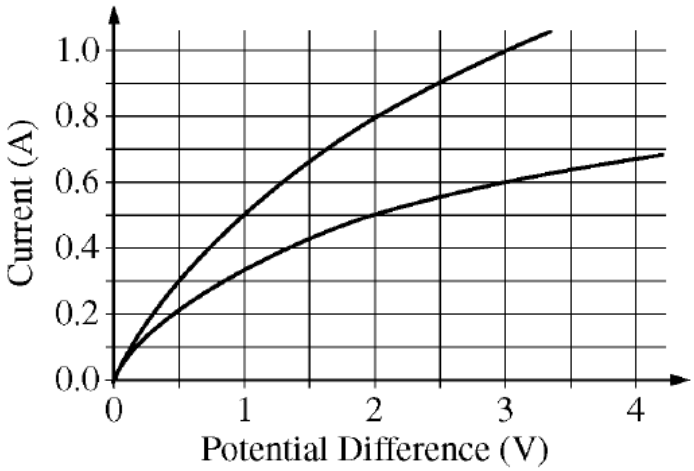
(A) Putting the ammeter at Meter 1, the voltmeter at Meter 2, and varying the resistance at Resistor 1 by swapping in different resistors

(B) Putting the ammeter at Meter 1, the voltmeter at Meter 2, and varying the resistance at Resistor 2 by swapping in different resistors

(C) Putting the voltmeter at Meter 1, the ammeter at Meter 2, and varying the resistance at Resistor 1 by swapping in different resistors

(D) Putting the voltmeter at Meter 1, the ammeter at Meter 2, and varying the resistance at Resistor 2 by swapping in different resistors

38.



The above graph shows current as a function of potential difference for two different filament lamps. If the two lamps are connected in parallel to a 3.0 V battery, what is the total current supplied by the battery?

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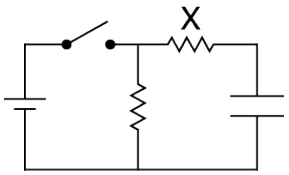
(A) 0.5 A

(B) 0.8 A

(C) 1.0 A

(D) 1.6 A

39.



The figure above shows a battery and an open switch connected to two resistors and a capacitor. The switch is closed, and the circuit reaches a steady state. Then the switch is opened again. Which of the following best describes the current in resistor X immediately before and immediately after the switch is opened again?

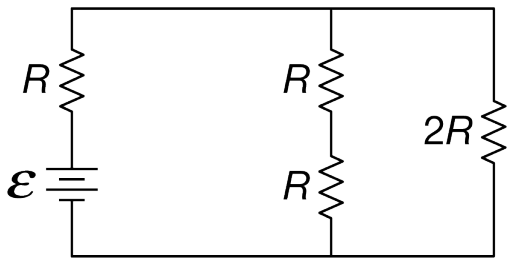
(A) Zero before and after

(B) Zero before and nonzero after

(C) Nonzero both before and after, but greater before

(D) Nonzero both before and after, but less before

40.



The circuit above contains four resistors with the resistances shown. What is the equivalent resistance of the circuit?

(A) $7R/5$

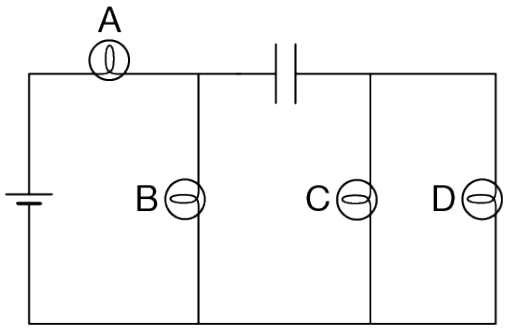
(B) $2R$

(C) $7R/2$

(D) $5R$

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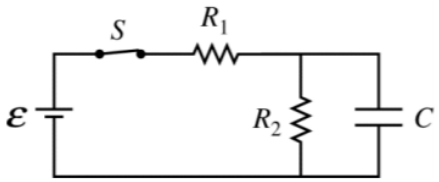
41.



The circuit shown above contains a capacitor, a battery, and four lightbulbs, A, B, C, and D, with the same resistance. The circuit has been connected for a long time. The capacitor is now removed from the circuit and replaced with a connecting wire. Which bulbs are dimmer at equilibrium in the second circuit compared to the first circuit?

(A) A
(B) B
(C) C
(D) D

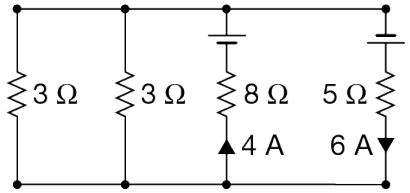
42.



The circuit shown above includes a battery, two resistors, and a capacitor with values as labeled. The current through the battery is I. Which of the following is a correct equation for potential differences in the circuit a long time after the switch is closed?

(A) $\varepsilon - IR_1 - IR_2 = 0$
(B) $\varepsilon + IR_1 + IR_2 = 0$
(C) $\varepsilon - IR_1 - IR_2 - C\varepsilon = 0$
(D) $\varepsilon - IR_1 - IR_2 + C\varepsilon = 0$

43.



In the circuit above, the current in the 8 Ω resistor is 4 A toward the top of the page and the current in the 5 Ω resistor is 6 A toward the bottom of the page. What is the current in each of the 3 Ω resistors?

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(A) 1 A toward the top of the page

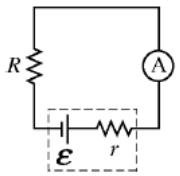
(B) 2 A toward the top of the page

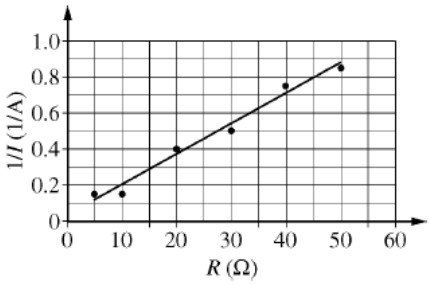
(C) 5 A toward the top of the page

(D) 5 A toward the bottom of the page

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44.





Students are given some resistors with various resistances, a battery with internal resistance, and an ammeter. They are asked to determine the emf ε and internal resistance *r* of the battery using just this equipment. Working with the circuit shown above, they insert each resistor into the circuit and measure the current *I* in the circuit each time they insert a resistor. From their data, the students generate a graph of 1/*I* as a function of the resistance *R* of each resistor, as shown above.

(a) i. Write an algebraic equation describing the circuit that includes ε , *R, r,* and *I.*

ii. Use your equation and the graph to calculate the emf of the battery and the internal resistance of the battery.

The students are now given a voltmeter and a new resistor X to use with the resistors, battery, and ammeter they already have. They are asked to determine whether resistor X is ohmic.

(b) i. Using standard symbols for circuit elements, as in the previously shown circuit, draw a diagram of a circuit that the students could use to determine whether resistor X is ohmic, including the appropriate placement of the meters. Clearly label your diagram.

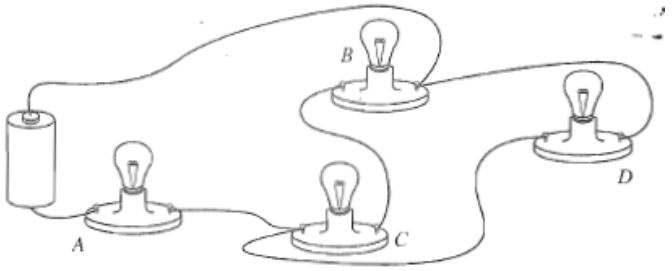
ii. Describe the procedure you would use with your circuit to get enough data to determine whether resistor X is ohmic.

iii. What would you graph using your data, and what would you look for on your graph to determine whether resistor X is ohmic?

(c) Would your procedure or data analysis in part (b) need to be different if the internal resistance of the battery was nonohmic? Justify your answer.

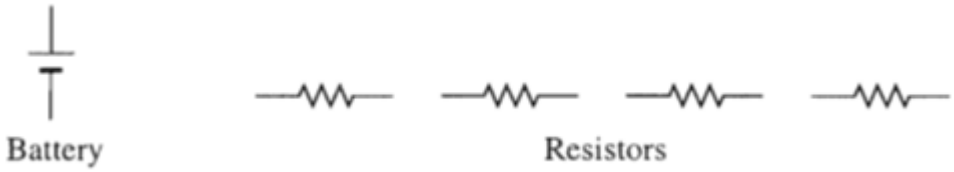
WorkBook Unit 11

45.



In the circuit shown above, *A*, *B, C,* and *D* are identical lightbulbs. Assume that the battery maintains a constant potential difference between its terminals (i.e., the internal resistance of the battery is assumed to be negligible) and the resistance of each lightbulb remains constant.

(a) Draw a diagram of the circuit in the box below, using the following symbols to represent the components in your diagram. Label the resistors *A*, *B, C,* and *D* to refer to the corresponding lightbulbs.



Draw your diagram in this box only.

(b) List the bulbs in order of their brightnesses, from brightest to least bright. If any two or more bulbs have the same brightness, state which ones. Justify your answer.

(c) Bulb *D* is then removed from its socket.

i. Describe the change in the brightness, if any, of bulb *A* when bulb *D* is removed from its socket. Justify your answer.

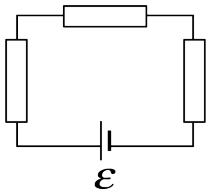
ii. Describe the change in the brightness, if any, of bulb *B* when bulb *D* is removed from its socket. Justify your answer.

WorkBook Unit 11

46. This question is a long free-response question. Show your work for each part of the question.

(12 points, suggested time 25 minutes)

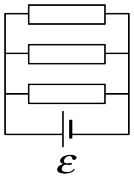
Students have a battery of emf ε with negligible internal resistance and many identical resistive heating elements and will connect a circuit so the heating elements can be used to heat some water. Each of the heating elements has resistance R.



(a) First the students consider connecting the heating elements in series with the battery, as shown above. One student says that to heat the water faster, they should use as many heating elements as possible.

i. Write an equation for the power P dissipated by a circuit containing N heating elements in series, in terms of ε, R, N, and physical constants, as appropriate.

ii. Does your equation support or refute the student's claim about the time it takes to heat the water? Justify your answer.



(b) Another student suggests that connecting the heating elements in parallel might be a better idea. The student says that a certain number of elements in parallel will heat the water faster than the same number of elements in series, because the current in the parallel circuit will be less.

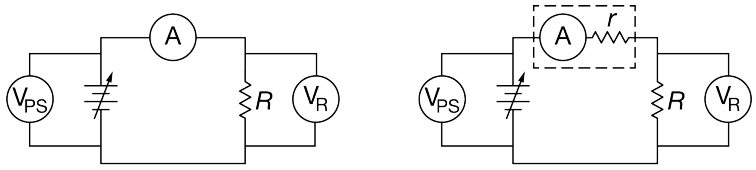
i. Write an equation for the power P dissipated by a circuit containing N heating elements in parallel, in terms of ε, R, N, and physical constants, as appropriate.

ii. Does your equation support or refute the student's claim about the time it takes to heat the water? Justify your answer.

iii. Is the student's claim about the current correct? Justify your answer.

WorkBook Unit 11

47. This question is a long free-response question. Show your work for each part of the question.



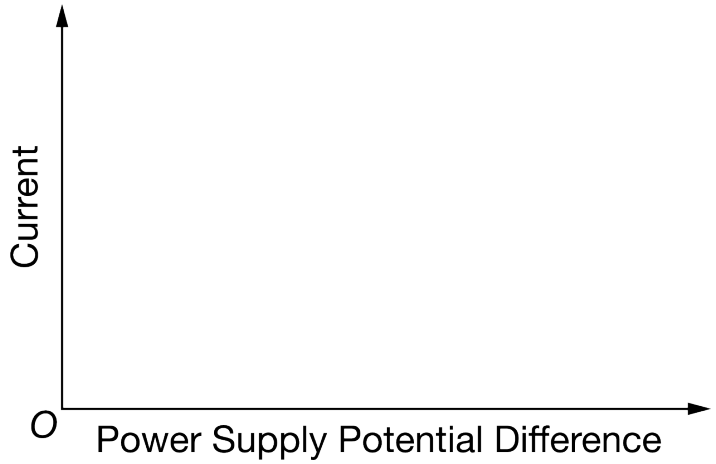
Circuit 1 Circuit 2

(12 points, suggested time 25 minutes)

The two circuits shown above contain an ideal variable power supply, an ohmic resistor of resistance R, an ammeter A, and two voltmeters $\mathrm{V_{PS}}$ and $\mathrm{V_R}$. In circuit 1 the ammeter has negligible resistance, and in circuit 2 the ammeter has significant internal ohmic resistance r. The potential difference of the power supply is varied, and measurements of current and potential difference are recorded.

(a) The axes below can be used to graph the current measured by the ammeter as a function of the potential difference measured across the power supply. On the axes, do the following.

Sketch a possible graph for circuit 1 and label it 1.
Sketch a possible graph for circuit 2 and label it 2.



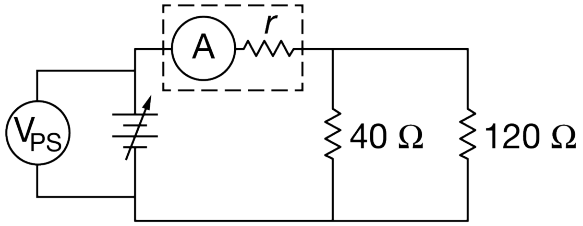
(b) Let ΔV_{PS} be the potential difference measured by voltmeter $\mathrm{V_{PS}}$ across the power supply, and let I be the current measured by the ammeter A. For each circuit, write an equation that satisfies conservation of energy, in terms of ΔV_{PS}, I, R, and r, as appropriate.

Circuit 1 Circuit 2

(c) Explain how your equations in part (b) account for any differences between graphs 1 and 2 in part (a).

(d) In circuit 2, $R = 40\ \Omega$. When voltmeter $\mathrm{V_{PS}}$ reads $3.0\ \mathrm{V}$, voltmeter $\mathrm{V_R}$ reads $2.5\ \mathrm{V}$. Calculate the internal resistance r of the ammeter.

(e) Voltmeter $\mathrm{V_R}$ in circuit 2 is replaced by a resistor with resistance $120\ \Omega$ to create circuit 3 shown below. Voltmeter $\mathrm{V_{PS}}$ still reads $3.0\ \mathrm{V}$.



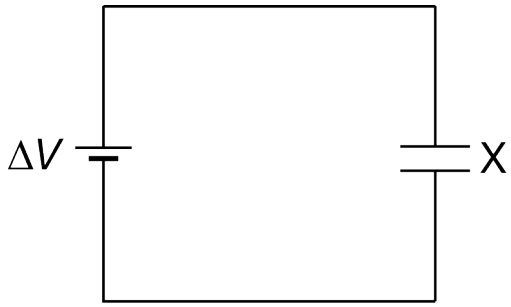
Circuit 3

i. Calculate the equivalent resistance R_{eq} of the circuit.

ii. Calculate the current in each of the resistors that are in parallel.

WorkBook Unit 11

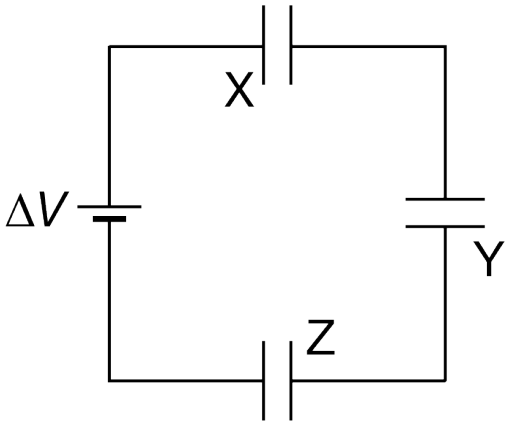
48. This question is a short free-response question. Show your work for each part of the question. (10 points, suggested time 15 minutes)



A battery with potential difference ΔV is connected to initially uncharged capacitor X of capacitance C_{X}. The magnitude of the charge on one plate of the capacitor after a long time is Q_{X}. Two other capacitors, Y and Z each with capacitance $\frac{C_{\mathrm{X}}}{2}$, are available to add to the circuit. All three capacitors are parallel-plate, air-filled capacitors.

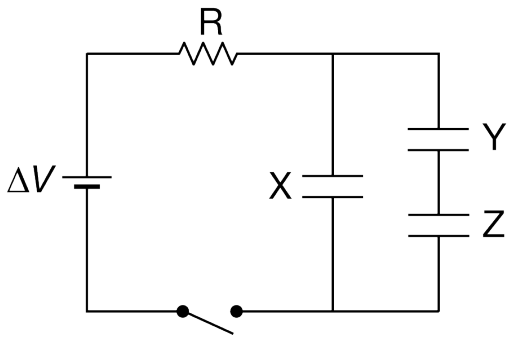
(a) Capacitor Y has the same plate separation as capacitor X. The area of a plate of capacitor X is A_{X}. What is the area of one of the plates of capacitor Y? Justify your answer.

Capacitors Y and Z are uncharged and are added to the circuit in series with capacitor X, as shown below.



(b) In terms of Q_{X}, what is the magnitude of the charge on one plate of each capacitor after a long time? Justify your answer.

WorkBook Unit 11



(c) The three uncharged capacitors are now connected in the circuit shown above. In a clear, coherent paragraph-length response, rank the potential difference across the resistor and each capacitor a long time after the switch is closed from least to greatest. Indicate explicitly if any values are the same. Include a justification of your rankings.

AP PHYSICS 2

UNIT 12

Magnetism and Electromagnetism

12–15%
AP EXAM WEIGHTING

~10–14
CLASS PERIODS

Conceptual Questions

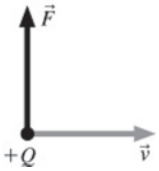
MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) If you were to cut a small permanent bar magnet in half, 1) _____
A) neither piece would be magnetic.
B) each piece would in itself be a smaller bar magnet with both north and south poles.
C) one piece would be a magnetic north pole and the other piece would be a south pole.
D) None of these statements is true.

2) A straight bar magnet is initially 4 cm long, with the north pole on the right and the south pole on the left. If you cut the magnet in half, the right half will 2) _____
A) contain only a south pole.
B) contain only a north pole.
C) contain a north pole on the right and a south pole on the left.
D) no longer contain any poles.

3) Which one of the following statements is correct? 3) _____
A) The north pole of a magnet points towards Earth's geographic south pole.
B) Earth's geographic south pole is the south pole of Earth's magnetic field.
C) Earth's geographic north pole is the north pole of Earth's magnetic field.
D) The north pole of a magnet points towards Earth's geographic north pole.
E) None of the above statements is correct.

4) A positive charge is moving to the right and experiences an upward magnetic force, as shown in the figure. In which direction must the magnetic field have a component? 4) _____



A) to the left
B) into the page
C) upward
D) to the right
E) out of the page

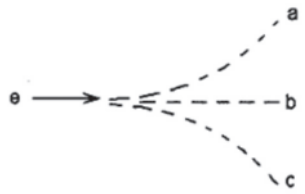
5) A charged particle that is moving in a static uniform magnetic field 5) _____
A) will always experience a magnetic force, regardless of its direction of motion.
B) may experience a magnetic force, but its speed will not change.
C) may experience a magnetic force which will cause its speed to change.
D) may experience a magnetic force, but its direction of motion will not change.
E) None of the above statements are true.

6) An electron moving along the $+x$–axis enters a magnetic field. If the electron experiences a magnetic deflection in the $-y$ direction, then the magnetic field must have a component 6) _____
A) along the $+y$–axis
B) along the $-z$–axis
C) along the $-y$–axis
D) along the $-x$–axis
E) along the $+z$–axis

7) An electron, moving west, enters a magnetic field. Because of this field the electron curves upward. We may conclude that the magnetic field must have a component 7) _____
A) towards the south.
B) upward.
C) towards the north.
D) downward.
E) towards the west.

8) A proton, moving east, enters a magnetic field. Because of this magnetic field the proton curves downward. We may conclude that the magnetic field must have a component 8) _____
A) upward.
B) towards the south.
C) downward.
D) towards the north.
E) towards the west.

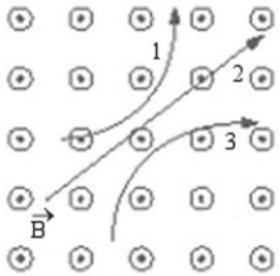
9) An electron is moving to the right, as shown in the figure. Suddenly it encounters uniform magnetic field pointing out of the page. Which one of the three paths shown will it follow in the field? 9) _____



A) path a B) path b C) path c

10) A proton is to orbit Earth at the equator using Earth's magnetic field to supply part of the necessary centripetal force. In what direction should the proton move? 10) _____
A) upward
B) southward
C) eastward
D) northward
E) westward

11) Three particles travel through a region of space where the magnetic field is out of the page, as shown in the figure. What are the signs of the charges of these three particles? 11) _____



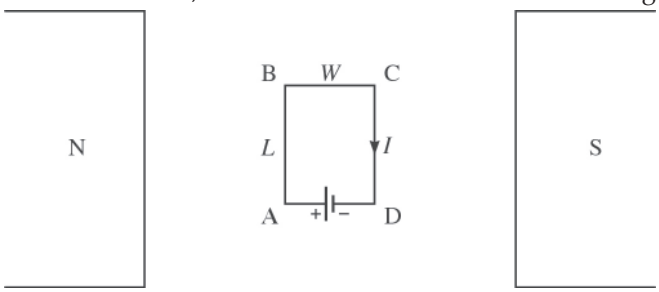
A) 1 is neutral, 2 is positive, and 3 is negative.
B) 1 is positive, 2 is neutral, and 3 is negative.
C) 1 is negative, 2 is neutral, and 3 is positive.
D) 1 is neutral, 2 is negative, and 3 is positive.
E) 1 is positive, 2 is negative, and 3 is neutral.

12) We observe that a moving charged particle experiences no magnetic force. From this we can definitely conclude that 12) _____
A) either no magnetic field exists or the particle is moving perpendicular to the field.
B) no magnetic field exists in that region of space.
C) either no magnetic field exists or the particle is moving parallel to the field.
D) the particle must be moving parallel to the magnetic field.
E) the particle is moving at right angles to the magnetic field.

13) A charged particle is observed traveling in a circular path of radius R in a uniform magnetic field. If the particle were traveling twice as fast, the radius of the circular path would be 13) _____
A) $4R$. B) $2R$. C) $8R$. D) $R/4$. E) $R/2$.

14) A proton, moving in a uniform magnetic field, moves in a circle perpendicular to the field lines and takes time T for each circle. If the proton's speed tripled, what would now be its time to go around each circle? 14) _____
A) T B) $9T$ C) $T/3$ D) $T/9$ E) $3T$

15) A rectangular coil, with corners labeled ABCD, has length L and width w. It is placed between the poles of a magnet, as shown in the figure. If there is a current I flowing through this coil in the direction shown, what is the direction of the force acting on section CD of this coil? 15) _____

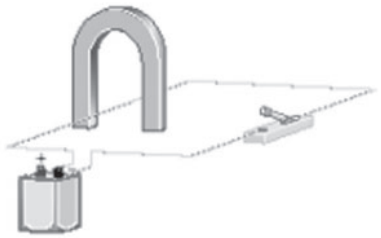


A) perpendicular to and out of the page
B) perpendicular to and into the page
C) in the opposite direction of the magnetic field
D) in the direction of the magnetic field
E) The force is zero.

16) A wire is carrying current vertically downward. What is the direction of the force on this wire due to Earth's magnetic field? 16) _____

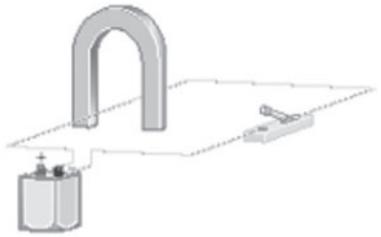
A) horizontally towards the east
B) horizontally towards the north
C) vertically upward
D) horizontally towards the west
E) horizontally towards the south

17) When the switch is closed in the circuit shown in the figure, the wire between the poles of the horseshoe magnet deflects upward. From this you can conclude that the left end of the magnet is 17) _____



A) a south magnetic pole.
B) a north magnetic pole.
C) There is not enough information given to answer the question.

18) For the horseshoe magnet shown in the figure, the left end is a north magnetic pole and the right end is a south magnetic pole. When the switch is closed in the circuit, which way will the wire between the poles of the horseshoe magnet initially deflect? 18) _____



A) to the right B) to the left C) downward D) upward

19) The magnetic force on a current–carrying wire in a magnetic field is the strongest when 19) _____

A) the current is perpendicular to the magnetic field lines.
B) the current is at a 0° angle with respect to the magnetic field lines.
C) the current is in the direction opposite to the magnetic field lines.
D) the current is in the direction of the magnetic field lines.
E) the current is at a 180° angle with respect to the magnetic field lines.

20) Two long parallel wires are placed side–by–side on a horizontal table. If the wires carry current in the same direction, 20) _____

A) the wires pull toward each other.
B) the wires push away from each other.
C) both wires are lifted slightly.
D) one wire is lifted slightly while the other wire is forced downward against the table's surface.

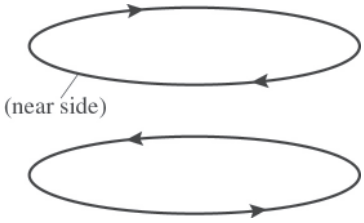
21) When two long parallel wires carry unequal currents, the magnitude of the magnetic force that one wire exerts on the other is F. If the current in both wires is now doubled, what is the magnitude of the new magnetic force on each wire? 21) _____

A) $8F$ B) $F\sqrt{2}$ C) $4F$ D) $16F$ E) $2F$

22) Two long parallel wires are placed side–by–side on a horizontal table and carry current in the same direction. The current in one wire is 20 A, and the current in the other wire is 5 A. If the magnetic force on the 20–A wire has magnitude F, what is the magnitude of the magnetic force on the 5–A wire? No external magnetic fields are present. 22) _____

A) $F/4$ B) $4F$ C) F D) $F/2$ E) $2F$

23) A ring with a clockwise current (as viewed from above the ring) is situated with its center directly above another ring, which has a counter–clockwise current, as shown in the figure. In what direction is the net magnetic force exerted on the top ring due to the bottom ring? 23) _____



A) upward
B) downward
C) to the right
D) to the left
E) The net force is zero.

24) An object is hung using a metal spring. If now a current is passed through the spring, what will happen to this system? 24) _____

A) The spring will begin to swing like a pendulum.
B) The weight will not move.
C) The spring will extend, lowering the weight.
D) The spring will contract, raising the weight.
E) None of these are true.

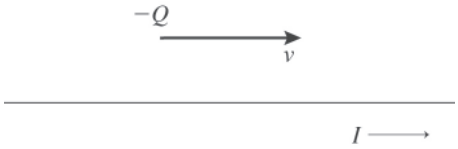
25) Two long parallel wires placed side–by–side on a horizontal table carry identical current straight toward you. From your point of view, the magnetic field at a point exactly between the two wires 25) _____

A) points away from you.
B) points downward.
C) points toward you.
D) points upward.
E) is zero.

26) A long, straight, horizontal wire carries current toward the east. A proton moves toward the east alongside and just south of the wire. What is the direction of the magnetic force on the proton? 26) _____

A) toward the east.
B) downward
C) toward the north
D) toward the south
E) upward

27) A negatively charged particle $-Q$ is moving to the right, directly above a wire having a current I flowing to the right, as shown in the figure. In what direction is the magnetic force exerted on the particle due to the current? 27) _____

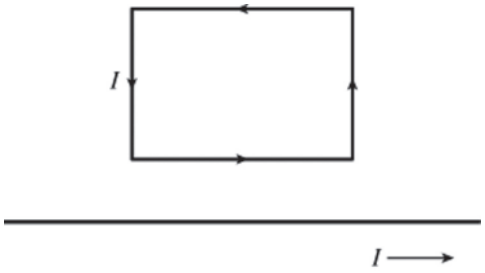


A) out of the page
B) downward
C) upward
D) into the page
E) The magnetic force is zero because the velocity is parallel to the current.

28) The magnetic field at a distance of 2 cm from a long straight current–carrying wire is 4 μT. What is the magnetic field at a distance of 4 cm from this wire? 28) _____

A) 6 μT B) 4 μT C) 10 μT D) 2 μT E) 8 μT

29) A long straight wire has a constant current flowing to the right. A rectangular metal loop is situated above the wire, and also has a constant current flowing through it, as shown in the figure. Which one of the following statements is true? 29) _____



A) The net magnetic force on the rectangle is zero, but there is a nonzero torque on the rectangle.
B) The net magnetic force on the rectangle is zero, and the net torque on it is zero.
C) The net magnetic force on the rectangle is downward, and the net torque on it is zero.
D) The net magnetic force on the rectangle is upward, and there is also a nonzero torque on the rectangle.
E) The net magnetic force on the rectangle is downward, and there is also a nonzero torque on the rectangle.

30) Consider an ideal solenoid of length L, N windings, and radius b (L is much longer than b). A current I is flowing through the wire windings. If the radius of the solenoid is doubled to $2b$, but all the other quantities remain the same, the magnetic field inside the solenoid will 30) _____

A) become one–fourth as strong as initially.
B) become one–half as strong as initially.
C) become twice as strong as initially.
D) remain the same.
E) become four times as strong as initially.

31) When you double the number of windings in an ideal solenoid while keeping all other parameters (radius, length and current) fixed, the magnetic field at the center of the solenoid will 31) _____
A) double.
B) be reduced by a factor of one–fourth.
C) triple.
D) quadruple.
E) be reduced by a factor of one–half.

32) When a ferromagnetic material is placed in an external magnetic field, the net magnetic field of its magnetic domains becomes 32) _____
A) smaller. B) larger. C) zero.

33) A flat coil is in a uniform magnetic field. The magnetic flux through the coil is greatest when the plane of its area is 33) _____
A) at 45° with the magnetic field.
B) perpendicular to the magnetic field.
C) parallel to the magnetic field.

34) According to Faraday's law, a coil in a strong magnetic field must have a greater induced emf in it than a coil in a weak magnetic field. 34) _____
A) True B) False

35) A coil of wire containing N turns is in an external magnetic field that is perpendicular to the plane of the coil and it steadily changing. Under these circumstances, an emf ε is induced in the coil. If the rate of change of the magnetic field and the number of turns in the coil are now doubled (but nothing else changes), what will be the induced emf in the coil? 35) _____
A) $\varepsilon/2$ B) ε C) 2ε D) $\varepsilon/4$ E) 4ε

36) According to Lenz's law, the induced current in a circuit always flows to oppose the external magnetic flux through the circuit. 36) _____
A) True B) False

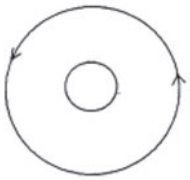
37) According to Lenz's law, the induced current in a circuit always flows to oppose the external magnetic field through the circuit. 37) _____
A) True B) False

38) A coil lies flat on a horizontal tabletop in a region where the magnetic field points straight down. The magnetic field disappears suddenly. When viewed from above, what is the direction of the induced current in this coil as the field disappears? 38) _____
A) There is no induced current in this coil.
B) clockwise
C) clockwise initially, then counterclockwise before stopping
D) counterclockwise

39) A long straight wire lies on a horizontal table and carries an ever–increasing current toward the north. Two coils of wire lie flat on the table, one on either side of the wire. When viewed from above, the direction of the induced current in these coils is 39) _____

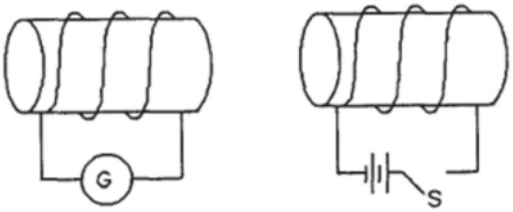
A) clockwise in both coils.
B) clockwise in the east coil and counterclockwise in the west coil.
C) counterclockwise in the east coil and clockwise in the west coil.
D) counterclockwise in both coils.

40) An outer metal ring surrounds an inner metal ring, as shown in the figure. The current in the outer ring is counterclockwise and decreasing. What is the direction of the induced current in the inner ring? 40) _____



A) clockwise
B) counterclockwise
C) There is no induced current in the inner ring.

41) Two solenoids are close to each other, as shown in the figure, with the switch S open. When the switch is suddenly closed, which way will the induced current flow through the galvanometer in the left–hand solenoid? 41) _____



A) from right to left
B) from left to right
C) There will be no induced current through the galvanometer.

42) A circular coil lies flat on a horizontal surface. A bar magnet is held fixed above the center of the coil with its north pole pointing downward. What is the direction of the induced current in the coil, as viewed from above? 42) _____

A) clockwise
B) counterclockwise
C) There is no current in the coil.

43) A circular coil of copper wire is lying flat on a horizontal table. A bar magnet is held above the center of the coil with its south pole downward. The magnet is released from rest and falls toward the coil. As viewed from above, what is the direction of the current induced in the coil as the magnet approaches the coil? 43) _____

A) clockwise
B) counterclockwise
C) No current is induced in the coil.
D) An emf but no current is induced in the coil.

44) A bar magnet is oriented above a copper ring, as shown in the figure. If the magnet is kept fixed while the ring is dropped, what is the direction of the current induced in the ring, as viewed from *below* the ring? 44) _____



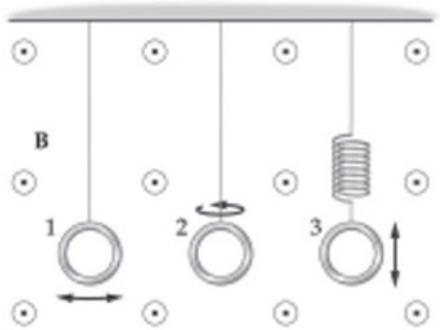
A) There is no current in the ring.
B) counterclockwise
C) clockwise

45) A bar magnet is oriented above a copper ring, as shown in the figure. If the magnet is pulled upward, what is the direction of the current induced in the ring, as viewed from above? 45) _____



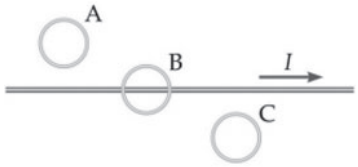
A) There is no current in the ring.
B) clockwise
C) counterclockwise

46) The three loops of wire shown in the figure are all hanging in the same uniform magnetic field $\vec{B}$ that is perpendicular to the page and does not vary with time. Loop 1 swings back and forth like the bob in a pendulum, Loop 2 rotates about a vertical axis, and Loop 3 oscillates up and down at the end of a spring. Which loop (or loops) will have an emf induced in them? 46) _____



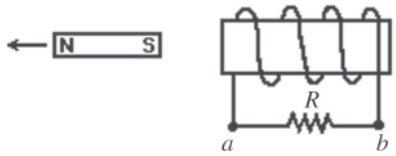
A) Loop 1
B) Loop 2
C) Loop 3
D) Loops 1 and 3
E) Loops 2 and 3

47) The wire in the figure carries a current I that is increasing with time at a constant rate. The wire and the three loops are all in the same plane. What is true about the currents induced in each of the three loops shown? 47) _____



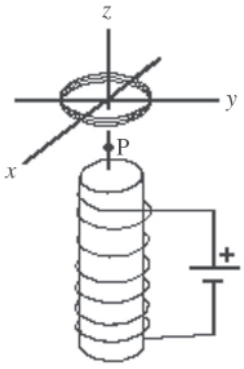
A) The currents are counterclockwise in all three loops.
B) Loop A has clockwise current, loop B has no induced current, and loop C has counterclockwise current.
C) The currents are clockwise in all three loops.
D) Loop A has counterclockwise current, loop B has no induced current, and loop C has clockwise current.
E) No current is induced in any loop.

48) In the figure, a bar magnet moves away from the solenoid. The direction of the induced current through the resistor R is 48) _____



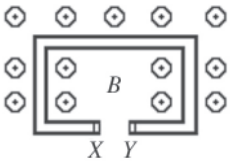
A) from b to a. B) from a to b. C) No current is induced.

49) As shown in the figure, a coil of wire is placed in the xy–plane, centered on the z–axis. A solenoid is centered on the z–axis and is carrying a steady dc current. Which one of the following actions will *not* result induce an emf in the coil? 49) _____



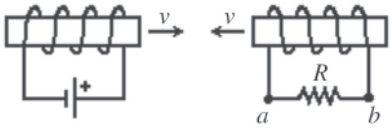
A) Rotate the coil about the z–axis.
B) Change the current in the solenoid.
C) Rotate the coil about the x–axis.
D) Move the coil toward point P.
E) Rotate the coil about the y–axis.

50) As shown in the figure, a C–shaped conductor is in a uniform magnetic field B out of the page, and this field is increasing. What is the polarity of the induced emf in terminals X and Y? 50) _____



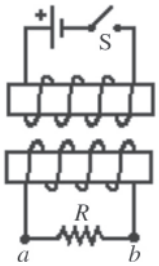
A) X is positive and Y is negative.
B) X and Y are at the same potential.
C) Y is positive and X is negative.

51) As shown in the figure, a battery supplies a steady current to the solenoid on the left. The two solenoids are moving toward each other with speeds v. The direction of the induced current through the resistor R is 51) _____



A) from b to a. B) from a to b. C) No current is induced.

52) As shown in the figure, two solenoids are side by side. The switch S is initially open. When S is suddenly closed, the direction of the induced current through the resistor R is 52) _____



A) from b to a. B) from a to b. C) No current is induced.

53) An ac generator consisting of a coil with N turns rotating at angular speed ω in an external magnetic field B produces a maximum emf ε_0.If you now decrease the number of turns in the coil to $N/2$ and at the same time cause it to rotate twice as fast, what will be the new maximum emf if you do not change any of the other quantities? 53) _____

A) ε_0 B) $4\varepsilon_0$ C) $2\varepsilon_0$ D) $\varepsilon_0/4$ E) $\varepsilon_0/2$

54) A transformer is a device used to 54) _____

A) transform an alternating current into a direct current.
B) transform a direct current into an alternating current.
C) increase or decrease a dc voltage.
D) increase or decrease an ac voltage.

55) A transformer is a device that normally 55) _____

A) operates only on ac.
B) operates on either dc or ac.
C) operates only on dc.

56) If the secondary coil of a transformer contains more loops than the primary coil, then it is a 56) _____

A) step–up transformer. B) step–down transformer.

57) Coil A is close to, but not touching, coil B. If a steady counterclockwise 100–A current flows in coil A, what can you say about the induced current in coil B? 57) _____

A) There will be no induced current in coil B.
B) The current in coil B will be clockwise and less than 100 A, but not zero.
C) The current in coil B will be equal to 100 A clockwise.
D) The current in coil B will be counterclockwise and less than 100 A, but not zero.
E) The current in coil B will be equal to 100 A counterclockwise.

58) An ideal solenoid has a self–inductance *L*. If you now double its radius and its length, but do not change the number of coils, what will its self–inductance be? 58) _____

A) *L*/2 B) 4*L* C) *L* D) 2*L* E) *L*/4

59) Inductors cannot store energy because they cannot store charge. 59) _____

A) True B) False

60) A resistor and an inductor are connected in series to a battery. The time constant for the circuit represents the time required for the current to reach 60) _____

A) 25% of the maximum current.
B) 75% of the maximum current.
C) 100% of the maximum current.
D) 63% of the maximum current.
E) 37% of the maximum current.

61) A resistor and an inductor are connected in series to a battery. The battery is suddenly removed from the circuit and replaced by a wire to complete the circuit. The time constant for of the new circuit represents the time required for the current to decrease to 61) _____

A) 75% of the original value.
B) 37% of the original value.
C) 25% of the original value.
D) 63% of the original value.
E) zero.



WorkBook Unit 12

1.



The figure above shows a bar magnet with two metal paper clips hanging from it. Which of the following explains why the bottom paper clip is attracted to the top paper clip?

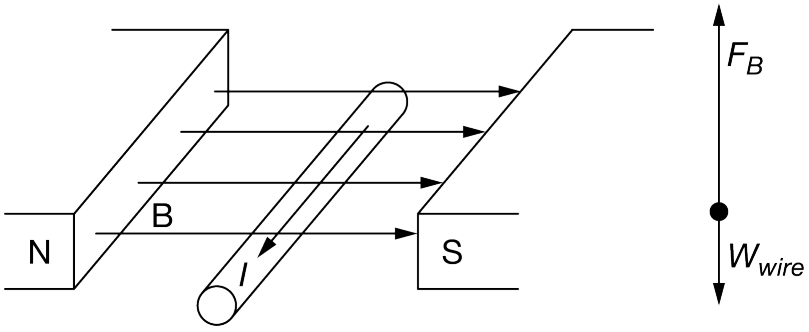
(A) The top clip becomes a permanent magnet as soon as it touches the bar magnet.

(B) The magnetic field of the bar magnet flows through the top paper clip like a current, holding the bottom paper clip in place.

(C) The top paper clip's magnetic domains are temporarily aligned, making the clip magnetic while it touches the bar magnet.

(D) The magnetic field of the bar magnet extends into the region of the bottom paper clip and holds the clip in place.

2.



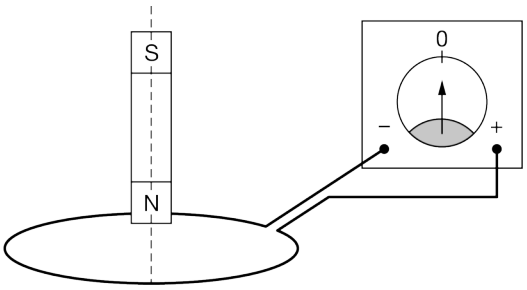
A uniform horizontal segment of wire with length ℓ and mass m carries a current I . It is placed in a uniform magnetic field of magnitude B that is horizontal and perpendicular to the wire, as shown above. A student draws the force diagram shown. What is the acceleration of the wire?

(A) $I\ell B$

(B) $I\ell B/m$

(C) $I\ell B - mg$

(D) $(I\ell B/m) - g$



A loop of wire is connected to an ammeter, as shown above, and held in place. The ammeter has a scale with zero in the middle and a needle that moves away from zero in either direction to measure the magnitude and direction of any current in the loop. A bar magnet is held above the center of the loop and released from rest.

3. Which of the following best describes the action of the ammeter needle as the magnet falls through the loop?

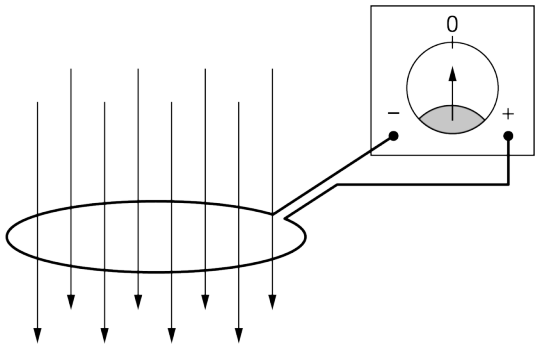
(A) The ammeter needle remains at zero.

(B) The ammeter needle deflects in one direction as the magnet approaches, passes through, and moves away from the loop.

(C) The ammeter needle deflects in one direction as the north end of the magnet approaches the loop, returns to zero when the center of the magnet is in the loop, and deflects in the same direction when the south end of the magnet is moving away from the loop.

(D) The ammeter needle deflects in one direction as the north end of the magnet approaches the loop, returns to zero when the center of the magnet is in the loop, and deflects in the other direction when the south end of the magnet is moving away from the loop.

4.



The bar magnet is removed, and the loop is placed in a region with a uniform constant magnetic field, as shown above. The loop then undergoes simple harmonic motion, moving up and down vertically. Which of the following statements correctly describes the resulting induced current, if any?

(A) The induced current is nonzero and constant.

(B) The induced current oscillates at the same frequency as the loop.

(C) The induced current oscillates at twice the frequency as the loop.

(D) The induced current is zero.

5. Is there a force exerted by the magnet on the loop as the magnet falls through it? Why or why not?

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(A) No. Because the loop is not magnetized, it will not be affected by the magnet.

(B) No. Because the loop is not moving, there are no forces on it.

(C) Yes. Because the magnetic field due to current in the loop exerts a force on the magnet, the magnet also exerts a force on the loop.

(D) Yes. Because gravity acts on the loop, the magnet must exert a force on the loop to keep it at rest.

6.

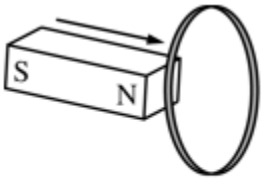


Figure 1

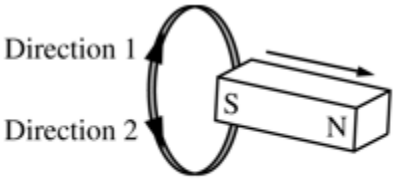


Figure 2

A bar magnet is pushed from left to right completely through a loop of metal wire, as shown in the figures above. The two possible directions for current in the loop are also shown. Which of the following indicates the directions of the induced current in the loop, if any, as the magnet is moving through the positions shown?

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(A)

Figure 1	Figure 2
Direction 1	Direction 1

(B)

Figure 1	Figure 2
Direction 1	Direction 2

(C)

Figure 1	Figure 2
Direction 2	Direction 1

(D)

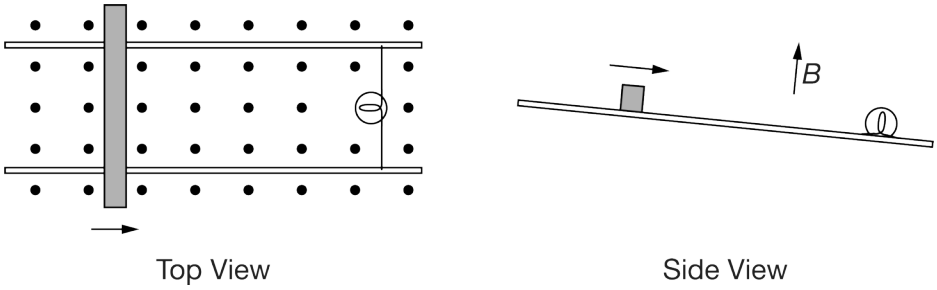
Figure 1	Figure 2
Direction 2	Direction 2

(E)

Figure 1	Figure 2
None (no current)	None (no current)

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7.



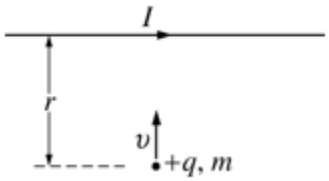
The system in the figure above consists of a conducting bar moving with negligible friction along a pair of stationary parallel conducting rails that are connected to a light bulb, forming a circuit. The rails are slightly tilted from horizontal, and the system is in a uniform magnetic field B that is perpendicular to the plane of the rails and pointing up. As the bar slides downward along the rails, a current is induced that causes the bulb to light. Which of the following statements is true about the energy in the system or the forces exerted on the sliding bar?

(A) The gravitational potential energy of the circuit-magnetic field-Earth system is converted only to kinetic energy of the sliding bar.

(B) The gravitational potential energy of the circuit-magnetic field-Earth system increases due to the work done by the magnetic field on the moving bar.

(C) The motion of the bar is determined by the magnetic force due to the induced current and a component of the force of gravity, which are in the same direction.

(D) The motion of the bar is determined by the magnetic force due to the induced current and a component of the force of gravity, which are in opposite directions.



A particle of charge +*q* and mass *m* moves toward a long wire that carries a current *I*. At the instant shown above, the particle is a distance *r* from the wire and is moving perpendicularly toward the wire with speed *v*. There is a magnetic force on the particle as a result of the current.

8. What is the direction of the magnetic force on the particle at the instant shown?

(A) Toward the wire

(B) Toward the left

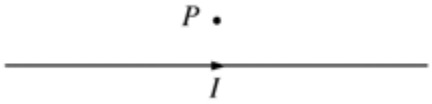
(C) Toward the right

(D) Out of the plane of the page

(E) Into the plane of the page

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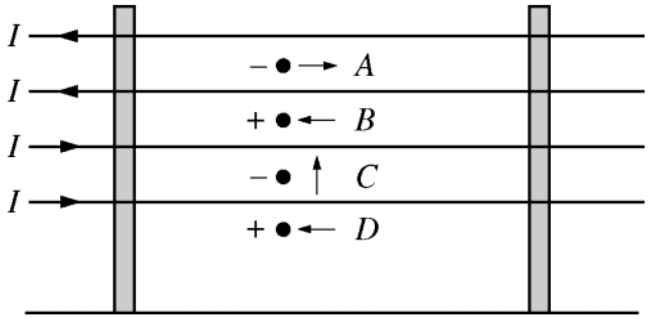
9.



A long, straight wire in the plane of the page carries a current *I* toward the right, as shown above. What is the direction of the magnetic field at point *P*, which is in the plane of the page and above the wire?

(A) Out of the page
(B) Into of the page
(C) Toward the left
(D) Toward the right
(E) Toward the wire

10.

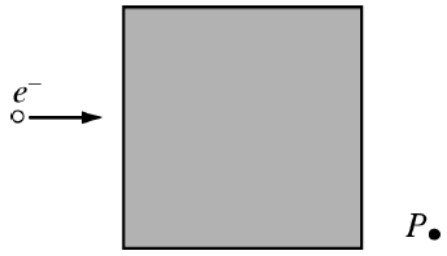


Four horizontal wires are arranged on vertical wooden poles, as shown in the figure above. The wires are equally spaced and have equal currents *I* in the directions indicated in the figure. The dots represent four charged dust particles moving in the plane of the wires. The sign of the charge on each particle and its direction of motion at a particular instant are shown. Which of the dust particles has a magnetic force exerted on it in the downward direction at this instant?

(A) A
(B) B
(C) C
(D) D

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11.



An electron is moving to the right, as shown in the figure above. It passes through the shaded region, which contains a magnetic field. The electron travels along a path that takes it through point P. The gravitational force on the electron is negligible. What is the direction of the magnetic field?

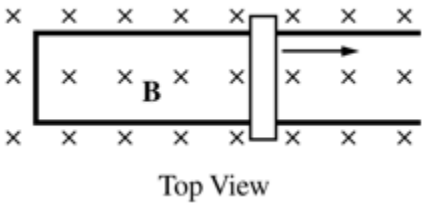
(A) Into the page

(B) Out of the page

(C) Toward the top of the page

(D) Toward the bottom of the page

12.



Top View

The figure above shows a metal bar that is supported by two sections of a fixed, conducting U-shaped horizontal rail. The rod and rail are located in a region of magnetic field **B** directed into the page. The bar is moving to the right at a constant speed. Which of the following is true of the horizontal forces acting on the bar?

(A) There are no forces.

(B) There is only a magnetic force.

(C) There is only an external applied force.

(D) There is both a magnetic force and an external applied force.

(E) Nothing can be said about the forces without knowing the magnitude of the magnetic field.

13. Two parallel wires, each carrying a current *I*, repel each other with a force *F*. If both currents are doubled, the force of repulsion is

(A) 2F

(B) $2\sqrt{2}F$

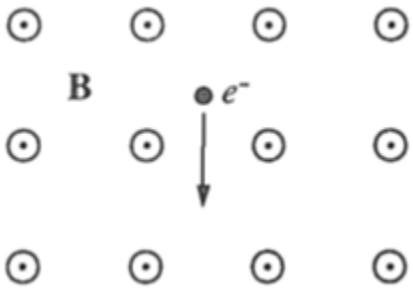
(C) 4F

(D) $4\sqrt{2}F$

(E) 8F

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14.



An electron is in a uniform magnetic field **B** that is directed out of the plane of the page, as shown above. When the electron is moving in the plane of the page in the direction indicated by the arrow, the force on the electron is directed

(A) toward the right

(B) out of the page

(C) into the page

(D) toward the top of the page

(E) toward the bottom of the page

15. A proton moving to the right in the plane of the page with speed v enters a magnetic field of magnitude B directed toward the top of the page. What is the direction of the initial magnetic force that is exerted on the proton?

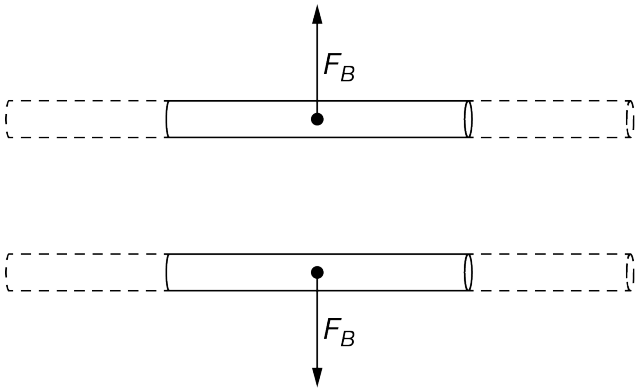
(A) Toward the top of the page

(B) Into the page

(C) Out of the page

(D) No direction, because there is no magnetic force on the proton

16.



The figure above shows the magnetic forces acting on two parallel horizontal wires. Which of the following statements about the currents in the wires is correct?

(A) Both currents are to the right

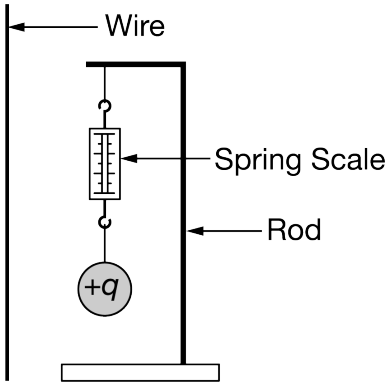
(B) Both currents are to the left.

(C) Both currents are in the same direction, but it could be either right or left.

(D) The currents are in opposite directions.

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17.



A sphere with net charge $+q$ hangs vertically from a spring scale, as shown above. There is a vertical wire to the left of the sphere. The current in the wire can be switched on or off. Students want to determine the direction of the current in the wire when it is switched on using only the apparatus shown above. Which of the following measurements could be used to determine the direction of the current?

(A) With the current initially off, turn the current on and look for any change in the spring scale reading.

(B) With the current initially off, turn the current on and look for any change in distance between the sphere and the wire.

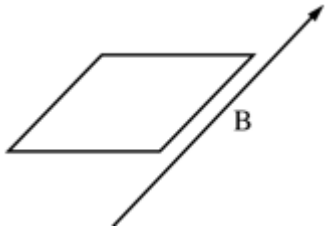
(C) With the current on, move the apparatus with the charged sphere toward the wire and look for any change in the spring scale reading.

(D) With the current on, move the wire toward the charged sphere and look for any change in distance between the sphere and the rod.

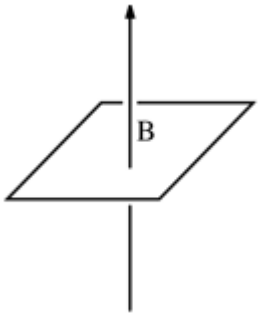
18. A square loop is located in a region containing a magnetic field of magnitude *B*. The following figures show five possible directions of the magnetic field relative to the loop. For which of the directions is the magnetic flux through the loop greatest?

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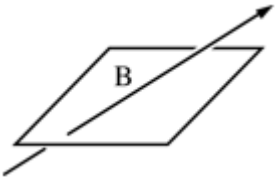
(A)



(B)



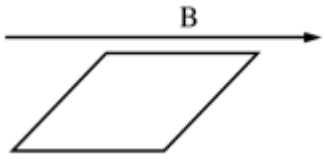
(C)



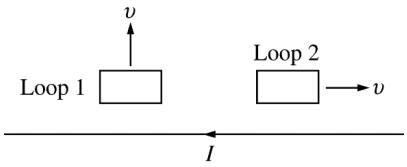
(D)



(E)



19.



Two conducting wire loops move near a very long, straight conducting wire that carries a current *I*. When the loops are in the positions shown above, they are moving in the directions shown with the same constant speed *v*. Assume that the loops are far enough apart that they do not affect each other. Which of the following is true about the induced electric currents, if any,in the loops?

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(A)

Loop 1	Loop 2
No current	No current

(B)

Loop 1	Loop 2
No current	Counterclockwise direction

(C)

Loop 1	Loop 2
Clockwise direction	No Current

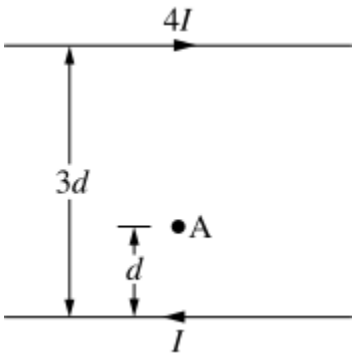
(D)

Loop 1	Loop 2
Clockwise direction	Clockwise direction

(E)

Loop 1	Loop 2
Counterclockwise direction	Clockwise direction

20.



Two long parallel wires, each in the plane of the page, are separated by a distance $3d$, as shown. The top wire carries current $4I$ directed to the right, and the bottom wire carries current I directed to the left. Point A is also in the plane of the page, located a distance d above the bottom wire. Which of the following correctly represents the magnitude of the magnetic field due to the wires at Point A?

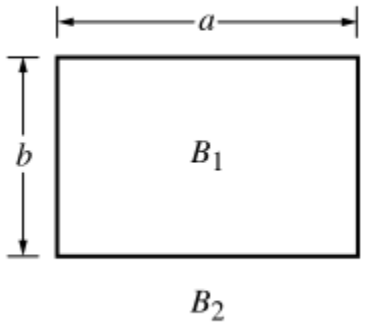
WorkBook Unit 12

(A) $\frac{1}{2}\frac{\mu_0 I}{\pi d}$

(B) $\frac{3}{4}\frac{\mu_0 I}{\pi d}$

(C) $\frac{3}{2}\frac{\mu_0 I}{\pi d}$

(D) $\frac{5}{2}\frac{\mu_0 I}{\pi d}$



The conducting loop shown above is in a region containing two uniform magnetic fields that point out of the page. The magnitude of the field inside the loop is B_1, and the magnitude of the field outside the loop is B_2. The magnitude of each field is decreasing at a constant rate.

21. Which of the following represents the magnetic flux through the loop while the magnitudes of the fields are decreasing?

(A) $B_1 ab$

(B) $B_2 ab$

(C) $(B_1 - B_2)ab$

(D) $(B_1 + B_2)ab$

22. Which of the following is true of the magnitude of the induced current in the loop while the magnitudes of the fields decrease?

(A) It decreases.

(B) It increases.

(C) It has a constant nonzero value.

(D) It is zero.

WorkBook Unit 12

23.



The figure above shows a uniform magnetic field directed into the page and two spheres that enter the field with velocities in the plane of the page. Sphere 1 has mass m, charge $+q$, speed v, and is moving to the right. Sphere 2 has mass $2m$, charge $+2q$, speed $2v$, and is moving to the left. Which of the following correctly compares the magnitudes of the magnetic forces F on the spheres due to the field?

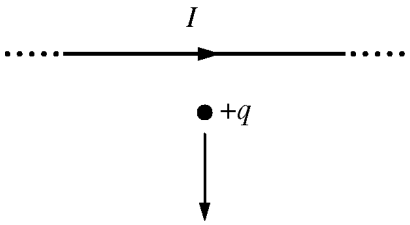
(A) $F_2 = F_1/2$

(B) $F_2 = 2F_1$

(C) $F_2 = 4F_1$

(D) $F_2 = 8F_1$

24.



The figure above shows a long conducting wire that lies in the plane of the page and carries an electric current *I* toward the right. At the instant shown, a positive point charge +*q* is in the plane of the page and moving toward the bottom of the page. What is the direction of the magnetic force on the point charge at that instant?

(A) Into the page

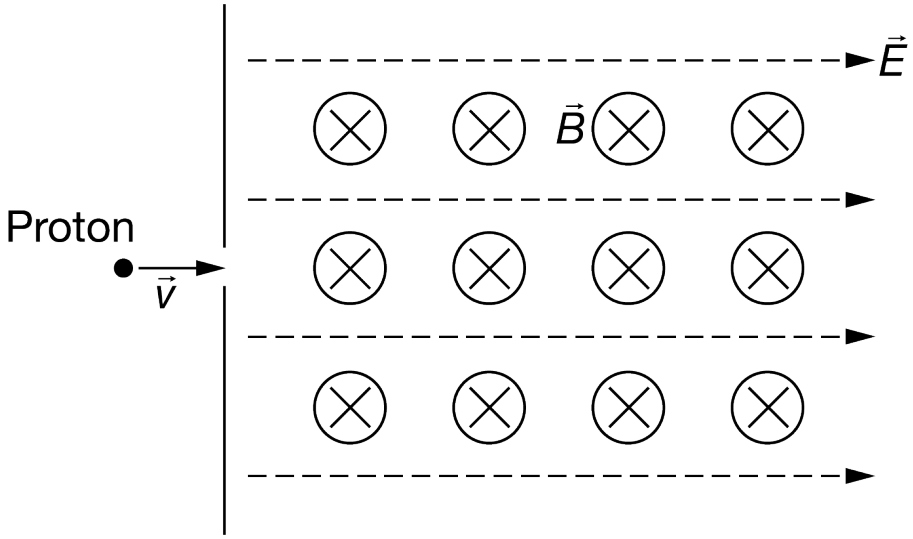
(B) Out of the page

(C) Toward the right

(D) Toward the left

WorkBook Unit 12

25.



A proton traveling horizontally with velocity $\vec{v}$ enters a region with a uniform electric field of magnitude E to the right and a uniform magnetic field of magnitude B into the page, as shown above. At the instant it enters the region, what is the magnitude and direction of the proton's acceleration?

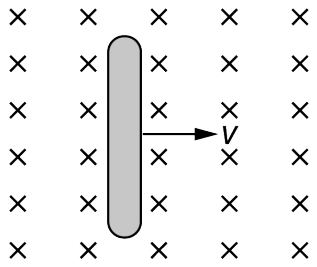
(A) $\frac{eE}{m_p}$ to the right

(B) $\frac{evB}{m_p}$ upward

(C) $\frac{e}{m_p}\sqrt{E^2+(vB)^2}$ diagonally up and to the right

(D) $\frac{e}{m_p}\left(E+vB\right)$ diagonally up and to the right

26.



An initially neutral and non-polarized conducting rod is moving to the right at a constant speed v through a large region that has a uniform magnetic field, as shown above. Which of the following statements correctly describes the motion of the particles within the rod and the resulting charge separation in the rod caused by this motion?

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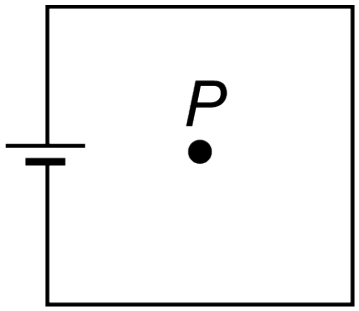
(A) Positively charged particles within the rod moved downward, causing the bottom of the rod to have a positive charge and the top to have a negative charge.

(B) Negatively charged particles moved upward, causing the bottom of the rod to have a positive charge and the top to have a negative charge.

(C) Positively charged particles moved upward, causing the top of the rod to have a positive charge and the bottom of the rod to have a negative charge.

(D) Negatively charged particles moved downward, causing the top of the rod to have a positive charge and the bottom of the rod to have a negative charge.

27.



A circuit consists of a battery and some wire with significant resistance. The wire is bent so that the circuit is in the shape of a square, and the circuit is aligned in the plane of the page, as shown above. Which of the following best describes the direction of the magnetic field near point P ?

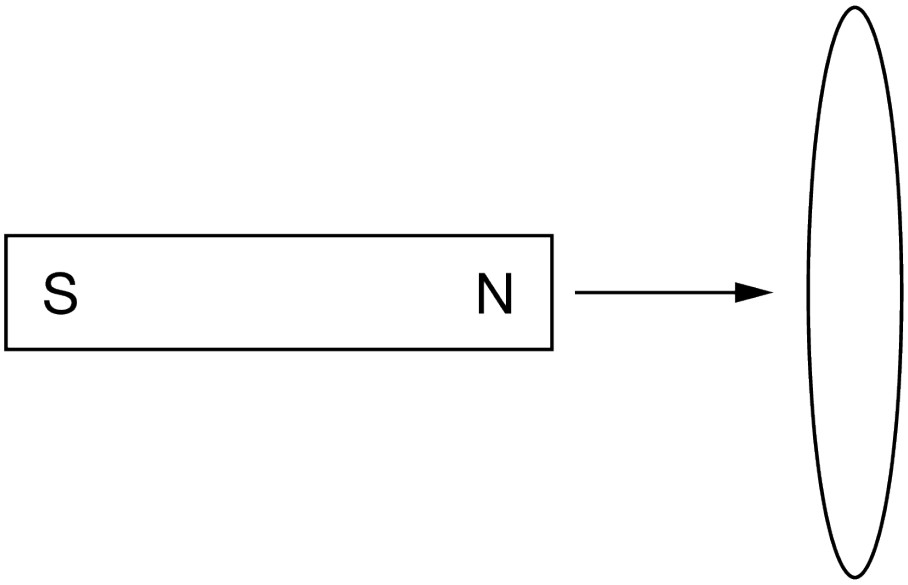
(A) Clockwise around point P

(B) Out of the page

(C) Into the page

(D) The field has no direction because the magnitude of the field is zero.

28.



A bar magnet moves toward the center of a loop of wire, as shown above. Which of the following statements correctly describes the magnetic fields from the bar magnet and the loop as the magnet moves toward the loop?

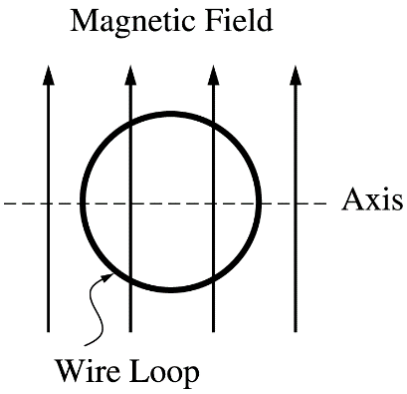
WorkBook Unit 12

(A) The bar magnet and the loop both create monopole fields.
(B) The bar magnet creates a monopole field and the loop creates a dipole field.
(C) The bar magnet creates a dipole field and the loop creates a monopole field.
(D) The bar magnet and the loop both create dipole fields.

29. A 100-turn wire coil has a resistance of 400 Ω and a cross-sectional area of 0.01 m^2. The coil is placed in a magnetic field directed parallel to the axis of the coil. If the field changes at a rate of -0.4 T/s, what is the current induced in the coil?

(A) 1×10^{-5} A
(B) 1×10^{-4} A
(C) 1×10^{-3} A
(D) 1×10^{-2} A
(E) 1×10^{-1} A

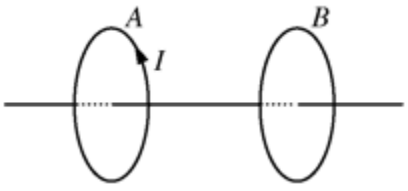
30.



A wire loop is rotated in a uniform magnetic field about an axis perpendicular to the field, as shown above. How many times is the induced current in the loop reversed if the loop makes 3 complete revolutions from the position shown?

(A) One
(B) Two
(C) Three
(D) Six
(E) Twelve

31.



The two circular wire loops represented above are coaxial. If the current *I* in loop *A* is increasing linearly with respect to time, which of the following is true of the induced current in loop *B* ?

(A) It is in the same direction as that in loop *A* and also increasing with respect to time.

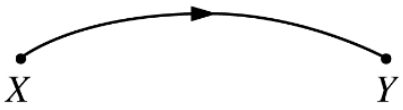
(B) It is in the opposite direction of that in loop *A* and also increasing with respect to time.

(C) It is in the same direction as that in loop *A* and constant with respect to time.

(D) It is in the opposite direction of that in loop *A* and constant with respect to time.

(E) None of the above

32.



A charged particle is in a region that contains both an electric and a magnetic field. The kinetic energy of the particle is 9 mJ at point *X* and 11 mJ at point *Y* in the figure above. If the only forces present are electromagnetic, what is the work done by the electric force on the particle as it moves from *X* to *Y* along the path shown?

(A) $0\,\mu J$

(B) $1\,\mu J$

(C) $2\,\mu J$

(D) $11\,\mu J$

(E) $20\,\mu J$

33. An electron with speed $3 \times 10^3\ \mathrm{m/s}$ is moving in a circle perpendicular to a $0.001\ \mathrm{T}$ uniform magnetic field. What is the acceleration of the electron?

(A) $1.9 \times 10^{-12}\ \mathrm{m/s^2}$

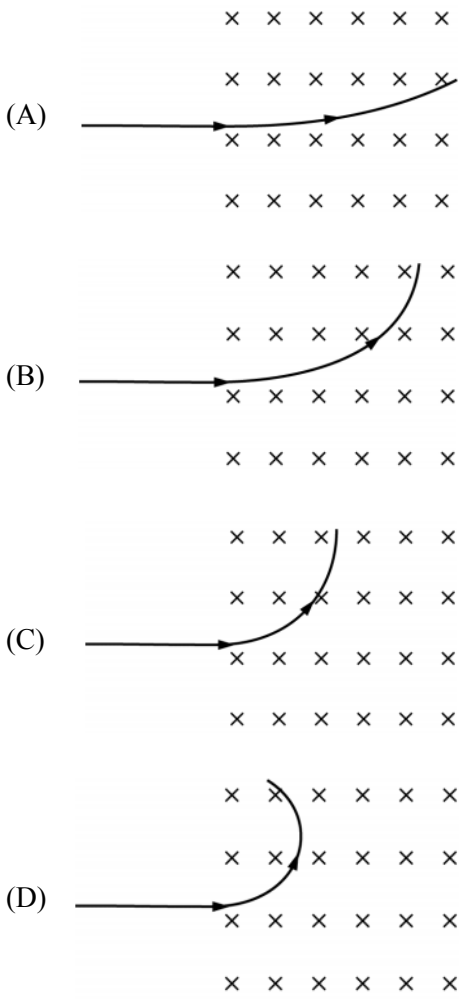
(B) $1.7 \times 10^{-5}\ \mathrm{m/s^2}$

(C) $1.8 \times 10^{8}\ \mathrm{m/s^2}$

(D) $5.3 \times 10^{11}\ \mathrm{m/s^2}$

34. Each of the figures below shows the path of a charged particle moving in the plane of the page in a magnetic field that is perpendicular to the page. If the mass, speed, and charge of the particles are the same, in which case does the field have the greatest magnitude?

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An isolated, neutral lambda particle (A^0) is moving to the right with speed v. It then decays into a proton and a pion ($A^0 \rightarrow p^+ + \pi^-$). The following are the masses of the three particles:

Lambda: 1115.7 MeV/c^2

Proton: 938.3 MeV/c^2

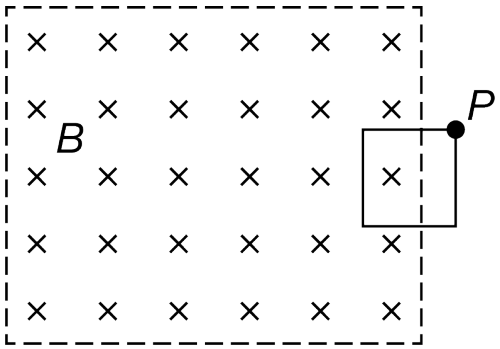
Pion: 139.6 MeV/c^2

35. At some later time, the proton and pion are both moving to the right in the plane of the page when they enter a magnetic field directed out of the page. Which of the following describes the directions of the magnetic forces on the proton and pion at the instant they enter the field?

(A) Proton: toward the top of the page
Pion: toward the top of the page

(B) Proton: toward the top of the page
Pion: toward the bottom of the page

(C) Proton: toward the bottom of the page Pion: toward the bottom of the page



The figure above shows a region containing a magnetic field of magnitude B directed into the page. A small loop of wire with a resistance R is initially at rest in the plane of the page. The loop is partially in the field, enclosing an area A of the field.

36. While the loop is at rest in the position shown, the magnitude of the magnetic field is gradually increased by an amount ΔB over a time Δt. If an amount of charge Q passes through point P while the field is changing, what is ΔB ?

(A) QR

(B) QR/A

(C) $QR\,\Delta t\,/A$

(D) $QR/A\,\Delta t$

37. After a time Δt the magnetic field is kept constant. To induce a counterclockwise current in the loop while the field is constant, in what direction should the loop be moved?

(A) Into the page

(B) Out of the page

(C) To the right

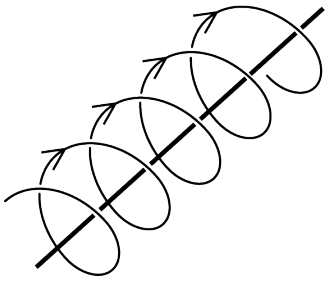
(D) To the left

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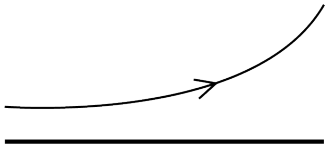
38. A student is analyzing results from several experiments where electrons were projected at various orientations near wires that could carry current. Which of the following, where the heavy line is the wire, shows a path for the electrons that would correspond to an experiment where the electrons were projected parallel to a current-carrying wire?

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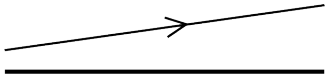
(A)



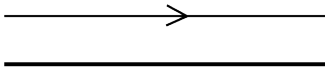
(B)



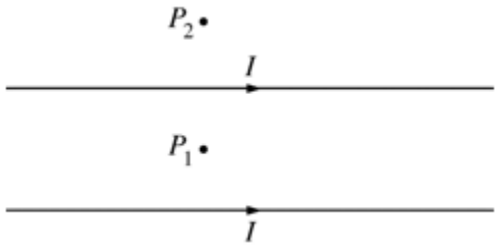
(C)



(D)



39.



Two long, parallel, straight wires in the plane of the page each carry a current I to the right, as shown above. Points P_1 and P_2 are in the plane of the page, with P_1 being midway between the wires. Which of the following is true of the net magnetic field at the two points that result from the two currents?

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(A)

Point P_1	Point P_2
Directed out of page	Directed into pag

(B)

Point P_1	Point P_2
Directed out of page	Directed out of page

(C)

Point P_1	Point P_2
Has magnitude zero	Directed into page

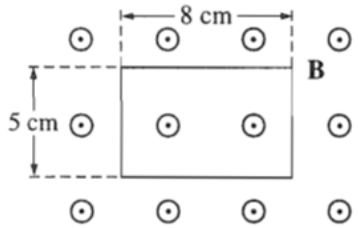
(D)

Point P_1	Point P_2
Has magnitude zero	Directed out of page

(E)

Point P_1	Point P_2
Directed into page	Directed into page

40.



A rectangular wire loop is at rest in a uniform magnetic field **B** of magnitude 2 **T** that is directed out of the page. The loop measures 5 cm by 8 cm, and the plane of the loop is perpendicular to the field, as shown above. The total magnetic flux through the loop is

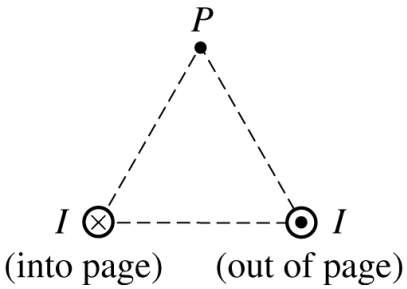
(A) zero
(B) 2×10^{-3} T■m^2
(C) 8×10^{-3} T■m^2
(D) 2×10^{-1} T■m^2
(E) 8×10^{-1} T■m^2

41. A circular loop is placed in a magnetic field of unknown orientation. The initial magnetic flux through the loop is nonzero. The loop is then rotated 180° about an axis in the plane of the loop and through the loop's center. Which of the following correctly compares the magnitude and sign of the magnetic flux after the loop has been rotated to those of the original flux?

	Magnitude of flux	**Sign of flux**
A	The same	The same
B	The same	Opposite sign
C	Less	The same
D	Less	Opposite sign

(A) A
(B) B
(C) C
(D) D

42.



Current *I* flows into and out of the page in two parallel, long, straight wires that are fixed at two vertices of an equilateral triangle, as shown above. The net magnetic field at the third vertex (point *P*) due to these currents is in which of the following directions?

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(A) 

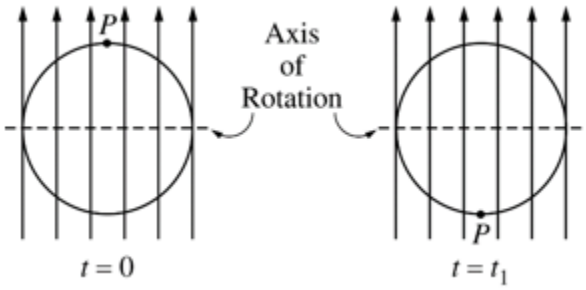
(B) 

(C)

(D)

(E) None of the above

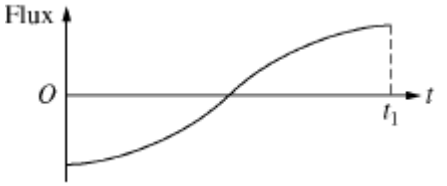
43.



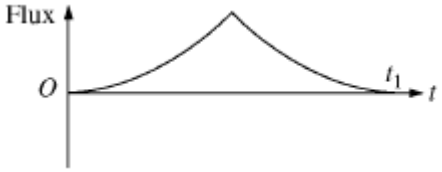
A circular loop of wire in a uniform magnetic field rotates at a constant rotational speed about the axis shown. At time $t = 0$ the plane of the loop is parallel to the field with point P at the top as shown above left. At time $t = t_1$ the loop has rotated 180° so that point P is at the bottom as shown above right. Which of the following graphs best represents the magnetic flux through the loop as a function of time t from 0 to t_1 ?

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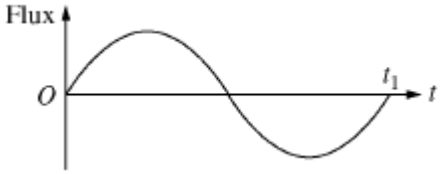
(A)



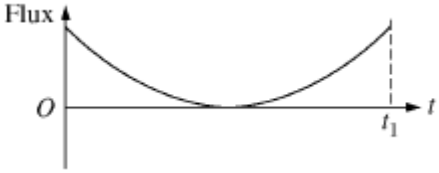
(B)



(C)



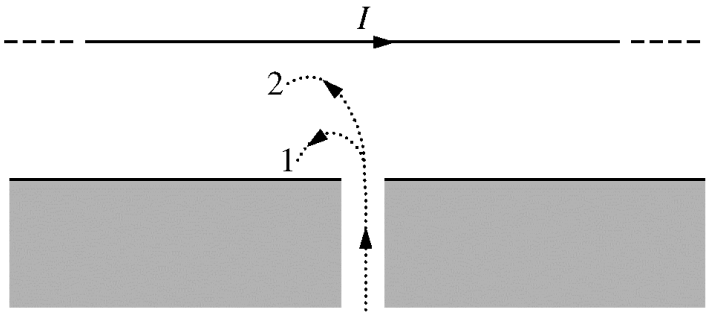
(D)



(E)



44.



Note: Figure not drawn to scale.

A scientist is studying the radioactive decay of an element and wants to determine the sign of some of the particles emitted. The scientist lets a beam of the particles, all with the same speed, pass through an opening in a shield and enter the field created by a wire carrying a constant current. Paths 1 and 2 represent the observed paths of the particles. Based on the paths, which of the following could be the types of particles?

(A)

Path 1 Particle	Path 2 Particle
Electron	Alpha

(B)

Path 1 Particle	Path 2 Particle
Alpha	Electron

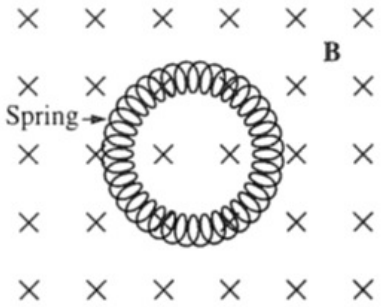
(C)

Path 1 Particle	Path 2 Particle
Positron	Alpha

(D)

Path 1 Particle	Path 2 Particle
Alpha	Positron

45.



A metal spring has its ends attached so that it forms a circle. It is placed in a uniform magnetic field, as shown above. Which of the following will NOT cause a current to be induced in the spring?

(A) Changing the magnitude of the magnetic field

(B) Increasing the diameter of the circle by stretching the spring

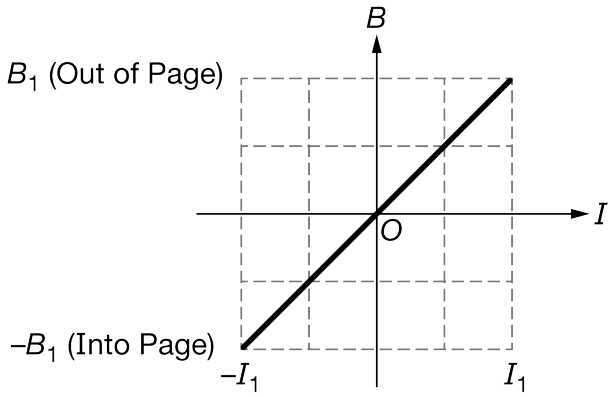
(C) Rotating the spring about a diameter

(D) Moving the spring parallel to the magnetic field

(E) Moving the spring in and out of the magnetic field

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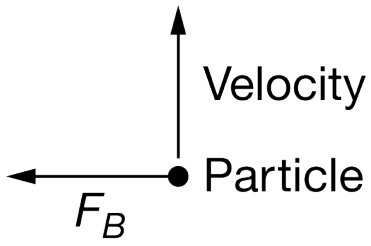
46. This question is a long free-response question. Show your work for each part of the question.



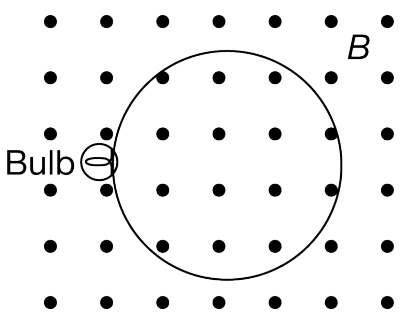
Graph 1

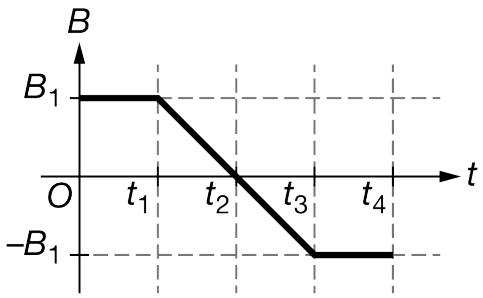
(12 points, suggested time 25 minutes)

An electromagnet produces a magnetic field that is uniform in a certain region and zero outside that region. The graph above represents the field as a function of the current in the electromagnet, with positive field directed out of the page and negative field directed into the page.



(a) The current in the electromagnet is set at $0.5I_1$. When a charged particle in the region moves toward the top of the page, the force exerted on it by the field is F_B toward the left, as shown above. What changes to the current in the electromagnet could make the magnitude of the force exerted on the particle equal to $2F_B$ and the direction of the force to the right? Support your answer using physics principles.

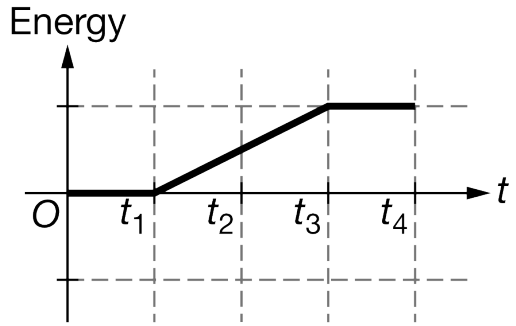




Graph 2

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A circuit is made by connecting an ohmic lightbulb of resistance R and a circular loop of area A made of a wire with negligible resistance. The circuit is placed with the plane of the loop perpendicular to the field of the electromagnet, as shown above on the left. The magnetic field changes as a function of time, as shown in Graph 2. The bulb dissipates energy during the interval $t_1 < t < t_3$. Graph 3 below shows the cumulative energy dissipated by the bulb (the total energy dissipated since $t = 0$) as a function of time.



Graph 3

(b) The original bulb is replaced by a new ohmic lightbulb with a greater resistance, but everything else stays the same. How would the cumulative energy graph for the new bulb be different, if at all, from Graph 3 above? Support your answer using physics principles.

(c) The new lightbulb is removed and replaced by the original lightbulb. The magnetic field now changes from $2B_1$ to $-2B_1$ during the same interval $t_1 < t < t_3$. A new cumulative energy graph is created for this situation. How would the new graph be different, if at all, from Graph 3 ? Support your answer using physics principles.

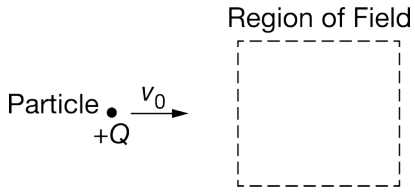
(d) A student derives the following expression for the cumulative energy dissipated by the original bulb during the interval $t_1 < t < t_3$ and with the original change in magnetic field shown in Graph 2.

$$\text{Energy} = \frac{A^2 B_1 R}{4(t_3 - t_1)}$$

Whether or not the equation is correct, does the functional dependence of cumulative energy on the elapsed time $(t_3 - t_1)$ make physical sense? Support your answer using physics principles.

WorkBook Unit 12

47. This question is a short free-response question. Show your work for each part of the question.

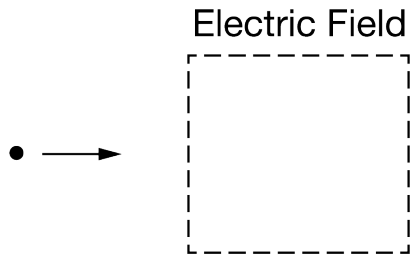


(10 points, suggested time 20 minutes)

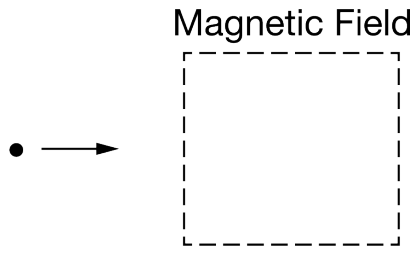
The figure above shows a particle with positive charge $+Q$ traveling with a constant speed v_0 to the right and in the plane of the page. The particle is approaching a region, shown by the dashed box, that contains a constant uniform field. The effects of gravity are negligible.

(a)

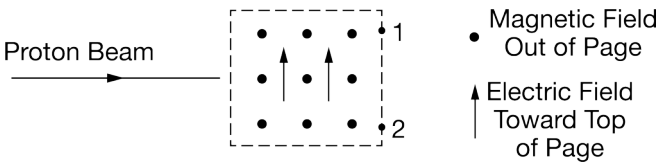
i. On the figure below, draw a possible path of the particle in the region if the region contains only an electric field directed toward the bottom of the page.



ii. On the figure below, draw a possible path of the particle in the region if the region contains only a magnetic field directed out of the page.



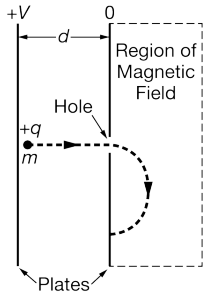
iii. For which of the previous situations is the motion more similar to that of a projectile in only a gravitational field near Earth's surface, and why?



(b) Another region of space contains an electric field directed toward the top of the page and a magnetic field directed out of the page. Both fields are constant and uniform. A horizontal beam of protons with a variety of speeds enters the region, as shown above. Protons exit the region at a variety of locations, including points 1 and 2 shown on the figure. In a coherent, paragraph-length response, explain why some protons exit the region at point 1 and others exit at point 2. Use physics principles to explain your reasoning.

WorkBook Unit 12

48. This question is a long free-response question. Show your work for each part of the question. (15 points, suggested time 25 minutes)



A potential difference V exists between two parallel plates that are separated by a distance d. A particle of mass m and charge $+q$ is accelerated from rest through the potential difference, and as a result moves between the plates in a straight line in the plane of the page, as shown above. The particle reaches the right-hand plate with a speed v, passes through a hole in the plate, and enters a region containing a uniform magnetic field. The particle then follows the semicircular path shown. Gravitational effects are negligible.

(a) Would the speed v of the particle be greater, smaller, or the same if the magnitude of the potential difference was greater? Explain your answer.

(b) A student derives the equation $v = \frac{2qV}{md}$ for the speed of the particle. Whether or not the equation is correct, is it consistent with your answer to part (a)? Explain your reasoning.

(c) A student equates the following two expressions to determine how an increase in the magnitude of the particle's charge affects the radius of its path in the field.

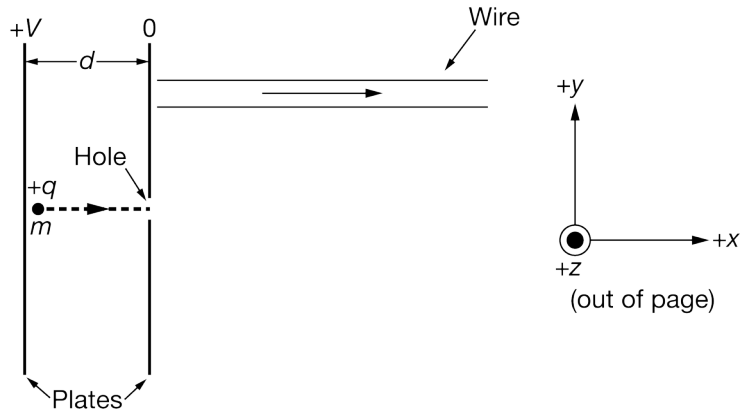
$\frac{mv^2}{r}$ for the magnitude of the net force on an object in uniform circular motion with radius r
qvB for the magnitude of the force on a charged particle in a magnetic field of magnitude B

Describe the relationship between the particle's charge and the radius of its path indicated by these expressions, and indicate how an increase in charge affects the radius.

(d) Another student writes the equation $r = \sqrt{\frac{2Vm}{(B^2+qV)}}$ for the radius of the path. Whether or not the equation is correct, is it consistent with your answer to part (c)? Explain your reasoning.

(e) If the particle had a charge of $-q$, would its motion be the same? Justify your answer.

WorkBook Unit 12

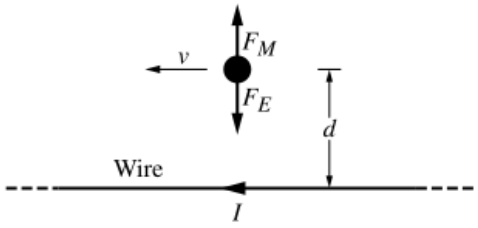


The region of the magnetic field is replaced by a wire that is in the page and has a current directed toward the right. The positively charged particle still moves in a straight line between the plates.

(f) With respect to the coordinate axes shown, in what direction is the acceleration of the particle at the instant it passes through the hole? Explain your reasoning.

WorkBook Unit 12

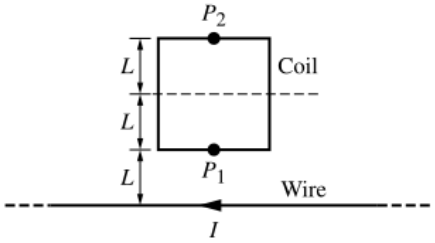
49. This question is a short free-response question. Show your work for each part of the question.



(10 points, suggested time 20 minutes)

At the instant shown above, a negatively charged object is moving to the left with constant velocity v near a long, straight wire that has a current I directed to the left. The region contains a uniform electric field of magnitude E, and the charged object is at a distance d from the wire. The figure shows the electric and magnetic forces, F_E and F_M, respectively, exerted on the charged object.

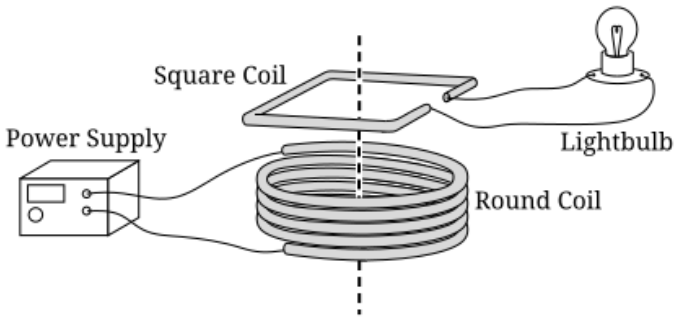
(a) Derive an expression for v in terms of E, d, I, and physical constants, as appropriate.



(b) The charged object is removed, and a square coil with side length $2L$ is placed near the long, straight wire, as shown above. The bottom of the coil is a distance L from the wire. The magnitude of the magnetic field due to the current in the wire is $3B_0$ at point P_1 and B_0 at point P_2.

i. Write an "X" at a location on the figure where the magnitude of the magnetic field is $2B_0$. Briefly justify your reasoning.

ii. Over a time interval of $2.0\ \text{s}$, the current in the wire is decreased. The initial magnetic flux through the coil is $5.0 \times 10^{-5}\ \text{T} \cdot \text{m}^2$ and the final magnetic flux through the coil is $1.0 \times 10^{-5}\ \text{T} \cdot \text{m}^2$. The coil has a total resistance of $10\ \Omega$. Calculate the magnitude of the average current in the coil during the $2.0\ \text{s}$ time interval.

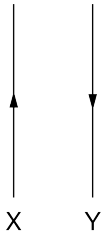


The wire is removed and the square coil is positioned so that the coil is directly above and concentric with a round coil of wire connected to a power supply. A part of the square coil is removed and a lightbulb is connected to the coil, as shown above.

(c) During a short time interval, the current in the power supply is constantly increasing. Use physics principles to explain why the lightbulb is lit during the entire time interval.

WorkBook Unit 12

50. This question is a short free-response question. Show your work for each part of the question.



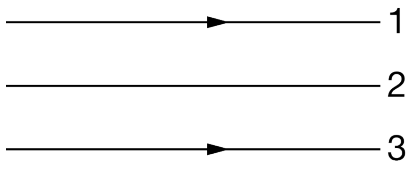
Note: Figures not drawn to scale.

(10 points, suggested time 20 minutes)

The figure above shows two straight wire segments, X and Y, that are 3.0 mm apart. Each segment has a length 0.85 m and carries a current of 5.0 A, but the currents are in opposite directions, as indicated by the arrows in the figure.

(a) Calculate the magnitude of the magnetic force segment X exerts on segment Y. Indicate the direction of the force.

(b) Determine the magnitude and direction of the magnetic force segment Y exerts on segment X.



(c) Three long, parallel, current-carrying wires are shown above. The currents in wires 1 and 3 are toward the right. The magnitude and direction of the current in wire 2 are unknown. The net force on wire 3 is zero. In a coherent paragraph-length response, describe the sources of the individual forces on wire 3 that result in a zero net force. Indicate the direction of the current in wire 2. Include a comparison of the magnitudes of the currents in wires 1 and 2.

WorkBook Unit 12

51. Show all your work for each part of the question. The parts within the question may not have equal weight.

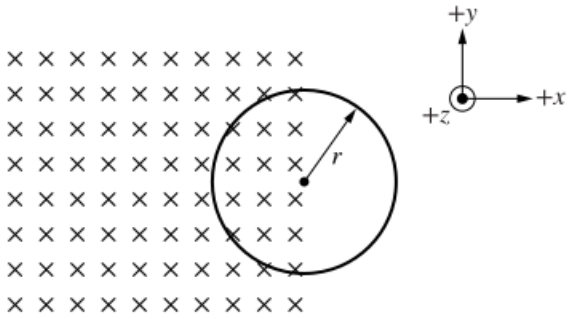


Figure 1

A circular conducting loop of radius r is held at rest in the xy-plane, as shown in Figure 1. The resistance of the loop is R. Half of the loop is in a region with a uniform magnetic field of magnitude B_0 that is directed into the page ($-z$-direction). During a time interval Δt, the magnitude of the magnetic field is increased at a constant rate from B_0 to $4B_0$.

(a) Express your answers for (a)(i) and (a)(ii) in terms of r, R, B_0, Δt, and physical constants, as appropriate.

i. **Derive** an expression for the emf $\mathcal{E}$ induced in the loop during the interval Δt. Begin your derivation by writing a fundamental physics principle or an equation from the reference booklet.

ii. **Derive** an expression for the total energy dissipated by the loop during the interval Δt. Begin your derivation by writing a fundamental physics principle or an equation from the reference booklet.

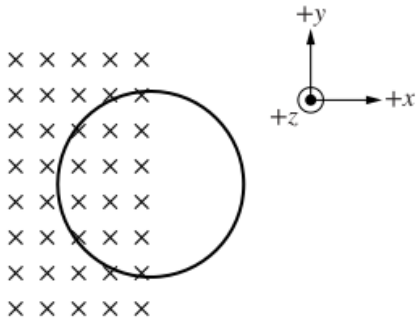


Figure 2

WorkBook Unit 12

iii. Complete the following tasks on Figure 2.

Draw an arrow labeled I to indicate the direction of the current in the loop during the interval Δt.

Draw an arrow labeled F to indicate the direction of the net magnetic force exerted on the loop during the interval Δt.

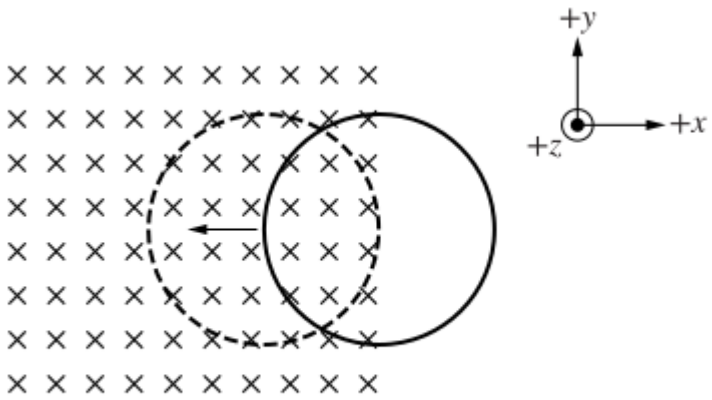


Figure 3

The magnitude of the magnetic field is now held constant while the loop is pulled in the $-x$-direction by an external force until the loop is completely inside the region with the magnetic field, as indicated by the dashed line in Figure 3.

(b) On both blanks below, **write** "yes" or "no" to indicate whether an emf is induced in the loop when the loop is partially in the region with the magnetic field and when the loop is completely in the region.

_____When the loop is partially in the region

_____When the loop is completely in the region

Justify your answers using physics principles.

AP PHYSICS 2

UNIT 13

Geometric Optics

12–15%
AP EXAM WEIGHTING

~8–12
CLASS PERIODS

Conceptual Questions

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Which one of the following is *not* an electromagnetic wave? 1) _____
A) gamma rays
B) ultraviolet
C) radio waves
D) infrared
E) sound waves

2) Which one of the following lists gives the correct order of the electromagnetic spectrum from *low to high frequencies*? 2) _____
A) radio waves, microwaves, infrared, visible, ultraviolet, x–rays, gamma rays
B) radio waves, infrared, x–rays, microwaves, ultraviolet, visible, gamma rays
C) radio waves, ultraviolet, x–rays, microwaves, infrared, visible, gamma rays
D) radio waves, infrared, microwaves, ultraviolet, visible, x–rays, gamma rays
E) radio waves, microwaves, visible, x–rays, infrared, ultraviolet, gamma rays

3) In an electromagnetic wave in free space, the electric and magnetic fields are 3) _____
A) perpendicular to one another and parallel to the direction of wave propagation.
B) parallel to one another and perpendicular to the direction of wave propagation.
C) parallel to one another and parallel to the direction of wave propagation.
D) perpendicular to one another and perpendicular to the direction of wave propagation.

4) Which one of the following lists gives the correct order of the electromagnetic waves from *longer wavelength to shorter wavelength*? 4) _____
A) radio waves, microwaves, visible, x–rays, infrared, ultraviolet, gamma rays
B) radio waves, ultraviolet, x–rays, microwaves, infrared, visible, gamma rays
C) radio waves, microwaves, infrared, visible, ultraviolet, x–rays, gamma rays
D) radio waves, infrared, x–rays, microwaves, ultraviolet, visible, gamma rays
E) radio waves, infrared, microwaves, ultraviolet, visible, x–rays, gamma rays

5) Which one of the following types of electromagnetic wave travels through space the fastest? 5) _____
A) radio waves
B) ultraviolet
C) infrared
D) microwaves
E) They all travel through space at the same speed.

6) For an electromagnetic wave in free space having an electric field of amplitude E and a magnetic field of amplitude B, the ratio of B/E is equal to 6) _____
A) c^2 B) $1/c$ C) c D) $\sqrt{c}$ E) $1/c^2$



7) Which of the following statements about electromagnetic waves in free space are true? (There could be more than one correct choice.) 7) _____

A) The electric and magnetic fields have equal amplitudes.
B) The frequency of the electric field is higher than the frequency of the magnetic field.
C) The electric field carries the same mount of energy as the magnetic field.
D) The electric field carries more energy than the magnetic field.
E) The frequency of the magnetic field is the same as the frequency of the electric field.

8) Two light beams of different frequency but the same intensity fall on a black (totally absorbing) surface, striking perpendicular to the surface. Which of the following statements are true? (There could be more than one correct choice.) 8) _____

A) Painting the surface white would not affect the pressure on it due to these beams.
B) The light beams exert no pressure on the surface because light is just energy.
C) The high–frequency beam exerts more pressure on the surface.
D) Both beams exert the same pressure on the surface.
E) If the surface were painted white (totally reflecting), the pressure on it would be less than when it was black.

9) A light beam has speed c in vacuum and speed v in a certain plastic. The index of refraction n of this plastic is 9) _____

A) $n = cv$. B) $n = (v/c)^2$. C) $n = c/v$. D) $n = (c/v)^2$. E) $n = v/c$.

10) If the index of refraction of a material is 2, this means that light travels 10) _____

A) 2 times as fast in vacuum as it does in the material.
B) 1/2 as fast in air as it does in the material.
C) 2 times as fast in the material as it does in air.
D) 2 times as fast in the material than it does in vacuum.
E) 2 times as fast in air as it does in vacuum.

11) Light enters air from water. The angle of refraction will be 11) _____

A) less than the angle of incidence.
B) greater than the angle of incidence.
C) equal to the angle of incidence.

12) Light enters glass from air. The angle of refraction will be 12) _____

A) less than the angle of incidence.
B) equal to the angle of incidence.
C) greater than the angle of incidence.

13) Light goes from material having a refractive index of n_1 into a material with refractive index n_2. If the refracted light is bent away from the normal, what can you conclude about the indices of refraction? 13) _____

A) $n_1 = n_2$ B) $n_1 < n_2$ C) $n_1 > n_2$ D) $n_1 \leq n_2$ E) $n_1 \geq n_2$

14) A beam of light traveling in air strikes a glass slab at an angle of incidence less than 90°. After entering the glass slab, what does the beam of light do? (There could be more than one correct choice.) 14) _____

A) It slows down.
B) It bends away from the normal at the point of contact.
C) It bends closer to the normal at the point of contact.
D) I follows the same path as before it struck the glass.
E) It follows the normal to the glass slab.

15) The critical angle for a beam of light passing from water into air is 48.8°. This means that all light rays with an angle of incidence in the water that is greater than 48.8° will be 15) _____

A) totally reflected.
B) totally absorbed by the water.
C) totally transmitted.
D) partially reflected and partially transmitted.

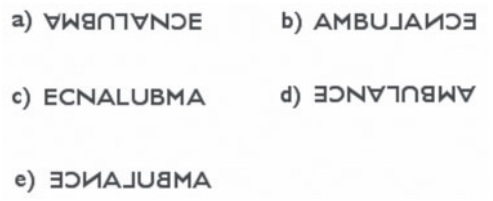
16) Which one of the following sets of characteristics describes the image formed by a plane mirror? 16) _____

A) virtual and larger than the object
B) virtual and inverted
C) real and upright
D) real and inverted
E) virtual and upright

17) The image formed in a plane mirror is 17) _____

A) at a shorter distance behind the mirror than the distance the object is in front of the mirror.
B) at the same distance behind the mirror as the object is in front of the mirror.
C) at a larger distance behind the mirror than the distance the object is in front of the mirror.
D) at the same distance in front of the mirror as the object is in front of the mirror.
E) a real image behind the mirror.

18) You may have seen ambulances on the street with the letters of the word AMBULANCE written on the front of them, in such a way as to appear correctly when viewed in your car's rear-view mirror. (See the figure.) How do the letters appear when you look directly at the ambulance (not through the mirror)? 18) _____

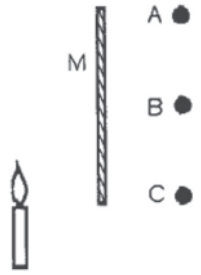


A) a B) b C) c D) d E) e

19) Which one of the following numbers is the correct magnification produced by a plane mirror? 19) _____

A) 3/2 B) 1 C) 1/4 D) 2 E) 1/2

20) A lighted candle is placed a short distance from a plane mirror, as shown in the figure. At which location will the image of the flame appear to be located? 20) _____



A) at A
B) at B
C) at C
D) at M (at the mirror)

21) Suppose you place an object in front of a concave mirror. Which of the following statements *must* be true? (There could be more than one correct choice.) 21) _____
A) No matter where you place the object, the image of the object will always be virtual and upright.
B) The image of the object will always be smaller than the object.
C) No matter where you place the object, a real image of the object will be formed.
D) If you position the object between the mirror and the focal point of the mirror, its image must be upright and virtual.
E) The image of the object will always be inverted.

22) Which statements about images are correct? (There could be more than one correct choice.) 22) _____
A) Mirrors always produce real images because they reflect light.
B) A virtual image cannot be viewed by the unaided eye.
C) A virtual image cannot be formed on a screen.
D) A virtual image cannot be photographed.
E) A real image must be erect.

23) A negative magnification for a mirror means that 23) _____
A) the image is inverted, and the mirror is convex.
B) the image is upright, and the mirror could be either concave or convex.
C) the image is inverted, and the mirror could be either concave or convex.
D) the image is upright, and the mirror is convex.
E) the image is inverted, and the mirror is concave.

24) The focal length of a concave mirror has a magnitude of 20 cm. What is its radius of curvature? 24) _____
A) –20 cm B) 10 cm C) 20 cm D) 40 cm E) –40 cm

25) The focal length of a convex mirror is has a magnitude of 20 cm. What is its radius of curvature? 25) _____
A) 40 cm B) –20 cm C) 20 cm D) –40 cm E) –10 cm

26) If a spherical concave mirror has a radius of curvature R, its focal length is 26) _____
A) $R/4$. B) $2R$. C) $R/2$. D) $4R$. E) R.

27) If a spherical convex mirror has a radius of curvature R, the magnitude of its focal length is 27) _____
A) R. B) $R/4$. C) $2R$. D) $R/2$. E) $4R$.

28) An object is placed to the left of a spherical mirror in front of the mirror. If the image of the object is formed on the right side of the mirror, which of the following statements *must* be true? (There could be more than one correct choice.) 28) _____

A) The image is upright and the mirror must be convex.
B) The image is upright and virtual.
C) The mirror could be either concave or convex.
D) The image is inverted and real, and the mirror must be convex.
E) The image is upright and the mirror must be concave.

29) Which of the following statements about spherical mirrors is correct? (There could be more than one correct choice.) 29) _____

A) A convex mirror always produces an upright image.
B) A concave mirror always produces a virtual image.
C) A convex mirror always produces a real image.
D) A concave mirror always produces a real image.
E) A convex mirror always produces a virtual image.

30) A convex mirror has a focal length of magnitude *f*. An object is placed in front of this mirror at a point $\frac{2}{3}f$ from the face of the mirror. The image will appear 30) _____

A) upright and enlarged.
B) inverted and enlarged.
C) upright and reduced.
D) behind the mirror.
E) inverted and reduced.

31) If an object is placed at the center of curvature of concave mirror, the image formed by the mirror is located 31) _____

A) at the focal point.
B) at the center of curvature.
C) at infinity.
D) between the center of curvature and the focal point.
E) out beyond the center of curvature.

32) If you stand in front of a convex mirror, at the same distance from it as its focal length, 32) _____

A) you will see your image and you will appear larger than you.
B) you will see your image and you will appear smaller than you.
C) you won't see your image because there is none.
D) you will see your image and it will be the same size as you except upside down.
E) you won't see your image because it is focused at infinity.

33) An upright object is 50 cm from a concave mirror of radius 60 cm. The character of the image is 33) _____

A) virtual and upright
B) real and inverted
C) virtual and inverted
D) real and upright

34) A beam of light that is parallel to the principal axis strikes a convex mirror. What happens to the reflected beam of light? 34) _____

A) It appears to be coming from a point between the focal point and the center of curvature of the mirror.
B) It is perpendicular to the principal axis.
C) It appears to be coming from the focal point on the other side of the mirror.
D) It also is parallel to the principal axis.
E) It appears to be coming from the center of curvature of the mirror.

35) Which of the following terms describe lenses that are thinner at the center than at the edges? (There could be more than one correct choice.) 35) _____

A) diverging lenses
B) concave lenses
C) convex lenses
D) converging lenses

36) Which of the following terms describe lenses that are thicker at the center than at the edges? (There could be more than one correct choice.) 36) _____

A) diverging lenses
B) concave lenses
C) convex lenses
D) converging lenses

37) A convex lens has focal length *f*. If an object is located at "infinity" (very far away), the image formed is located at a distance from the lens 37) _____

A) of 2*f*.
B) of *f*.
C) between *f* and 2*f*.
D) of infinity.
E) between the lens and *f*.

38) A convex lens has focal length *f*. If an object is placed at a distance of 2*f* from the lens on the principal axis, the image is located at a distance from the lens 38) _____

A) of infinity.
B) of *f*.
C) between the lens and *f*.
D) of 2*f*.
E) between *f* and 2*f*.

39) If a object is placed between a convex lens and its focal point, the image formed is 39) _____

A) real and upright.
B) real and inverted.
C) virtual and upright.
D) virtual and inverted.

40) Starting from very far away, an object is moved closer and closer to a converging lens, eventually reaching the lens. What happens to its image formed by that lens? (There could be more than one correct choice.) 40) _____

A) The image eventually changes from real to virtual.
B) The image gets closer and closer to the lens.
C) The image keeps getting larger and larger.
D) The image gets farther and farther from the lens.
E) The image eventually changes from virtual to real.

41) Starting from very far away, an object is moved closer and closer to a diverging lens, eventually reaching the lens. What happens to its image formed by that lens? (There could be more than one correct choice.) 41) _____

A) The image gets farther and farther from the lens.
B) The image eventually changes from real to virtual.
C) The image keeps getting larger and larger.
D) The image gets closer and closer to the lens.
E) The image always remains virtual.

42) Which of following statements about the image formed by a single converging lens are true? (There could be more than one correct choice.) 42) _____

A) The image is always virtual.
B) The image is always real.
C) The image is always upright.
D) The image is always inverted.
E) None of the above choices are correct.

43) If a single lens forms a virtual image of an object, then 43) _____

A) the lens must be a diverging lens.
B) the lens could be either a diverging or a converging lens.
C) the lens must be a converging lens.
D) the image must be inverted.
E) the object must be between the lens and its focal point.

44) When light travels from air into water, 44) _____

A) its velocity changes, but its frequency and wavelength do not change.
B) its frequency changes, but its velocity and wavelength do not change.
C) its wavelength changes, but its velocity and frequency do not change.
D) its velocity, wavelength and frequency all change.
E) its velocity and wavelength change, but its frequency does not change.

45) Monochromatic coherent light shines through a pair of slits. If the distance between these slits is decreased, which of the following statements are true of the resulting interference pattern? (There could be more than one correct choice.) 45) _____

A) The distance between the maxima decreases.
B) The distance between the maxima stays the same.
C) The distance between the minima stays the same.
D) The distance between the maxima increases.
E) The distance between the minima increases.

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1.

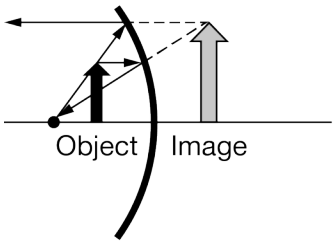


Figure 1

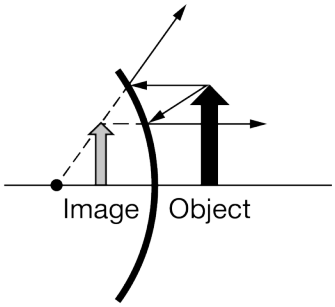


Figure 2

The figures shown illustrate the images formed when a real object is placed in front of a spherical mirror at a distance that is less than the magnitude of the focal length. Which of the following is a correct conclusion from the two figures shown?

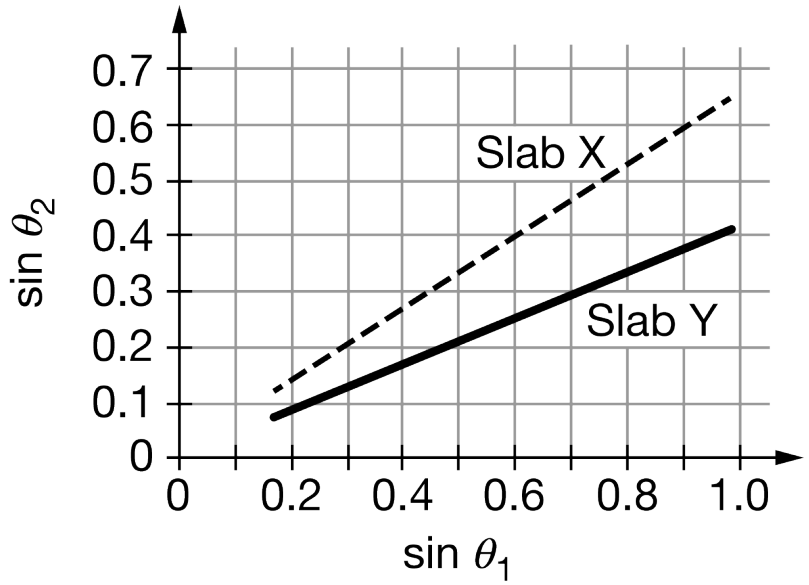
(A) The image formed by the concave mirror is virtual and enlarged, while the image formed by the convex mirror is real and reduced in size.

(B) The image formed by the concave mirror is real and reduced in size, while the image formed by the convex mirror is virtual and enlarged.

(C) The image formed by both concave and convex mirrors is virtual, but the image formed by the concave mirror is enlarged, while the image formed by the convex mirror is reduced in size.

(D) The image formed by both concave and convex mirrors is enlarged, but the image formed by the concave mirror is virtual, while the image formed by the convex mirror is real.

2.



A laser beam is shone on two different slabs of material, and data on the incident angle θ_1 and refracted angle θ_2 of the ray is recorded. The graph shows a best-fit line to the data for each of the slabs. Which of the following is correct about the indices of refraction n of the materials?

(A) $n_X > n_Y$

(B) $n_X = n_Y$

(C) $n_X < n_Y$

(D) Either $n_X > n_Y$ or $n_X < n_Y$, but which is true cannot be determined from the graph.

3. On an optical bench, a student places a small screen and a light source $54\ \text{cm}$ apart. The student then slowly moves a lens along the optical bench from the light source toward the screen. The student measures distances of $18\ \text{cm}$ and $36\ \text{cm}$ between the lens and the light source at two locations where a clear image is produced on the screen. What kind of lens did the student use, and what is the focal length of the lens?

(A) A converging lens with a focal length of $12\ \text{cm}$

(B) A converging lens with a focal length of $36\ \text{cm}$

(C) A diverging lens with a focal length of $12\ \text{cm}$

(D) A diverging lens with a focal length of $36\ \text{cm}$

4. A dentist uses a concave mirror with radius R to examine a tooth. The dentist holds the mirror a distance $R/4$ from the tooth. The image observed by the dentist is

(A) virtual, inverted, and smaller than the tooth

(B) virtual, upright, and larger than the tooth

(C) real, inverted, and larger than the tooth

(D) real, upright, and smaller than the tooth

5. Which of the following indicates the object distance s_o for which a spherical concave mirror of focal length f produces an upright image?

(A) $s_o < f$

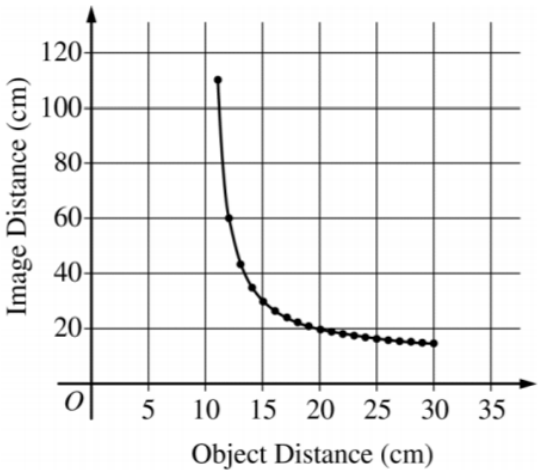
(B) $s_o = f$

(C) $2f > s_o > f$

(D) $s_o > 2f$

WorkBook Unit 13

6.



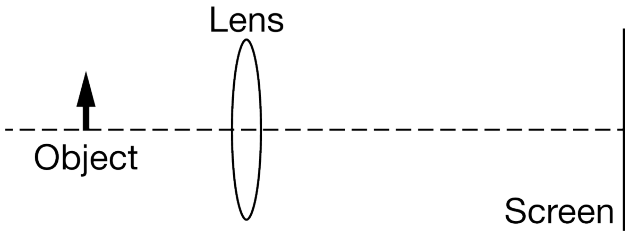
Some students collect data to determine the focal length of a concave mirror. Using a small candle as the object, they move a piece of white cardboard until they see a focused image of the candle flame on it. The students measure the distances of the candle and the image from the mirror. A graph of the data is shown above. Which of the following is closest to the focal length of the mirror?

(A) 10 cm

(B) 15 cm

(C) 20 cm

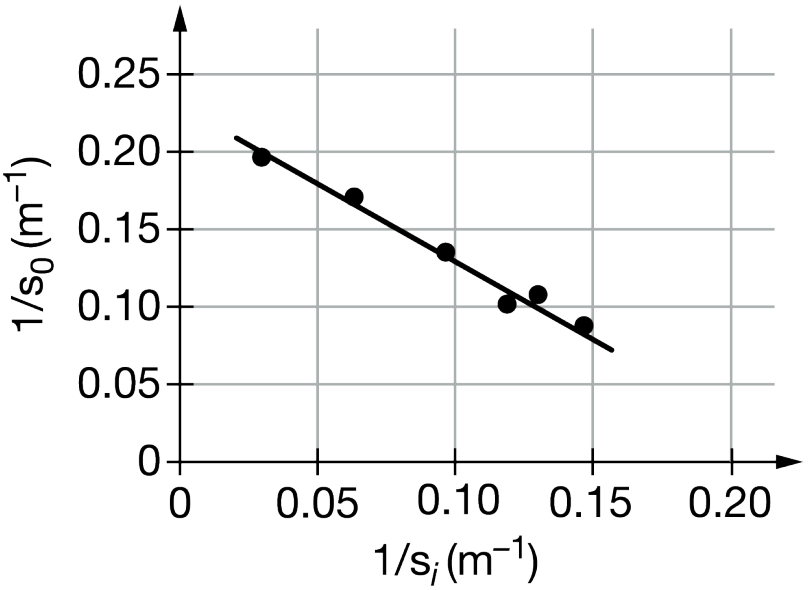
(D) 30 cm



The figure shows a converging glass lens in air, an object to the left of the lens, and a screen to the right of the lens.

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7.



A student places the object at various locations to the left of the lens and moves the screen to observe focused images. The student uses the data for object distance s_o and image distance s_i to create the graph, drawing a best-fit line to the data. Which of the following represents the focal length of the lens?

(A) The magnitude of the line's slope

(B) The inverse of the magnitude of the line's slope

(C) The intercept of the line with the horizontal axis

(D) The inverse of the intercept of the line with the vertical axis

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8.

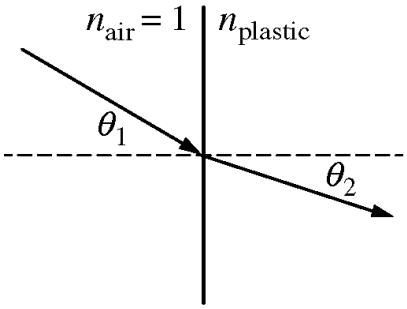
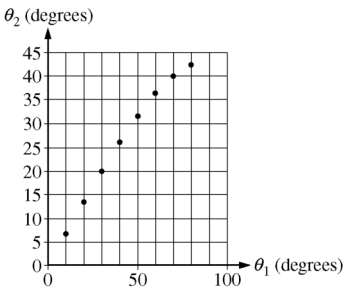
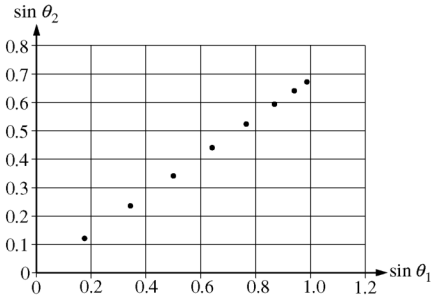


Figure 1

A student performs an experiment to determine the index of refraction of a plastic block. The student shines a laser beam from air ($n_{air} = 1$) into a piece of plastic ($n_{plastic}$) at an angle of incidence θ_1 and measures the angle of refraction θ_2, as shown in Figure 1 above. Measurements are taken as θ_1 is increased. From the data the student produces the two graphs shown below.



Graph 1



Graph 2

Which of the following is equal to the index of refraction of the plastic block.

(A) The slope of the best-fit line for graph 1

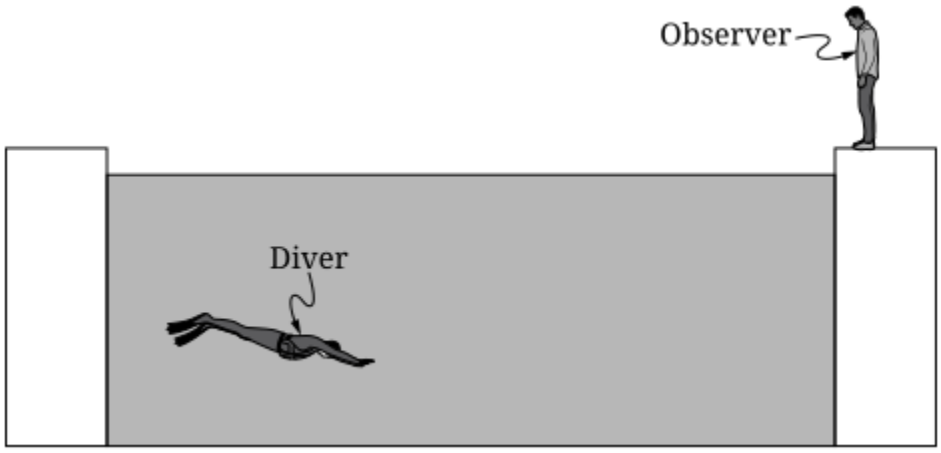
(B) The slope of the best-fit line for graph 2

(C) The inverse of the slope of the best-fit line for graph 1

(D) The inverse of the slope of the best-fit line for graph 2

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9.



A diver is swimming underwater in a pool filled with water. An observer is standing at the edge of the pool, as shown. Will the observer always be able to see the diver no matter where the diver is in the pool? Why or why not?

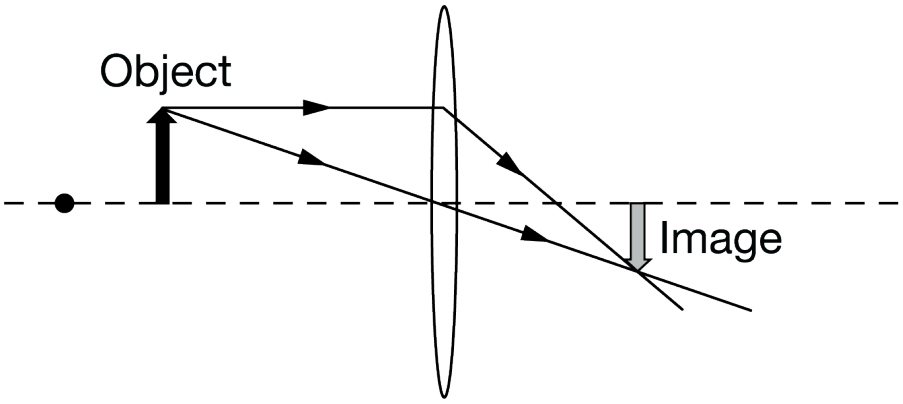
(A) Yes, because there will always be a light ray originating from the diver that will refract toward the observer and reach the observer.

(B) Yes, because for every position of the diver, there is a straight-line path connecting the diver and the observer.

(C) No, because light passing from a medium with high index of refraction to low index of refraction will always totally reflect at the boundary between the two media.

(D) No, because for some positions of the diver some of the light rays from the diver will totally reflect at the water-air boundary.

10.



The figure shows a ray diagram for an object near a lens. How will the height of the image change if the object is moved to the left toward the position of the dot?

(A) It will not change.

(B) It will decrease.

(C) It will increase.

(D) It cannot be determined without knowing the ratio of the image distance to the focal length.

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11.

θ_i (degrees)	θ_r (degrees)
0	0
16	10.8
32	20.4
48	30.1
64	37.9
80	39.8

A student measures the incident angle θ_i and the refracted angle θ_r of a beam of light that travels through air and then enters a material of unknown index of refraction. The student's data for several angles are shown in the table above. Which of the following procedures should the student follow so that the slope of a best-fit line will be the index of refraction of the material?

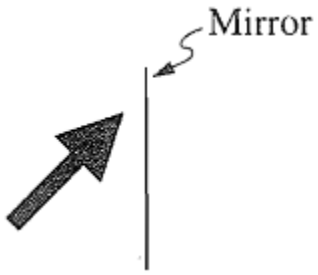
(A) Graph θ_r as a function of θ_i.

(B) Graph θ_i as a function of θ_r.

(C) Graph $\sin\ \theta_r$ as a function of $\sin\ \theta_i$.

(D) Graph $\sin\ \theta_i$ as a function of $\sin\ \theta_r$

12.

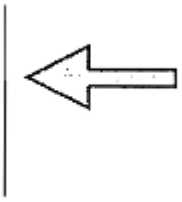


An object, slanted at an angle of 45°, is placed in front of a vertical plane mirror, as shown above. Which of the following shows the apparent position and orientation of the object's image?

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(A) 

(B) 

(C) 

(D) 

(E) 

13. An object is placed in front of a converging thin lens at a distance from the center of the lens equal to half the focal length. Compared to the object, the image is

(A) upright and larger
(B) upright and smaller
(C) inverted and larger
(D) inverted and smaller
(E) inverted and the same size

14. A student holds a lens $40\ \text{cm}$ from an object and observes an enlarged image on a screen on the opposite side of the lens. Which of the following could be the type of lens and its focal length?

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(A)

Type of Lens	Focal Length
Diverging	30 cm

(B)

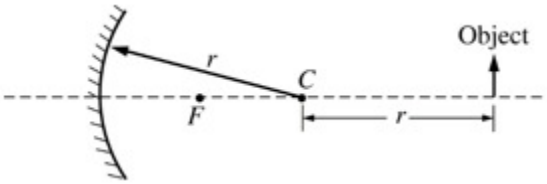
Type of Lens	Focal Length
Diverging	50 cm

(C)

Type of Lens	Focal Length
Converging	30 cm

(D)

Type of Lens	Focal Length
Converging	50 cm



The concave spherical mirror of radius r shown above has focal point F and center of curvature C. An object is placed at a distance $2r$ from the mirror.

15. Where will the image be formed?

(A) To the left of the mirror
(B) Between F and the mirror
(C) Between F and C
(D) Between C and the object
(E) To the right of the object

16. Which of the following correctly describes the image?

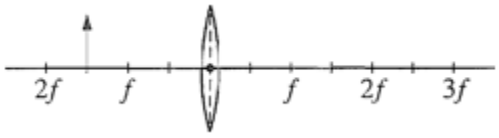
(A) Real, inverted, smaller than the object
(B) Real, inverted, larger than the object
(C) Real, upright, smaller than the object
(D) Virtual, upright, smaller than the object
(E) Virtual, inverted, larger than the object

---

17. A student holds a large concave mirror of focal length 1.0 m that faces a teacher. Which of the following describes the teacher's image as the student approaches the teacher from a distance of 2.0 m to a distance of 0.5 m?

(A) It continuously increases in size.
(B) It continuously decreases in size.
(C) It is first upright, then it is upside down.
(D) It is first upside down, then it is upright.
(E) It is always smaller than the teacher.

18.



An object is placed at a distance of 1.5*f* from a converging lens of focal length *f,* as shown above. What type of image is formed and what is its size relative to the object?

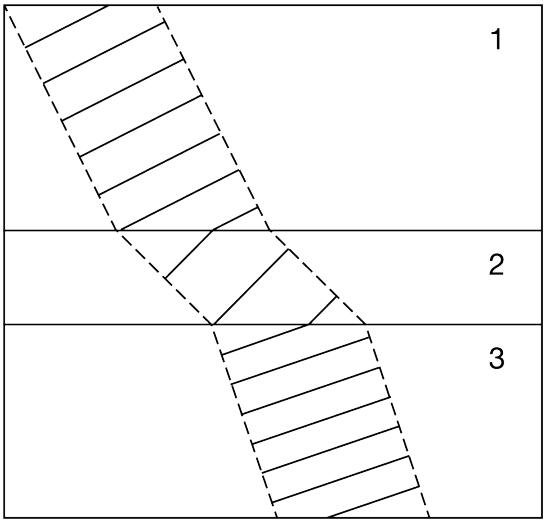
(A) **Type:** Virtual
Size: Larger
(B) **Type:** Virtual
Size: Same size
(C) **Type:** Virtual
Size: Smaller
(D) **Type:** Real
Size: Larger
(E) **Type:** Real
Size: Smaller

19. A student in a classroom sees a tree outside that is far from the classroom window and uses a concave mirror to form an image of the tree on a screen. Which of the following best describes the image?

(A) Real and near the center of curvature of the mirror
(B) Real and near the focal point of the mirror
(C) Virtual and near the center of curvature of the mirror
(D) Virtual and near the focal point of the mirror

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20.



A beam of light travels through materials 1, 2, and then 3, as shown. Wave fronts of the beam within each material are represented as parallel lines. Which of the following is a correct statement about the propagation speed of light in these materials?

(A) The propagation speed in material 2 is equal to the propagation speed in materials 1 and 3, because the propagation speed of light is c.

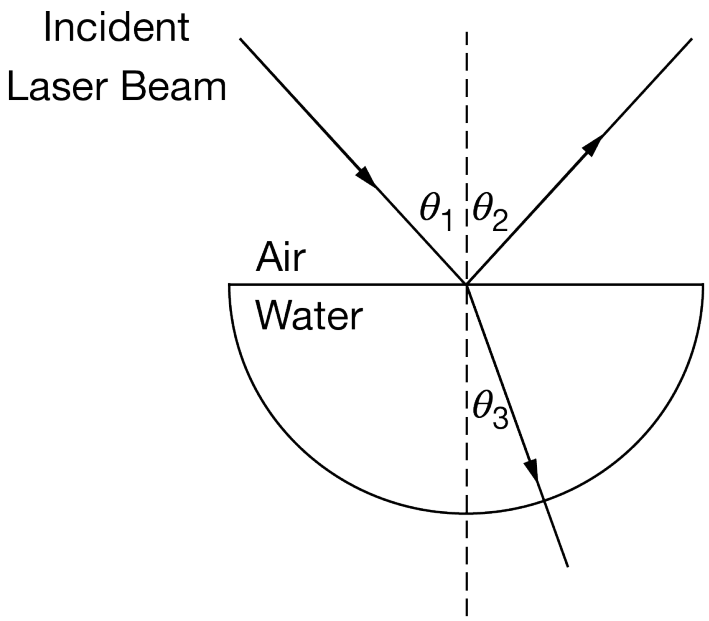
(B) The propagation speed in material 2 is higher than the propagation speed in materials 1 and 3, because the wave fronts are farther apart.

(C) The propagation speed in material 2 is lower than the propagation speed in materials 1 and 3, because the angle of the beam entering and leaving material 2 is steeper than within this material.

(D) The propagation speed of material 3 cannot be compared with that of material 1, because these materials are not adjacent to one another.

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21.



The figure shows a clear hemispherical tank filled with water. Students are to experimentally determine the index of refraction of the water. The students shine a laser toward the flat top of the water at an angle θ_1, as shown. They measure the angle θ_2 of the reflected beam and the angle θ_3 of the refracted beam. Which of the following is the best graph for analyzing the data to determine the index of refraction of the water?

(A) Graph θ_2 as a function of θ_1, draw a best-fit line, and find the intercept of the line with the θ_2-axis.

(B) Graph θ_3 as a function of θ_1, draw a best-fit line, and find the intercept of the line with the θ_3-axis.

(C) Graph $\sin\theta_3$ as a function of $\sin\theta_1$, draw a best-fit line, and find the intercept of the line with the $\sin\theta_3$-axis.

(D) Graph $\sin\theta_3$ as a function of $\sin\theta_1$, draw a best-fit line, and find the slope of the line.

22. A student places a lens $40\ \text{cm}$ from an object and observes an enlarged image on a screen on the opposite side of the lens. Which of the following could be the type of lens and the focal length of the lens?

(A) A diverging lens with a focal length of $30\ \text{cm}$

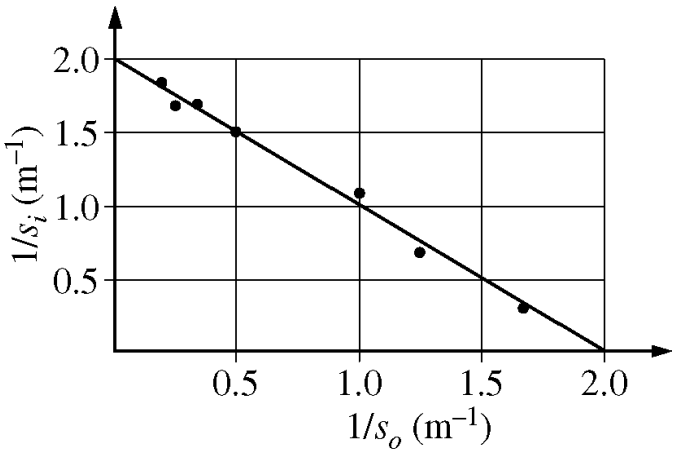
(B) A diverging lens with a focal length of $50\ \text{cm}$

(C) A converging lens with a focal length of $30\ \text{cm}$

(D) A converging lens with a focal length of $50\ \text{cm}$

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Refer to the material below to answer the following questions.



A group of students collected data using a lens. They varied the distance s_o of an object from the lens and measured the image distance s_i. The figure above is their graph of the inverse of the image distance as a function of the inverse of the object distance.

23. What is the magnitude of the image's magnification when the object is placed 2 m from the lens?

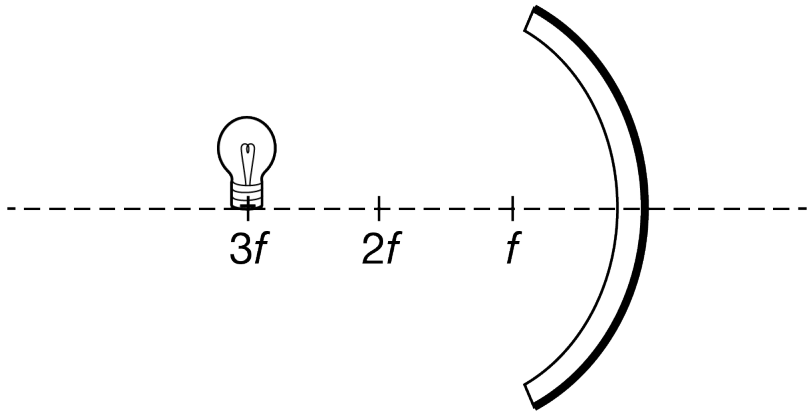
(A) 1/3

(B) 1

(C) 3

(D) The magnification is undefined because the image is an infinite distance from the lens.

24.



A lightbulb is placed a distance $3f$ from a concave mirror with focal length f, as shown above. What is the distance of the image from the mirror, and how does the size of the image compare with the size of the object?

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(A)

Distance of Image from Mirror	Size of Image
$\frac{2}{3}f$	Larger than object

(B)

Distance of Image from Mirror	Size of Image
$\frac{2}{3}f$	Smaller than object

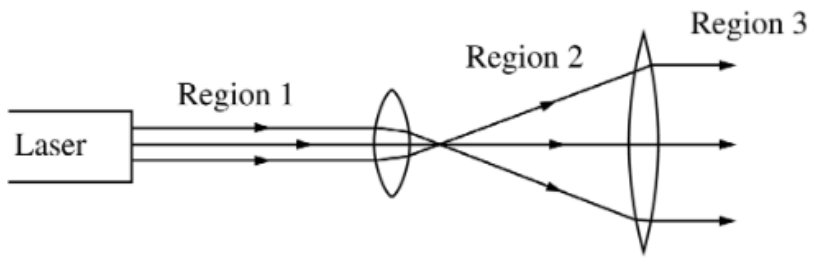
(C)

Distance of Image from Mirror	Size of Image
$\frac{3}{2}f$	Larger than object

(D)

Distance of Image from Mirror	Size of Image
$\frac{3}{2}f$	Smaller than object

25.

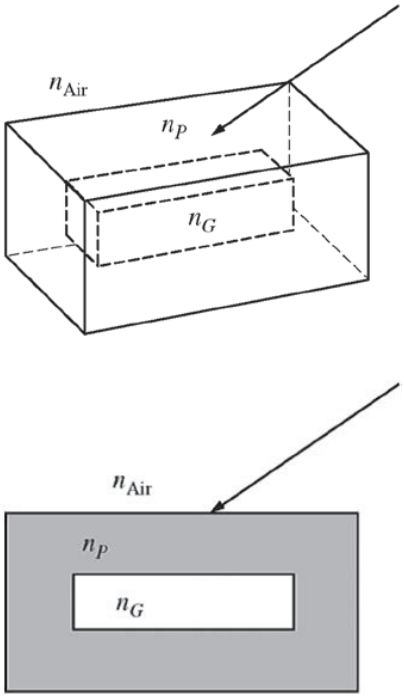


As shown above, two lenses are arranged so that a narrow laser beam of parallel rays spreads into a wider beam of parallel rays. If the lenses are interchanged, the light in region 3 will be a beam of

(A) parallel rays narrower than the beam in region 1
(B) parallel rays wider than the beam in region 1
(C) parallel rays about the same width as in region 1
(D) diverging rays
(E) converging rays

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26.



Students in a lab group are given a plastic block with a hollow space in the middle, as shown in the figures above. The index of refraction n_P of the plastic is known. The hollow space is filled with a gas, and the students are asked to collect the data needed to find the index of refraction n_G of the gas. The arrow represents a light beam that they shine into the plastic. They take the following set of measurements:

Angle of incidence of the light in the air above the plastic block	30°
Angle of refraction of the beam as it enters the plastic from the air	45°
Angle of refraction of the beam as it enters the plastic from the gas	45°

The three measurements are shared with a second lab group. Can the second group determine a value of n_G from only this data?

(A) Yes, because they have information about the beam in air and in the plastic above the gas.

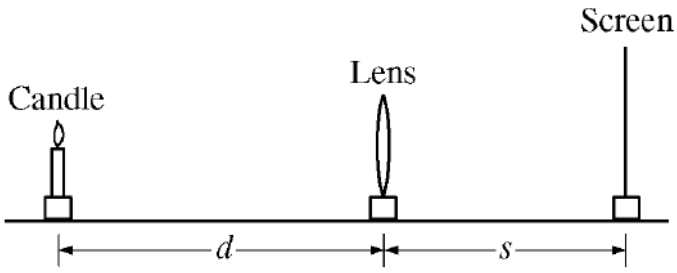
(B) Yes, because they have information about the beam on both sides of the gas.

(C) No, because they need additional information to determine the angle of the beam in the gas.

(D) No, because they do not have multiple data points to analyze.

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27.



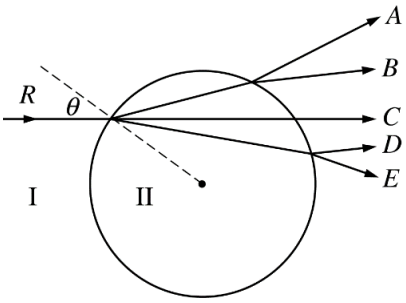
In the experimental setup shown in the figure above, a thin lens stands between a candle and a screen. A student moves the candle to several positions relative to the lens and then adjusts the position of the screen until an image is formed. Each time, the student records the distance d from the candle to the lens and the distance s from the lens to the screen. Which of the following procedures will allow the student to determine the focal length of the lens?

(A) Graph s as a function of d. The focal length will be the vertical intercept.

(B) Graph s as a function of d. The focal length will be the slope of the line.

(C) Graph 1/s as a function of 1/d . The focal length will be the inverse of the vertical intercept.

(D) Graph 1/s as a function of 1/d . The focal length will be the inverse of the slope of the line.



A light ray R in medium I strikes a sphere of medium II with angle of incidence Θ, as shown above. The figure shows five possible subsequent paths for the light ray.

28. Which path is possible if medium I is air and medium II is glass?

(A) A

(B) B

(C) C

(D) D

(E) E

29. Which path is possible if medium I is glass and medium II is air?

(A) A
(B) B
(C) C
(D) D
(E) E

---

30. When light passes from air into water, the frequency of the light remains the same. What happens to the speed and the wavelength of light as it crosses the boundary in going from air into water?

(A)

Speed	Wavelength
Increases	Remains the same Increases

(B)

Speed	Wavelength
Remains the same	Decreases

(C)

Speed	Wavelength
Remains the same	Remains the same

(D)

Speed	Wavelength
Decreases	Increases

(E)

Speed	Wavelength
Decreases	Decreases

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31.

Object Distance from Lens	Orientation of Image	Type of Image	Size of Image Compared to Object
5 cm	Upright	Virtual	Larger
10 cm	Upright	Virtual	Larger
15 cm	Inverted	Real	Larger
25 cm	Inverted	Real	Smaller

A student uses a convex lens to create images and collects the data shown above. The focal length of the lens is in which of the following ranges?

(A) 0–5 cm

(B) 5–10 cm

(C) 10–15 cm

(D) 15–25 cm

32. When a light wave passes from air into glass, quantities that remain constant include which of the following?

I. Frequency

II. Wavelength

III. Speed

(A) I only

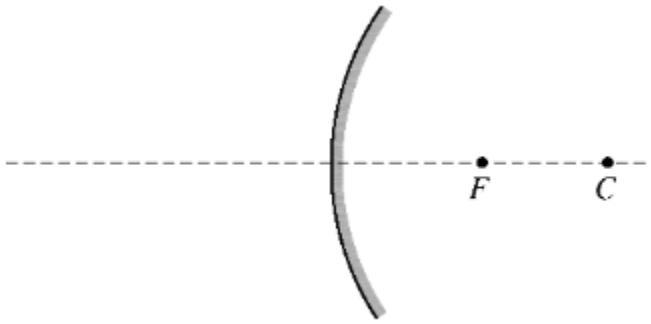
(B) II only

(C) I and III only

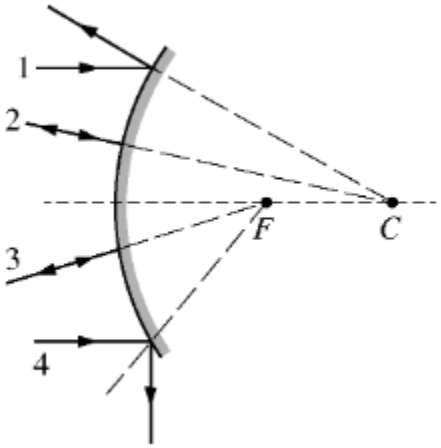
(D) II and III only

(E) I, II, and III

WorkBook Unit 13



The figure above shows a convex mirror, its focal point *F*, and its center of curvature *C*.



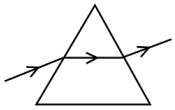
33. Which of the rays shown above is correct for this mirror?

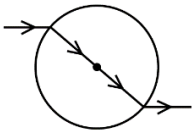
(A) 1
(B) 2
(C) 3
(D) 4

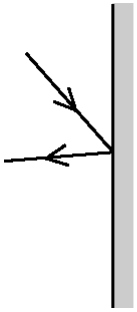
34. An object is located 2 m to the left of the mirror. If point C is 30 cm from the mirror, what is the approximate location of the object's image?

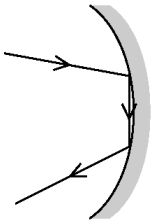
(A) 26 cm to the right of the mirror
(B) 14 cm to the right of the mirror
(C) 16 cm to the left of the mirror
(D) 35 cm to the left of the mirror

35. A student working with glass objects and silvered mirrors traces light rays in the laboratory. Which of the following ray tracings is drawn most correctly?

(A)  Triangular Glass Prism

(B)  Glass Sphere

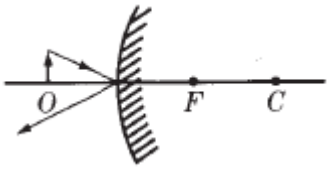
(C)  Plane Mirror

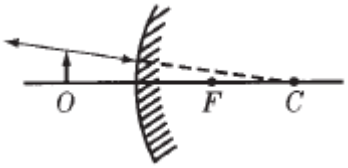
(D)  Concave Mirror

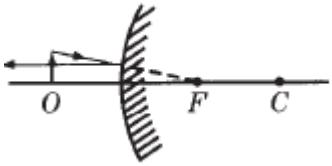
(E)  Converging Lens

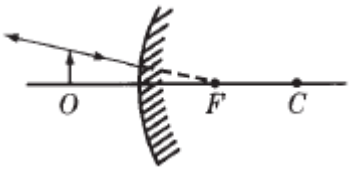
36. An object *O* is in front of a convex mirror. The focal point of the mirror is labeled *F* and the center of curvature is labeled *C*. The direction of the reflected ray is correctly illustrated in all of the following EXCEPT which diagram?

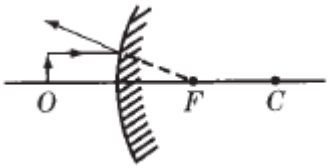
WorkBook Unit 13

(A) 

(B) 

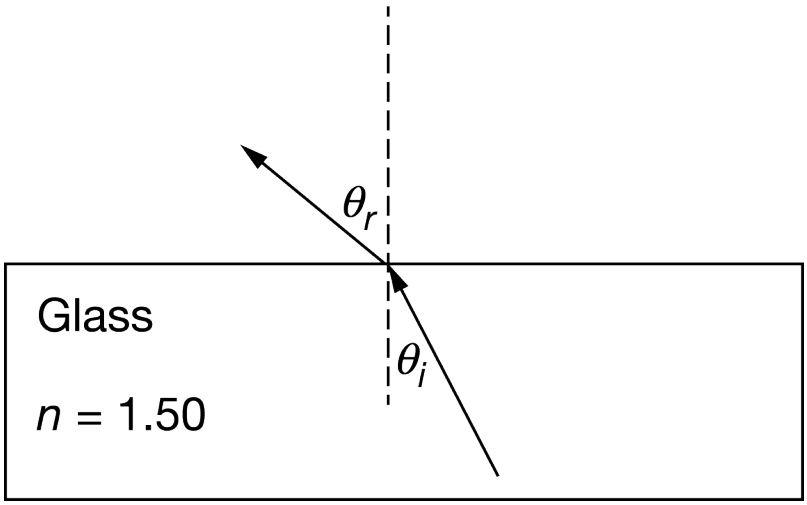
(C) 

(D) 

(E) 

WorkBook Unit 13

37.



In an optics experiment, students are given a laser light source, eye protection, and a glass block $(n = 1.50)$. The students want to find the critical angle (angle of incidence above which total internal reflection occurs) at the glass-air boundary. They adjust the angle of incidence θ_i of the laser beam in the glass and record the angle of refraction θ_r in air, as shown in the figure. The results are shown in the table. Which of the following refinements to the procedure would best allow the students to more accurately estimate the critical angle?

θ_i	θ_r
0°	0°
15°	23°
30°	49°
45°	None
60°	None

(A) Reverse the direction of the laser beam and have it go from air into the glass. Repeat the measurements of incident and refracted angles.

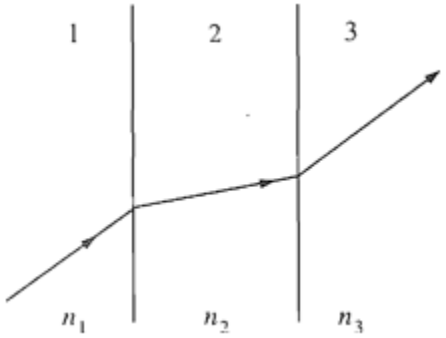
(B) Use smaller increments of θ_i between 30° and 45°.

(C) Have two groups independently perform the original procedure using the same incident angles and then average the results.

(D) Take measurements at 75° and 90°.

WorkBook Unit 13

38.



A light ray passes through substances 1, 2, and 3, as shown above. The indices of refraction for these three substances are n_1, n_2, and n_3, respectively. Ray segments in 1 and in 3 are parallel. From the directions of the ray, one can conclude that

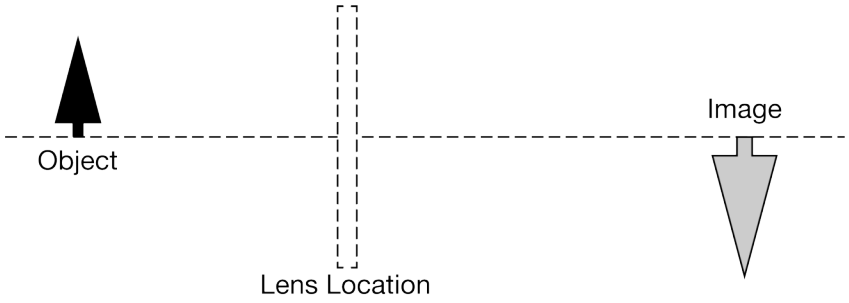
(A) n_3 must be the same as n_1

(B) n_2 must be less than n_1

(C) n_2 must be less than n_3

(D) n_1 must be equal to 1.00

(E) all three indices must be the same



The figure above shows an object near a thin lens and the resulting image formed by the lens.

39. Which of the following correctly describes the type of lens and image?

WorkBook Unit 13

(A)

Lens	Image
Converging	Virtual

(B)

Lens	Image
Converging	Real

(C)

Lens	Image
Diverging	Virtual

(D)

Lens	Image
Diverging	Real

---

40. Students design an experiment to determine the focal length f of a converging lens. The students use a lit candle as an object and project its image onto an index card on the other side of the lens. The distance s_o from the candle to the lens and the distance s_i from the card to the lens are recorded for several values of s_o. Are these data sufficient to calculate f, and why or why not?

(A) No, because the magnification must also be measured for each value of s_o.

(B) No, because f will be different for each value of s_o.

(C) Yes, because $\frac{1}{s_i}$ can be plotted as function of $\frac{1}{s_o}$ and the y-intercept used to calculate f.

(D) Yes, because $\frac{1}{s_i}$ can be plotted as function of $\frac{1}{s_o}$ and the slope can be used to calculate f.

41. This question is a long free-response question. Show your work for each part of the question.

(12 points, suggested time 25 minutes)

Students use a lightbulb, a screen, and a meterstick to determine the focal length of a concave mirror from a graph.

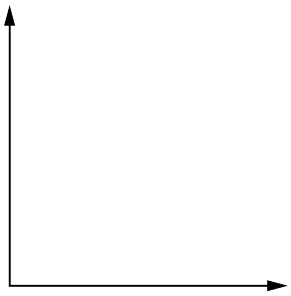
(a)

i. Describe a procedure that could be used to collect appropriate data and the measurements to be taken.

- Label the axes below with the quantities to be graphed.
- Draw a clearly labeled diagram of the experimental setup.
- Describe the procedure and indicate the quantities to be measured.

Label the Axes

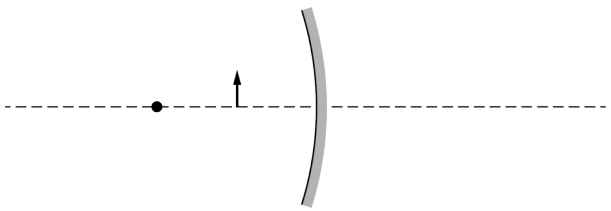
Describe the Procedure



Draw a Diagram

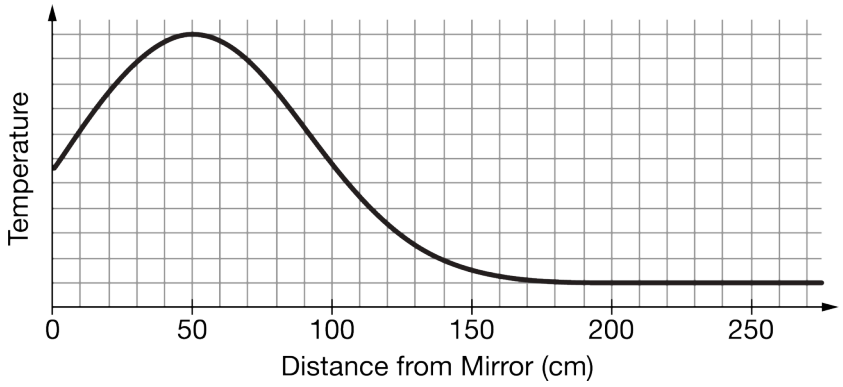
ii. Describe how the focal length can be determined from the graph.

(b) The figure below represents one arrangement of a concave mirror and an object. The dot represents the focal point of the mirror and the arrow represents the object. Draw a ray diagram to illustrate the formation of the image that includes at least two rays and the resulting image.



(c) Some students take the concave mirror outside and hold it with the concave side facing the Sun. The students want to locate the image of the Sun. They measure the temperature of the air in front of the mirror along the principal axis and graph the air temperature as a function of distance from the mirror. The graph below is their

best-fit curve to the data.



i. Based on the curve, at what distance from the mirror would you expect to find the image of the Sun?

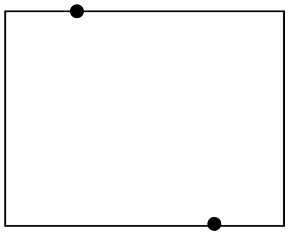
ii. Use a relevant equation to explain how your response to (c)(i) relates to the focal length of the mirror.

42. This question is a long free-response question. Show your work for each part of the question.

(12 points, suggested time 25 minutes)

A group of students is asked to determine the index of refraction of a plastic block. They have a laser pointer mounted on a protractor. The laser can be pivoted and the angle of incidence of the laser on the block can be measured. However, the laser beam is not visible inside the plastic block. Only the spots on the surface of the block where the laser enters and exits are visible.

(a) The rectangle below represents the plastic block. The laser beam enters at the dot on the top of the block and exits at the dot on the bottom. On the figure, indicate all the distance measurements needed to determine the index of refraction of the block. Justify why the measurements are useful to determine the index of refraction. You may add other lines to the figure to assist in your justification.



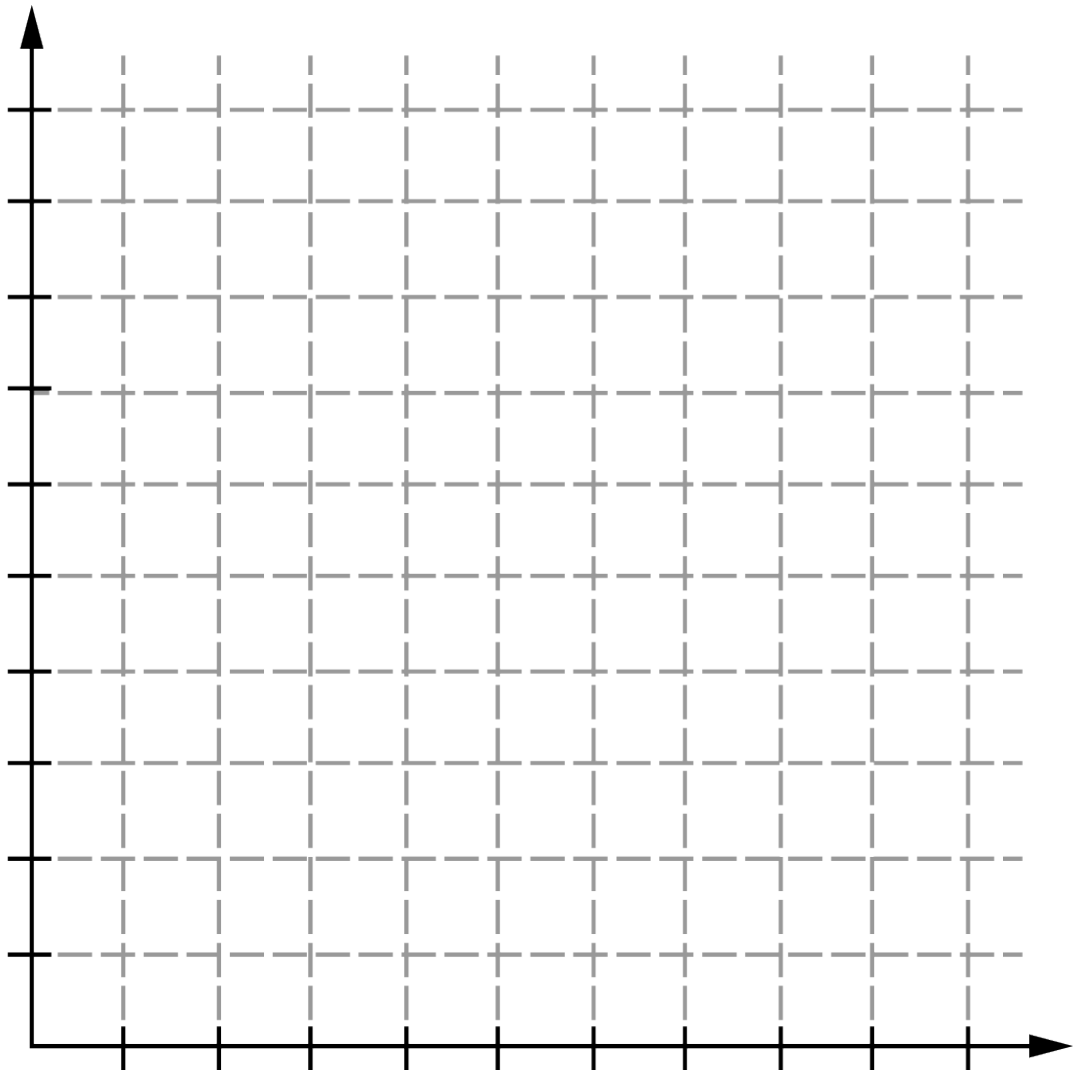
The students obtain the data in the table.

θ_{air}	$\sin\theta_{\text{air}}$	θ_{plastic}	$\sin\theta_{\text{plastic}}$
20	0.34	11	0.19
30	0.50	17	0.30
40	0.64	26	0.44
50	0.77	31	0.51
60	0.87	33	0.55
70	0.94	40	0.64

(b)

i. On the axes below, plot data that will allow determination of the index of refraction of the plastic from a best-fit line. Be sure to label and scale the axes. Draw a best-fit line that could represent the data.

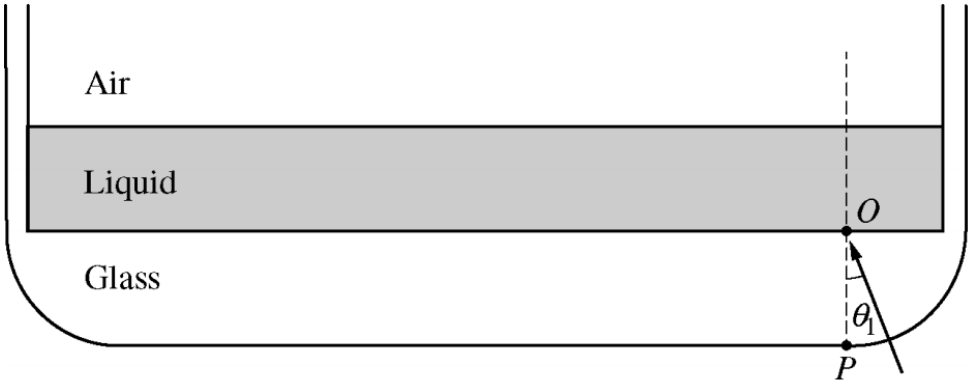
WorkBook Unit 13



ii. Determine the index of refraction from the graph.

(c) Blocks of plastic 1 and plastic 2, with indices of refraction n_1 and n_2, respectively, are placed in contact with each other. A laser beam in plastic 1 is incident on the boundary with plastic 2. Using the model of light as it crosses the boundary between the plastics, determine an expression for the ratio λ_1/λ_2 of the wavelengths of the light in the two plastics in terms of n_1, n_2, and physical constants as appropriate.

43.

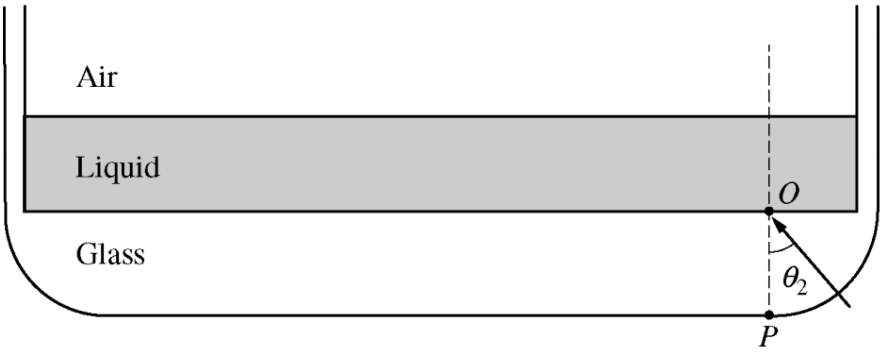


The figure above shows a cross section of a drinking glass (index of refraction 1.52) filled with a thin layer of liquid (index of refraction 1.33). The bottom corners of the glass are circular arcs, with the bottom right arc centered at point *O*. A monochromatic light source placed to the right of point *P* shines a beam aimed at point *O* at an angle of incidence θ. The flat bottom surface of the glass containing point *P* is frosted so that bright spots appear where light from the beam strikes the bottom surface and does not reflect. When $\theta = \theta_1$, two bright spots appear on the bottom surface of the glass. The spot closer to point *P* will be referred to as *X*; the spot farther from *P* will be referred to as *Y*. The location of spot *X* and that of spot *Y* both change as θ is increased.

a. In a coherent paragraph-length answer, describe the processes involved in the formation of spots *X* and *Y* when $\theta = \theta_1$. Include an explanation of why spot *Y* is located farther from point *P* than spot *X* is and what factors affect the brightness of the spots.

b. When θ is increased to θ_2 , one of the spots becomes brighter than it was before, due to total internal reflection.

i. On the figure below, draw a ray diagram that clearly and accurately shows the formation of spots *X* and *Y* when $\theta = \theta_2$.

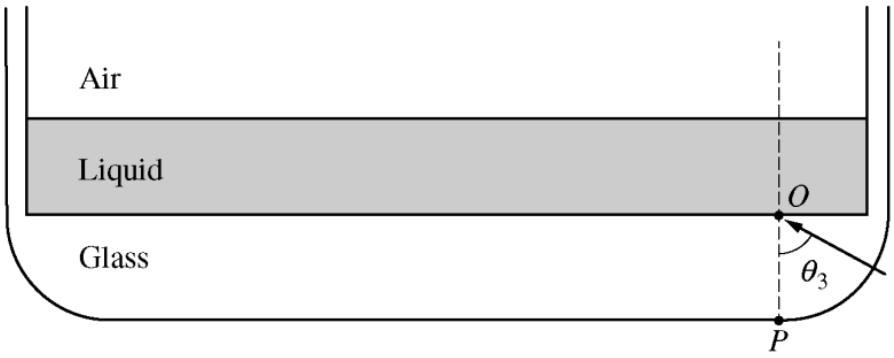


ii. Which spot, *X* or *Y*, becomes brighter than it was before due to total internal reflection? Explain

your reasoning.

c. When θ is further increased to θ_3 , one of the spots disappears entirely.

i.

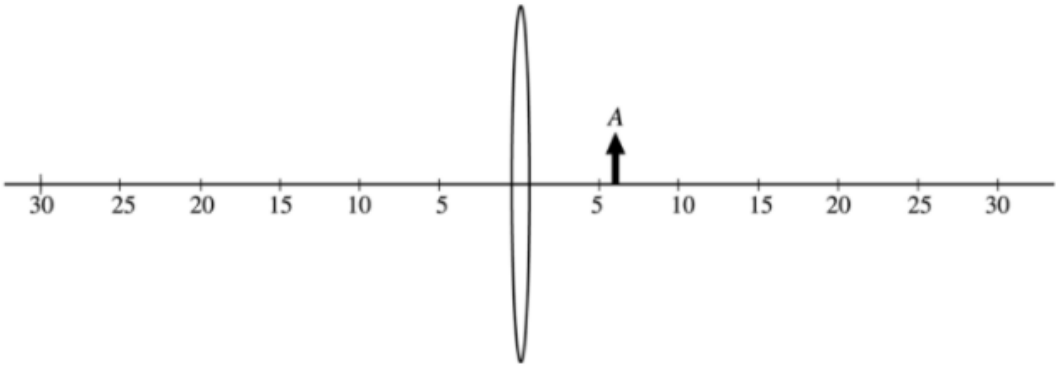


ii. Indicate which spot, *X* or Y, disappears. Explain your reasoning in terms of total internal reflection.

WorkBook Unit 13

44. A thin converging lens of focal length 10 cm is used as a simple magnifier to examine an object *A* that is held 6 cm from the lens.

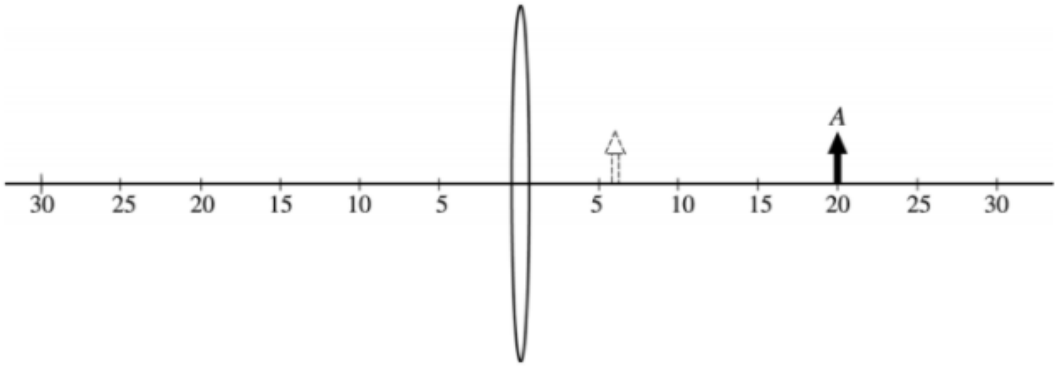
(a) On the figure below, draw a ray diagram showing the position and size of the image formed.



(b) State whether the image is real or virtual. Explain your reasoning.

(c) Calculate the distance of the image from the center of the lens.

(d) Calculate the ratio of the image size to the object size.



(e) The object *A* is now moved to the right from x=6 cm to a position of x=20 cm, as shown above. Describe the image position, size, and orientation when the object is at x=20 cm.

WorkBook Unit 13

45. Show all your work for each part of the question. The parts within the question may not have equal weight.

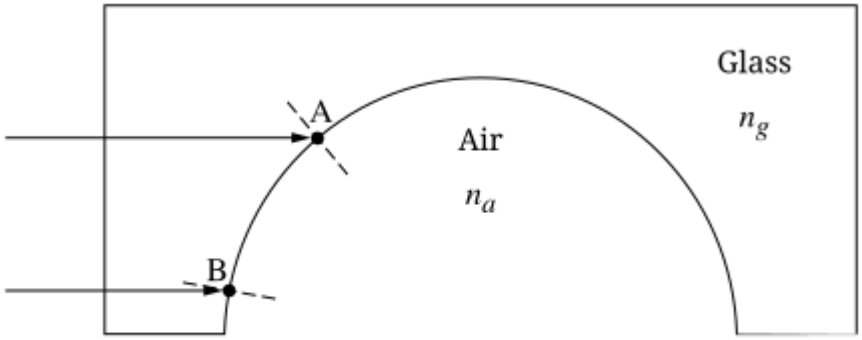


Figure 1

A glass block has a semicircular section removed, as shown in Figure 1. The air has an index of refraction $n_a = 1$ and the glass has an index of refraction $n_g > 1$. Two rays of monochromatic light of frequency f_0 travel parallel to each other in the glass block and are incident on the glass-air boundary at points A and B, as shown. Both rays refract at the glass-air boundary.

(a) On the diagram of the block in Figure 2 below, **sketch** the paths of the rays in the air after the rays exit the glass at points A and B.

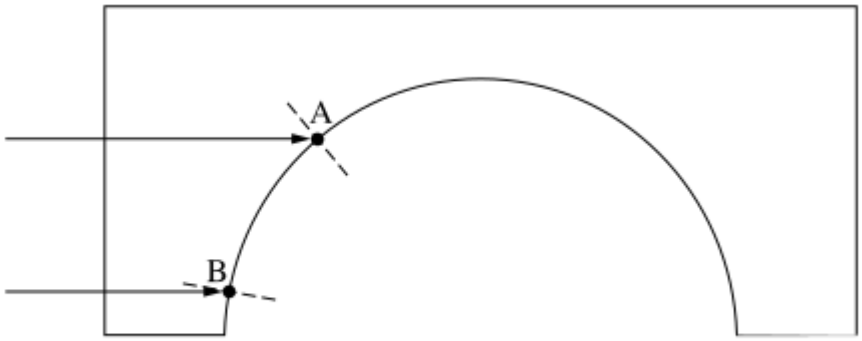


Figure 2

(b) **Derive** an equation for the change in wavelength $\Delta\lambda$ of the light when it passes from the glass into the air. Express your answer in terms of f_0, n_g, and fundamental constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference booklet.

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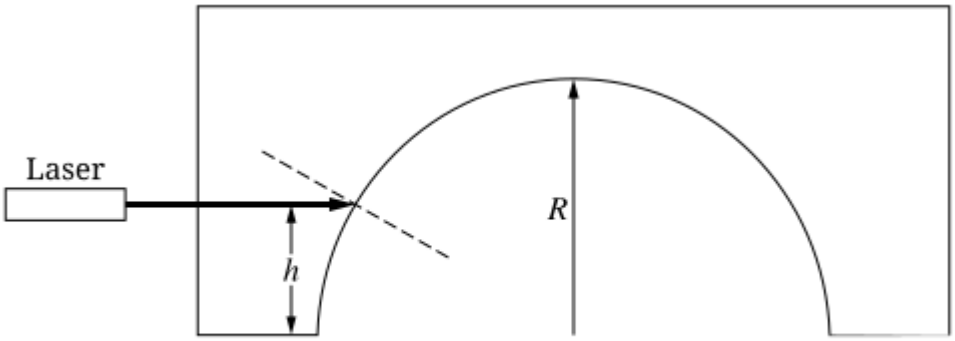


Figure 3

(c) A laser is used to shine a horizontal beam of light on the glass block, as shown in Figure 3. The height h of the beam is varied and the angle θ_r of the refracted beam to the normal is measured for several values of h, and the value of $\sin\theta_r$ is calculated.

On the axes below in Figure 4, **sketch** $\sin\theta_r$ as a function of h.

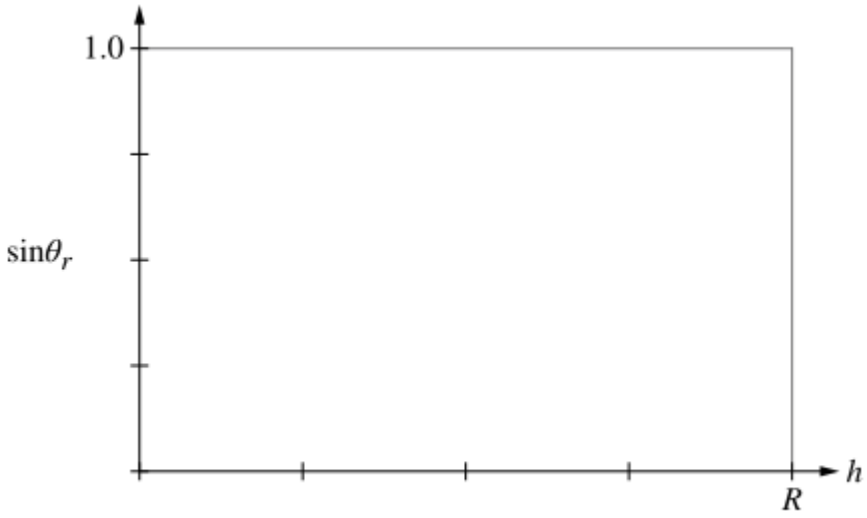


Figure 4

(d) **Indicate** whether the sketch drawn in part (c) is consistent with the rays drawn in part (a). Briefly **justify** your response.

AP PHYSICS 2

UNIT 14

Waves, Sound, and Physical Optics

12–15%
AP EXAM WEIGHTING

~14–23
CLASS PERIODS

54. The x and y coordinates of a point each execute simple harmonic motion. The result might be a circular orbit if:

A. the amplitudes are the same but the frequencies are different
B. the amplitudes and frequencies are both the same
C. the amplitudes and frequencies are both different
D. the phase constants are the same but the amplitudes are different
E. the amplitudes and the phase constants are both different

55. The x and y coordinates of a point each execute simple harmonic motion. The frequencies are the same but the amplitudes are different. The resulting orbit might be:

A. an ellipse
B. a circle
C. a parabola
D. a hyperbola
E. a square

56. For an oscillator subjected to a damping force proportional to its velocity:

A. the displacement is a sinusoidal function of time.
B. the velocity is a sinusoidal function of time.
C. the frequency is a decreasing function of time.
D. the mechanical energy is constant.
E. none of the above is true.

57. Five particles undergo damped harmonic motion. Values for the spring constant k, the damping constant b, and the mass m are given below. Which leads to the smallest rate of loss of mechanical energy?

A. $k = 100\,\mathrm{N/m},\ m = 50\,\mathrm{g},\ \ b = 8\,\mathrm{g/s}$
B. $k = 150\,\mathrm{N/m},\ m = 50\,\mathrm{g},\ \ b = 5\,\mathrm{g/s}$
C. $k = 150\,\mathrm{N/m},\ m = 10\,\mathrm{g},\ \ b = 8\,\mathrm{g/s}$
D. $k = 200\,\mathrm{N/m},\ m = 8\,\mathrm{g},\ \ b = 6\,\mathrm{g/s}$
E. $k = 100\,\mathrm{N/m},\ m = 2\,\mathrm{g},\ \ b = 4\,\mathrm{g/s}$

58. A sinusoidal force with a given amplitude is applied to an oscillator. To maintain the largest amplitude oscillation the frequency of the applied force should be:

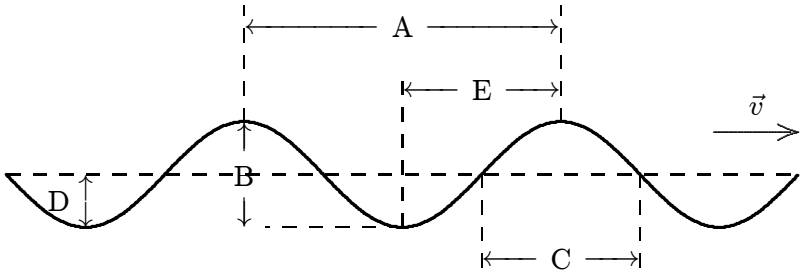
A. half the natural frequency of the oscillator
B. the same as the natural frequency of the oscillator
C. twice the natural frequency of the oscillator
D. unrelated to the natural frequency of the oscillator
E. determined from the maximum speed desired

Chapter 16: WAVES — I

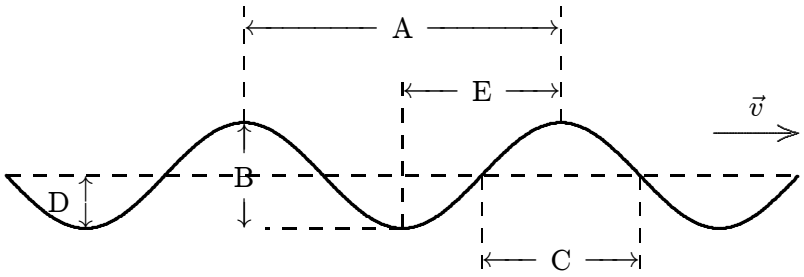
1. For a transverse wave on a string the string displacement is described by $y(x,t) = f(x - at)$, where f is a given function and a is a positive constant. Which of the following does NOT necessarily follow from this statement?

   A. The shape of the string at time $t = 0$ is given by $f(x)$.
   B. The shape of the waveform does not change as it moves along the string.
   C. The waveform moves in the positive x direction.
   D. The speed of the waveform is a.
   E. The speed of the waveform is x/t.

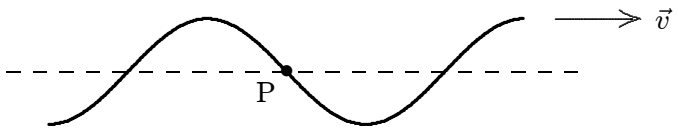
2. A sinusoidal wave is traveling toward the right as shown. Which letter correctly labels the amplitude of the wave?



3. A sinusoidal wave is traveling toward the right as shown. Which letter correctly labels the wavelength of the wave?

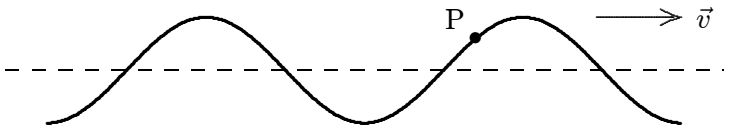


20. The transverse wave shown is traveling from left to right in a medium. The direction of the instantaneous velocity of the medium at point P is:



A. $\uparrow$
B. $\downarrow$
C. $\rightarrow$
D. $\nearrow$
E. no direction since $v = 0$

21. A wave traveling to the right on a stretched string is shown below. The direction of the instantaneous velocity of the point P on the string is:



A. $\uparrow$
B. $\downarrow$
C. $\rightarrow$
D. $\nearrow$
E. no direction since $v = 0$

22. Sinusoidal waves travel on five different strings, all with the same tension. Four of the strings have the same linear mass density, but the fifth has a different linear mass density. Use the mathematical forms of the waves, given below, to identify the string with the different linear mass density. In the expressions x and y are in centimeters and t is in seconds.

A. $y(x,t) = (2\,\text{cm})\sin(2x - 4t)$
B. $y(x,t) = (2\,\text{cm})\sin(4x - 10t)$
C. $y(x,t) = (2\,\text{cm})\sin(6x - 12t)$
D. $y(x,t) = (2\,\text{cm})\sin(8x - 16t)$
E. $y(x,t) = (2\,\text{cm})\sin(10x - 20t)$

23. Any point on a string carrying a sinusoidal wave is moving with its maximum speed when:

A. the magnitude of its acceleration is a maximum
B. the magnitude of its displacement is a maximum
C. the magnitude of its displacement is a minimum
D. the magnitude of its displacement is half the amplitude
E. the magnitude of its displacement is one-fourth the amplitude

37. Three separate strings are made of the same material. String 1 has length L and tension τ, string 2 has length $2L$ and tension 2τ, and string 3 has length $3L$ and tension 3τ. A pulse is started at one end of each string. If the pulses start at the same time, the order in which they reach the other end is:

A. 1, 2, 3
B. 3, 2, 1
C. 2, 3, 1
D. 3, 1, 2
E. they all take the same time

38. A long string is constructed by joining the ends of two shorter strings. The tension in the strings is the same but string I has 4 times the linear mass density of string II. When a sinusoidal wave passes from string I to string II:

A. the frequency decreases by a factor of 4
B. the frequency decreases by a factor of 2
C. the wave speed decreases by a factor of 4
D. the wave speed decreases by a factor of 2
E. the wave speed increases by a factor of 2

39. Two identical but separate strings, with the same tension, carry sinusoidal waves with the same frequency. Wave A has a amplitude that is twice that of wave B and transmits energy at a rate that is _____ that of wave B.

A. half
B. twice
C. one-fourth
D. four times
E. eight times

40. Two identical but separate strings, with the same tension, carry sinusoidal waves with the same frequency. Wave A has an amplitude that is twice that of wave B and transmits energy at a rate that is _____ that of wave B.

A. half
B. twice
C. one-fourth
D. four times
E. eight times

54. A wave on a string is reflected from a fixed end. The reflected wave:

A. is in phase with the original wave at the end
B. is 180° out of phase with the original wave at the end
C. has a larger amplitude than the original wave
D. has a larger speed than the original wave
E. cannot be transverse

55. A standing wave:

A. can be constructed from two similar waves traveling in opposite directions
B. must be transverse
C. must be longitudinal
D. has motionless points that are closer than half a wavelength
E. has a wave velocity that differs by a factor of two from what it would be for a traveling wave

56. Which of the following represents a standing wave?

A. $y = (6.0\,\text{mm})\sin[(3.0\,\text{m}^{-1})x + (2.0\,\text{s}^{-1})t] - (6.0\,\text{mm})\cos[(3.0\,\text{m}^{-1})x + 2.0]$
B. $y = (6.0\,\text{mm})\cos[(3.0\,\text{m}^{-1})x - (2.0\,\text{s}^{-1})t] + (6.0\,\text{mm})\cos[(2.0\,\text{s}^{-1})t + 3.0\,\text{m}^{-1})x]$
C. $y = (6.0\,\text{mm})\cos[(3.0\,\text{m}^{-1})x - (2.0\,\text{s}^{-1})t] - (6.0\,\text{mm})\sin[(2.0\,\text{s}^{-1})t - 3.0]$
D. $y = (6.0\,\text{mm})\sin[(3.0\,\text{m}^{-1})x - (2.0\,\text{s}^{-1})t] - (6.0\,\text{mm})\cos[(2.0\,\text{s}^{-1})t + 3.0\,\text{m}^{-1})x]$
E. $y = (6.0\,\text{mm})\sin[(3.0\,\text{m}^{-1})x] + (6.0\,\text{mm})\cos[(2.0\,\text{s}^{-1})t]$

57. When a certain string is clamped at both ends, the lowest four resonant frequencies are 50, 100, 150, and 200 Hz. When the string is also clamped at its midpoint, the lowest four resonant frequencies are:

A. 50, 100, 150, and 200 Hz
B. 50, 150, 250, and 300 Hz
C. 100, 200, 300, and 400 Hz
D. 25, 50, 75, and 100 Hz
E. 75, 150, 225, and 300 Hz

58. When a certain string is clamped at both ends, the lowest four resonant frequencies are measured to be 100, 150, 200, and 250 Hz. One of the resonant frequencies (below 200 Hz) is missing. What is it?

A. 25 Hz
B. 50 Hz
C. 75 Hz
D. 125 Hz
E. 225 Hz

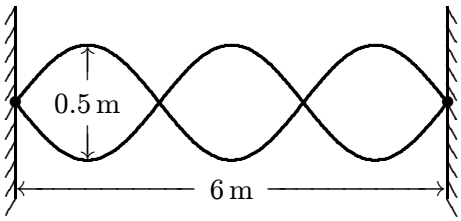
59. Two traveling waves $y_1 = A\sin[k(x - vt)]$ and $y_2 = A\sin[k(x + vt)]$ are superposed on the same string. The distance between the adjacent nodes is:

A. vt/π
B. $vt/2\pi$
C. $\pi/2k$
D. π/k
E. $2\pi/k$

60. If λ is the wavelength of each of the component sinusoidal traveling waves that form a standing wave, the distance between adjacent nodes in the standing wave is:

A. $\lambda/4$
B. $\lambda/2$
C. $3\lambda/4$
D. λ
E. 2λ

61. A standing wave pattern is established in a string as shown. The wavelength of one of the component traveling waves is:



A. 0.25 m
B. 0.5 m
C. 1 m
D. 2 m
E. 4 m

62. Standing waves are produced by the interference of two traveling sinusoidal waves, each of frequency 100 Hz. The distance from the second node to the fifth node is 60 cm. The wavelength of each of the two original waves is:

A. 50 cm
B. 40 cm
C. 30 cm
D. 20 cm
E. 15 cm

11. The largest number of beats per second will be heard from which pair of tuning forks?
    A. 200 and 201 Hz
    B. 256 and 260 Hz
    C. 534 and 540 Hz
    D. 763 and 774 Hz
    E. 8420 and 8422 Hz

12. Two stationary tuning forks (350 and 352 Hz) are struck simultaneously. The resulting sound is observed to:
    A. beat with a frequency of 2 beats/s
    B. beat with a frequency of 351 beats/s
    C. be loud but not beat
    D. be Doppler shifted by 2 Hz
    E. have a frequency of 702 Hz

13. When listening to tuning forks of frequency 256 Hz and 260 Hz, one hears the following number of beats per second:
    A. zero
    B. 2
    C. 4
    D. 8
    E. 258

14. Two identical tuning forks vibrate at 256 Hz. One of them is then loaded with a drop of wax, after which 6 beats/s are heard. The period of the loaded tuning fork is:
    A. 0.006 s
    B. 0.005 s
    C. 0.004 s
    D. 0.003 s
    E. none of these

15. Which of the following properties of a sound wave determine its "pitch"?
    A. Amplitude
    B. Distance from source to detector
    C. Frequency
    D. Phase
    E. Speed

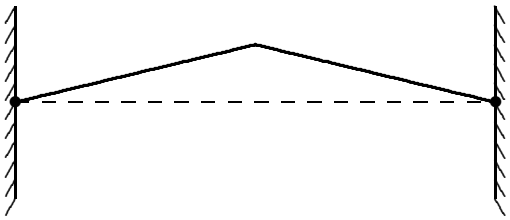
26. If the length of a piano wire (of given density) is increased by 5%, what approximate change in tension is necessary to keep its fundamental frequency unchanged?

A. Decrease of 10%
B. Decrease of 5%
C. Increase of 5%
D. Increase of 10%
E. Increase of 20%

27. A piano wire has a length of 81 cm and a mass of 2.0 g. If its fundamental frequency is to be 394 Hz, its tension must be:

A. 0.32 N
B. 63 N
C. 130 N
D. 250 N
E. none of these

28. A stretched wire of length 1.0 m is clamped at both ends. It is plucked at its center as shown. The three longest wavelengths in the wire are (in meters):



A. 4, 2, 1
B. 2, 1, 0.5
C. 2, 0.67, 0.4
D. 1, 0.5, 0.33
E. 1, 0.67, 0.5

29. Two identical strings, A and B, have nearly the same tension. When they both vibrate in their fundamental resonant modes, there is a beat frequency of 3 Hz. When string B is tightened slightly, to increase the tension, the beat frequency becomes 6 Hz. This means:

A. that before tightening A had a higher frequency than B, but after tightening, B has a higher frequency than A
B. that before tightening B had a higher frequency than A, but after tightening, A has a higher frequency than B
C. that before and after tightening A has a higher frequency than B
D. that before and after tightening B has a higher frequency than A
E. none of the above

44. An organ pipe with both ends open is 0.85 m long. Assuming that the speed of sound is 340 m/s, the frequency of the third harmonic of this pipe is:

A. 200 Hz
B. 300 Hz
C. 400 Hz
D. 600 Hz
E. none of these

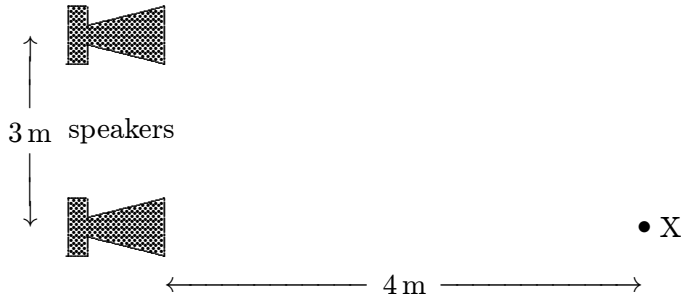
45. The "A" on a trumpet and a clarinet have the same pitch, but the two are clearly distinguishable. Which property is most important in enabling one to distinguish between these two instruments?

A. Intensity
B. Fundamental frequency
C. Displacement amplitude
D. Pressure amplitude
E. Harmonic content

46. The valves of a trumpet and the slide of a trombone are for the purpose of:

A. playing short (staccato) notes
B. tuning the instruments
C. changing the harmonic content
D. changing the length of the air column
E. producing gradations in loudness

47. Two small identical speakers are connected (in phase) to the same source. The speakers are 3 m apart and at ear level. An observer stands at X, 4 m in front of one speaker as shown. If the amplitudes are not changed, the sound he hears will be most intense if the wavelength is:



A. 1 m
B. 2 m
C. 3 m
D. 4 m
E. 5 m

46) Monochromatic coherent light shines through a pair of slits. If the wavelength of the light is decreased, which of the following statements are true of the resulting interference pattern? (There could be more than one correct choice.) 46) _____

A) The distance between the maxima decreases.
B) The distance between the minima stays the same.
C) The distance between the maxima stays the same.
D) The distance between the minima increases.
E) The distance between the minima decreases.

47) In a double–slit interference experiment you are asked to use laser light of different wavelengths and determine the separation between adjacent maxima. You observe that this separation is greatest when you illuminate the double slit with 47) _____

A) red light.
B) green light.
C) blue light.
D) yellow light.
E) The separation is the same for all wavelengths.

48) Two light sources are said to be *coherent* if they are 48) _____

A) of the same amplitude, and maintain a constant phase difference.
B) of the same frequency.
C) of the same frequency and amplitude.
D) of the same frequency, and maintain a constant phase difference.

49) Two beams of coherent light start out at the same point in phase and travel different paths to arrive at point P. If the maximum constructive interference is to occur at point P, the two beams must travel paths that differ by 49) _____

A) a whole number of half–wavelengths.
B) an odd number of half–wavelengths.
C) a whole number of wavelengths.

50) Two beams of coherent light start out at the same point in phase and travel different paths to arrive at point P. If the maximum destructive interference is to occur at point P, the two beams must travel paths that differ by 50) _____

A) an odd number of half–wavelengths.
B) a whole number of half–wavelengths.
C) a whole number of wavelengths.

51) If a sheet containing two very thin slits is heated (without damaging it), what happens to the angular location of the first–order interference minimum? 51) _____

A) It moves toward the centerline.
B) It doesn't change.
C) It moves away from the centerline.

52) In a single–slit diffraction experiment, the width of the slit through which light passes is reduced. What happens to the width of the central bright fringe in the resulting diffraction pattern? 52) _____

A) It becomes wider. B) It becomes narrower. C) It stays the same.

53) A single–slit diffraction pattern is formed on a distant screen. Assuming the angles involved are small, by what factor will the width of the central bright spot on the screen change if the wavelength of the illuminating light is doubled? 53) _____
A) It will become eight times as large.
B) It will be cut in half.
C) It will be cut to one–quarter its original size.
D) It will become four times as large.
E) It will double.

54) A single–slit diffraction pattern is formed on a distant screen. Assuming the angles involved are small, by what factor will the width of the central bright spot on the screen change if the slit width is doubled? 54) _____
A) It will be cut to one–quarter its original size.
B) It will become eight times as large.
C) It will be cut in half.
D) It will double.
E) It will become four times as large.

55) If a sheet containing a single slit is heated (without damaging it) and therefore expands, what happens to the angular location of the first–order diffraction minimum? 55) _____
A) It moves toward the centerline.
B) It doesn't change.
C) It moves away from the centerline.

56) If a sheet containing a single thin slit is heated (without damaging it) and therefore expands, what happens to the width of the central bright diffraction region on a distant screen? 56) _____
A) It doesn't change. B) It gets narrower. C) It gets wider.

57) What principle is responsible for light spreading as it passes through a narrow slit? 57) _____
A) polarization B) diffraction C) dispersion D) refraction

58) Radio waves are diffracted by large objects such as buildings, whereas light is not noticeably diffracted. Why is this? 58) _____
A) The wavelength of light is much smaller than the wavelength of radio waves.
B) Radio waves are unpolarized, whereas light is normally polarized.
C) Radio waves are coherent and light is usually not coherent.
D) Radio waves are polarized, whereas light is usually unpolarized.
E) The wavelength of light is much greater than the wavelength of radio waves.

59) Which of the following changes would *increase* the separation between the bright fringes in the diffraction pattern formed by a diffraction grating? 59) _____
A) Increase the wavelength of the light used.
B) Immerse the apparatus in water.
C) Increase the separation between the slits.
D) All of these.
E) None of these.

60) Light of the same wavelength passes through two diffraction gratings. One grating has 4000 lines/cm, and the other one has 6000 lines/cm. Which grating will spread the light through a larger angle in the first–order pattern? 60) _____
A) the 6000–line grating
B) the 4000–line grating
C) Both gratings spread the light the same.

61) If a diffraction grating is heated (without damaging it) and therefore expands, what happens to the angular location of the first–order maximum? 61) _____
A) It doesn't change.
B) It moves away from the centerline.
C) It moves toward the centerline.

62) Light in a frozen block of ice reflects off the ice–air interface at the surface of the block. What phase shift does it undergo? 62) _____
A) 270°
B) 0°
C) 180°
D) 90°
E) It does not undergo any phase shift.

63) Light reflects off the surface of Lake Superior. What phase shift does it undergo? 63) _____
A) 90°
B) 180°
C) 270°
D) 0°
E) It does not undergo any phase shift.

64) When a beam of light that is traveling in glass strikes an air boundary at the surface of the glass, there is 64) _____
A) no phase change in the reflected beam.
B) a 60° phase change in the reflected beam.
C) a 90° phase change in the reflected beam.
D) a 45° phase change in the reflected beam.
E) a 180° phase change in the reflected beam.

65) When a beam of light that is traveling in air is reflected by a glass surface, there is 65) _____
A) no phase change in the reflected beam.
B) a 45° phase change in the reflected beam.
C) a 180° phase change in the reflected beam.
D) a 90° phase change in the reflected beam.
E) a 60° phase change in the reflected beam.

66) When a beam of light, originally traveling in air, enters a piece of glass having an index of refraction of 3/2, its frequency 66) _____
A) increases by a factor of 3/2.
B) is unaffected.
C) is reduced to 2/3 its original value.

67) When a beam of light, originally traveling in air, enters a piece of glass having an index of refraction of 3/2, its wavelength 67) _____
A) is unaffected.
B) is reduced to 2/3 its original value.
C) increases by a factor of 3/2.

68) When a beam of light, originally traveling in air, enters a piece of glass having an index of refraction of 3/2, its speed 68) _____
A) is reduced to 2/3 its original value.
B) increases by a factor of 3/2.
C) is unaffected.

69) The colors on an oil slick are caused by reflection and 69) _____
A) ionization.
B) interference.
C) diffraction.
D) polarization.
E) refraction.

70) If unpolarized light of intensity I_0 passes through an ideal polarizer, what is the intensity of the emerging light? 70) _____
A) $I_0/4$ B) $I_0/16$ C) $I_0/2$ D) $I_0/\sqrt{2}$ E) I_0

71) For a beam of light, the direction of polarization is defined as 71) _____
A) the direction of the electric field's vibration.
B) the beam's direction of travel.
C) the direction of the magnetic field's vibration.
D) the direction that is perpendicular to both the electric and magnetic field vectors.



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1. Light from a source that produces a single frequency passes through a single slit *A*. The diffraction pattern on a screen is observed. Slit A is then replaced by slit *B*, and the new pattern is observed to have fringes that are more closely spaced than those in the first pattern. Which of the following is a possible explanation for why the spacings are different?

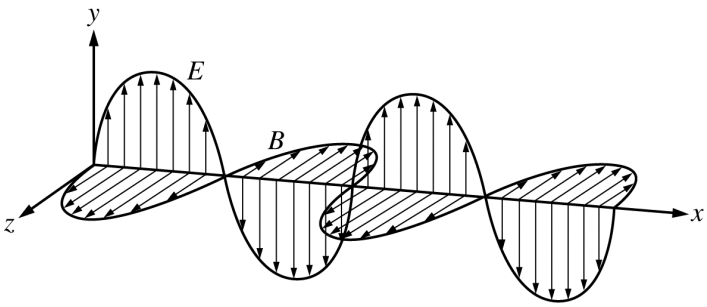
(A) Slit A is wider than slit *B*.

(B) Slit *B* is wider than slit *A*.

(C) The distance between the light source and the slit is greater for slit *A* than for slit *B*.

(D) The distance between the light source and the slit is greater for slit *B* than for slit A.

2.



The figure above shows a model of an electromagnetic wave, where *E* is the electric field and *B* is the magnetic field. In what direction is the energy of the wave transmitted?

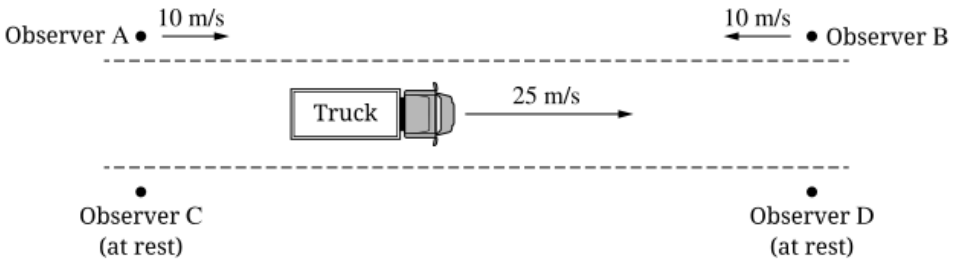
(A) Along the *x*-axis only

(B) Along the *y*-axis only

(C) Along the *z*-axis only

(D) In a direction that is at a nonzero angle to each of the axes

3.



A truck is traveling to the east with a speed of $25\ \mathrm{m/s}$ relative to the road while emitting a loud sound with a single frequency. The figure shows four different observers and indicates their locations and velocities relative to the road. Which observer hears a sound with the highest frequency?

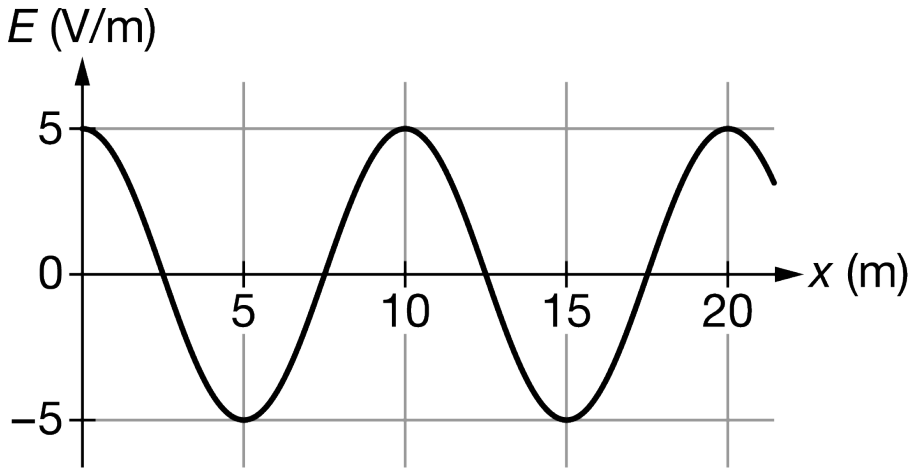
(A) A

(B) B

(C) C

(D) D

4.



The graph above shows the electric field E of an electromagnetic wave as a function of position x at a particular instant. Which of the following equations, where all quantities are assumed to have units consistent with the graph, best represents the graph?

(A) $E = 10\ \cos(2\pi x/10)$

(B) $E = 10\cos(2\pi x/5)$

(C) $E = 5\cos(2\pi x/10)$

(D) $E = 5\cos(2\pi x/5)$

5. A red laser beam ($\lambda = 675\ \text{nm}$) is incident on two very narrow slits that are spaced $10\ \text{mm}$ apart. The interference pattern is observed on a distant screen. A green laser ($\lambda = 500\ \text{nm}$) is then used instead. Which of the following claims correctly compares characteristics of the dark fringes observed with the green laser to those observed with the red laser?

(A) Narrower and farther apart

(B) Narrower and closer together

(C) Broader and farther apart

(D) Broader and closer together

6. To investigate the propagation of light and sound waves, a scientist proposes an experiment to set off a bright and loud explosion on the Moon and then use sensitive instruments on the surface of Earth to measure the arrival times of the light and sound waves from the explosion. Which of the following statements correctly predicts and explains the results of this experiment?

(A) The light will be detected a very short time after the explosion. The sound will be detected a few seconds later because sound propagates at the same speed as light in the vacuum of space but slower than light in Earth's atmosphere.

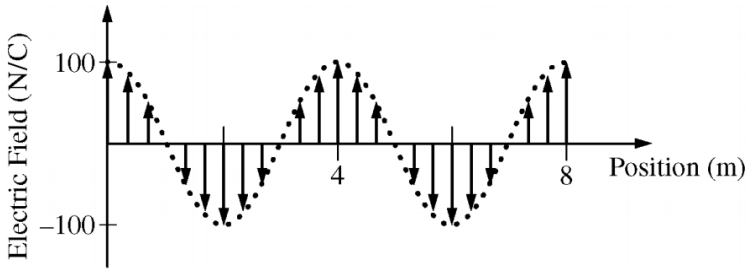
(B) The light will be detected a very short time after the explosion. The sound will be detected much later because sound propagates much slower than light for the entire distance from the Moon to Earth.

(C) The light will be detected a very short time after the explosion. The sound will never be detected because sound waves cannot propagate in the vacuum of space between the Moon and Earth.

(D) Neither the light nor the sound will ever be detected because neither can propagate in the vacuum of space between the Moon and Earth.

WorkBook Unit 14

7.



The figure above represents the electric field *E* as a function of position *x* due to a particular radio wave at time *t* = 0 *s*. Which of the following equations correctly describes the electric field?

(A) $E=(100N/C)\cos(\pi x/1m)$

(B) $E=(100N/C)\cos(\pi x/2m)$

(C) $E=(100N/C)\cos(\pi x/4m)$

(D) $E=(100N/C)\sin(\pi x/4m)$

8. A wave on a string approaches a piece of cardboard and passes through a narrow slit in the cardboard. The wave has amplitude A as it approaches the slit, which is much longer than A . The wave on the other side of the slit has amplitude $\frac{A}{3}$. Which of the following could explain why the amplitude $\frac{A}{3}$ on the other side of the slit is smaller?

(A) The wave is polarized vertically and the slit is aligned horizontally, so the amplitude would be reduced.

(B) The wave is polarized vertically and the slit is aligned vertically, but the finite width of the slit means only some of the wave will pass through the slit.

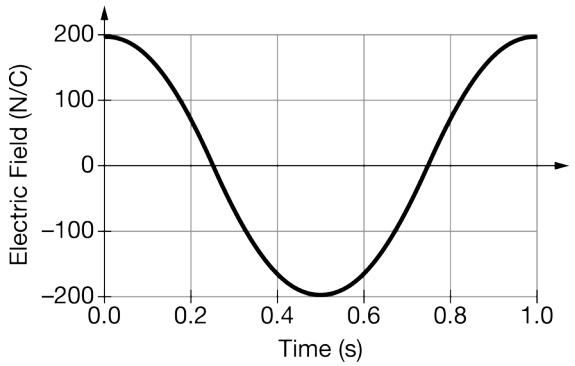
(C) The wave is polarized horizontally and the slit is aligned vertically, so only vertical components of the wave pass through the slit.

(D) The wave is polarized at some angle between horizontal and vertical, and the slit is aligned vertically, so only the vertical component of the wave will pass through the slit.

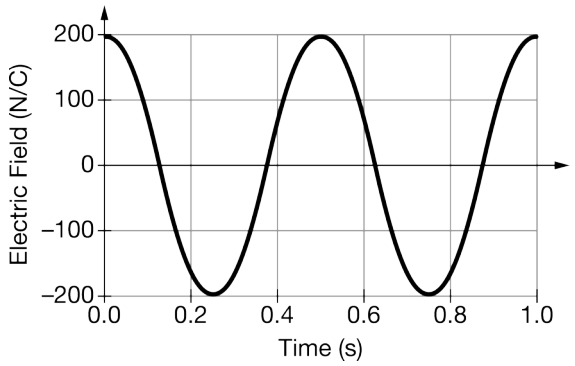
9. Which of the following graphs depicts the equation $E(t) = (200\ \mathrm{N/C})\cos(4\pi t)$, where E is the electric field strength at time t ?

WorkBook Unit 14

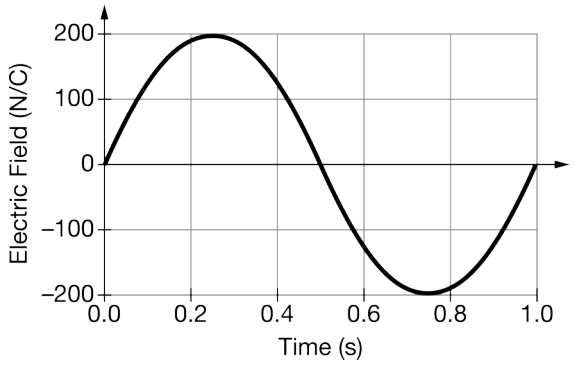
(A)



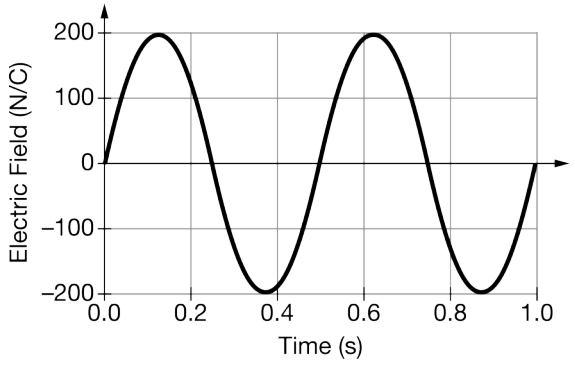
(B)



(C)

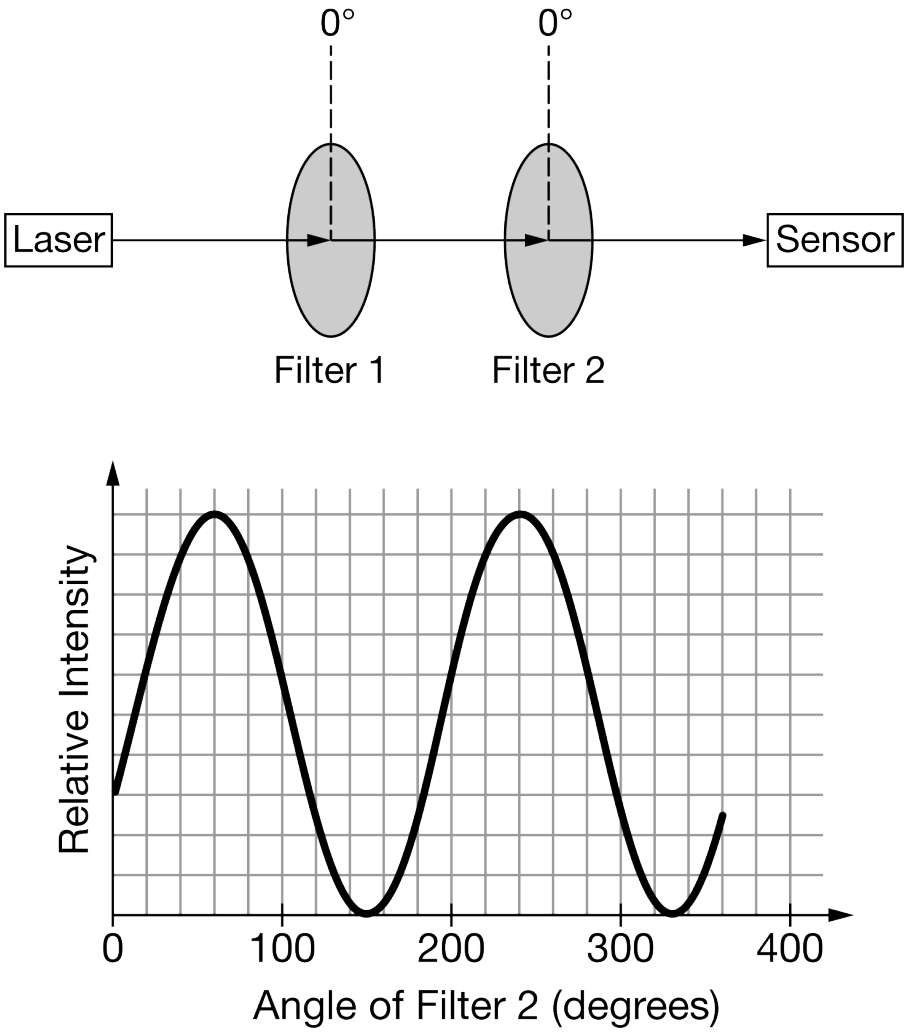


(D)



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10.



A laser is directed toward a pair of identical linear polarizing filters with unknown orientation. The intensity of laser light that passes through both filters is measured by a sensor. In order for the orientation of the filters to be determined, filter 2 is rotated through 360° from its initial orientation. The graph plots the resulting intensity data as a function of the angle of rotation. Which of the following explains how to determine the orientation of filter 1 ?

(A) The filters each spin the wave by the same amount, and maximum intensity is measured when filter 2 is rotated 60°, so filter 1 must be oriented 30° from vertical.

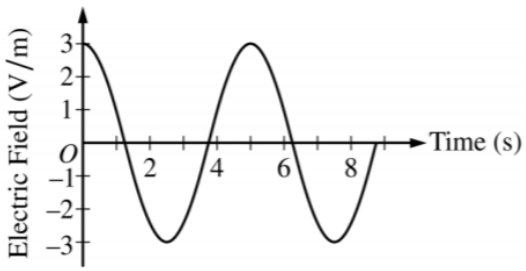
(B) Minimum intensity occurs when the polarizers are perpendicular to one another, and minimum intensity is measured when filter 2 is rotated 150°, so filter 1 must be oriented 60° from vertical.

(C) Maximum intensity occurs when the polarizers are parallel to one another, and maximum intensity is measured when filter 2 is rotated 60°, so filter 1 must be oriented 120° from vertical.

(D) The measured light intensity goes through two complete cycles as filter 2 is rotated through 360°, so filter 1 must be oriented 180° from vertical.

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11.



The graph above represents the magnitude E of the electric field at a point in space as a function of time t. Which of the following is a correct equation for the field if E is in V/m and t is in s ?

(A) $E = 3\cos(2\pi t / 5)$

(B) $E = 3\cos(2\pi t / 3)$

(C) $E = 5\cos(2\pi t / 5)$

(D) $E = 5\cos(2\pi t / 3)$

12. A ray of light has a wavelength of 800 nm in air. The light ray enters a material in which its wavelength is 600 nm. Assuming the index of refraction for air is 1.00, the index of refraction of the material is most nearly

(A) 0.75

(B) 1.00

(C) 1.25

(D) 1.33

(E) 1.50

13. Two waves are created in air and move toward outer space. Wave X cannot travel in the vacuum of outer space, but wave Y can. What types of waves are X and Y ?

WorkBook Unit 14

(A)

Wave X	Wave Y
Electromagnetic	Mechanical

(B)

Wave X	Wave Y
Electromagnetic	Electromagnetic

(C)

Wave X	Wave Y
Mechanical	Mechanical

(D)

Wave X	Wave Y
Mechanical	Electromagnetic

14. In a movie set in space, a planet explodes, simultaneously producing a bright flash of light and a loud boom. The director of the movie wants to depict how people in a spaceship that is speeding away from the planet would perceive the explosion. Which of the following best explains what the people should perceive?

(A) The light and sound should both be perceived but with a lower frequency than was produced, because the spaceship is moving away from the source.

(B) The light should be perceived but the sound should not, because light is an electromagnetic wave while sound is a mechanical wave.

(C) The light and sound should be perceived simultaneously, because light and sound travel the same distance from the explosion to the spaceship.

(D) The light should be perceived before the sound, because light propagates faster than sound.

15.



Two identical glass plates, whose circular surfaces are carefully polished to be very flat, are arranged as shown above. At one side the pieces touch, and a thin rod is placed between them at the other side to form a wedge of air. Monochromatic light shines on the top plate, creating an interference pattern of light and dark regions. Which of the following best describes this pattern?

WorkBook Unit 14

(A) Concentric circles centered on the center of the plates

(B) Concentric circles whose centers are close to the rod

(C) Lines parallel to the rod

(D) Lines perpendicular to the rod

(E) A random pattern

16.

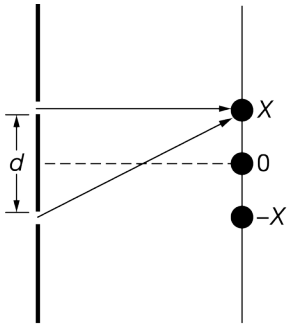


Figure 1

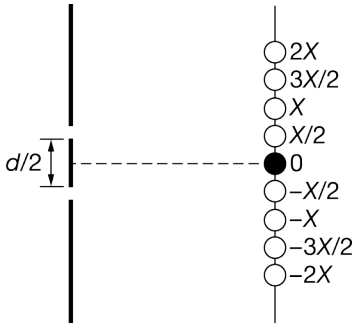


Figure 2

Light from a coherent source passes through a pair of narrow slits spaced a distance d apart and produces a diffraction pattern with bright spots that are spaced a distance X apart, as shown in Figure 1. The slits are then replaced by a pair that are spaced a distance $d/2$ apart, as shown in Figure 2. Which of the following most accurately describes the spacing of the bright spots in the interference pattern produced by the new slits?

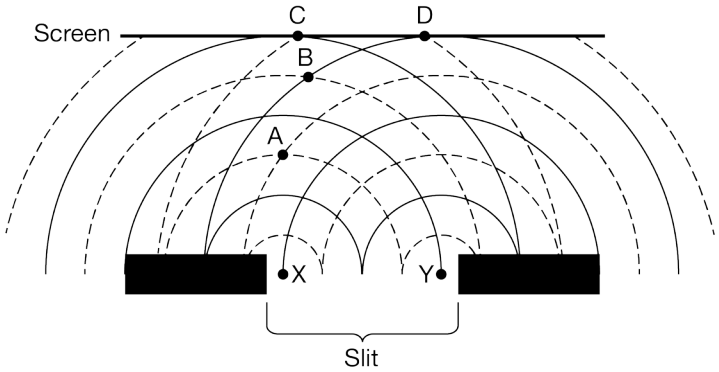
(A) $2X$

(B) $3X/2$

(C) X

(D) $X/2$

17.



Light incident on a single slit forms an interference pattern on a distant screen. Each point on the wave front moving through the slit acts like a point source. The crests and troughs for waves originating from two points, X and Y, are shown by solid and dashed lines, respectively. Which of the following statements about the diagram is true?

WorkBook Unit 14

(A) Destructive interference will occur at point A.
(B) Constructive interference will occur at point B.
(C) A bright fringe will be formed at point C on the screen.
(D) A dark fringe will be formed at point D on the screen.

An astronaut on the International Space Station (ISS) sends a message in the form of a wave to Earth.

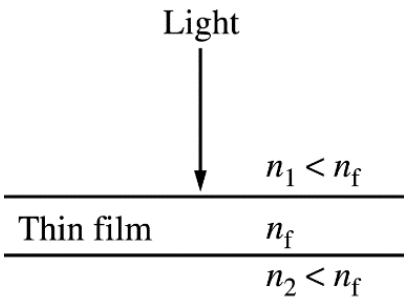
18. The astronaut wants to use the wave with the longest wavelength to send the signal. Which of the following is equivalent to the requirement of the longest wavelength?

(A) The smallest speed
(B) The greatest speed
(C) The lowest frequency
(D) The highest frequency

19. Which of the following types of waves could NOT be used by the astronaut to transmit the message?

(A) Radio wave
(B) Sound wave
(C) Microwave
(D) A red laser beam

20.



A thin film with index of refraction n_f separates two materials, each of which has an index of refraction less than n_f. A monochromatic beam of light is incident normally on the film, as shown above. If the light has wavelength λ within the film, maximum constructive interference between the incident beam and the reflected beam occurs for which of the following film thicknesses?

(A) 3λ
(B) 2λ
(C) λ
(D) $\frac{\lambda}{2}$
(E) $\frac{\lambda}{4}$

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21. Light from a laser shines on a diffraction grating, creating an interference pattern on a screen. The grating is then moved farther from the screen. How does the spacing between maxima in the interference pattern for the farther distance compare to that for the closer distance?

(A) The spacing is the same.

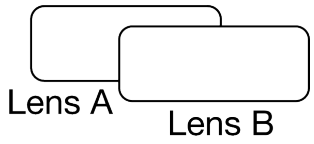
(B) The spacing is smaller for the farther distance.

(C) The spacing is larger for the farther distance.

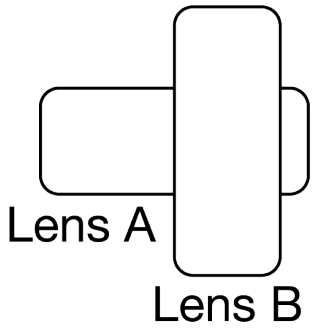
(D) The spacing cannot be compared without knowing the wavelength of the laser light.

22. A student has three sunglass lenses. Lenses A and B have identical polarization patterns, and lens C is not polarized. Which of the following arrangements of lenses would have the most light blocked where the lenses overlap?

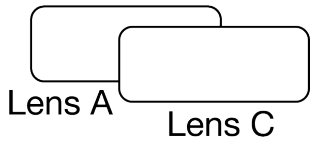
(A)



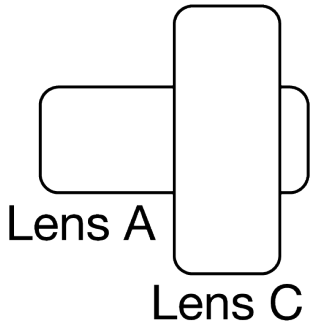
(B)



(C)

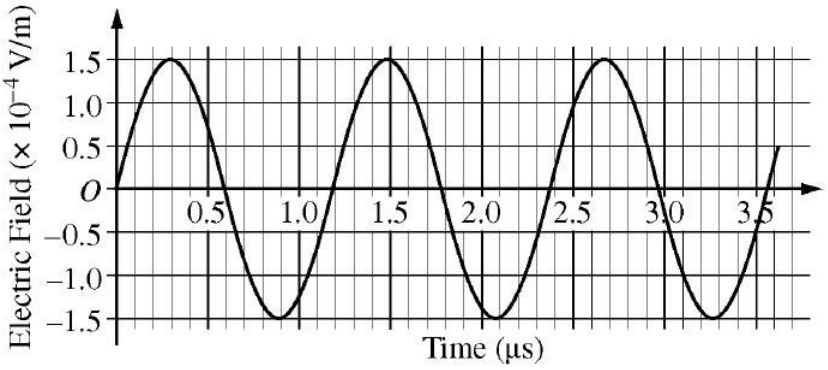


(D)



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You and a friend are traveling in a car. You tune the car radio to a station of frequency 850 kHz. The graph below represents the electric field strength of the radio wave at a given position as a function of time.



23. Which of the following best represents the electric field strength E measured in V/m as a function of time t measured in μs?

(A) $E = (3.0 \times 10^{-4})\sin(10.6t)$

(B) $E = (3.0 \times 10^{-4})\sin(5.32t)$

(C) $E = (1.5 \times 10^{-4})\sin(10.6t)$

(D) $E = (1.5 \times 10^{-4})\sin(5.32t)$

24. Both the sound and radio waves are present inside the car. Which of the following is true about these waves?

(A) Both waves are longitudinal.

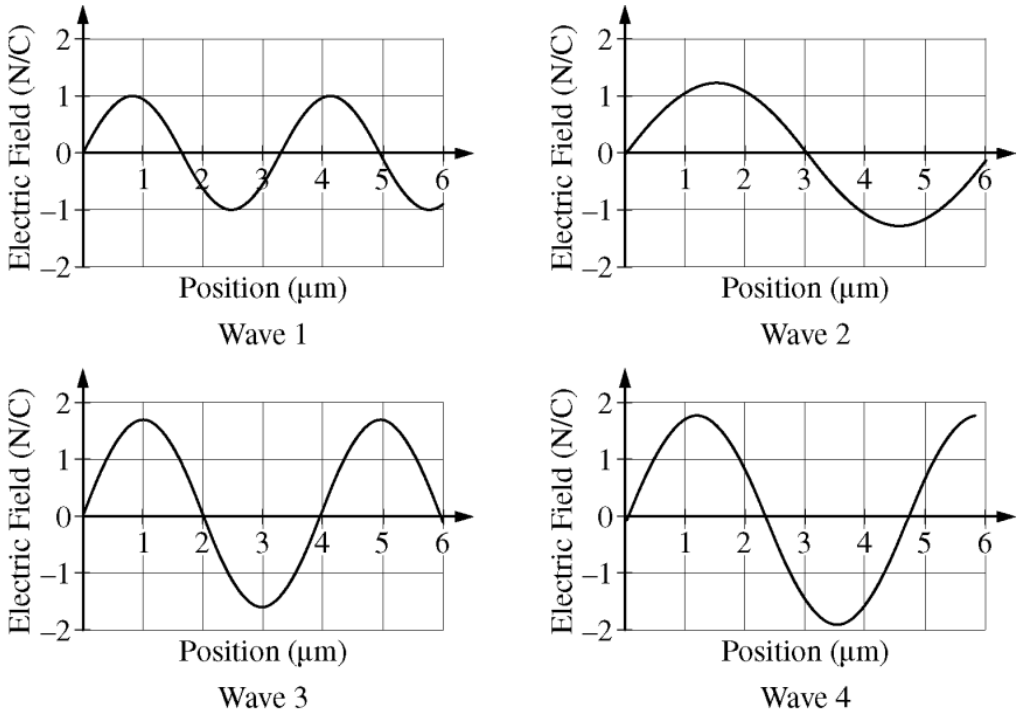
(B) Both waves can be polarized.

(C) Both waves travel at the speed of light.

(D) Sound waves require a medium in which to propagate and radio waves do not.

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25.



The graphs above show the electric field as a function of position at some instant of time for four different electromagnetic plane waves. Which of the following correctly ranks the frequency f of the waves?

(A) $f_1 > f_2 > f_3 > f_4$

(B) $f_1 > f_3 > f_4 > f_2$

(C) $f_2 > f_4 > f_3 > f_1$

(D) $f_4 > f_3 > f_2 > f_1$

26. A satellite in space uses a receiver to record a rocket being launched from Earth's surface. Which of the following correctly predicts what the receiver will measure as the rocket is launched?

(A) The receiver will measure only sound waves from the rocket.

(B) The receiver will measure only the light waves from the rocket.

(C) The receiver will measure both the sound waves and the light waves from the rocket.

(D) The receiver will not measure the sound waves or the light waves.

27. A student is given a loudspeaker with a square opening and asked to make a change in the dimensions of the opening so that the sound wave is more spread out vertically and narrowed horizontally. Which of the following is the correct use of the principle of diffraction to accomplish the desired result?

(A) The task is impossible since diffraction affects only electromagnetic radiation and very short wavelengths.

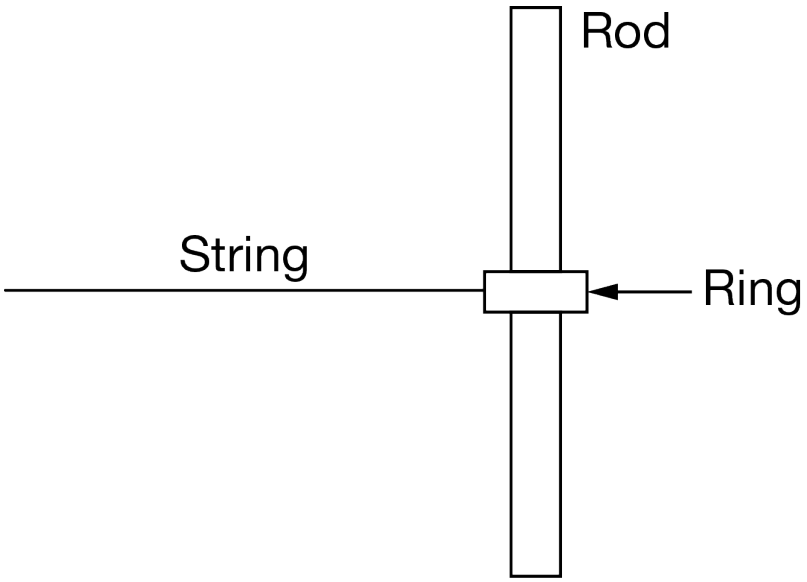
(B) Make the opening into a rectangle with a longer vertical dimension and a shorter horizontal dimension.

(C) Make the opening into a rectangle with a longer horizontal dimension and a shorter vertical dimension.

(D) Keep the opening in the shape of a square, but reduce both the horizontal and vertical dimensions.

WorkBook Unit 14

28.



Amplitude on String (cm)	**Amplitude of Ring** (cm)
2.0	0
3.0	2.5
4.0	3.8
5.0	5.0
6.0	4.5

A student holds one end of a string. The other end of the string is attached to a metal ring, as shown, that can slide freely on a vertical rod. The student creates a transverse wave that travels along the string. The student changes the amplitude of the wave and measures the amplitude of the motion of the ring. The data are shown in the table. Based on this data, what can be concluded about the polarization of the wave on the string?

(A) The wave was vertically polarized.

(B) The wave was horizontally polarized.

(C) The wave was polarized at some angle between vertical and horizontal that was the same for all trials.

(D) The polarization of the wave had various orientations.

29. Which of the following types of electromagnetic radiation have the shortest wavelength?

(A) Gamma rays

(B) Microwaves

(C) Radio waves

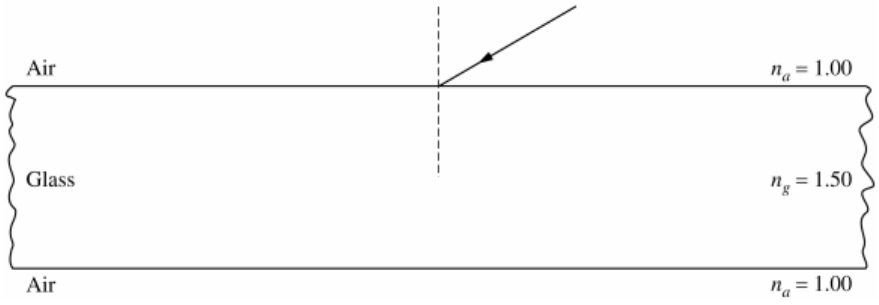
(D) Visible light rays

(E) X-rays

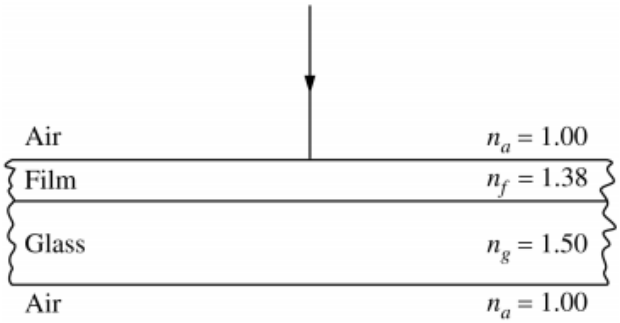
WorkBook Unit 14

30. A sheet of glass has an index of refraction $n_g = 1.50$. Assume that the index of refraction for air is $n_a = 1.00$.

(a) Monochromatic light is incident on the glass sheet, as shown in the figure below, at an angle of incidence of 60°. On the figure, sketch the path the light takes the first time it strikes each of the two parallel surfaces. Calculate and label the size of each angle (in degrees) on the figure, including angles of incidence, reflection, and refraction at each of the two parallel surfaces shown.



(b) Next a thin film of material is to be tested on the glass sheet for use in making reflective coatings. The film has an index of refraction $n_f = 1.38$. White light is incident normal to the surface of the film as shown below. It is observed that at a point where the light is incident on the film, light reflected from the surface appears green ($\lambda = 525\ nm$).



i. What is the frequency of the green light in air?

ii. What is the frequency of the green light in the film?

iii. What is the wavelength of the green light in the film?

iV. Calculate the minimum thickness of film that would produce this green reflection.

WorkBook Unit 14

31. Show all your work for each part of the question. The parts within the question may not have equal weight.

A laser emits monochromatic light of wavelength λ that is incident on a diffraction grating with slits separated by a distance d. An interference pattern is produced on a screen a distance L from the grating. The first maximum of the pattern is a distance x from the central maximum.

(a) The setup is then placed underwater. In the new diffraction pattern, the first maximum is a distance x_n from the central maximum. **Indicate** whether the distance x_n is greater than, less than or equal to the original distance x. **Justify** your answer without citing or manipulating equations.

(b) The water has an index of refraction of n. **Derive** an equation for the ratio $\frac{x_n}{x}$ in terms of n and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference booklet.

(c) **Indicate** whether your equation in part (b) is consistent with your reasoning from part (a)? **Justify** your answer.

AP PHYSICS 2

UNIT 15

Modern Physics

12–15%
AP EXAM WEIGHTING

~14–22
CLASS PERIODS

Conceptual Questions

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) As the temperature of a blackbody increases, what happens to the peak wavelength of the light it radiates? 1) _____
   A) It gets longer.
   B) It gets shorter.
   C) The wavelength is not affected by the temperature of the object.

2) Two identical metal bars are heated up until they are both glowing. One of them is "red hot" and the other is "blue hot." Which one is hotter, the one that glows red or the one that glows blue? 2) _____
   A) the red one
   B) the blue one
   C) We cannot tell without knowing more about the two bars.

3) A photon of blue light and a photon of red light are traveling in vacuum. The photon of blue light 3) _____
   A) has a longer wavelength than a photon of red light and travels with the same speed.
   B) has a longer wavelength than a photon of red light and travels with a greater speed.
   C) has a smaller wavelength than a photon of red light and travels with the same speed.
   D) has a smaller wavelength than a photon of red light and travels with a greater speed.

4) Increasing the *brightness* of a beam of light without changing its color will increase 4) _____
   A) the speed of the photons.
   B) the wavelength of the photons.
   C) the number of photons per second traveling in the beam.
   D) the frequency of the light.
   E) the energy of each photon.

5) If the wavelength of a photon is doubled, what happens to its energy? 5) _____
   A) It stays the same.
   B) It is reduced to one–fourth of its original value.
   C) It is reduced to one–half of its original value.
   D) It is increased to four times its original value.
   E) It is doubled.

6) Which of the following actions will increase the energy of a photon? (There could be more than one correct choice.) 6) _____
   A) Increase its speed.
   B) Decrease its wavelength.
   C) Decrease its frequency.
   D) Increase its wavelength.
   E) Increase its frequency.

7) Light of a given wavelength is used to illuminate the surface of a metal, however, no photoelectrons are emitted. In order to cause electrons to be ejected from the surface of this metal you should 7) _____
   A) use light of a longer wavelength.
   B) use light of the same wavelength but decrease its intensity.
   C) use light of a shorter wavelength.
   D) use light of the same wavelength but increase its intensity.



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8) A beam of light falling on a metal surface is causing electrons to be ejected from the surface. If we now double the frequency of the light, which of the following statements are correct? (There could be more than one correct choice.) 8) _____

A) Twice as many photons hit the metal surface as before.
B) The speed of the ejected electrons doubles.
C) The number of electrons ejected per second doubles.
D) The kinetic energy of the ejected electrons doubles.
E) None of the above things occur.

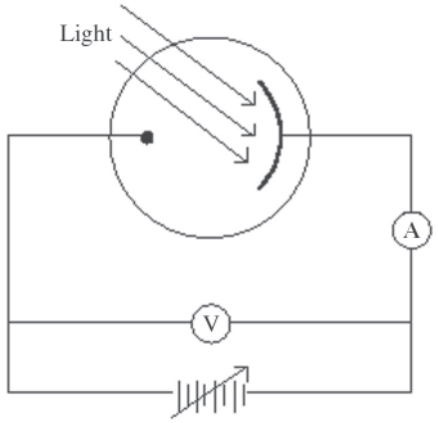
9) Monochromatic light falls on a metal surface and electrons are ejected. If the intensity of the light is increased, what will happen to the ejection rate and maximum energy of the electrons? 9) _____

A) same rate; greater maximum energy.
B) same rate; same maximum energy.
C) greater rate; greater maximum energy.
D) greater rate; same maximum energy.

10) When the surface of a metal is exposed to blue light, electrons are emitted. If the intensity of the blue light is increased, which of the following things will also increase? 10) _____

A) the maximum kinetic energy of the ejected electrons
B) the work function of the metal
C) the number of electrons ejected per second
D) the time lag between the onset of the absorption of light and the ejection of electrons
E) all of the above

11) Monochromatic light is incident on a metal surface, and the ejected electrons give rise to a current in the circuit shown in the figure. The maximum kinetic energy of the ejected electrons is determined by applying a reverse ('stopping') potential, sufficient to reduce the current in the ammeter to zero. If the intensity of the incident light is increased, how will the required stopping potential change? 11) _____



A) It will remain unchanged.
B) It will decrease.
C) It will increase.

12) A photon scatters off of a stationary electron. Which of the following statements about the photon are true? (There could be more than one correct choice.) 12) _____

A) Its wavelength increases due to the scattering.
B) Its energy does not change.
C) Its frequency increases due to the scattering.
D) Its frequency decreases due to the scattering.
E) Its wavelength decreases due to the scattering.

13) Photon A has twice the momentum of photon B as both of them are traveling in vacuum. Which statements about these photons are correct? (There could be more than one correct choice.) 13) _____
A) The wavelength of photon A is twice as great as the wavelength of photon B.
B) The wavelength of photon B is twice as great as the wavelength of photon A.
C) Photon A is traveling twice as fast as photon B.
D) Both photons have the same wavelength.
E) Both photons have the same speed.

14) If the frequency of a light beam is doubled, what happens to the momentum of the photons in that beam of light? 14) _____
A) It is reduced to one–fourth of its original value.
B) It is increased to four times its original value.
C) It stays the same.
D) It is halved.
E) It is doubled.

15) If the wavelength of a light beam is doubled, what happens to the momentum of the photons in that light beam? 15) _____
A) It is halved.
B) It is reduced by one–fourth of its original value.
C) It is increased to four times its original value.
D) It is doubled.
E) It stays the same.

16) If the wavelength of a photon is the same as the de Broglie wavelength of an electron, which one has the greater momentum? 16) _____
A) The electron because it has more mass.
B) The photon because it is traveling faster.
C) They both have the same momentum.

17) If the wavelength of a photon in vacuum is the same as the de Broglie wavelength of an electron, which one is traveling faster through space? 17) _____
A) The electron because it has more mass.
B) The photon because photons always travel through space faster than electrons.
C) They both have the same speed.

18) If a proton and an electron have the same de Broglie wavelengths, which one is moving faster? 18) _____
A) the electron
B) the proton
C) They both have the same speed.

19) Which of the following actions will increase the de Broglie wavelength of a speck of dust? (There could be more than one correct choice.) 19) _____
A) Decrease its speed.
B) Increase its mass.
C) Decrease its momentum.
D) Increase its speed.
E) Decrease its mass.

20) Protons are being accelerated in a particle accelerator. When the energy of the protons is doubled, their de Broglie wavelength will 20) _____

A) decrease by a factor of 2.
B) increase by a factor of 4.
C) decrease by a factor of $\sqrt{2}$.
D) increase by a factor of 2.
E) increase by a factor of $\sqrt{2}$.

21) A proton and an electron are both accelerated to the same final speed. If λ_p is the de Broglie wavelength of the proton and λ_e is the de Broglie wavelength of the electron, then 21) _____

A) $\lambda_p = \lambda_e$. B) $\lambda_p > \lambda_e$. C) $\lambda_p < \lambda_e$.

22) To which of the following values of n does the longest wavelength in the Balmer series correspond? 22) _____

A) 3
B) 5
C) 7
D) 1
E) ∞ (very large)

23) To which of the following values of n does the shortest wavelength in the Balmer series correspond? 23) _____

A) 7
B) 1
C) 3
D) 5
E) ∞ (very large)

24) The Balmer series is formed by electron transitions in hydrogen that 24) _____

A) end on the $n = 2$ shell.
B) are between the $n = 1$ and $n = 3$ shells.
C) end on the $n = 1$ shell.
D) begin on the $n = 2$ shell.
E) begin on the $n = 1$ shell.

25) The Lyman series is formed by electron transitions in hydrogen that 25) _____
A) begin on the $n = 1$ shell.
B) begin on the $n = 2$ shell.
C) end on the $n = 1$ shell.
D) end on the $n = 2$ shell.
E) are between the $n = 1$ and $n = 3$ shells.

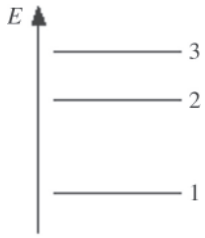
26) The Paschen series is formed by electron transitions that 26) _____
A) end on the $n = 1$ shell.
B) begin on the $n = 1$ shell.
C) end on the $n = 3$ shell.
D) begin on the $n = 3$ shell.
E) end on the $n = 2$ shell.

27) Hydrogen atoms can emit four spectral lines with visible colors from red to violet. These four visible lines emitted by hydrogen atoms are produced by electrons 27) _____
A) that end up in the ground state.
B) that start in the $n = 2$ level.
C) that end up in the $n = 3$ level.
D) that start in the ground state.
E) that end up in the $n = 2$ level.

28) When an electron jumps from an orbit where $n = 4$ to one where $n = 2$ 28) _____
A) two photons are absorbed.
B) a photon is absorbed.
C) a photon is emitted.
D) two photons are emitted.
E) None of the given answers are correct.

29) The energy difference between adjacent orbit radii in a hydrogen atom 29) _____
A) decreases with increasing values of n.
B) increases with increasing values of n.
C) varies randomly with increasing values of n.
D) remains constant for all values of n.

30) The figure shows part of the energy level diagram of a certain atom. The energy spacing between levels 1 and 2 is twice that between 2 and 3. If an electron makes a transition from level 3 to level 2, the radiation of wavelength λ is emitted. What possible radiation wavelengths might be produced by other transitions between the three energy levels? 30) _____



A) both $\lambda/2$ and $\lambda/3$
B) only 2λ
C) both 2λ and 3λ
D) only $\lambda/2$

31) An ionized atom having Z protons has had all but one of its electrons removed. If E is the total energy of the ground state electron in atomic hydrogen, then what is the total energy of the remaining electron in the ionized atom? 31) _____

A) Z^2E B) E/Z C) E/Z^2 D) ZE E) E

32) The distance between adjacent orbits in a hydrogen atom 32) _____

A) varies randomly with increasing values of n.
B) increases with increasing values of n.
C) decreases with increasing values of n.
D) remains constant for all values of n.

33) Which of the following statements are true for the Bohr model of the atom? (There could be more than one correct choice.) 33) _____

A) The spacing between all the electron shells is the same.
B) As we look at higher and higher electron shells, they get closer and closer together, but the difference in energy between them gets greater and greater.
C) There is no general pattern in the spacing of the shells or their energy differences.
D) The energy difference between all the electron shells is the same.
E) As we look at higher and higher electron shells, they get farther and farther apart, but the difference in energy between them gets smaller and smaller.

34) The primary reason that very large nuclei are unstable is due to 34) _____

A) the cumulative repulsive force of the protons.
B) the repulsive force between the neutrons and the protons.
C) the cumulative attractive force between the protons and the orbiting electrons.
D) the extreme weakness of the gravitational attraction of the protons.

35) Atomic nuclei that are all isotopes of an element all have the same 35) _____

A) number of neutrons. B) number of protons.
C) mass. D) number of nucleons.

36) All nuclei of a given element have the same number of 36) _____

A) nucleons. B) protons + neutrons.
C) neutrons. D) protons.

37) An yttrium isotope has 39 protons and 50 neutrons in its nucleus. Which one of the following symbols accurately represents this isotope? 37) _____

A) $^{89}_{50}\mathrm{Y}$ B) $^{89}_{39}\mathrm{Y}$ C) $^{39}_{50}\mathrm{Y}$ D) $^{39}_{89}\mathrm{Y}$ E) $^{50}_{39}\mathrm{Y}$

38) The mass number of an atom is equal to the number of what particles in the nucleus? 38) _____
A) nucleons B) positrons. C) neutrons D) protons E) electrons

39) The atomic number and mass number for calcium–39 are 20 and 39, respectively. How many neutrons are in one atom? 39) _____
A) 59 B) 1 C) 19 D) 20 E) 39

40) In a $^{93}_{41}\text{Nb}$ *nucleus*, how many protons, neutrons, and electrons are found there? 40) _____
A) 41, 52, 0. B) 93, 41, 93. C) 41, 52, 93. D) 52, 41, 41. E) 41, 52, 41.

41) The symbol for a certain isotope of radium is $^{226}_{88}\text{Ra}$. How many nucleons are there in the nucleus of this isotope? 41) _____
A) 226 B) 138 C) 214 D) 88 E) 314

42) What is true of an element of atomic number 15 that has an isotope of mass number 32? (There could be more than one correct choice.) 42) _____
A) the number of neutrons in the nucleus is 17.
B) the number of protons in the nucleus is 15.
C) the number of neutrons in the nucleus is 15.
D) the number of nucleons in the nucleus is 47.
E) the number of protons in the nucleus is 32.

43) The atomic mass unit is defined as 43) _____
A) the mass of a hydrogen–1 atom.
B) the mass of a proton.
C) one twelfth the mass of a carbon–12 atom.
D) the mass of a carbon–12 nucleus.
E) the mass of an electron.

44) To a reasonable approximation, the density of a nucleus 44) _____
A) varies unpredictably for different values of Z.
B) is independent of the mass number.
C) is greater for large–Z nuclei.
D) is smaller for large–Z nuclei.

45) A stable nucleus contains many protons very close to each other, all positively charged. Why do the protons not fly apart due to mutual Coulomb repulsion? 45) _____
A) An attractive nuclear force in the nucleus counteracts the effect of the Coulomb forces.
B) There are an equal number of electrons in the nucleus which neutralize the protons.
C) The Coulomb force does not operate within nuclei.
D) The gravity of the protons and neutrons overcomes their repulsion at such close distances.
E) The neutrons in the nucleus shield the protons from each other.

46) The two strongest forces that act between protons in a nucleus are 46) _____
A) the electrostatic and the gravitational forces.
B) the strong nuclear and the electrostatic forces.
C) the weak nuclear and electromagnetic forces.
D) the weak nuclear and the gravitational forces.

47) Which of the following statements is *not* true of the strong nuclear force? 47) _____
A) The nuclear force does not depend on charge.
B) The nuclear force affects both neutrons and protons.
C) For two protons that are very close together, the nuclear force and the electric force have about the same magnitudes.
D) A nucleon in a large nucleus interacts via the nuclear force only with nearby nucleons, not with ones far away in the nucleus.
E) The nuclear force has a short range, of the order of nuclear dimensions.

48) The main reason that there is a limit to the size of a stable nucleus is 48) _____
A) the weakness of the electrostatic force.
B) the weakness of the gravitational force.
C) the limited range of the gravitational force.
D) the short range nature of the strong nuclear force.

49) In massive stars, three helium nuclei fuse together, forming a carbon nucleus, and this reaction heats the core of the star. The net mass of the three helium nuclei must therefore be 49) _____
A) higher than that of the carbon nucleus.
B) the same as that of the carbon nucleus because mass is always conserved.
C) the same as that of the carbon nucleus because energy is always conserved.
D) less than that of the carbon nucleus.

50) A fusion reaction releases energy because the binding energy of the resulting nucleus 50) _____
A) is less than the binding energy of the original nuclei.
B) is released in the process.
C) is absorbed in the process.
D) is greater than the binding energy of the original nuclei.
E) is equal to the binding energy of the original nuclei.

51) The binding energy per nucleon of a nucleus 51) _____
A) decreases steadily as we go to heavier elements.
B) increases steadily as we go to heavier elements.
C) has a minimum near iron in the periodic table and increases after that for heavier elements.
D) is approximately constant throughout the periodic table, except for very light nuclei.
E) has a maximum near iron in the periodic table and decreases after that for heavier elements.

52) An α particle is also known as 52) _____
A) a positron.
B) an electron.
C) a helium nucleus.
D) a high-energy photon.

53) A β^- particle is also known as 53) _____
A) an electron.
B) a helium nucleus.
C) a positron.
D) a high-energy photon.

54) A gamma ray is also known as 54) _____
A) a helium nucleus.
B) an electron.
C) a positron.
D) a high–energy photon.

55) In beta–minus decay 55) _____
A) a proton is emitted.
B) a neutron is emitted.
C) an electron decays into another particle.
D) an electron is emitted.
E) a proton is transformed into a neutron.

56) During β^+ decay 56) _____
A) a proton is transformed to a neutron.
B) the number of nucleons decreases.
C) a neutron is transformed to a proton.
D) a neutron is ejected from the nucleus.
E) a proton is ejected from the nucleus.

57) A radioactive isotope of atomic number Z emits a beta–minus particle, and then the daughter nucleus emits a gamma ray. What is the atomic number of the resulting nucleus after both processes? 57) _____
A) $Z - 1$ B) $Z - 3$ C) $Z - 2$ D) $Z + 1$

58) When a β^- particle is emitted from an unstable nucleus, the atomic number of the nucleus 58) _____
A) increases by 2.
B) increases by 1.
C) decreases by 1.
D) decreases by 2.
E) does not change.

59) A radioactive isotope of atomic number Z emits an alpha particle, and the daughter nucleus then emits a beta–minus particle. What is the atomic number of the resulting nucleus? 59) _____
A) $Z - 3$ B) $Z - 1$ C) $Z - 2$ D) $Z + 1$

60) Polonium–216 decays to lead–212 by emitting what kind of nuclear radiation? 60) _____
A) gamma
B) beta–plus
C) beta–minus
D) x–rays.
E) alpha

61) When an unstable nucleus decays by emitting an alpha particle, the atomic number Z of the nucleus 61) _____
A) decreases by 4.
B) increases by 4.
C) decreases by 2.
D) increases by 2.
E) remains unchanged.

62) Isotope A has a decay constant of 0.723 s^{-1} and isotope B has a decay constant of 0.514 s^{-1}. Which isotope has a longer half–life? 62) _____

A) isotope A B) isotope B

63) The half–life of cobalt–60 is 5.3 years, while that of strontium–90 is about 29 years. Suppose you have samples of both isotopes, and that they initially contain equal numbers of atoms of these isotopes. How will the activities (number of decays per second) of the samples compare? 63) _____

A) The activity of the cobalt–60 sample will be greater.
B) The activities will be equal.
C) The activities cannot be compared without more information.
D) The activity of the strontium–90 sample will be greater.

64) The half–life of cobalt–60 is 5.3 years, while that of strontium–90 is about 29 years. Suppose you have samples of both isotopes, and that they initially have the same activity (number of decays per second). What must be true of the numbers of cobalt–60 and strontium–90 nuclei in these samples? 64) _____

A) There are equal numbers of cobalt–60 and strontium–90 nuclei.
B) There are more cobalt–60 than strontium–90 nuclei.
C) There are more strontium–90 than cobalt–60 nuclei.
D) It is not possible to compare numbers of nuclei without knowing the masses of the samples.

65) What happens to the half–life of a radioactive substance as it decays? 65) _____

A) It decreases at an exponential rate. B) It increases.
C) It remains constant. D) It decreases at a constant rate.

66) What happens to the half–life of a radioactive substance as we increase its temperature? 66) _____

A) It decreases at an exponential rate. B) It does not change.
C) It increases. D) It decreases at a constant rate.

67) Two radioactive isotopes, X and Y, both decay to stable products. The half–life of X is about a day, while that of Y is about a week. Suppose a radioactive sample consists of a mixture of these two nuclides. If the mixture is such that the activities arising from X and Y are initially equal, then a few days later the activity of the sample will be due 67) _____

A) predominantly to X. B) entirely to Y.
C) to X and Y equally. D) predominantly to Y.

68) Suppose the half–life of an isotope is 2 days. You purchase 10 grams of the isotope, but it was produced in a laboratory 4 days before it was delivered to you. How much of this isotope will you have 3 days after it was delivered to you? 68) _____

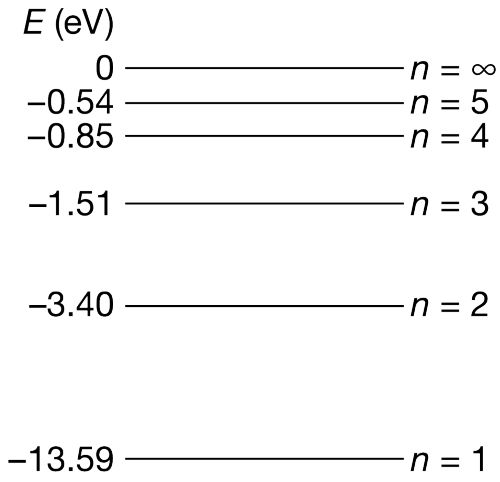
A) 1.25 grams
B) 2.5 grams
C) less than 1.25 grams
D) more than 1.25 grams but less than 2.5 grams
E) more than 2.5 grams but less than 5 grams

69) Carbon–14 decays by β^- emission. What nucleus is left after this decay? 69) _____

A) nitrogen–14
B) carbon–15
C) carbon–13
D) nitrogen–15
E) carbon–14

WorkBook Unit 15

1.



The figure shows some energy levels of hydrogen. If the magnitude of the elementary charge were $10e$ instead of e, how would the ground state energy of hydrogen be affected?

(A) It would have a much larger magnitude.

(B) It would have a much smaller magnitude.

(C) It would be the same.

(D) It would depend on which energy state the atom is actually in.

2. In a demonstration of the double slit experiment, a high velocity stream of dense spheres is fired at a slit , and another stream is fired at an adjacent identical slit. For each stream, all spheres pass straight through the slit, then hit the same location on a screen. Next, two streams of electrons are fired at two slits next to one another. On a screen beyond the slits, a diffraction pattern is formed. Which situation does classical mechanics explain, and which situation does quantum mechanics explain?

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(A)

Spheres	Electrons
Classical mechanics	Classical mechanics

(B)

Spheres	Electrons
Classical mechanics	Quantum mechanics

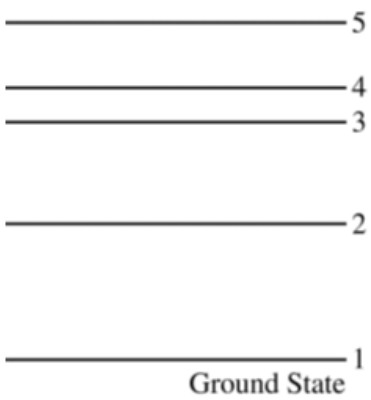
(C)

Spheres	Electrons
Quantum mechanics	Classical mechanics

(D)

Spheres	Electrons
Quantum mechanics	Quantum mechanics

3.



The energy states for an electron in a hypothetical atom are represented above. A cold sample of gas of such atoms is illuminated by a brief flash of light with a continuous spectrum. The atoms are then observed to emit light with discrete wavelengths. Which of the following transitions produces the longest wavelength light?

(A) 5 to 4

(B) 4 to 3

(C) 3 to 2

(D) 2 to 1

WorkBook Unit 15

4. The Sun's energy production is due to the fusion of ^{1}H into ^{4}He . How does the mass of four ^{1}H nuclei ($4m_H$) compare with the mass of one ^{4}He nucleus (m_{He})?

(A) $4m_H = m_{He}$

(B) $4m_H < m_{He}$

(C) $4m_H > m_{He}$

(D) It cannot be determined without knowing the amount of energy released.

5. A city uses energy at a rate of 2×10^9 W. If a total of 0.5 kg of mass is completely converted to energy, for about how much time would the released energy supply the city?

(A) 6×10^3 hours

(B) 4×10^5 hours

(C) 8×10^{10} hours

(D) 3×10^{22} hours

6. An unstable nucleus of $^{40}_{19}$K undergoes a β^- decay. What are the atomic number and nucleon number of the resulting nucleus?

(A) The atomic number is 18, and the nucleon number is 39.

(B) The atomic number is 18, and the nucleon number is 40.

(C) The atomic number is 19, and the nucleon number is 40.

(D) The atomic number is 20, and the nucleon number is 40.

7. A deuterium nucleus ^2_1H interacts with a lithium nucleus ^7_3Li. Which of the following could NOT possibly occur as the end products of this reaction because of violation of conservation of charge or mass number?

(A) $^4_2He + ^3_1H + ^2_1H$

(B) $2\,^3_1H + ^1_1H + 2\,^1_0n$

(C) $2\,^4_2He + ^1_0n$

(D) $^7_3Li + ^2_1H$

(E) $^8_3Li + ^1_1H$

Cosmic rays bombard Earth's upper atmosphere, creating subatomic particles called muons, which then stream toward the surface of Earth at nearly the speed of light. The muon is not a nucleon. It is an unstable fundamental particle with most properties the same as those of an electron but with a larger mass. In a frame of reference at rest with respect to the muon, the muon decays with a half-life of approximately 2.2×10^{-6}s.

8. Which of the following could be the decay products of a negatively charged muon?

(A) An electron and a positron

(B) A neutron and an electron

(C) Two electrons and a proton

(D) An electron, a neutrino, and an antineutrino

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9. The Bohr model of the hydrogen atom proposed that the waves of the orbiting electron form a standing wave pattern. Which of the following describes the application of this model that led to discrete energy levels for the atom?

(A) The energy of the electron depends on its mass and velocity, which together determine the energy of the states.

(B) A whole number of wavelengths must fit in the circumference of the electron's orbit, which limits the size of the orbit to certain values.

(C) The physical size of the electron affects its energy and thus its de Broglie wavelength, which limits the number of possible energy states.

(D) The ratio of the mass of the electron to that of the proton affects the electron's energy and thus its de Broglie wavelength, which limits the number of possible energy states.

10. A canister and the hydrogen gas it contains are at 100°C. The canister is placed in a vacuum, and the temperature of the canister and gas begins to decrease. Which of the following statements of reasoning best explains how the canister-gas system loses energy?

(A) High-energy hydrogen molecules collide with lower-energy molecules and the walls inside the canister, losing energy during the collisions.

(B) The molecules collide with the walls of the canister, causing the canister molecules to vibrate and carry energy from the canister to the canister's surroundings.

(C) Energy is released from the canister as infrared radiation that can travel through the vacuum, causing a decrease in the average energy of the canister and the molecules.

(D) Energy is released from the canister and travels through the vacuum by convection, causing a decrease in the average energy of the canister and the molecules.

11. An electron moves with speed *v*. If the de Broglie wavelength of the electron doubles, then the new speed of the electron is

(A) *v*/4

(B) *v*/2

(C) *v*

(D) 2*v*

(E) 4*v*

12. Each of the following atoms is stripped of all orbital electrons and they are all placed in the same electric field. Which ionized atom in the field is acted on by the strongest electrostatic force?

(A) ${}^{12}_{6}C$

(B) ${}^{12}_{7}N$

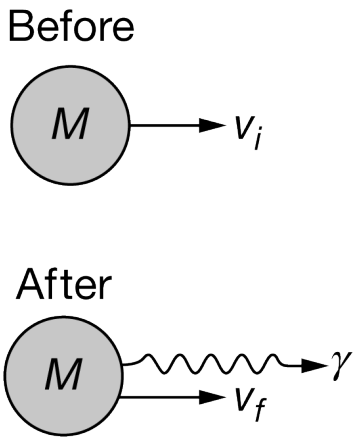
(C) ${}^{16}_{8}O$

(D) ${}^{19}_{8}O$

(E) ${}^{17}_{9}F$

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13.



A thorium nucleus of mass $M = 3.85 \times 10^{-25}$ kg is traveling in the $+x$-direction with $v_i = 600$ m/s. The thorium nucleus then undergoes gamma decay, in which a gamma ray photon is emitted with wavelength $\lambda_\gamma = 0.004$ nm, in the $+x$-direction as shown in the figure. Which of the following gives the best estimate for the final velocity v_f of the thorium nucleus after the photon is emitted?

(A) 170 m/s

(B) 430 m/s

(C) 600 m/s

(D) 1030 m/s

14. Which of the following situations would result in a diffraction pattern with the greatest spacing between bright fringes?

(A) A beam of high-velocity electrons, each with de Broglie wavelength λ, incident on an opening of width λ

(B) A beam of high-velocity electrons, each with de Broglie wavelength λ, incident on an opening of width 10λ

(C) A beam of high-velocity electrons, each with de Broglie wavelength 10λ, incident on an opening of width λ

(D) A beam of high-velocity electrons, each with de Broglie wavelength 10λ, incident on an opening of width 10λ

15. An electron beam interacts with a crystal lattice, and a diffraction pattern is observed using a detector. Which of the following changes will cause the spacing between the observed diffraction maxima to increase?

(A) Use electrons with a smaller speed.

(B) Use an electron beam that has a greater intensity but the same electron speed.

(C) Move the detector closer to the crystal.

(D) Use a crystal with a greater lattice spacing.

16. A beam consisting of electrons with energy E is incident on a single slit X. The diffraction pattern is observed. Slit X is then replaced by slit Y, and the new diffraction pattern is observed to have fringes that are farther apart than those in the first pattern. Which of the following is a possible explanation for why the spacings are different?

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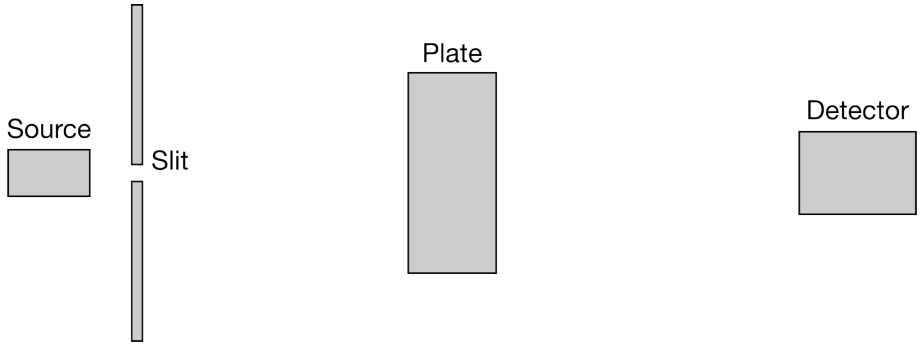
(A) Slit X is narrower than slit Y.

(B) Slit Y is narrower than slit X.

(C) The distance between the electron source and the slit is less for slit X than for slit Y.

(D) The distance between the electron source and the slit is greater for slit Y than for slit X.

17.



A beam of electrons with the same de Broglie wavelength is directed through a narrow slit whose width is about the same size as the electrons' wavelength. Electrons travel toward a detector that measures electron speed. The detector initially measures the speed of the electrons as v_0. A thick lead plate is then placed halfway between the source and detector, as shown, that absorbs all the electrons that hit it. Which of the following predictions is correct about how the new electron speed v_1 measured by the detector compares with v_0?

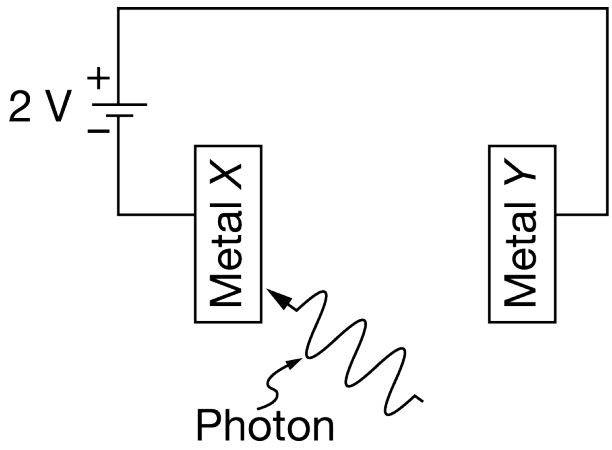
(A) $v_1 < v_0$. Some electrons diffracted by the slit will also be diffracted by the plate, and they will lose energy due to their interaction with the plate.

(B) $v_1 > v_0$. Some electrons diffracted by the slit will also be diffracted by the plate, and only the fastest electrons will arrive at the detector.

(C) $v_1 = v_0$. Some electrons diffracted by the slit will also diffract around the plate, without affecting the electron speed.

(D) $v_1 = 0$, because the plate will prevent all the electrons from reaching the detector.

18.



A photon with energy $4\,\mathrm{eV}$ is incident on metal X, which has a work function of $3\,\mathrm{eV}$. Metal X is electrically connected to metal Y through a $2\,\mathrm{V}$ voltage supply with the polarity shown in the figure above. What is the maximum kinetic energy an emitted electron could have when it reaches metal Y ?

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(A) $1\,\text{eV}$
(B) $2\,\text{eV}$
(C) $3\,\text{eV}$
(D) $5\,\text{eV}$

19. For the hydrogen atom, how does the potential energy U of the atom change as the energy level n increases?

(A) The potential energy always increases.
(B) The potential energy always decreases.
(C) The potential energy increases, then decreases.
(D) The potential energy decreases, then increases.
(E) The potential energy does not change.

20. Which of the following identifies the force that keeps protons from repelling each other in a stable nucleus?

(A) Gravitational force
(B) Strong force
(C) Magnetic force
(D) Electric force

21. In each of the nuclear reactions given below, a product is indicated by the letter X. In which of the reactions does the letter X represent a positron? (A positron is a positively charged particle with the same mass and magnitude of charge as an electron.)

(A) ${}^{14}_{7}\text{N} + {}^{1}_{1}\text{H} \rightarrow {}^{15}_{8}\text{O} + \text{X}$

(B) ${}^{15}_{8}\text{O} \rightarrow {}^{15}_{7}\text{N} + \text{X} + \nu$

(C) ${}^{15}_{7}\text{N} + {}^{1}_{1}\text{H} \rightarrow {}^{12}_{6}\text{C} + \text{X}$

(D) ${}^{12}_{6}\text{C} + \text{X} \rightarrow {}^{13}_{7}\text{N} + \gamma$

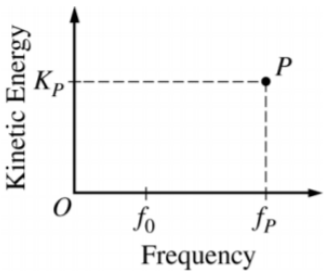
22. ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + X + \gamma$

Which of the following predictions is correct about the identity of product X that is created by the nuclear process shown?

(A) An α particle
(B) A β^{+} particle
(C) A β^{-} particle
(D) A neutron

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23.



When light of frequency f_P is incident on a metal with threshold frequency f_0, electrons are ejected with maximum kinetic energy *KP* . This is represented in the graph above by point *P*. How would the location of point *P* on the graph change if the intensity of the same light source were increased?

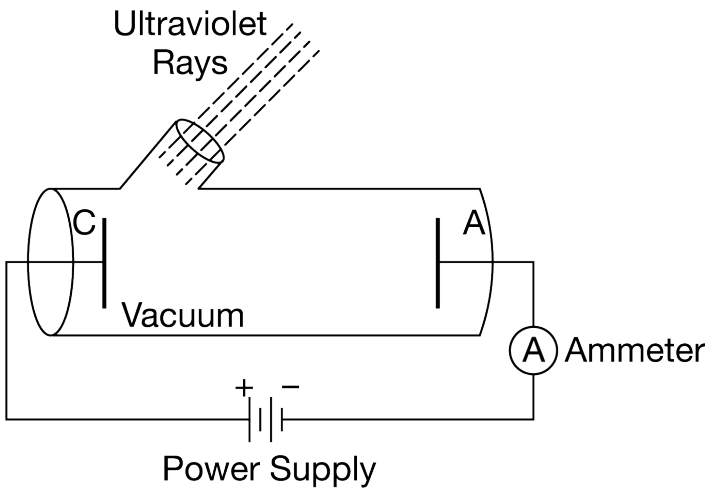
(A) *P* would not move.

(B) *P* would move vertically up.

(C) *P* would move horizontally to the right.

(D) *P* would move diagonally up and to the right

24.



A student prepares the setup shown in the figure and then shines low-intensity ultraviolet light on metal plate C. When no potential difference is applied to the circuit, the ammeter still registers electric current. How will the ammeter reading be affected if the potential difference is gradually increased?

(A) The current will increase as $V_{power\ supply}$ is increased.

(B) The current will decrease and will stay zero after a certain $V_{power\ supply}$ value.

(C) The current will decrease and reach zero, and as $V_{power\ supply}$ is further increased the current will increase again.

(D) The current will stay the same regardless of how much $V_{power\ supply}$ is applied.

The energy levels of the Bohr model for the atom can be expressed mathematically as $E_n = -13.6\frac{Z^2}{n^2}$ eV, where Z is the atomic number, and n is the quantum number. This model is reasonably accurate for hydrogen and for singly ionized helium.

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25. Which of the following describes the change in kinetic energy K of the electron and the change in potential energy U of the singly ionized helium atom when the electron transitions from the ground state to the first excited state?

(A)

K	U
Increases	Increases

(B)

K	U
Increases	Decreases

(C)

K	U
Decreases	Increases

(D)

K	U
Decreases	Decreases

---

26. Which of the following properties of atoms helps explain the existence of discrete energy levels of atoms with one electron?

(A) Electrons orbit the nucleus in elliptical orbits.

(B) An atomic nucleus exerts an attractive force on the electron.

(C) Electrons are fundamental particles, while nucleons are made up of quarks.

(D) The radius of the electron's orbit is an integer multiple of the electron's de Broglie wavelength.

27. $^{214}_{83}Bi \rightarrow X + \beta^{+}$

In the decay represented above, what is the nucleon number of particle X?

(A) 82

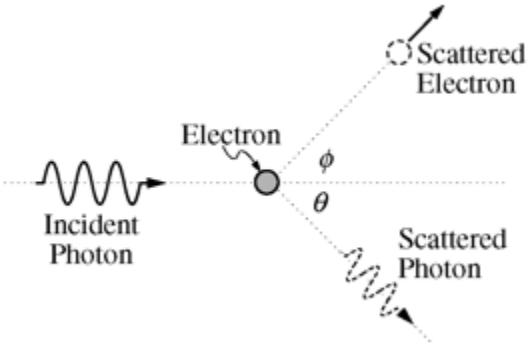
(B) 83

(C) 213

(D) 214

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28.



A photon scatters from an electron that is initially at rest, as shown above. Which of the following is true of the wavelength and frequency of the scattered photon compared to those of the incident photon?

(A)

Wavelength	Frequency
Greater	Less

(B)

Wavelength	Frequency
Greater	The same

(C)

Wavelength	Frequency
The same	Less

(D)

Wavelength	Frequency
Less	Greater

(E)

Wavelength	Frequency
Less	The same

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An atom of polonium (Po-216) is moving slowly enough that it can be considered to be at rest. The Po-216 undergoes alpha decay and becomes lead (Pb-212), via the reaction ${}^{216}_{84}Po \rightarrow {}^{212}_{82}Pb + {}^{4}_{2}\alpha$. After the decay, the lead atom is moving to the left with speed v_{Pb} , and the alpha particle is moving to the right with speed v_a. The masses of the three isotopes involved in the decay are given below.

$$M_{\text{Po-216}} = 216.001915 \text{ u}$$

$$M_{\alpha} = 4.002603 \text{ u}$$

$$M_{\text{Pb-212}} = 211.991898 \text{ u}$$

29. The half-life of Po-216 is 1/7 s. What is the probability that any particular Po-216 atom will decay within one second?

(A) 0 %
(B) 7 %
(C) 99 %
(D) 100 %

30. How much energy is released in the decay?

(A) None
(B) 0.00741 MeV
(C) 6.90 MeV
(D) 9.31 MeV

---

31. A radioactive nucleus initially has *Z* protons and *N* neutrons. It emits a beta minus particle and then a gamma-ray photon. How many protons and neutrons does the remaining nucleus contain?

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(A)

Protons	Neutrons
Z-2	*N*-2

(B)

Protons	Neutrons
Z-1	*N*

(C)

Protons	Neutrons
Z-1	*N*+1

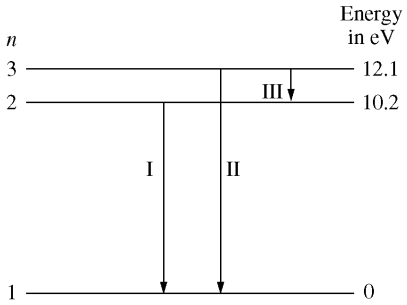
(D)

Protons	Neutrons
Z	N-2

(E)

Protons	Neutrons
Z+1	*N*-1

32.



The figure above shows a portion of the energy level diagram for hydrogen. Possible transitions among the energy levels are represented by arrows I, II, and III. Which of the following lists the transitions in order of their associated wavelengths of light, from shortest wavelength to longest wavelength?

(A) I, II, III

(B) I, III, II

(C) II, I, III

(D) II, III, I

(E) III, I, II

33. Which of the following is true about the momentum of photons?

(A) Photons have no rest mass, so they have no momentum.

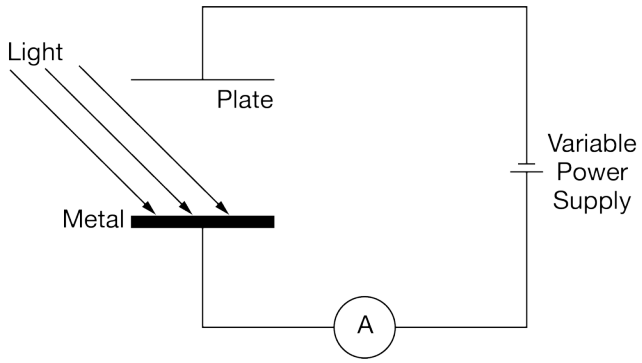
(B) Photons all travel at the speed of light, so they all have the same momentum.

(C) The momentum of a photon is directly proportional to its energy.

(D) The momentum of a photon is directly proportional to its wavelength.

(E) The momentum of a photon is inversely proportional to the square root of its frequency.

34.



The figure shows a sketch of a photoelectric-effect experiment. Light incident on a metal ejects electrons from the metal's surface with a range of energies. The electrons hit the plate and a current is created in the circuit. The variable power supply is used to create a potential difference between the metal and the plate. There is a potential difference for which no electrons reach the plate and the current goes to zero. Which of the following justifies the use of this potential difference to determine the maximum kinetic energy of the electrons?

(A) The potential difference changes the minimum amount of energy needed to release an electron from the material.

(B) The potential difference changes the amount of energy imparted to the electrons by the incident light.

(C) The gain in the electrons' potential energy caused by the potential difference equals their loss in kinetic energy.

(D) The gain in the electrons' potential energy caused by the potential difference equals the energy of the photons of incident light.

35. $^{238}_{92}\mathrm{U}$ undergoes a series of reactions in which it emits eight $^{4}_{2}He$ nuclei and six electrons. What is the isotope that results from this series of reactions?

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(A) $^{222}_{66}\mathrm{Dy}$

(B) $^{206}_{81}\mathrm{Tl}$

(C) $^{206}_{82}\mathrm{Pb}$

(D) $^{207}_{82}\mathrm{Pb}$

(E) $^{214}_{88}\mathrm{Ra}$

36. When a californium (Cf) nucleus undergoes spontaneous fission, the following two reactions occur, where X and Y are unidentified nuclei.

$^{252}_{98}\mathrm{Cf} \rightarrow \mathrm{X} + {}^{108}_{44}\mathrm{Ru} + 4{}^{1}_{0}\mathrm{n}$
$\mathrm{X} \rightarrow \mathrm{Y} + {}^{0}_{-1}\beta$

Which nucleus in the pair of equations has the greatest neutron-to-proton ratio?

(A) Cf

(B) Y

(C) Ru

(D) X

37. Electrons are accelerated through a potential difference ΔV and are then incident on a crystal. The resulting diffraction pattern is observed on a screen. If the potential difference is then decreased, the spacing between maxima of the diffraction pattern increases. Does this increase in spacing support the wave model of matter, and why or why not?

(A) Yes, because the lower ΔV would yield a lower electron momentum, which would yield a shorter de Broglie wavelength, which would increase the spacing between maxima.

(B) Yes, because the lower ΔV would yield a lower electron momentum, which would yield a longer de Broglie wavelength, which would increase the spacing between maxima.

(C) No, because the formation of a diffraction pattern is a particle phenomenon.

(D) No, because if the electrons could be described as waves they could not interact with the crystal, which consists of particles.

38. A mid-sized state uses about $1.8 \times 10^{18}\ \mathrm{J}$ of energy per year. Approximately how long could the state be powered by the conversion of $50\ kg$ of mass to energy?

(A) 2.5 years

(B) 20 years

(C) 1000 years

(D) 10^8 years

39. A nuclear power plant uses nuclear fission to convert mass into heat energy, which is then converted into electrical energy. Estimate the order of magnitude of the amount of mass converted into energy in one year in a typical nuclear power plant generating an average output electrical power of $300\ \mathrm{MW}$. Assume that 35% of the energy obtained from the fission is converted to electric energy.

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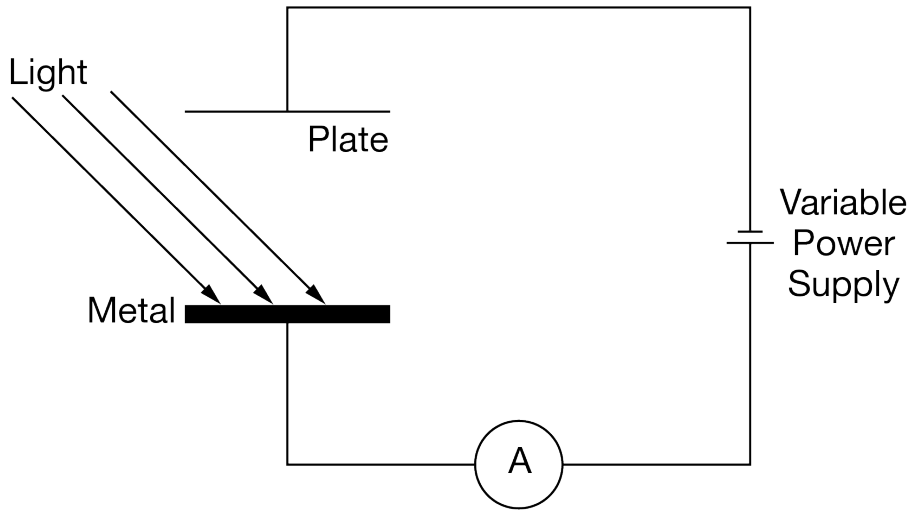
(A) $10^{-1}\,\text{kg}$

(B) $10^{5}\,\text{kg}$

(C) $10^{8}\,\text{kg}$

(D) $10^{16}\,\text{kg}$

40.



In an experiment to investigate the photoelectric effect, monochromatic light is shined on a metal plate. Electrons emitted from the metal move across a vacuum to a collector plate, as shown in the diagram. As the potential difference of the power supply increases, the number of electrons detected by the ammeter decreases. The potential difference is increased until it reaches V_s, at which no electrons reach the collector plate. When the potential difference is set to V_s , which of the following claims is correct about the emitted electrons?

(A) The electric field between the two plates exerts a force on the emitted electrons, so they remain suspended between the two plates.

(B) Electrons that were already emitted now accumulate on the collector plate, repelling the newly emitted electrons, so no new electrons are collected.

(C) The electric field between the two plates does negative work on the emitted electrons, so they never reach the collector plate.

(D) The potential at the metal plate overwhelms the kinetic energy of the emitted electrons, so no electrons are emitted.

41. Quantities that are conserved in all nuclear actions include which of the following?

I. Electric charge

II. Number of nuclei

III. Number of protons

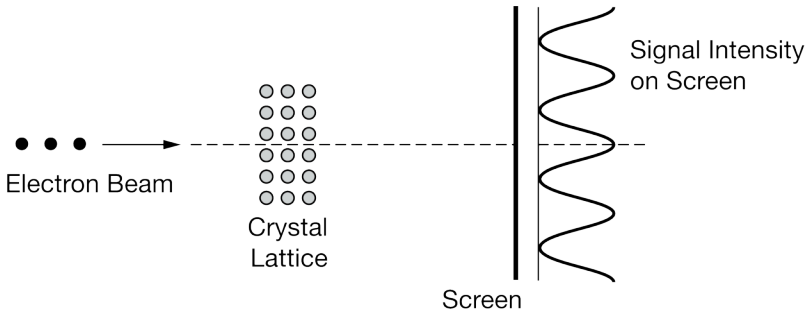
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(A) I only
(B) II only
(C) I and III only
(D) II and III only
(E) I, II, and III

42. Radioactive element X has a half-life of approximately 50 days. Radioactive element Y has a half-life of approximately 100 days. A 0.8 g sample of element X and a 0.8 g sample of element Y are weighed and examined. Some time later, both samples are measured again and 0.2 g of element X remains. How much of element Y remains?

(A) $0.1\,\text{g}$
(B) $0.2\,\text{g}$
(C) $0.4\,\text{g}$
(D) $0.8\,\text{g}$

43.



An electron beam is projected at a crystal, as shown in the diagram. The signal on the other side of the crystal is recorded on a screen. A student claims this is evidence that the electrons can exhibit wave properties. Which of the following statements justifies the claim?

(A) A signal from the electrons passes through the crystal.
(B) The pattern of signal intensity on the screen has bright and dark areas.
(C) The pattern of the signal intensity on the screen is similar to the interference pattern of electromagnetic radiation passing through narrow slits.
(D) The spread of the pattern of the signal intensity on the screen is different for crystals of different interatomic spacing.

$2{}^{3}_{2}\text{He} \rightarrow 2{}^{1}_{1}\text{H} + {}^{4}_{2}\text{He}$

A scientist researching nuclear fusion is studying the reaction above in which two helium-3 nuclei react to form two hydrogen nuclei and a helium-4 nucleus. The scientist determines that the total mass of the two helium-3 nuclei is 1.00156×10^{-26} kg and that the total mass of the two hydrogen nuclei and the helium-4 nucleus is 9.9596×10^{-27} kg.

44. The scientist notes that this reaction involves a high-energy gamma ray photon. In what way is the photon associated with this reaction?

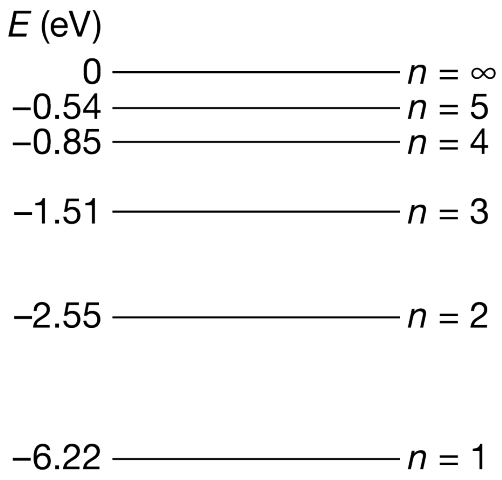
(A) An electron transitions to a higher allowed energy state and absorbs a photon of this energy difference.

(B) An electron transitions to a lower allowed energy state and emits a photon of this energy difference.

(C) A nucleus transitions to a higher energy state and absorbs a photon of this energy difference.

(D) A nucleus transitions to a lower energy state and emits a photon of this energy difference.

45.



The figure shows some of the energy levels for a hypothetical atom. A photon of wavelength $487\ \text{nm}$ has an energy of $2.55\ \text{eV}$. Which of the following correctly indicates whether the absorption spectrum of a collection of these atoms will have a dark line at $487\ \text{nm}$ and describes a feature of the model that indicates this? (The absorption spectrum does not include any process that ionizes the atom.)

(A) Yes. The atom has energy levels that have magnitudes less than $2.55\ \text{eV}$.

(B) Yes. The atom has an energy level that is exactly the same energy as a $487\ \text{nm}$ photon.

(C) No. The atom does not have a transition between levels with an energy difference equal to the energy of a $487\ \text{nm}$ photon.

(D) No. The atom has only one level that is an integer multiple of $2.55\ \text{eV}$.

46. $^{263}_{106}\text{Sg} \rightarrow {}^{259}_{104}\text{Rf} + {}^{4}_{2}\text{He}$

Consider the decay of Seaborgium-263 $(\text{mass} = 263.118322\ \text{u})$ that forms Rutherfordium-259 $(\text{mass} = 259.105637\ \text{u})$ and an alpha particle (a helium nucleus, $\text{mass} = 4.001506\ \text{u}$), as shown in the reaction. How much energy is released in the reaction?

(A) $10.41\ \text{MeV}$

(B) $3725.40\ \text{MeV}$

(C) $237,501.95\ \text{MeV}$

(D) $241,237.76\ \text{MeV}$

47. If photons of light of frequency *f* have momentum *p,* photons of light of frequency 2*f* will have a momentum of

(A) $2p$

(B) $\sqrt{2}p$

(C) p

(D) $\frac{p}{\sqrt{2}}$

(E) $\frac{1}{2}p$

48. A beam of electrons, all with the same speed, is incident on a crystal whose atoms are at a fixed spacing. The electrons then strike a detection screen, resulting in a diffraction pattern with alternating areas of minimum/ maximum intensity similar to a double slit interference pattern. If the potential difference used to accelerate the electrons is increased, which of the following predictions is correct about the spacing between the maxima change?

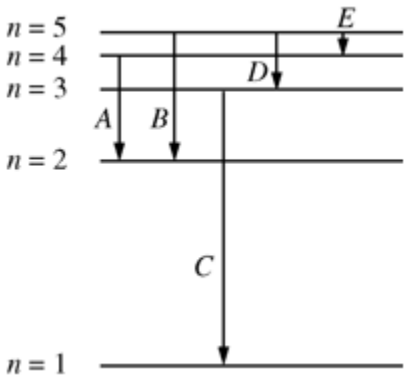
(A) The spacing decreases because as potential difference increases, so do the kinetic energy and momentum of the electrons, resulting in a longer wavelength.

(B) The spacing decreases because as potential difference increases, so do the kinetic energy and momentum of the electrons resulting in a shorter wavelength.

(C) The spacing increases because as potential difference increases, so do the kinetic energy and momentum of the electrons, resulting in a longer wavelength.

(D) The spacing increases because as potential difference increases, so do the kinetic energy and momentum of the electrons resulting in a shorter wavelength.

49.



The figure above shows part of the energy level diagram for a hypothetical atom. Which of the transitions indicated produces the most energetic photons?

(A) *A*

(B) *B*

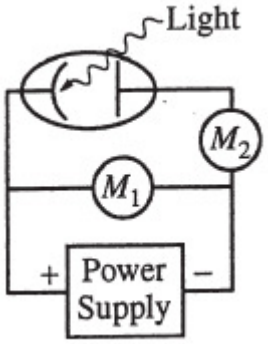
(C) *C*

(D) *D*

(E) *E*

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50.



A student performs a photoelectric effect experiment in which light of various frequencies is incident on a photosensitive metal plate. This plate, a second metal plate, and a power supply are connected in a circuit, which also contains two meters, M_1 and M_2, as shown above.

The student shines light of a specific wavelength λ on the plate. The voltage on the power supply is then adjusted until there is no more current in the circuit, and this voltage is recorded as the stopping potential V_s.

The student then repeats the experiment several more times with different wavelengths of light. The data, along with other values calculated from it, are recorded in the table below.

λ (m)	4.00×10^{-7}	4.25×10^{-7}	4.50×10^{-7}	4.75×10^{-7}
V_S (volts)	0.65	0.45	0.30	0.15
f (Hz)	7.50×10^{14}	7.06×10^{14}	6.67×10^{14}	6.32×10^{14}
K_{max} (eV)	0.65	0.45	0.30	0.15

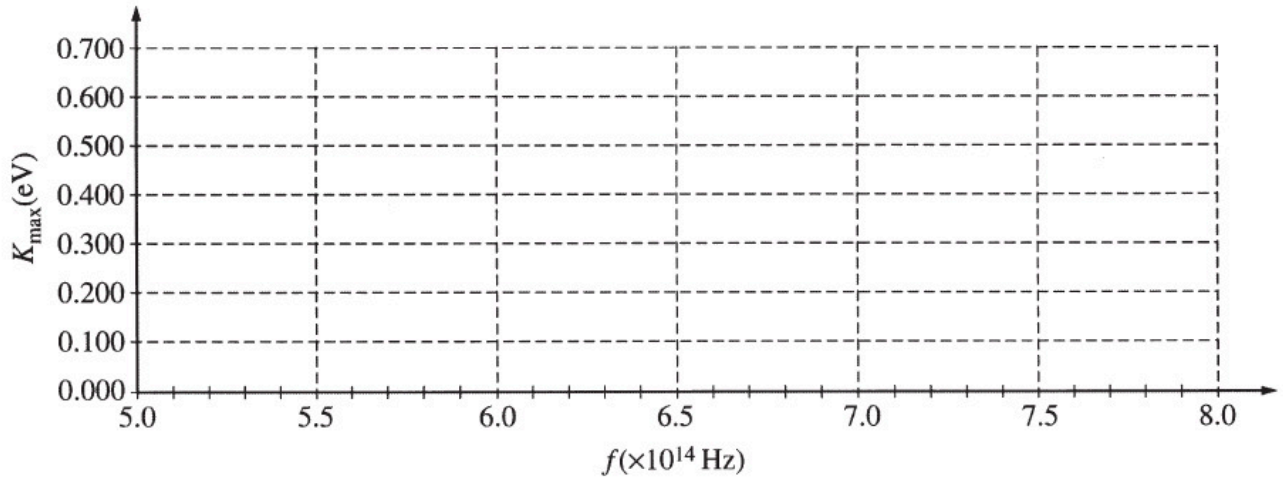
(a) Indicate which meter is used as an ammeter and which meter is used a voltmeter by checking the appropriate spaces below.

	M1	*M2*
Ammeter	___	___
Voltmeter	___	___

(b) Use the data above to plot a graph of K_{max} versus f on the axes below, and sketch a best-fit line through the data.

(c) Use the best-fit line you sketched in part (b) to calculate an experimental value for Planck's constant.

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(d) If the student had used a different metal with a lager work function, how would the graph you sketched in part (b) be different? Explain your reasoning.

51. Show all your work for each part of the question. The parts within the question may not have equal weight.

Monochromatic light of wavelength λ_0 is incident on a metal with work function ϕ. Electrons are ejected from the metal and have a maximum speed v_0.

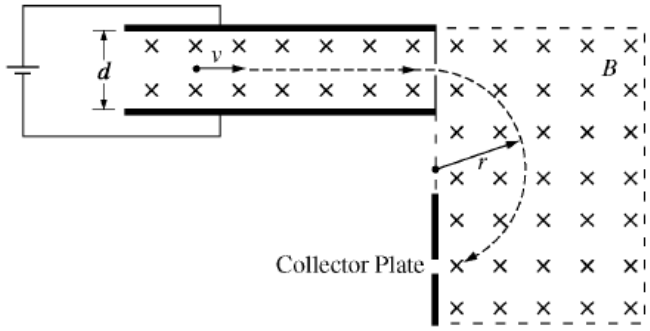
(a) When monochromatic light with wavelength greater than λ_0 is incident on the metal, electrons are still ejected. **Indicate** whether the maximum speed of the ejected electrons will be less than, equal to, or greater than v_0. **Justify** your response.

(b) **Derive** an expression for the maximum speed v_0 of the electrons when the original light of wavelength λ_0 is incident on the metal in terms of λ_0, ϕ, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference booklet.

(c) **Indicate** whether the equation derived in part (b) agrees with the justification in part (a). Briefly **justify** your reasoning.

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52.



A particle with unknown mass and charge is projected into the apparatus shown above. The particle moves with constant speed v as it passes undeflected through a pair of parallel plates, as shown above. The plates are separated by a distance *d*, and a constant potential difference *V* is maintained between them. A uniform magnetic field of magnitude *B* directed into the page exists both in the region between the plates and in the region to the right of the plates that is enclosed by the dashed lines. In the region to the right of the plates, the particle's path is circular with radius *r*. Assume the effects of gravity are negligible compared to other forces.

(a) Explain why the particle moves through the parallel plates undeflected in terms of the forces exerted on the particle.

(b) What is the sign of the charge on the particle? Justify your answer.

A magnetic field of 0.30 T is applied with the plate separation at 5.0×10^{-3}m . Singly ionized particles with various speeds enter the region between the plates, and only those with speed 2.0×10^{6} m/s are undeflected as they pass between the plates. These particles then reach the collector plate a distance of 0.42 m below the point at which they left the region between the parallel plates.

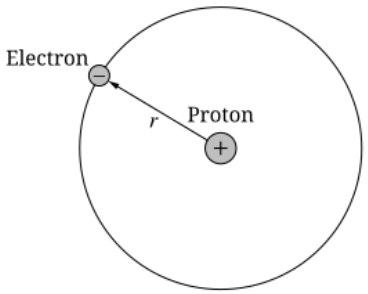
(c) Based on your explanation in part (a), derive an algebraic expression for the potential difference that must be applied to produce the motion of the undeflected particles. Use that expression to calculate the numerical value of the potential difference.

(d) By analyzing the circular part of the motion, derive an algebraic expression for the mass of the particles. Use that expression to calculate a numerical value for the mass.

(e) A scientist wants to use the apparatus to separate singly ionized atoms of ^{12}C and ^{14}C in order to use the ^{14}C in radiocarbon dating. Describe how the motion of the two isotopes of carbon in both regions of the apparatus leads to their separation, appropriately relating your description to the algebraic equations you wrote in parts (c) and (d).

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53. This question is a long free-response question. Show your work for each part of the question.



Note: Figure not drawn to scale.

(12 points, suggested time 25 minutes)

A hydrogen atom can be modeled as an electron in a circular orbit of radius r about a stationary proton, as shown above. The gravitational force between the proton and electron is negligible compared to the electrostatic force between them.

(a) Derive an equation for the speed v of the electron in terms of r and physical constants, as appropriate.

(b) Derive an equation for the total energy of the atom in terms of r and physical constants, as appropriate.

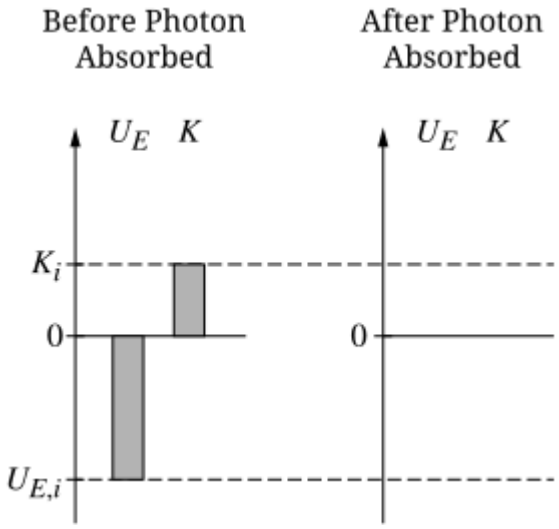
(c) When the hydrogen atom absorbs a photon, the electron moves to an orbit with a larger radius and the total energy of the atom increases. Is your equation for the energy derived in part (b) consistent with this description of the model of a hydrogen atom absorbing a photon? Explain why the equation is or is not consistent.

(d) Experiments show that a hydrogen atom can absorb a photon of frequency 3.2×10^{15} Hz.

i. Calculate the energy of a photon with this frequency.

ii. A student claims that when a hydrogen atom absorbs a photon at this frequency, the energy could be converted into mass, adding an electron to the atom. Calculate the amount of energy needed to create a particle with the mass of an electron and determine whether or not there is sufficient energy gained by the atom to add another electron.

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iii. The left bar chart in the figure above is complete and represents the initial electric potential energy $U_{E,i}$ of the atom and the initial kinetic energy K_i of the electron before the photon is absorbed. In the space provided on the right, draw a bar chart to represent a possible final electric potential energy of the atom and final kinetic energy of the electron.

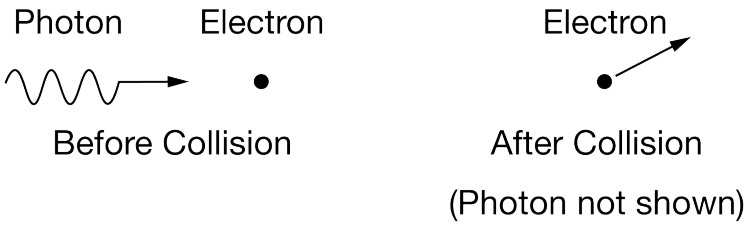
54. This question is a short free-response question. Show your work for each part of the question.

(10 points, suggested time 20 minutes)

Light and matter can be modeled as waves or as particles. Some phenomena can be explained using the wave model, and others can be explained using the particle model.

(a) Calculate the speed, in $\mathrm{m\ s}$, of an electron that has a wavelength of $5.0\ \mathrm{nm}$.

(b) The electron is moving with the speed calculated in part (a) when it collides with a positron that is at rest. A positron is a particle identical to an electron except that its charge is positive. The two particles annihilate each other, producing photons. Calculate the total energy of the photons.



(c) A photon approaches an electron at rest, as shown above on the left, and collides elastically with the electron. After the collision, the electron moves toward the top of the page and to the right, as shown above on the right, at a known speed and angle. In a coherent, paragraph-length response, indicate a possible direction for the photon that exists after the collision and its frequency compared to that of the original photon. Describe the application of physics principles that can be used to determine the direction of motion and frequency of the photon that exists after the collision.

55. A monochromatic source emits a 2.5 mW beam of light of wavelength 450 nm.

(a) Calculate the energy of a photon in the beam.

(b) Calculate the number of photons emitted by the source in 5 minutes. The beam is incident on the surface of a metal in a photoelectric-effect experiment. The stopping potential for the emitted electron is measured to be 0.86 V.

(c) Calculate the maximum speed of the emitted electrons.

(d) Calculate the de Broglie wavelength of the most energetic electrons.